GEK-40401

Design Manual

PM1000* CONTROL

Featuring FLO-THRU* Logic For Plastic Processing Machines

IC 7966

*Trademark of General Electric Company

GENERAL ELECTRIC COMPANY
INDUSTRIAL CONTROL PRODUCTS DEPARTMENT
SALEM, VA. 24153

GEK-40401

Design Manual

PM1000* CONTROL

Featuring FLO-THRU* Logic For Plastic Processing Machines

IC 7966

The information contained herein is intended to assist users and dealers in the servicing of control furnished by the General Electric Company. It does not purport to cover all details or variations in equipment or provide for every possible contingency to be met in connection with installation, operation or maintenance.

Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purpose, the matter should be referred to the machinery manufacturer through his normal service channels.

Information contained in these instructions is Proprietary Information of the General Electric Company. The dissemination or use of this information for any other purpose than for the installation, operation and/or maintenance of the General Electric Company equipment for which these instructions are provided is prohibited except by express permission of the General Electric Company.

GENERAL ELECTRIC

GEK-40401

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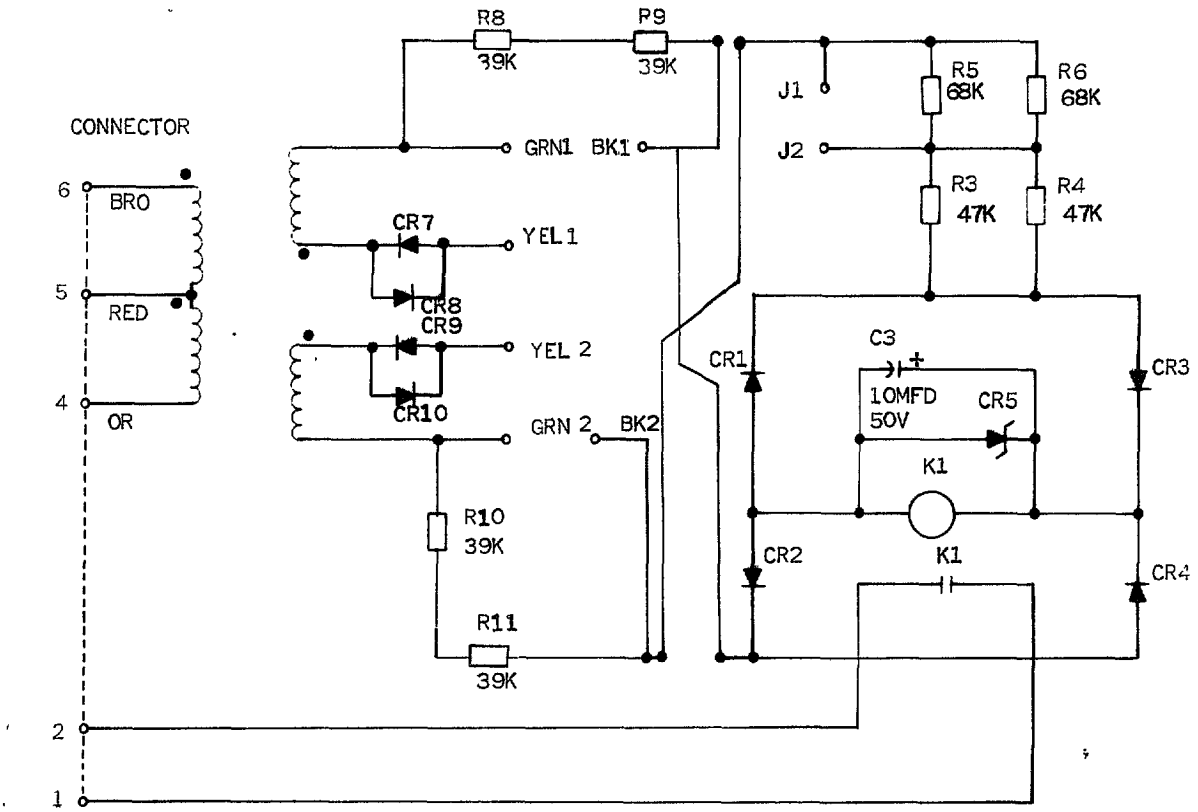
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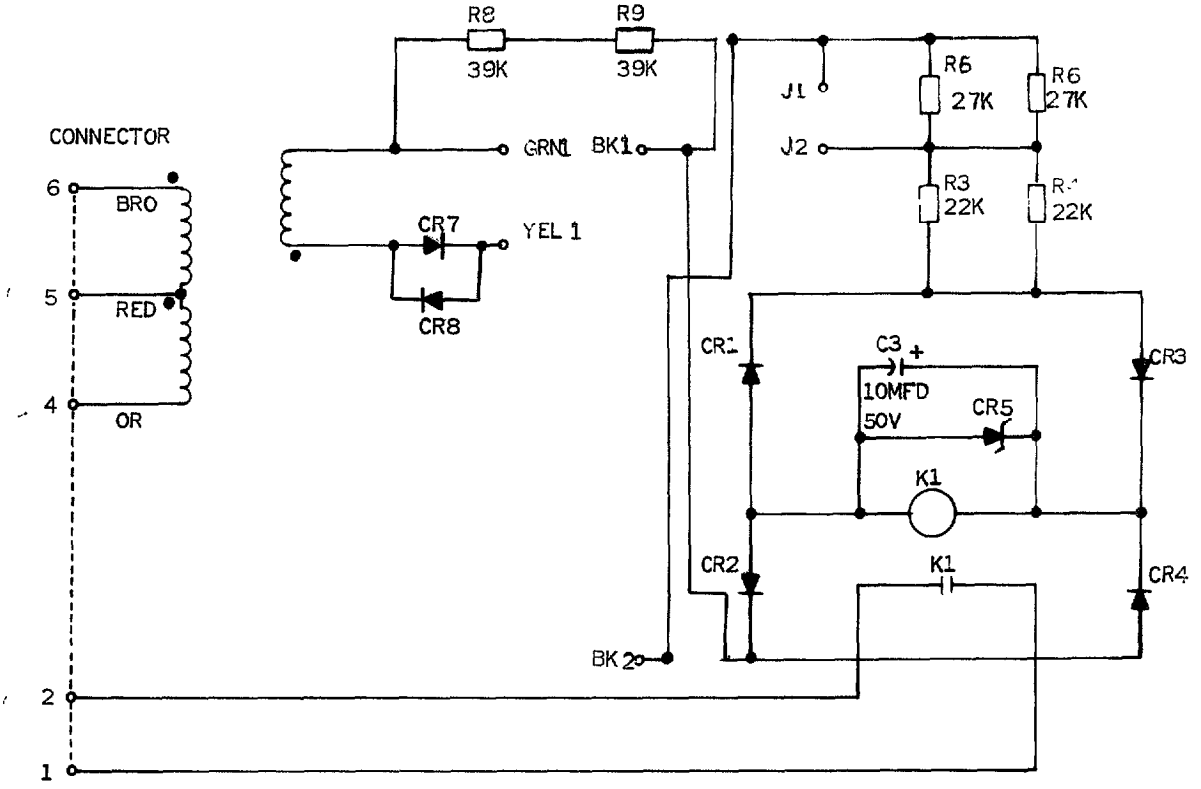
FORM IC 2390 (11-67)



THIS CARD IS USED AS A PART OF HEATER DRIVER IC3625D (UNIVERSAL DRIVERS).

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

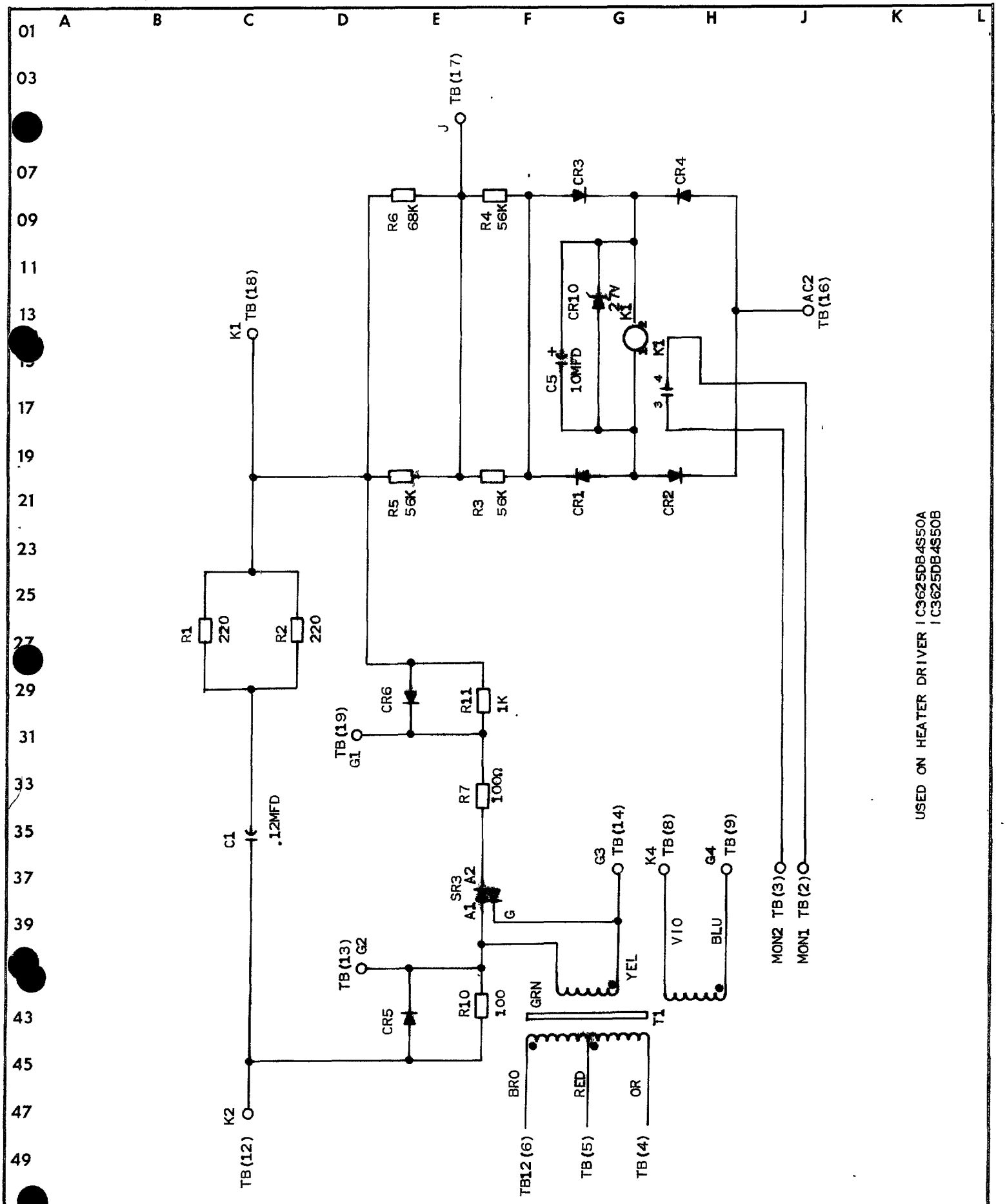
REV 1 REVISED TITLE 10/15/70 EAH	REV 4	REV 5	PRINTS TO DL58, DL1	APPROVALS 6/N/70	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV 2A WRIGHT JULY 16, 71	ISSUED 6-10-70	2530 CHVI VN	FIRST MADE FOR PLASTI-MET		ELEMENTARY DIAGRAM HEAT DRIVER AUXILIARY (UNIVERSAL)
REV 3 WMH 5-8-72	MADE BY JL SCOTT		IC NO		IC3621GHDA1
					CONT. ON SH FL. SH NO. 3



THIS CARD IS USED AS A PART OF HEATER DRIVER IC3625D (SINGLE DRIVERS).

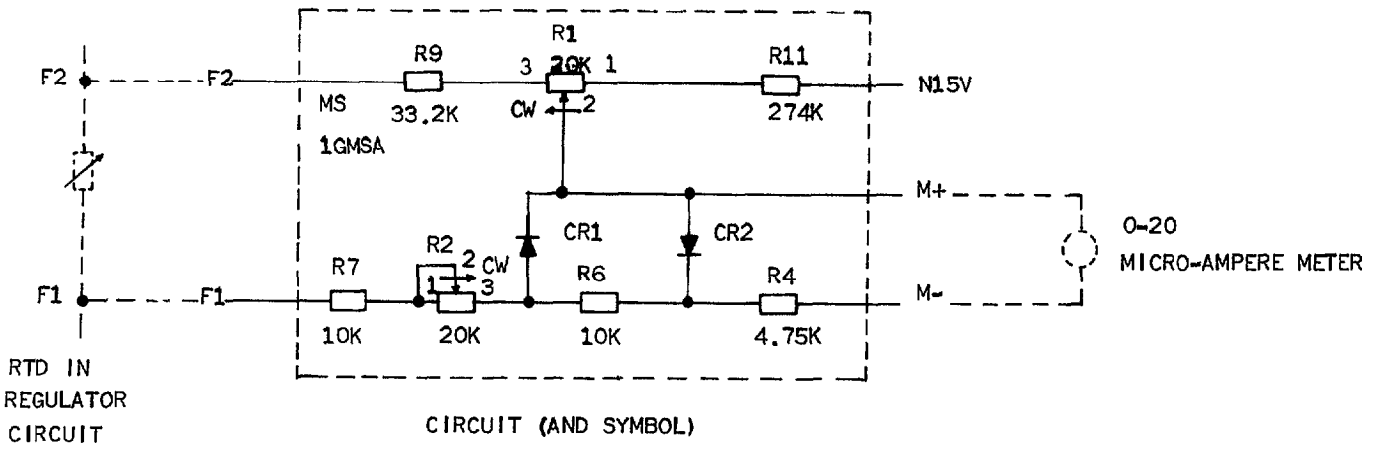
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV 1 REVISED PAGE 10/19/70 B.H.	REV 4	REV 5	PRINTS TO DL1	APPROVED 7/19/71	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV 2 A WRIGHT JULY 16-71	ISSUED 6-10-70	6V P6A	FIRST MADE FOR PLASTA-MATIC		ELEMENTARY DIAGRAM HEATER DRIVER AUXILIARY (SINGLE)
REV 3 WRIGHT 3-5-72	MADE BY JL SCOTT	1338	I C NO		I C 3 6 2 1 G H D B 1 CONT. ON SH. FL. SH. NO. 3



PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV 1	REV. 4	REV 5	PRINTS TO DL1, P6A	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM 50 AMP HEATER DRIVER
REV. 2	ISSUED Feb. 1, 71. 1338			FIRST MADE FOR PLASTIC MOLD		IC3621GHDC1
REV 3	MADE BY RON MUCKENFUSS A1F(4)			I.C NO.		CONT. ON SH. FL. SH. NO. 3.1



CIRCUIT (AND SYMBOL)

FUNCTION AND THEORY OF OPERATION

THE PLASTIC MOLDING REGULATOR, IC3625RK, PROVIDES A CURRENT THROUGH ITS ASSOCIATED FEEDBACK RESISTOR (RTD). THE RESISTANCE OF THE RTD VARIES DIRECTLY WITH ITS TEMPERATURE, SO THERE IS A VOLTAGE ACROSS THE RTD WHICH INCREASES WITH TEMPERATURE. THIS CIRCUIT CALIBRATES A METER TO READ THE TEMPERATURE OF THE RTD FEEDBACK AS A FUNCTION OF THE VOLTAGE DEVELOPED ACROSS IT IN THE REGULATOR CIRCUIT. POTENTIOMETER R1 SETS THE METER TO NULL BALANCE WHEN THE RTD IS AT A TEMPERATURE OF 250 DEGREES FAHRENHEIT. RHEOSTAT R2 THEN CALIBRATES THE METER SENSITIVITY SO THE METER READS FULL SCALE WHEN THE RTD IS AT 800 DEGREES FAHRENHEIT.

THE REGULATOR RANGE IS 250 TO 800 DEGREES AND THE READOUT IS DESIGNED COMPATIBLE TO THE REGULATOR RANGE. IF READING BELOW REGULATOR RANGE IS DESIRED, IT CAN BE SECURED BY SWITCHING METER POLARITY. CR1 AND CR2 PROTECT THE METER SUCH THAT METER WILL BE SUBJECT TO LESS THAN TEN TIMES FULL SCALE CURRENT FOR 15V APPLIED EITHER DIRECTION) BETWEEN F1 AND F2.

INPUT TERMINALS (F1 AND F2). CONNECT ACROSS RTD FEEDBACK IN REGULATOR CIRCUIT.

SUPPLY - (N15V) SHOULD BE SECURED FROM N15 CONNECTION TO REGULATOR CIRCUIT, ELIMINATING DISTRIBUTION EFFECTS. SUPPLY CURRENT IN EACH CIRCUIT IS APPROXIMATELY 50 MICROAMPERES SO RUNNING ONE SUPPLY LEAD FOR SEVERAL METERS IS NO PROBLEM; HOWEVER, SUPPLY LEAD FROM REGULATOR SHOULD ONLY BE USED FOR SCALING CIRCUIT.

METER IS 20 MICROAMPERE FULL SCALE, 1500 TO 4000 OHMS. SCALE GRADUATION FOR REGULAR AND BELOW RANGE SCALES LISTED BELOW AS SCALES A AND B. SCALE A, IS 250 TO 800 DEGREES F(REGULAR). SCALE B IS 250 TO 0 DEGREES F (BELOW RANGE) AND APPLIES WITH METER POLARITY REVERSED. GRADUATION, IN PERCENT FULL SCALE AS SHOWN BELOW:

(SEE SHEET 3.1)

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV 1 AIM/// 2-4-71	REV 4	REV 5	PRINTS TO P6A, 1338	APPROVALS	ELEMENTARY DIAGRAM: METER SCALING CIRCUIT
REV 2	ISSUED 1-29-71 DL1, 3T A1F(4), A6F		FIRST MADE FOR PLASTIC MOLD	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	IC3621GMSA1
REV 3	MADE BY W M HURD		IC NO		CONT. ON SH. 3.1 SH. NO. 3.0

FORM 1C 2390 (08-68)

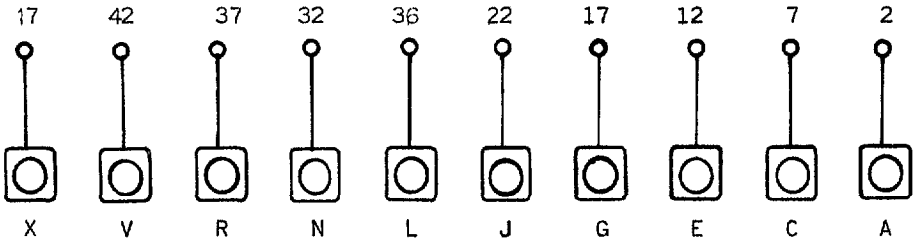
SCALE A				SCALE B	
TEMP. °F	%FS	TEMP. °F	%FS	TEMP. °F	%FS
250	0	550	55.52	250	0
300	9.42	600	64.54	200	9.48
350	18.77	650	73.50	150	19.05
400	28.06	700	82.40	100	28.68
450	37.28	750	91.23	50	38.38
500	46.43	800	100.00	0	48.15

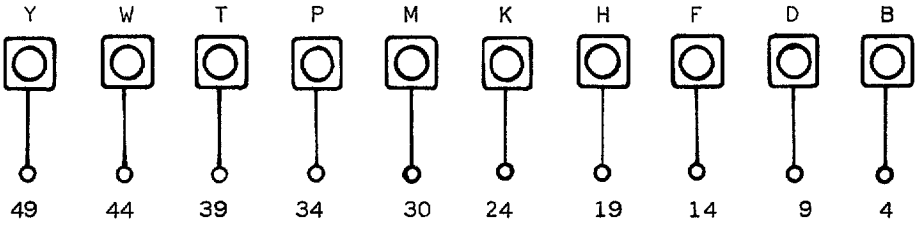
METER CALIBRATION - IS PERFORMED WITH SYSTEM CONNECTION COMPLETE, EXCEPT WITH PRECISION RESISTORS (±0.1%) SUBSTITUTED FOR THE RTD FEEDBACK SENSOR. PROCEDURE AS FOLLOWS:

1. SET REGULATOR FOR 250°F AND CONNECT 147.5 OHM RESISTOR IN PLACE OF RTD. SET R1 FOR 250°F ON METER.
2. SET REGULATOR FOR 800°F AND CONNECT 267.3 OHM RESISTOR IN PLACE OF RTD. SET R2 FOR 800°F ON METER.
3. RETEST 1 AND 2 TO VERIFY CORRECT SETTING.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV 1	REV 4	REV 5	PRINTS TO DL1, P6A	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM: METER SCALING CIRCUIT
REV 2	ISSUED 1-29-71		3T, 1338 A1F(4)	FIRST MADE FOR PLASTIC MOLD	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	I C 3 6 2 1 G M S A 1
REV. 3	MADE BY W M HURD			I C NO		CONT. ON SH FL. SH NO 3.1





THE PURPOSE OF THE UNIVERSAL COMPONENT BOARD IS TO PROVIDE THE MECHANICAL MOUNTING OF AXIAL LEAD ELECTRONIC COMPONENTS OF THE PASSIVE TYPE USED AS SELECTED DEVICES IN SYSTEM DESIGN AND IN FIELD MODIFICATIONS TO THAT DESIGN. THESE COMPONENTS ARE NORMALLY USED TO MODIFY, OR CONTROL THE PERFORMANCE OF A STANDARD CIRCUIT AS REQUIRED FOR SPECIFIC SYSTEM PERFORMANCE.

THE MECHANICAL CONFIGURATION OF THE UNIVERSAL COMPONENT BOARD IS COMPATIBLE WITH THE NORMAL MOUNTING OF STANDARD FUNCTIONAL CARDS. THEREFORE, IT IS SUITABLE FOR MOUNTING IN THE IDENTICAL MANNER.

APPLICATION DATA:

1. COMPONENT SIZE
   - A. BODY SIZE - (AXIAL LEAD COMPONENTS ONLY)
     1. LENGTH - 1.8 INCHES MAXIMUM
     2. WIDTH - .4 INCHES MAXIMUM
     3. HEIGHT - .70 INCHES MAXIMUM
   - B. LEAD LENGTH - (END TO END) 2.20 INCHES MINIMUM.
2. VOLTAGE CLEARANCE (NEMA)
   - A. ACROSS COMPONENTS
     1. UNLIMITED POWER - 50 VOLTS MAXIMUM.
     2. LIMITED POWER (1500 VA MAX) - 50 VOLTS MAXIMUM.
   - B. BETWEEN CIRCUITS AND PINS ON THE CARD
     1. UNLIMITED POWER - 50 VOLTS MAXIMUM.
     2. LIMITED POWER (1500 VA MAX) - 50 VOLTS MAXIMUM.
3. CONDUCTOR CURRENT RATING
   - A. CONNECTOR CONTACTS - 5 AMPS MAXIMUM
   - B. CONDUCTOR RUNS - 3 AMPS MAXIMUM

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 Retyped Feb. 18, 69. ACW	REV. 2 DL1	2520 1338	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM UNIVERSAL COMPONENT BOARD
RELEASED Feb 18 1969	PRINTED TO P6A		FIRST MADE FOR REQ. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	IC3600SCBA1
MADE BY			TC NO.		CONT. ON SH. 2/ SH. NO. 3.0

4. POWER LEVEL ON COMPONENTS
   A. WITHOUT SPECIAL ENGINEERING CONSIDERATION - 1 WATT MAXIMUM PER BOARD
   B. WITH SPECIAL ENGINEERING CONSIDERATION
      1. MAXIMUM WATTS PER COMPONENT- 5 WATTS
      2. MAXIMUM SURFACE TEMPERATURE OF ANY DEVICE IN CONTACT WITH THE BOARD SURFACE IS 100 DEGREES CENTIGRADE.
      3. RADIATED TEMPERATURE FROM ANY COMPONENT MUST NOT SUBJECT BOARD TO MORE THAN 100 DEGREES CENTIGRADE.

5. ELEMENTARY SYMBOL
   A. THE CORRECT METHOD OF INDICATING COMPONENTS MOUNTED ON THIS CARD ON AN ELEMENTARY IS DESCRIBED IN DETAIL IN THE IPC NOMENCLATURE SPECIFICATION 68A991477.
   B. TERMINAL NUMBERS ON THE MASTER ELEMENTARY INDICATE LOCATION ON THE CARD.

6. MOUNTING
   A. COMPONENTS MUST BE MOUNTED BETWEEN TWO OPPOSITE TERMINAL CLAMPS.
   B. NO MORE THAN ONE WIRE OR COMPONENT LEAD MAY BE PLACED UNDER THE TERMINAL CLAMP WITHOUT SPECIAL ENGINEERING CONSIDERATIONS.

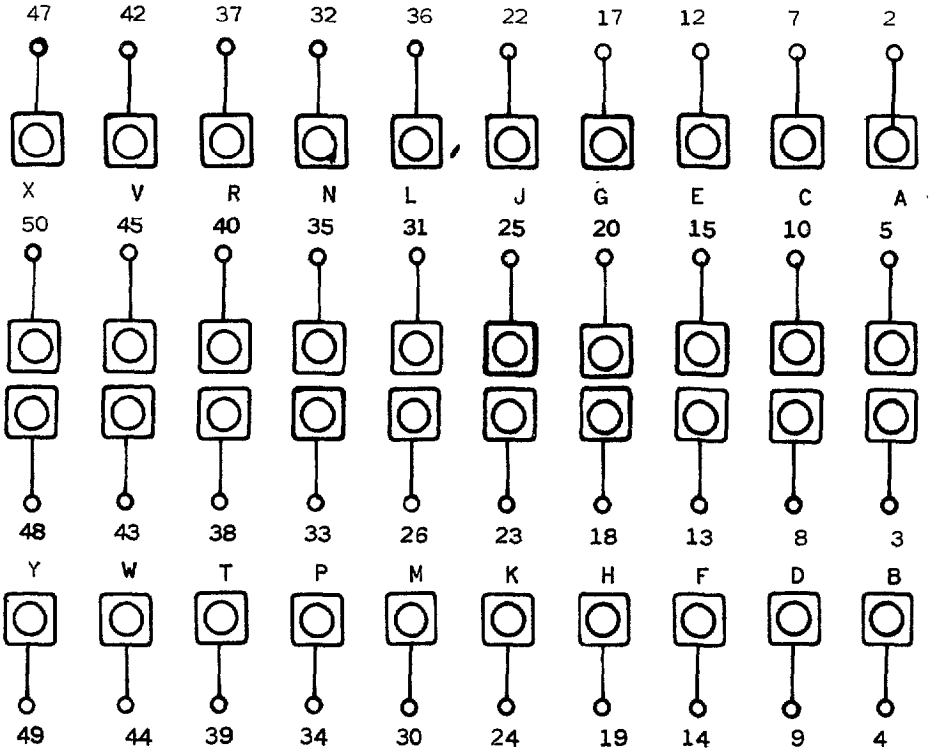
INTERNAL DEPARTMENT ORDERING INFORMATION AND TECHNIQUES

1. THIS UNIVERSAL COMPONENT BOARD, LESS COMPONENTS, MAY BE ORDERED BY THE IC CATALOG NUMBER - IC3600SCBA1

2. FOR THE CORRECT PROCEDURE IN ORDERING THIS COMPONENT BOARD WITH COMPONENTS REFER TO THE DEPARTMENT STANDARD ROUTINE.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 2 2520 UL1	APPROVALS J. H. Smith	GENERAL ELECTRIC	ELEMENTARY DIAGRAM UNIVERSAL COMPONENT BOARD	
ISSUED 4/14/65	PRINTS TO 1338	FIRST MADE FOR REQ. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	IC3600SCBA1	DR. NO. 3.1
MADE BY J. SMITH	P6A	I.C. NO.			



THE PURPOSE OF THE UNIVERSAL COMPONENT BOARD IS TO PROVIDE THE MECHANICAL MOUNTING OF AXIAL LEAD ELECTRONIC COMPONENTS OF THE PASSIVE TYPE USED AS SELECTED DEVICES IN SYSTEM DESIGN AND IN FIELD MODIFICATION TO THAT DESIGN. THESE COMPONENTS ARE NORMALLY USED TO MODIFY, OR CONTROL THE PERFORMANCE OF A STANDARD CIRCUIT AS REQUIRED FOR SPECIFIC SYSTEM PERFORMANCE.

THE MECHANICAL CONFIGURATION OF THE UNIVERSAL COMPONENT BOARD IS COMPATIBLE WITH THE NORMAL MOUNTING OF STANDARD FUNCTIONAL CARDS. THEREFORE, IT IS SUITABLE FOR MOUNTING IN THE IDENTICAL MANNER.

APPLICATION DATA:

1. COMPONENT SIZE
   - A. BODY SIZE - (AXIAL LEAD COMPONENTS ONLY)
     1. LENGTH - .5 INCHES MAXIMUM
     2. WIDTH - .4 INCHES MAXIMUM
     3. HEIGHT - .70 INCHES MAXIMUM
   - B. LEAD LENGTH- (END TO END) 2.20 INCHES MINIMUM.
   - C. THIS COMPONENT BOARD IS DESIGNED TO ACCOMMODATE 20 SUCH COMPONENTS

2. VOLTAGE CLEARANCE (NEMA)
   - A. ACROSS COMPONENTS
     1. UNLIMITED POWER- 50 VOLTS MAXIMUM.
     2. LIMITED POWER (1500 VA MAX)- 50 VOLTS MAXIMUM.
   - B. BETWEEN CIRCUITS AND PINS ON THE CARD
     1. UNLIMITED POWER- 50 VOLTS MAXIMUM
     2. LIMITED POWER (1500 VA MAX)- 50 VOLTS MAXIMUM

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 Retyped Feb.18,69 ACW	REV. 2 DL1	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	
RELEASED Feb. 18, 1969	PRINT NO. 1338	FIRST MADE FOR REQ. 2520	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	UNIVERSAL COMPONENT BOARD	
MADE BY J. SMITH	P6A	STANDARD LINE		IC3600SCBB1	SH. NO. 3.0
		I.C. NO.		CONT. ON SH. 3.1	

3. CONDUCTOR CURRENT RATING
   A. CONNECTOR CONTACTS- 5 AMPS MAXIMUM
   B. CONDUCTOR RUNS- 3 AMPS MAXIMUM

4. POWER LEVEL ON COMPONENTS
   A. WITHOUT SPECIAL ENGINEERING CONSIDERATION- 1 WATT MAXIMUM PER BOARD
   B. WITH SPECIAL ENGINEERING CONSIDERATION
      1. MAXIMUM WATTS PER COMPONENT- 5 WATTS
      2. MAXIMUM SURFACE TEMPERATURE OF ANY DEVICE IN CONTACT WITH THE BOARD SURFACE IS 100 DEGREES CENTIGRADE.
      3. RADIATED TEMPERATURE FROM ANY COMPONENT MUST NOT SUBJECT BOARD TO MORE THAN 100 DEGREES CENTIGRADE.

5. ELEMENTARY SYMBOL
   A. THE CORRECT METHOD OF INDICATING COMPONENTS MOUNTED ON THE CARD ON AN ELEMENTARY IS DESCRIBED IN DETAIL IN THE IPC NOMENCLATURE SPECIFICATION 68A991477.
   B. TERMINAL NUMBERS ON THE MASTER ELEMENTARY INDICATE LOCATION ON THE CARD.

6. MOUNTING
   A. COMPONENTS MUST BE MOUNTED BETWEEN TWO OPPOSITE TERMINAL CLAMPS.
   B. NO MORE THAN ONE WIRE OR COMPONENT LEAD MAY BE PLACED UNDER THE TERMINAL CLAMP WITHOUT SPECIAL ENGINEERING CONSIDERATIONS.

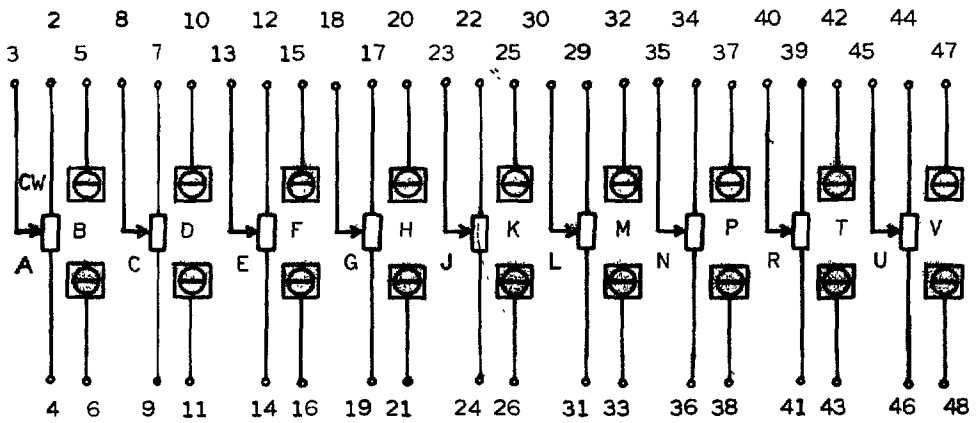
INTERNAL DEPARTMENT ORDERING INFORMATION AND TECHNIQUES

1. THIS UNIVERSAL COMPONENT BOARD, LESS COMPONENTS, MAY BE ORDERED BY THE IC CATALOG NUMBER- IC3600SCBB1.

2. FOR THE CORRECT PROCEEDURE IN ORDERING THIS COMPONENT BOARD WITH COMPONENTS REFER TO THE DEPARTMENT STANDARD ROUTINE.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 2 DL1	APPROVALS J. H. Smith	GENERAL ELECTRIC	ELEMENTARY DIAGRAM UNIVERSAL COMPONENT BOARD
ISSUED 4/16/63	PRINTS TO PGA	FIRST MADE FOR REL. 2520 STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	IC3600SCBB1 CONT. ON SH. FL.
MADE BY J. SMITH	1338	I.C. NO.		SH. NO. 3.1



THE PURPOSE OF THIS COMPONENT BOARD IS TO PROVIDE THE MECHANICAL MOUNTING OF AXIAL LEAD ELECTRONIC COMPONENTS OF THE PASSIVE TYPEUSED AS SELECTED DEVICES IN SYSTEM DESIGN AND IN FIELD MODIFICATIONS TO THAT DESIGN. THESE COMPONENTS ARE NORMALLY USED TO MODIFY, OR CONTROL THE PERFORMANCE OF, A STANDARD CIRCUIT AS REQUIRED FOR SPECIFIC SYSTEM PERFORMANCE.

THE MECHANICAL CONFIGURATION OF THE UNIVERSAL COMPONENT BOARD IS COMPATIBLE WITH THE NORMAL MOUNTING OF STANDARD FUNCTIONAL CARDS. THEREFORE, IT IS SUITABLE FOR MOUNTING IN THE IDENTICAL MANNER.

APPLICATION DATA.

1. COMPONENT SIZE AND TYPE
   - A. BODY SIZE- (AXIAL LEAD COMPONENTS ONLY)
     1. LENGTH- .8 INCHES MAXIMUM
     2. WIDTH- .55 INCHES MAXIMIM
     3. HEIGHT- .70 INCHES MAXIMUM
   - B. LEAD LENGTH-(END TO END) 2.20 INCHES MINIMUM.
   - C. THIS CARD WILL ALSO ACCOMMODATE 4 DIFFERENT TYPES OF POTENTIOMETERS. THEY ARE:
     1. 68A7100
     2. 68A7101
     3. 68A7105
     4. 68A7107
   - D. THIS CARD WILL ALSO ACCOMMODATE CARD FRONT. (SEE NOTE ON SHEET 3.1)
2. VOLTAGE CLEARANCE (NEMA)
   - A. ACROSS COMPONENTS
     1. UNLIMITED POWER - 50 VOLTS MAXIMUM.
     2. LIMITED POWER (1500 VA MAX)- 50 VOLTS MAXIMUM.
   - B. BETWEEN CIRCUITS AND PINS ON THE CARD
     1. UNLIMITED POWER- 50 VOLTS MAXIMUM.
     2. LIMITED POWER (1500 VA MAX)- 50 VOLTS MAXIMUM.
3. CONDUCTOR CURRENT RATING
   - A. CONNECTOR CONTACTS - 5 AMPS MAXIMUM
   - B. CONDUCTOR RUNS- 3 AMPS MAXIMUM

WH

REV 3 PDA 8-30-79
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
CHK'D

REV. A 12/11/6	DL1	1338	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM COMPONENT CARD
ISSUED	PRINTS TO K6A	FIRST MADE FOR REQ. STANDARD LINE			IC3600SCBN1 SHEET NO. 3.0
MADE BY	PXW	I.C. NO.			CHK'D ON SH. 3.1

4. POWER LEVEL ON COMPONENTS
   A. WITHOUT SPECIAL ENGINEERING CONSIDERATION- 1 WATT MAXIMUM PER BOARD
   B. WITH SPECIAL ENGINEERING CONSIDERATION-
      1. MAXIMUM WATTS PER COMPONENT- 5 WATTS
      2. MAXIMUM SURFACE TEMPERATURE OF ANY DEVICE IN CONTACT WITH THE BOARD SURFACE IS 100 DEGREES CENTIGRADE.
      3. RADIATED TEMPERATURE FROM ANY COMPONENT MUST NOT SUBJECT BOARD TO MORE THAN 100 DEGREES CENTIGRADE.

5. ELEMEMTARY SYMBOL
   A. THE CORRECT METHOD OF INDICATING COMPONENTS MOUNTED ON THIS CARD ON AN ELEMENTARY IS DESCRIBED IN DETAIL IN THE IPC NOMENCLATURE SPECIFICATION 68A991477.
   B. TERMINAL NUMBERS ON THE MASTER ELEMENTARY INDICATE LOCATION ON THE CARD.

6. MOUNTING
   A. COMPONENTS MUST BE MOUNTED BETWEEN TWO OPPOSITE TERMINAL CLAMPS.
   B. NO MORE THAN ONE WIRE OR COMPONENT LEAD MAY BE PLACED UNDER THE TERMINAL CLAMP WITHOUT SPECIAL ENGINEERING CONSIDERATIONS.

INTERNAL DEPARTMENT ORDERING INFORMATION AND TECHNIQUES

1. THIS UNIVERSAL COMPONENT BOARD, LESS COMPONENTS, MAY BE ORDERED BY THE IC CATALOG NUMBER- IC3600SCBN1

2. FOR THE CORRECT PROCEDURE IN ORDERING THIS COMPONENT BOARD WITH COMPONENTS REFER TO THE DEPARTMENT STANDARD ROUTINE.

NOTE: WHEN IT IS DESIRED TO USE A METAL CARD FRONT, THE FOLLOWING RESTRICTIONS MUST BE OBSERVED:

NO 68A7105 OR 68A7107 POTENTIOMETERS MAY BE USED IN POSITIONS A, C AND U.

NO 68A7125 POTENTIOMETER MAY BE USED IN POSITION U.

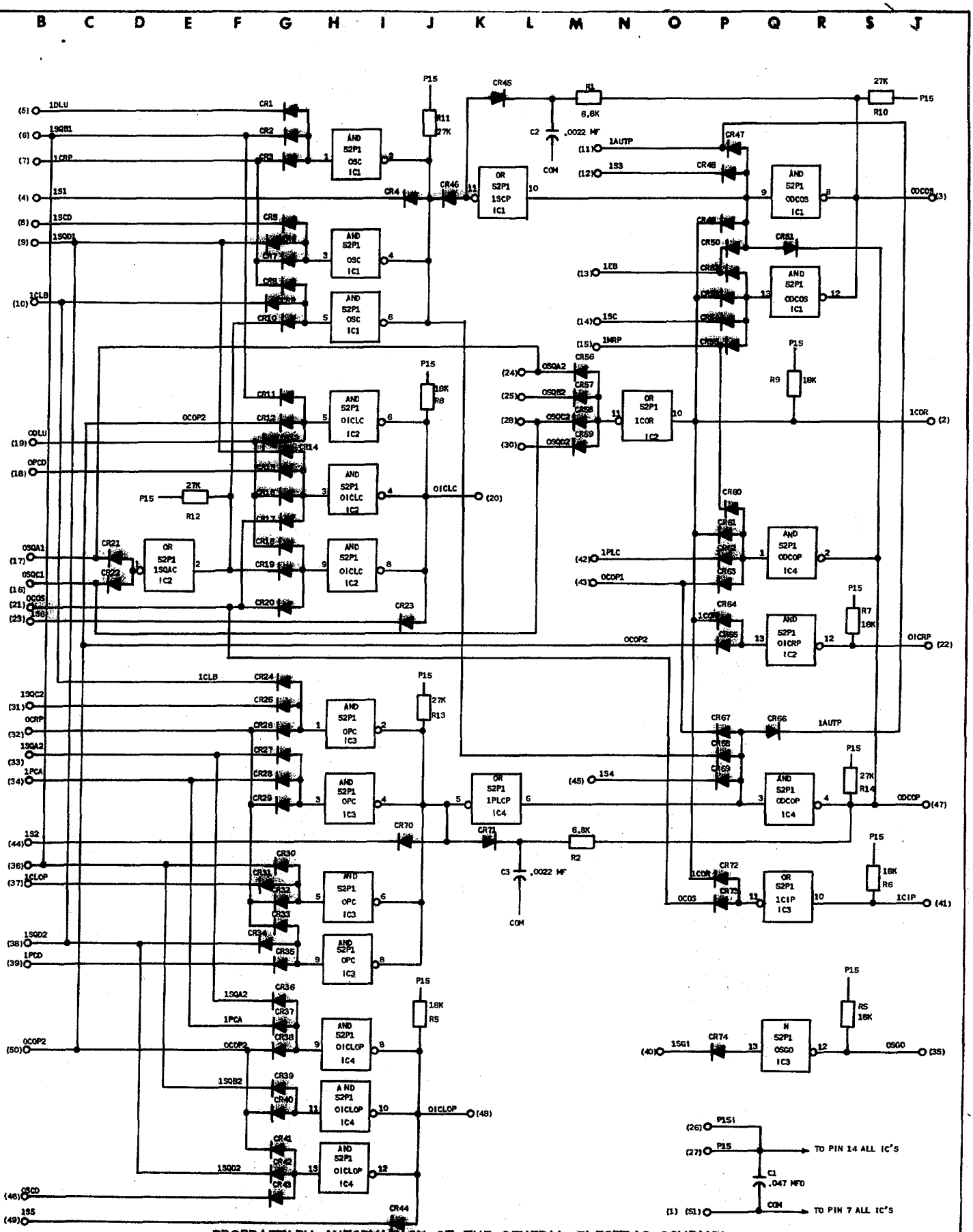
MECHANICAL INTERFERENCES WILL RESULT IF THIS IS VIOLATED.

WAY

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

CHVI

REV. 1 8-31-70 PDA	REV. 2 DLJ	APPROVALS J. H. Smith	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	
ISSUED 4/16/65	PRINTS TO 1338	FIRST MADE FOR REQ. STANDARD LINE	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	COMPONENT CARD	
MADE BY J.SMITH	P6A	IC NO.		IC3600SCBN1	DWG. NO.
				CONT. SHT. SL. FL	



PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV.1 A WRIGHT 16, 71	REV.4	REV.5	PRINTS TO DL1, P6A 1336.	APPROVALS L.B. Brown	GENERAL ELECTRIC
REV.2	ISSUED 1-19-71.		A1F(4)	FIRST MADE FOR PLASTIC MOLDING I.C. NO.	INDUSTRY CONTROL DEPT. SALEM, VA., USA
REV.3	MADE BY M A CONNER				

ELEMENTARY DIAGRAM
STANDARD CORE LOGIC

IC3622GCOA1

CONT. ON SH. 3.1 SH. NO. 3

CORE SEQUENCE - APPLICATION NOTES

I. GENERAL DESCRIPTION

A. THE STANDARD CORE SEQUENCE CARD CONTAINS THE CIRCUITRY TO PERFORM THE FOUR STANDARD SPI CORE SEQUENCES. AVAILABLE FEATURES ARE:

1. SELECTION OF CORE SEQUENCE A, B, C, D OR OFF.

A. SEQUENCE A
   1) RESET EJECTORS
   2) SET CORES IN CLAMP BACK POSITION
   3) CLOSE CLAMP
   4) INJECT
   5) OPEN CLAMP TO ADJUSTABLE STOP POSITION
   6) PULL CORES
   7) CONTINUE TO OPEN CLAMP
   8) EJECT

B. SEQUENCE B
   1) CLOSE CLAMP TO LOCK-UP
   2) SET CORES
   3) INJECT
   4) PULL CORES
   5) OPEN CLAMP
   6) EJECT

C. SEQUENCE C
   1) SET CORES IN BACK POSITION
   2) CLOSE CLAMP
   3) INJECT
   4) OPEN CLAMP
   5) PULL CORES IN BACK POSITION
   6) EJECT

D. SEQUENCE D
   1) CLOSE CLAMP - SET CORES WHILE CLOSING
   2) INJECT
   3) OPEN CLAMP - PULL CORES WHILE OPENING
   4) EJECT

2. MANUAL OR AUTOMATIC OPERATION

3. SELECTION OF CORE PULL BY SPRING RETURN OR BY A MOMENTARY OR MAINTAINED SOLENOID. (CORE SET IS USUALLY A MAINTAINED SOLENOID, HOWEVER, STANDARD CARD CAN BE USED WITH SPRING RETURN SET).

4. BAND OF OPERATION FOR SEQUENCE D USING AN ADJUSTABLE LIMIT SWITCH TO START SET OR PULL OPERATION AND ANOTHER LIMIT SWITCH TO STOP CLAMP MOTION IF THE SET OR PULL IS NOT COMPLETED.

5. COMPATABLE WITH THE STANDARD EJECTOR CIRCUIT.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL1. P6A	APPROVALS L.R. Brown	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM STANDARD CORE LOGIC
REV. 2	ISSUED 1-19-71	1338. A1F(4)	FIRST MADE FOR PLASTIC MOLD			IC3622GC0A1
REV. 3	MADE BY M A CONNER		I.C. NO.			CONT. ON SH. 3.2 SH. NO. 3.1

6. INHIBIT CIRCUIT FOR RECYCLE IF CORES ARE SELECTED AND NOT PULLED. SIGNAL IS ALSO USED FOR EJECT PERMISSIVE.

7. INJECTION INHIBIT IF CORES ARE NOT SET NO MATTER WHAT SEQUENCE.

8. A NEW SEQUENCE CAN BE ESTABLISHED USING SPARE INPUTS 1S1, 1S2, 1S5 AND 1S6.

9. SEQUENCE SELECTION INPUTS FOR PULL AND SET ARE SEPARATE BUS RUNS THOUGH TIED TOGETHER ON THE CARD. THIS PERMITS FUTURE SPLITTING SO THAT ONE SEQUENCE FOR SET AND ANOTHER FOR PULL COULD BE SELECTED.

10. CORES CAN BE MADE TO PULL ANYTIME THEY ARE NOT FULLY PULLED WITH CORES OFF, IF THE SIGNAL 1COR IS CONNECTED TO 1S2.

B. NOMENCLATURE FOR THE CORE SEQUENCE CIRCUIT INPUT AND OUTPUT SIGNALS IS CONSISTENT WITH THE STANDARDS FOR PLASTI-MATIC CONTROL (SEE APPLICATION SHEET DP-N1 OR THE BACK SHEET OF THE CONTROL ELEMENTARY DRAWING FOR DEFINITION.

C. PIN DESIGNATIONS FOR STANDARD OPERATION CARRY NOMENCLATURE FOR THE INPUTS REQUIRED. OPTION PINS LABELED SPARES (1S#) ARE AVAILABLE TO MODIFY STANDARD OPERATION.

II. OPERATION

SEQUENCE A, B, C, OR D MAY BE SELECTED FOR AUTOMATIC OPERATION. MANUAL OPERATION IS PERFORMED BY MANUAL CORE SELECTOR SWITCH TO SET OR PULL. IN MANUAL OPERATOR CONTROLS SEQUENCE.

A. AUTOMATIC OPERATION

1. INITIAL CONDITIONS
   - A. MACHINE SET FOR AUTOMATIC OPERATION (1AUTP)
   - B. EJECTORS ARE BACK.
   - C. SELECT CORE SEQUENCE A, B, C, D (1SQA, 1SQB, 1SQC, OR 1SQD)
   - D. CORES ARE PULLED (0COP2).

2. TO SET CORES IN ALL SEQUENCES, THE FOLLOWING CONDITIONS ARE NECESSARY:
   - A. IN AUTOMATIC (1AUTP IS A LOGIC "1").
   - B. EJECTORS ARE FULL BACK (1EB IS A LOGIC "1").
   - C. SPARE #3 (1S3) IS PERMISSIVE (LOGIC "1").
   - D. CYCLE RUN IS PERMISSIVE (1CRP IS A LOGIC "1").
   - E. A GIVEN CORE SEQUENCE IS SELECTED A, B, C, OR D (0SQA, 0SQC LOGIC "0" OR 1SQB, 1SQD LOGIC "1").
   - F. CLAMP IS IN THE APPROPRIATE POSITION TO SET THE CORES (CLAMP BACK FOR SEQUENCE "A" AND "C", CLAMP CLOSED FOR SEQUENCE "B", AND CLAMP CLOSING WHERE LIMIT SWITCH (SCD) IS TRIPPED (IS A LOGIC "1") FOR SEQUENCE D.

   WHEN ALL CONDITIONS LISTED ABOVE ARE MET, CORES WILL SET. WITH A MAINTAIN SET SOLENOID, CORES WILL REMAIN SET UNTIL INITIATED TO PULL.

3. ASSOCIATED WITH SETTING THE CORES WILL BE A SIGNAL 0ICLC. WHEN THIS SIGNAL IS A LOGIC "0" THE CLAMP WILL BE INHIBITED FROM CLOSING. THEREFORE, FOR

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL1. P6A	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM STANDARD CORE LOGIC
REV. 2	ISSUED 1-19-71	1338. A1F(4)	FIRST MADE FOR PLASTIC MOLD			IC3622GC0A1
REV. 3	MADE BY M A CONNER		I.C. NO.			CONT. ON SH. 3,3 SH. NO. 3,2

SEQUENCES A, B, C, AND D THE CLAMP WILL BE INHIBITED UNDER THE FOLLOWING CONDITIONS.

A. IN SEQUENCE "A" OR "C" THE CLAMP WILL NOT CLOSE UNTIL THE CORES ARE FULLY SET. WHEN THIS HAPPENS OCOS WILL GO TO A LOGIC "0" CAUSING OICLC TO GO TO A LOGIC "1", ALLOWING THE CLAMP TO CLOSE.

B. IN SEQUENCE "B" THE CLAMP WILL NOT CLOSE UNLESS THE CORES ARE FULLY PULLED (OCOP A LOGIC "0"). AT CLAMP CLOSED POSITION, ODLU BECOMES A LOGIC "0" WHICH KEEPS OICLC A LOGIC "1" AS CORES START TO SET AND OCOP BECOMES A LOGIC "1".

C. IN SEQUENCE "D" THE MOLD WILL CLOSE TO LIMIT SWITCH 1SCD WHICH TELLS THE CORES TO SET. CLAMP WILL CONTINUE TO CLOSE AS CORES ARE SETTING. HOWEVER, IF OPCD LIMIT SWITCH IS CONNECTED, AND THE CORES ARE NOT SET BY THE TIME THE CLAMP REACHES THIS LIMIT SWITCH, OPCD GOES TO LOGIC "1" CAUSING OICLC TO BECOME LOGIC "0" WHICH STOPS THE CLAMP FROM CLOSING. (THIS LIMIT SWITCH IS CONNECTED SO THAT THERE IS PROTECTION FOR CORES AND MOLD). YOU WILL NOTICE THAT ONCE THE MOLD HAS LOCKED-UP (ODLU A LOGIC "0") OICLC WILL NOT INHIBIT MOLD CLOSE.

4. TO PULL THE CORES IN ALL SEQUENCES THE FOLLOWING CONDITIONS ARE NECESSARY.

A. IN AUTOMATIC (1AUTP IS A LOGIC "1")
B. INITIATE CORES SET SIGNAL IS NO LONGER PRESENT (OSC IS A LOGIC "1")
C. SPARE #4 (1S4) IS PERMISSIVE (1S4 IS A LOGIC "1").
D. CORES ARE NOT PULLED (OCOP1 IS A LOGIC "1").
E. CYCLE RUN IS NO LONGER PERMISSIVE (OCRP IS A LOGIC "1").
F. CORE SEQUENCE A, B, C, OR D IS SELECTED.
G. CLMAP IS IN THE APPROPRIATE POSITION TO PULL THE CORES (CLAMP AT LIMIT SWITCH POSITION 1PCA IN SEQUENCE "A", CLAMP IS CLOSED FOR SEQUENCE "B", CLAMP IS BACK FOR SEQUENCE "C", AND CLAMP HAS TRIPPED 1PCD (BECOMES A LOGIC "1") LIMIT SWITCH ON OPENING FOR SEQUENCE "D").

WHEN ALL CONDITIONS LISTED ABOVE ARE MET, CORES WILL PULL. WITH A MOMENTARY PULL SOLENOID, CORES CONTINUE TO PULL UNTIL CORES ARE FULL OUT (OCOP1 IS A LOGIC "0"). FOR A MAINTAIN PULL SOLENOID, OCOP1 INPUT IS NOT CONNECTED, THEREFORE CORES WILL CONTINUE TO PULL UNTIL INITIATED TO SET.

5. ASSOCIATED WITH PULLING THE CORES IS THE SIGNAL OICLOP WHICH IS USED TO INHIBIT CLAMP OPENING WHEN OICLOP BECOMES A LOGIC "0". THEREFORE, THE CLAMP WILL BE INHIBITED WITH THE FOLLOWING CONDITIONS:

A. IN SEQUENCE "A" WITH LIMIT SWITCH 1PCA TRIPPED (LOGIC "1") CLAMP WILL NOT OPEN BEYOND 1PCA UNTIL THE CORES ARE FULLY PULLED (OCOP2 IS A LOGIC "0").
B. IN SEQUENCE "B" CLAMP WILL STAY FULLY CLOSED UNTIL CORES ARE PULLED (OCOP2 GO TO A LOGIC "0").
C. IN SEQUENCE "C" CLAMP WILL OPEN REGARDLESS OF CORE POSITION.
D. IN SEQUENCE "D" CLAMP WILL CONTINUE TO OPEN BEYOND 1PCD (LIMIT SWITCH TO START PULLING CORES) UNTIL OSCD IS TRIPPED (OSCD BECOMES A LOGIC "1"). IF CORES ARE PULLED (OCOP2 IS A LOGIC "0") BY THIS TIME, CLAMP CONTINUES TO OPEN. HOWEVER, IF CORES ARE NOT FULLY PULLED BY THE TIME OSCD BECOMES A LOGIC "1" THE CLAMP WILL STOP OPENING BECAUSE OICLC WILL BECOME LOGIC "0".

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL1. P6A	APPROVALS R.B. Brown	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM STANDARD CORE LOGIC
REV. 2	ISSUED 1-11-71	1338. A1F(4)	FIRST MADE FOR PLASTIC MOLD			I C 3 6 2 2 G C O A 1
REV. 3	MADE BY M A CONNER		I.C. NO.			CONT. ON SH. 3.4 SH. NO. 3.3

6. INJECTION PERMISSIVE 1CIP - REGARDLESS OF SEQUENCE SELECTED, BEFORE INJECTION CAN TAKE PLACE, THE CORES MUST BE SET (OCOS A LOGIC "0") OR THE CORES MUST BE OFF (1COR A LOGIC "0").

7. CYCLE RUN INHIBIT OICRP - REGARDLESS OF SEQUENCE SELECTED, BEFORE CYCLE RUN IS PERMISSIVE, THE CORES MUST BE PULLED (OCOP A LOGIC "0") OR THE CORES MUST BE OFF (1COR A LOGIC "0").

B. MANUAL OPERATION

1. INITIAL CONDITION
   A. MACHINE SET FOR MANUAL OPERATION (1MRP IS A LOGIC "1").
   B. CORES ARE SELECTED (1COR IS A "1") .

2. SET CORES
   A. EJECTORS ARE BACK (1EB IS LOGIC "1")
   B. SWITCH SET FOR SET CORES (1SC LOGIC "1"). THE SET CORES SOLENOID WILL BE ENERGIZED FOR THE TIME THE SWITCH IS HELD TO CORE SET POSITION.

3. PULL CORES
   A. CORES NOT PULLED (OCOP A LOGIC "1") IF CONNECTED.
   B. SWITCH SET FOR PULL CORES (1PLC A LOGIC "1"). THE PULL CORES SOLENOID WILL BE ENERGIZED FOR THE TIME THE SWITCH IS HELD OR UNTIL THE CORES ARE PULLED.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL1, P6A	APPROVALS C.B. Brown	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM STANDARD CORE LOGIC
REV. 2	ISSUED 1-1-71	1338, A1F(4)	FIRST MADE FOR PLASTIC MOLD			IC3622GC0A1
REV. 3	MADE BY M A CONNER		I.C. NO.			CONT. ON SH. 3.5 SH. NO. 3.4

INPUT LOADING IS TABULATED BELOW. 1 LOAD IS DEFINED AS 18K OHMS TO +15V.

INPUT	LOADS
OSQA1	2
OSQA2	2
OSQB2	1
OSQC1	2
OSQC2	2
OSQD2	1
1SQA2	2
1SQB1	4
1SQB2	4
1SQC2	1
1SQD1	4
1SQD2	4
1S1	3
1DLU	1
1CRP	3
1SCD	1
1CLB	2
OCOS	3
ODLU	3
OPCD	1
1S6	1
OCRP	4
1PCA	2
1S2	2
1CLOP	1
1PCD	1
OCOP1	2
OCOP2	5
OSCD	1
1S5	1
1S3	1
1EB	2
1SC	1
1MRP	2
1PLC	1
1AUTP	2
1S4	1
1SG1	1

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL1 P6A	APPROVALS R.B. Brown	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 2	ISSUED 1-19-71	1338 A1F(4)	FIRST MADE FOR PLASTIC MOLD		
REV. 3	MADE BY M A CONNER		I.C. NO.		

ELEMENTARY DIAGRAM: STANDARD CORE LOGIC

IC3622GC0A1

CONT. ON SH. 3.6 SH. NO. 3.5

OUTPUT DRIVE IS TABULATED BELOW:

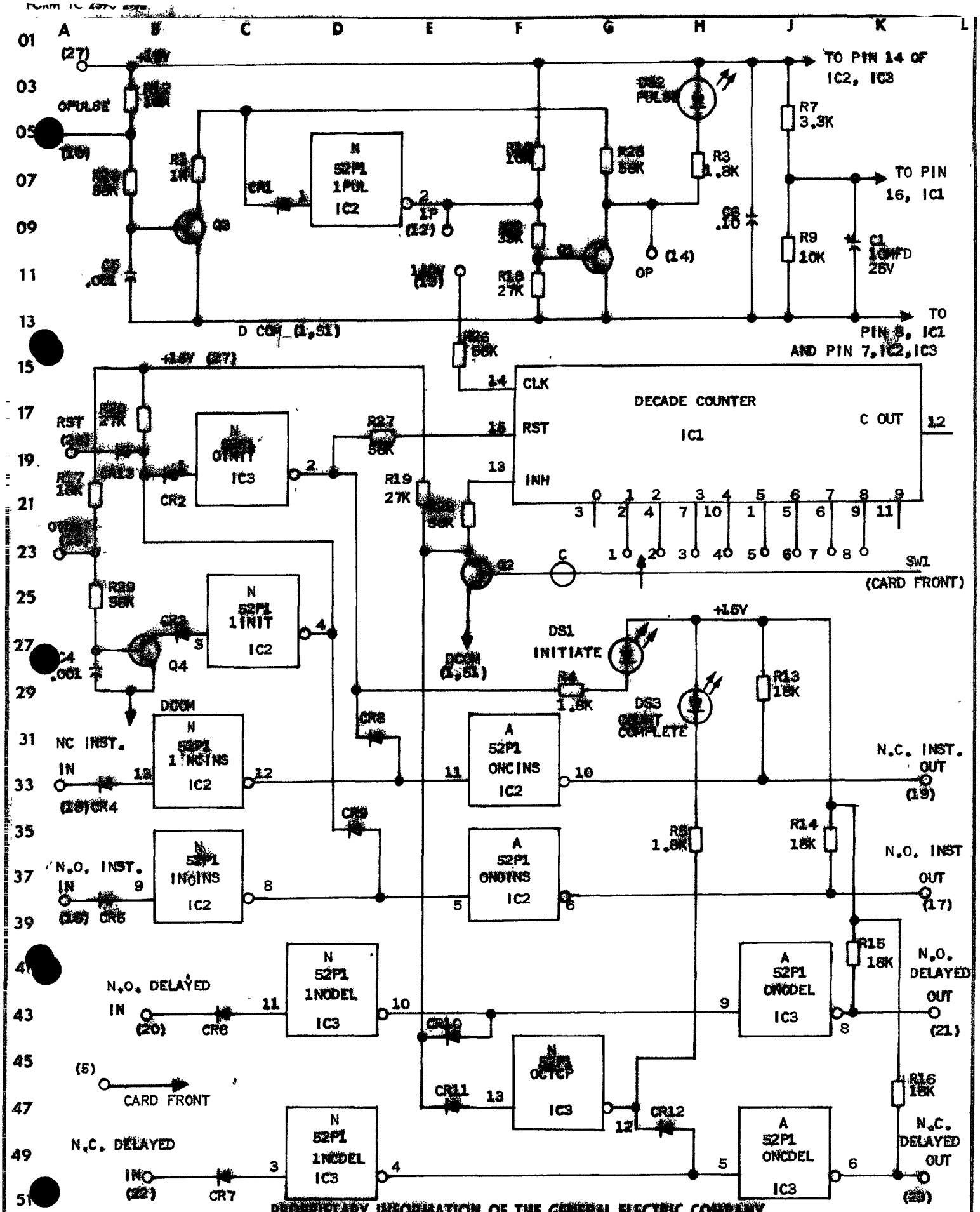
OUTPUT	LOADS
0DCOS	17
01CLC	17
0DCOP	16
01CLOP	17
1COR	12
1CIP	17
01CRP	17
0SGO	17

POWER SUPPLY REQUIREMENTS

+15V, 30 MA TYPICAL, 70 MA MAX.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV 1 ACW 11-15-71.	REV 4	REV. 5	PRINTS TO DL1 P6A	APPROVALS L.B.Brown	ELEMENTARY DIAGRAM STANDARD CORE LOGIC
REV. 2	ISSUED 1-19-71. 1338 A1F (4)			FIRST MADE FOR PLASTIC MOLD.	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 3	MADE BY M A CONNER			I.C. NO	I C 3 6 2 2 G C 0 A 1
					CONT. ON SH. FL. SH. NO. 3,6



PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. A IF (4)	REV. 5 3T	PRINTS TO A&E	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 2	ISSUED Aug 4, 1972	CRYIVN	FIRST MADE FOR		
REV. 3	MADE BY GARY HARRIS		I.C. NO.		

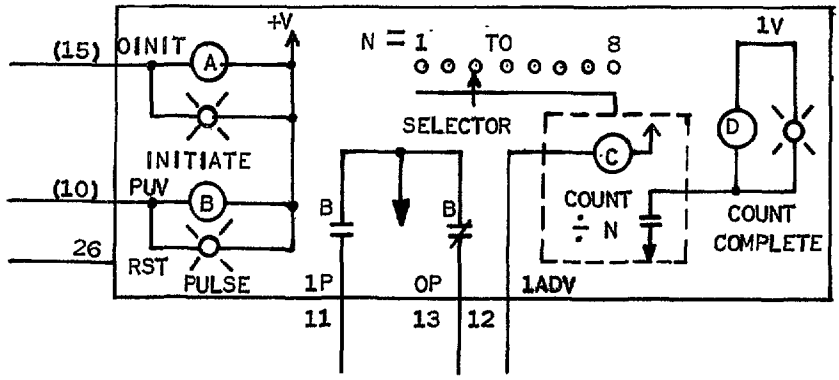
ELEMENTARY DIAGRAM
CYCLE COUNTER

I C 3 6 2 2 G D C B 1

CONT. ON SH. 3.1 SH. NO. 3.0

FORM IC 2570 2-72

SUGGESTED SYMBOL:





* DENOTES LOGIC LEVEL FLOW THROUGH SOLID STATE CONTACTS.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4 AIF (4)	REV. 5 3T 6V,DL1,P6A	PRINTS TO A6F, 2530,1338	APPROVALS K.BB4RE&	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 2	ISSUED Aug 4,197	CHVIVN WAY	FIRST MADE FOR PLAST. MOLD.		
REV. 3	MADE BY GARY HARRIS		I.C. NO.		

ELEMENTARY DIAGRAM
CYCLE COUNTER

I C 3 6 2 2 G D C B 1

CONT. ON SH 3.2 SH. NO. 3.1

WA4
CHVI
PRINTS TO

APPLICATION NOTES

THE CYCLE COUNTER IS USED WHEN IT IS DESIRED TO DELAY AN OUTPUT UNTIL A DESIRED NUMBER OF INPUT PULSES HAS BEEN APPLIED TO THE CIRCUIT. BOTH INSTANTANEOUS AND DELAYED OUTPUTS ARE AVAILABLE. THE NUMBER OF INPUT CYCLES CAN BE VARIED FROM 1 TO 8 BY MEANS OF A SELECTOR SWITCH ON THE CARD FRONT. THE OUTPUTS ARE SOLID STATE LOGIC LOAD CONTACTS COMPATIBLE WITH ALL OTHER IC3622G SERIES CARDS.

SPECIFICATIONS:

O INITIATE (15): CONNECTING TO COMMON WILL INITIATE THE COUNTER AND ALLOW IT TO COUNT, AN INDICATOR ON THE CARD FRONT LIGHTS WHEN THE COUNTER IS INITIATED. A 15V LEVEL OR OPEN CIRCUIT WILL RESET THE COUNTER, AND ANY INPUT PULSES WILL BE IGNORED. A RESET PULSE SHOULD BE AT LEAST 60 MICROSECONDS LONG. ONE LOAD.

O PULSE(10): EACH TIME THIS INPUT IS TIED TO COMMON "PULSE" IS CONSIDERED PICKED UP AND THE PULSE INDICATOR WILL LIGHT. THE COUNTER WILL ADVANCE ON PICK UP IF 1P IS JUMPERED TO 1ADV. ALTERNATELY, IF OP IS JUMPERED TO 1ADV, THE COUNTER WILL ADVANCE ON DROP OUT. INPUT PULSES MUST BE AT ZERO FOR AT LEAST 60 MICROSECONDS TO INSURE COUNTING, WITH 60 MICROSECONDS BETWEEN PULSES.

RST (26): USED AS A POWER APPLIED RESET, IF DESIRED. A GROUNDED INPUT WILL RESET THE COUNTER AND DROP OUT THE INSTANTANEOUS RELAY.

N.O. IST: (INPUT 16, OUTPUT 17) NORMALLY OPEN LOGIC LEVEL FLOW THROUGH CONTACTS. OPEN WHEN RESET, CLOSE INSTANTLY WHEN INITIATED.

N.C. IST: (INPUT 18, OUTPUT 19) NORMALLY CLOSED LOGIC LEVEL FLOW THROUGH CONTACT. CLOSED WHEN RESET, OPEN WHEN INITIATED.

N.O. DELAYED: (INPUT 20, OUTPUT 21) NORMALLY OPEN, CLOSE AFTER THE SET NUMBER OF CYCLES HAS BEEN COMPLETED. OPENS INSTANTLY WHEN RESET.

N.C. DELAYED: (INPUT 22, AUTPUT 23) NORMALLY CLOSED, OPENS AFTER THE SET NUMBER OF CYCLES HAS BEEN COMPLETED. CLOSES INSTANTLY WHEN RESET.

THE FOLLOWING CARD FRONT INDICATORS ARE AVAILABLE:

INITIATE - LIGHTS WHEN COUNTER IS ENABLED TO COUNT.

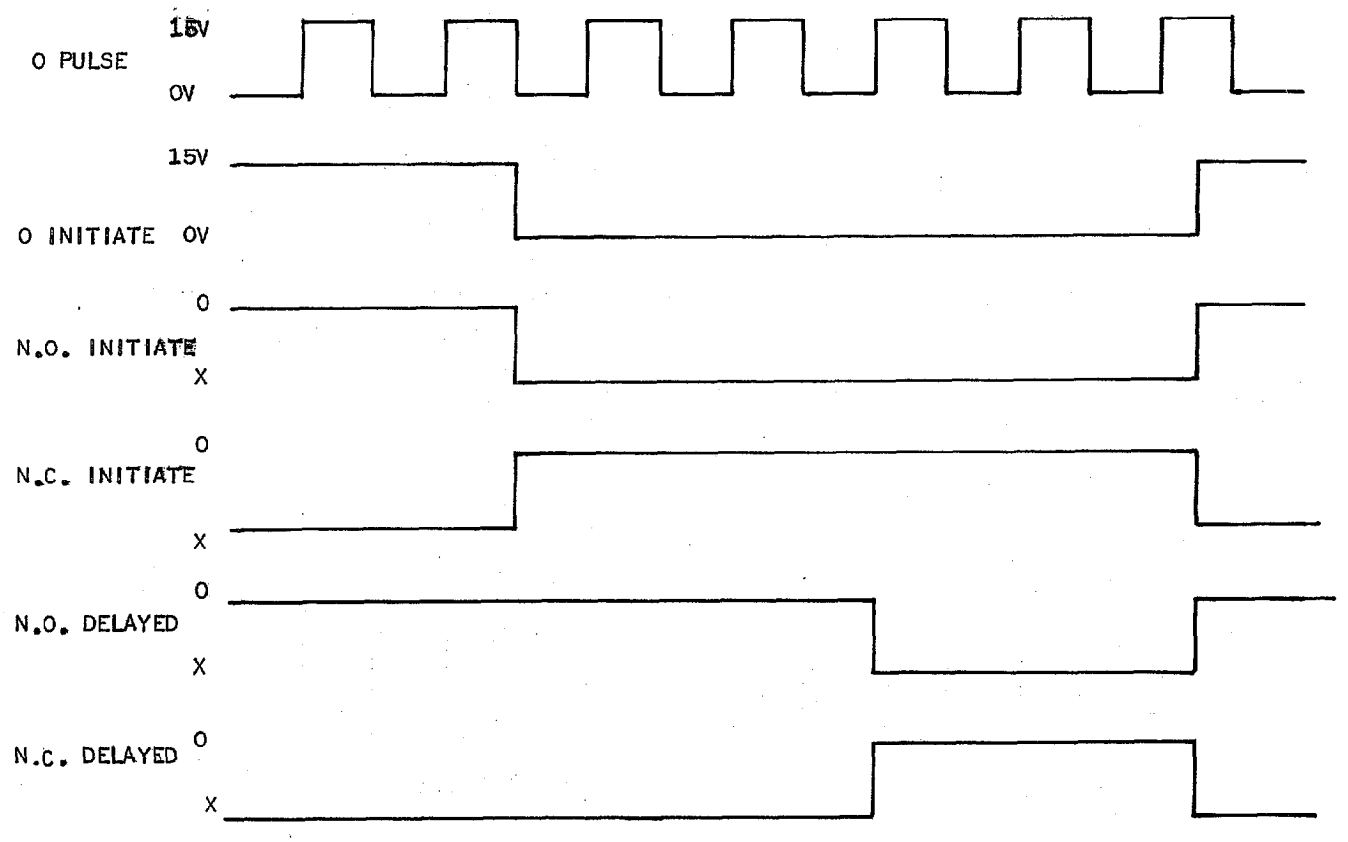
PULSE - LIGHTS EVERY TIME ONE INPUT PULSE IS APPLIED. IF 1P IS JUMPERED TO ADV, COUNT INCREM ENTS WHEN INDICATOR TURNS ON. IF OP IS JUMPERED TO ADV, COUNT INCRE M ENTS WHEN INDICATOR TURNS OFF.

COUNT COMPLETE - LIGHTS WHEN THE SET NUMBER OF INPUT PULSES HAS OCCURED.

FIGURE 1 SHOWS AN EXAMPLE TIMING SEQUENCE WITH OP JUMPERED TO 1ADV, AND COUNT SELECTOR AT 3.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4 A1F(4)	REV. 5 3T 6V,DL1,F8A	PRINTS TO ASF- 2530,1938	APPROVALS 2664 REL	ELEMENTARY DIAGRAM CYCLE COUNTER
REV. 2	ISSUED Aug 4, 1972 WAY		FIRST MADE FOR PLAST. MOLD.	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	I C 3 6 2 2 G D C B 1
REV. 3	MADE BY GARY HARRIS CHVIVN		I.C. NO.		CONT. ON SH. 3.3 SH. NO. 3.2



NOTE: 1) O DENOTES OPEN, X DENOTES CLOSED. WAVEFORMS ARE FOR CONTACT INPUTS TIED TO COMMON (TRUE INPUT SIGNAL)

2) EXAMPLE SHOWN FOR SELECTOR SWITCH SET TO 3

3) 1ADV CONNECTED TO OP

FIGURE 1

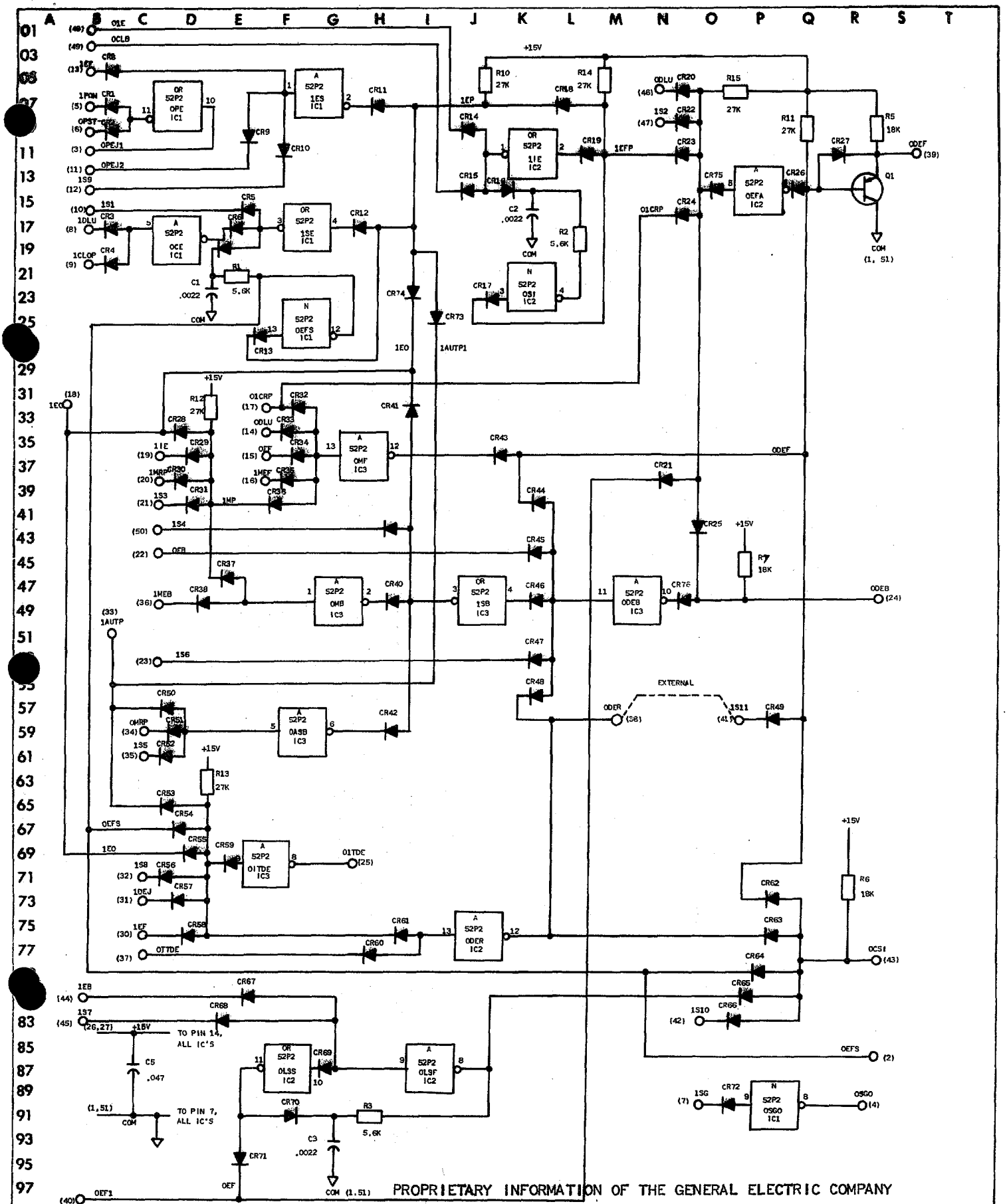
THE LOGIC RULES FOR THE LOGIC LEVEL OUTPUT CONTACTS ARE THE SAME AS DISCUSSED ON THE GLEB LOGIC RELAY AND ALL OTHER CARDS USING THE FLOW THROUGH LOGIC CONCEPT. POWER REQUIREMENTS:

+15V (27)
INITIATE OFF 25MA TYPICAL
COUNTING OR COUNT COMPLETE - 65MA TYPICAL, 80MA MAX.

THIS CARD HAS A CARD FRONT AND OCCUPIES TWO CARD ROW SPACES.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4 A1F(4)	REV. 5 3T 6V.DL1.PSA 2530,1336	PRINTS TO ASF	APPROVALS EBB 4 REB.	ELEMENTARY DIAGRAM CYCLE COUNTER
REV. 2	ISSUED Aug 4 1972 CHVIVN WAY		FIRST MADE FOR PLAST. MOLD.	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	IC3622GDCB1
REV. 3	MADE BY GARY HARRIS		I.C. NO.		CONT. ON SH. FL. SH. NO. 3.3



PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

DRAWN A. WRIGHT MAY 16, 71	REV 4	REV 5	PRINTS TO DL1. P6A 1338	APPROVALS	GENERAL ELECTRIC
REV 2	ISSUED 1-13-71		FIRST MADE FOR PLASTIC MOLDING		INDUSTRY CONTROL DEPT. SALEM, VA., U.S.A.
REV 3	MADE BY KENNEY C. COX		I.C. NO. 3622		

ELEMENTARY DIAGRAM

EJECTOR SEQUENCE LOGIC

I C 3 6 2 2 G E J A 1

CONT. ON SH. 3.1 SH. NO. 3

APPLICATION NOTES - HYDRAULIC EJECTOR

I. GENERAL DESCRIPTION

A. THE STANDARD EJECTORS CARD CONTAINS A UNIVERSAL CIRCUIT WITH INPUT AND OUTPUT LOGIC FLEXIBILITY TO PERFORM EJECTOR OPERATIONS. AVAILABLE FEATURES ARE:

1. CHOICE OF ONE OF FOUR (4) MODES OF OPERATION
   A. NORMAL SINGLE SHOT
   B. PULSATING EJECTORS (FULL OUT AND BACK REPETITIVE PULSING)
   C. DELAYED EJECTORS FORWARD
   D. COMBINATION OF ABOVE MODES
2. CHOICE OF SOLENOID OR SPRING RETURN SYSTEM
3. COMPATIBILITY WITH CIRCUIT FOR SPI STANDARD CORE SEQUENCE
4. SOLENOID RETURN SYSTEM MONITOR CIRCUIT CAN INSURE:
   A. IF IN AUTOMATIC MODE, EJECTORS REMAIN IN BACK POSITION UNTIL EJECT IS INITIATED
   B. EJECTORS REMAIN BACK CONTINUOUSLY WHEN TURNED OFF
5. EJECT INITIATED AT CLAMP BACK (OR OTHER POSITION SUCH AS SLOW OPEN) IF THE INITIATE SIGNAL SHOULD FAIL.
6. BAND OF OPERATION FOR MANUAL MODE BETWEEN THE FORWARD AND BACK LIMIT SWITCHES
7. INHIBIT CIRCUIT FOR MACHINE RECYCLE UNDER FOLLOWING CONDITIONS:
   A. IF EJECTORS HAVE NOT GONE FULL FORWARD
   B. IF PULSATING CYCLE IS RUNNING
   C. IF DELAY TIMER IS TIMING
   D. IF EITHER FORWARD OR BACK LIMIT SWITCH HAS FAILED
   E. SELECTED CONDITION SUCH AS IF EJECTORS ARE NOT BACK

B. NOMENCLATURE FOR THE EJECTOR CIRCUIT INPUT AND OUTPUT SIGNALS ARE CONSISTANT WITH THE STANDARDS FOR PLASTIMATIC CONTROL (SEE APPLICATION SHEET DP-N1 OR THE BACK SHEET OF THE CONTROL ELEMENTARY DRAWING FOR DEFINITIONS.)

C. PIN DESIGNATIONS FOR "PREPLANNED" OPTIONAL OPERATION CARRY NOMENCLATURE FOR THE INPUTS REQUIRED. IF A PLANNED OPTION IS NOT IMPLEMENTED ITS INPUTS MAY BE USED FOR OTHER OPTIONS.

II. RETURN SOLENOID - BASIC OPERATION

A. GENERAL

BASIC OPERATION IS DEFINED AS A SINGLE STROKE OF THE EJECTORS. A FORWARD AND A RETRACT SOLENOID AND AN EJECTORS FORWARD AND AN EJECTORS BACK LIMIT SWITCH ARE USED. BOTH AUTOMATIC AND MANUAL MODE ARE PROVIDED. AN INITIATE EJECTORS FORWARD SIGNAL IS REQUIRED FOR THE AUTOMATIC MODE. THIS SIGNAL CAN BE FROM AN EJECT POSITION LIMIT SWITCH OR FROM OTHER SOURCE SUCH AS CLAMP OPEN. OUTPUT SIGNALS ARE PROVIDED TO ENERGIZE THE FORWARD AND RETRACT SOLENOIDS. AN OUTPUT SIGNAL IS ALSO AVAILABLE TO INHIBIT THE NEXT MACHINE CYCLE UNTIL THE EJECTION STROKE IS COMPLETE.

B. AUTOMATIC MODE

1. INITIAL CONDITIONS
   A. EJECTORS ON (1EO)
   B. MACHINE IN AUTOMATIC RUN PERMISSIVE (1AUTP)
   C. EJECTORS NOT FORWARD (1EF1 A ZERO)

FORM IC 2219 (3-65)

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 Chgd sh.# from 2.2 to 3.1	REV. 2 DL1	APPROVED R E Sarus	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM EJECTOR SEQUENCE	
ISSUED Jan. 12, 71	PRINTS TO P&A	FIRST MADE FOR MFR. PLASTIC MOLDING		IC3622GEJA1	SH. NO. 3.1
MADE BY KENNEY C. COX	1338	I.C. NO.		CONT. OR SH. 3.2	

2. OPERATION

A. THE MACHINE WILL CYCLE NORMALLY TO THE END OF CURE TIME, THEN BEFORE DIE LOCK UP SIGNAL 1DLU IS LOST, IT AND THE CLAMP OPEN PERMISSIVE SIGNAL (1CLOP) WILL SET A SEAL CIRCUIT WHICH MAKES THE INITIATE EJECTORS CIRCUIT PERMISSIVE.

B. THE INITIATE EJECTORS SIGNAL (OIE) WILL SET A SEAL CIRCUIT THAT CALLS FOR EJECTORS FORWARD.

C. IF SAFETY CONDITIONS OF DIE NOT LOCKED UP (ODLU A ONE) AND CORES PERMISSIVE (OICRP) ARE MET THE DRIVE EJECTORS FORWARD SIGNAL (ODEF) WILL BE PROVIDED TO THE EJECTORS FORWARD SOLENOID DRIVER UNTIL THE FORWARD LIMIT SWITCH IS ACTIVATED. ODEF WILL INHIBIT THE DRIVE EJECTORS BACK SIGNAL (ODEB).

D. EJECTORS FORWARD LIMIT SWITCH SIGNAL (OEF1) WILL DE-ENERGIZE THE SOLENOID DRIVER AND THE 1EF1 SIGNAL WILL RESET THE SEAL CIRCUITS ESTABLISHED IN STEPS A AND B.

E. ONCE THE ODEF SIGNAL IS LOST THE EJECTORS NOT BACK SIGNAL (OEB A ONE) WILL INITIATE A DRIVE EJECTORS BACK SIGNAL (ODEB) TO THE EJECTORS RETRACT SOLENOID DRIVER. ODEB SIGNAL WILL CONTINUE UNTIL EJECTORS BACK LIMIT SWITCH IS ACTIVATED AND SIGNAL OEB IS A ZERO.

F. THE CYCLE START INHIBIT SIGNAL (OCSI) IS PROVIDED TO INHIBIT ANOTHER MACHINE CYCLE UNTIL THE EJECTORS ARE PERMISSIVE (SEC I.A.7)

C. MANUAL MODE

1. INITIAL CONDITIONS

A. EJECTORS ON (1EO)

B. MACHINE IS MANUAL RUN PERMISSIVE (1MRP)

C. CORES (IF PROVIDED) ARE PERMISSIVE (OICRP IS A ONE)

2. EJECT CYCLE

A. PERMISSIVE CONDITIONS CAN BE SELECTED BY CONNECTING INPUTS

1. OEF2 - INHIBIT ODEF ONCE EJECTORS ARE FORWARD
2. ODLU2 - INHIBIT ODEF WHEN DIES ARE LOCKED UP
3. 1IE - PERMIT MANUAL OPERATION ONLY AT THE INITIATE EJECTORS POSITION
4. 1S3 - OPTIONAL INPUT AVAILABLE

B. MANUAL EJECT FORWARD SIGNAL (1MEF) WILL INITIATE DRIVE EJECTORS FORWARD SIGNAL ODEF WITHIN LIMITS ESTABLISHED BY STEP A.

3. RETRACT CYCLE

A. PERMISSIVE CONDITIONS ARE MET

1. NOT EJECTING FORWARD (ODEF IS A ONE)
2. EJECTORS DELAYED EITHER COMPLETED TIMING OR NOT USED
3. OEB - INHIBIT EJECTORS BACK SIGNAL (ODEB) ONCE EJECTORS ARE BACK

B. OTHER PERMISSIVE CONDITIONS THAT CAN BE SELECTED BY CONNECTING INPUTS

1. 1S6 - OPTIONAL INPUT AVAILABLE
2. 1IE AND 1S3 SAME AS 2. A, 3. 4, 4.

C. MANUAL EJECTOR BACK SIGNAL (1MEB) WILL INITIATE DRIVE EJECTORS BACK SIGNAL (ODEB) WITHIN LIMITS ESTABLISHED BY STEP B.

III. DELAYED EJECTORS FORWARD - OPTIONAL OPERATION

A. GENERAL

OPERATION IS THE SAME AS FOR BASIC OPERATION EXCEPT THAT THE DRIVE EJECTORS BACK SIGNAL ODEB IS DELAYED BEYOND THE TIME REQUIRED FOR THE EJECTORS TO REACH THE EJECTORS FORWARD LIMIT SWITCH. EJECTOR CIRCUIT WILL PROVIDE A TIME DELAY INITIATE SIGNAL OITDE AND WILL INHIBIT THE ODEB SIGNAL UNTIL THE TIMER TIMED OUT SIGNAL OTTDE IS RECEIVED AS AN INPUT.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 Chg'd. Sh.# from 3.3 to 3.2	REV. 2	P6A	APPROVED RC Lacey	GENERAL ELECTRIC	ELEMENTARY DIAGRAM EJECTOR SEQUENCE
ISSUED Jan. 12, 71.	PRINTS TO DLI	FIRST MADE FOR MDL PLASTIC MOLDING		INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	IC3622GEJA1
MADE BY KENNEY C. COX	1338	LC NO.			CONT. DWG. SH. 3.3 SH. NO. 3.2

FORM IC 2219 (65)

B. INITIAL CONDITIONS

1. EJECTORS ON (1EO)
2. MACHINE SET FOR AUTOMATIC RUN PERMISSIVE (1AUTP)
3. PULSATING EJECTORS SELECTED (1PON)
4. JUMPER OPEJ1 TO OPEJ2 ADDED TO CONNECT IN CIRCUIT

C. OPERATION

1. THE PULSATING EJECTORS ON SIGNAL 1PON AND THE PULSATING CYCLE NOT COMPLETE (OPST A ONE) WILL PROVIDE AN OVERRIDE SIGNAL OF 1EF1 TO THE EJECTORS FORWARD CIRCUIT SO THE EJECTORS WILL KEEP CYCLING FORWARD UNTIL THE PULSATING CYCLE IS COMPLETE OPST GOES TO A ZERO AND PERMITS THE 1EF1 SIGNAL TO FUNCTION TO RESET THE SEAL CIRCUITS PROHIBITING EJECTORS FORWARD OPERATION UNTIL ESTABLISHED BY THE NEXT MACHINE CYCLE.

V. SPRING RETURN EJECTORS - OPTIONAL OPERATION

A. OPERATION IS THE SAME AS FOR RETURN SOLENOID EXCEPT THAT BY NOT HAVING THE SOLENOID THE ODEB SIGNAL IS NOT USED AND INPUTS FOR THAT PORTION OF THE ARE NOT REQUIRED. OEB SIGNAL OR 1S6 TO COMMON ASSURES THAT ODEB DOES NOT INTEFERE WITH LOGIC OPERATION.

B. OPERATION FOR DELAYED EJECTORS FORWARD REQUIRES ADDING JUMPER OPEP-1SS1 SO TIMER INITIATE SIGNAL MAINTAINS ODEF SIGNAL TO HOLD EJECTORS FORWARD AGAINST THE SPRING UNTIL THE DELAY TIME IS OVER.

C. OPERATION FOR PULSATING EJECTORS REQUIRES THE OEB SIGNAL TO INSURE FULL STROKE OPERATION.

V1. OPERATION WITHOUT FORWARD LIMIT SWITCH - OPTIONAL
OPERATION IS POSSIBLE WITHOUT THE FORWARD LIMIT SWITCH BY USING THE DELAY TIMER TIMED OUT SIGNAL AS THE EJECTORS FORWARD SIGNAL.

INPUT LOADING

PINS 5-16, 22, 23, 34-37, 42, 44, 45, 48, 49, 50 } 1 LOAD EACH

PINS 46, 47 - 2 LOADS EACH

PINS 17, 19, 20, 21, 30, 32, 40 - 3 LOADS EACH

PINS 18, 33 4 LOADS EACH

OUTPUT DRIVE

ODEF(39)	30 LOADS
ODEB(24)	16
OCSI(43)	13
OEFS(18)	13
OITDE(25)	18
OPEJ1(E)	18
1SGO(4)	18

POWER SUPPLY - +15 VOLTS AT 25MA

FORM IC 2219 (65)

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 dgl. sht. from 3.4 to 3.3-R	REV. 2 DL1	APPROVED R E Lewis	GENERAL ELECTRIC	ELEMENTARY DIAGRAM EJECTOR SEQUENCE	
ISSUED	PRINTS TO P6A	FIRST MADE FOR MFG. PLASTIC MOLDING	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	IC3622GEJA1	SH. NO. 3.3
MADE BY KENNEY C. COX	1338	I.C. NO.		CONT. ON SH.	

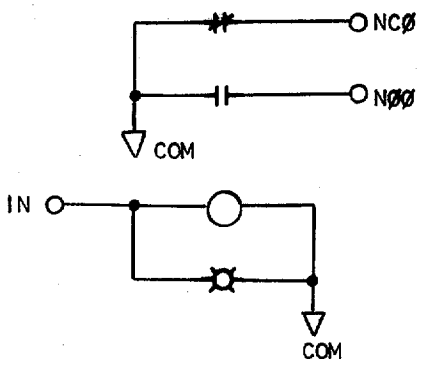


FIGURE A (SYMBOL FOR FIBURE B)

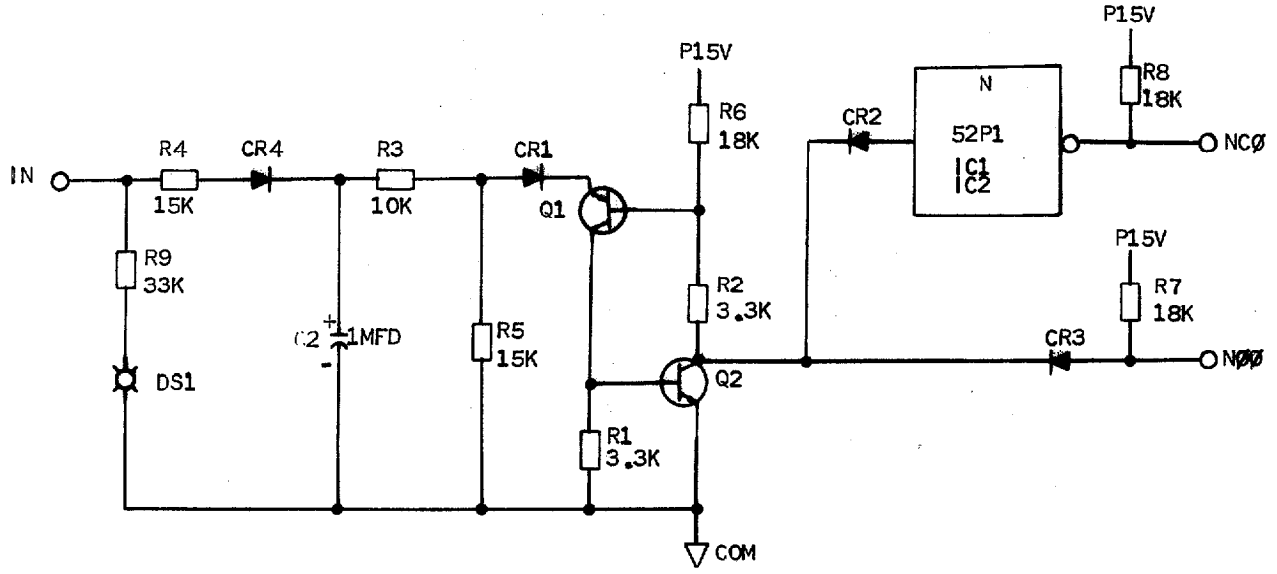
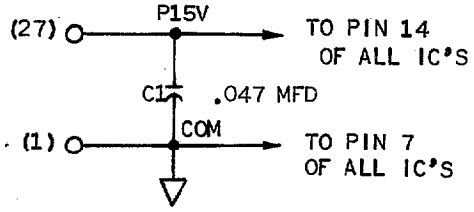


FIGURE B 8 CIRCUITS PER CARD



CKT.	IC	ICIN	ICOUT	IN	NCØ	NØØ
1	1	5	6	34	2	3
2	1	3	4	36	4	5
3	1	1	2	38	6	7
4	1	9	8	40	8	9
5	2	3	4	42	10	11
6	2	5	6	44	12	13
7	2	9	8	46	14	15
8	2	13	12	48	16	17

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO 6V,DL1,3T	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM: INPUT BUFFER
REV. 2	ISSUED 5/2/72		W AY P6A,1338 CHVIVN	FIRST MADE FOR PLASTIC MOLD	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	I C 3 6 2 2 G I B B 1
REV. 3	MADE BY D. M. MILLER		A6F,A1F(4)	I.C. NO.		CONT. ON SH. 3.1 SH. NO. 3.0

APPLICATION INFORMATION

FUNCTIONS AND THEORY OF OPERATIONS

THIS CARD FUNCTIONS IN RELAY TERMS AS SHOWN IN FIGURE A. WHEN THE LINE SUPPLY VOLTAGE IS APPLIED TO THE INPUTS THE COIL IS ENERGIZED, THE NORMALLY OPEN CONTACT CLOSES TO COMMON, THE NORMALLY CLOSED CONTACT OPENS FROM COMMON, AND THE LIGHT TURNS ON.

THE CARD CONVERTS THE LINE SUPPLY VOLTAGE INTO LOGIC LEVELS. WHEN LINE VOLTAGE IS APPLIED TO THE INPUT THE LIGHTS TURN ON AND A CURRENT LIMITED HALF-WAVE SIGNAL CHARGES THE CAPACITOR C2. WHEN THE VOLTAGE ON THE CAPACITOR REACHES THE HIGH THRESHOLD (NOMINALLY 27 VOLTS) Q1 WILL TURN ON TURNING Q2 ON. WHEN Q2 TURNS ON IT PULLS THE THRESHOLD DOWN (NOMINALLY 6 VOLTS) TURNING Q1 ON HARDER PROVIDING SNAP ACTION AND HYSTERESIS FOR THE BUFFER. WHEN Q2 IS TURNED ON NØØ IS A LOGIC "0" AND NCØ IS A LOGIC "1". WHEN OFF NØØ IS A LOGIC "1" AND NCØ A LOGIC "0".

SPECIFICATIONS

INPUTS

75 VAC TO 130 VAC TO ENERGIZE BUFFER.

ØVAC TO 20 VAC TO DE-ENERGIZE BUFFER.

5 MILLIAMPERES - MAXIMUM AVERAGE CURRENT

OUTPUTS

(NCØ,NØØ) LOGIC "0" - 1.0 VOLTS MAX.

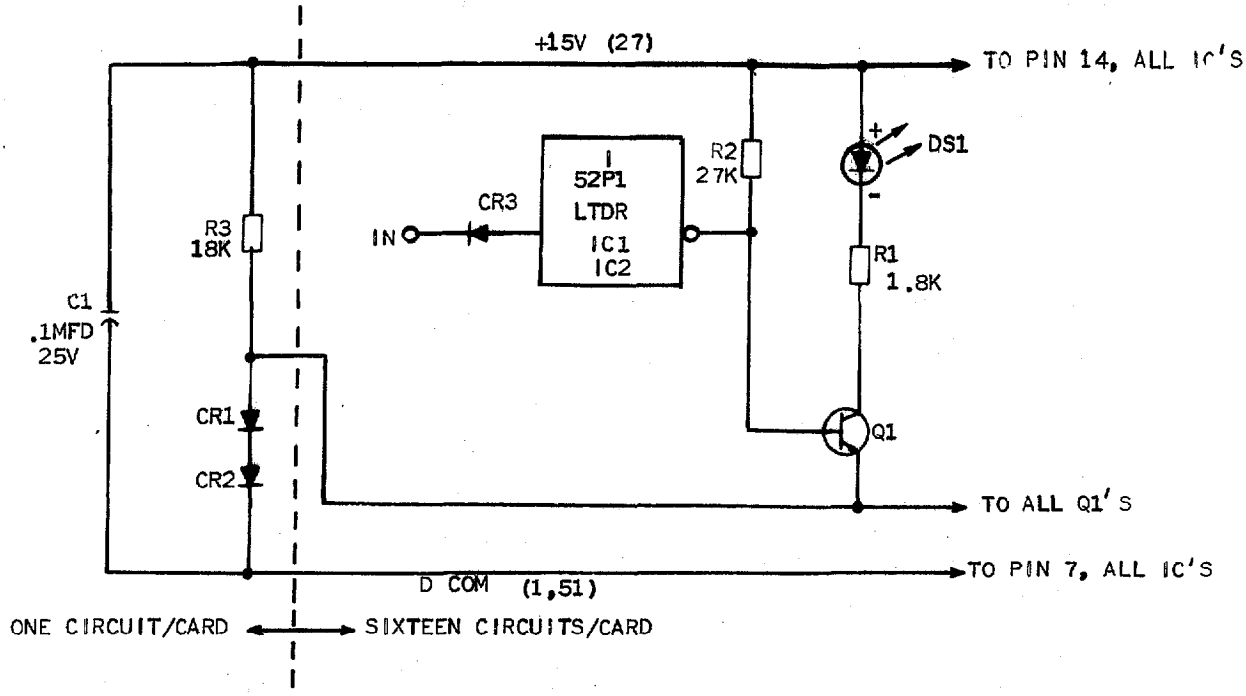
LOGIC "1" - 13.5 VOLTS MIN.

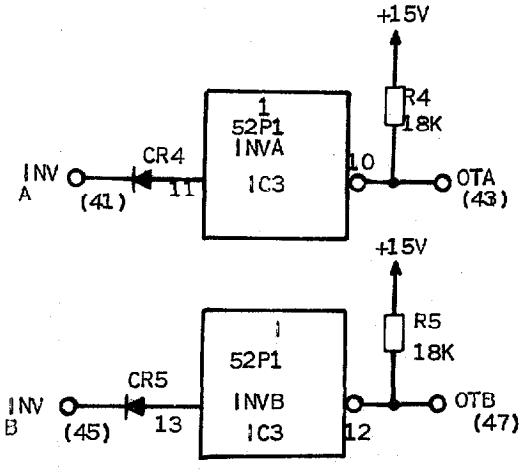
EACH OUTPUT CAN DRIVE 16 "ONE INPUTS" 18K LOADS TO P15V.

P15V - 14V TO 16V, 47 MA MAX, 24 MA TYPICAL.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV 1	REV 4	REV. 5	PRINTS TO 6V,DL1,3T	APPROVALS T.W. Logan	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM INPUT BUFFER
REV. 2	ISSUED 5/2/72		P6A,1338 WAY	FIRST MADE FOR PLASTIC MOLD		IC3622GIBB1
REV. 3	MADE BY D. M. MILLER		CHVIVN A6F,A,IF(4)	I.C. NO		CONT. ON SH FL. SH. NO 3.1





CIRCUIT CONNECTIONS

CIRCUIT	CARD INPUT	IC NUMBER	IC INPUT PIN	IC OUTPUT PIN
LT-1	2	IC1	1	2
LT-2	4	IC1	3	4
LT-3	6	IC1	5	6
LT-4	8	IC1	9	8
LT-5	10	IC1	11	10
LT-6	12	IC1	13	12
LT-7	14	IC2	1	2
LT-8	16	IC2	3	4
LT-9	18	IC2	5	6
LT-10	20	IC2	9	8
LT-11	30	IC2	11	10
LT-12	32	IC2	13	12
LT-13	34	IC3	1	2
LT-14	36	IC3	3	4
LT-15	38	IC3	5	6
LT-16	40	IC3	9	8

APPLICATION NOTES

THERE ARE SIXTEEN INDICATING LAMPS (LIGHT EMITTING DIODE TYPE) ON THIS CARD. THE LIGHT COMES ON WHEN THE INPUT IS TIED TO COMMON.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

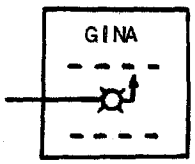
REV. 1	REV. 4	REV. 5	PRINTS TO A6F 3T, DL1	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM LED INDICATOR
REV. 2	ISSUED 1-17-72		P6A, A1F (4) 1338, 6V CAVITY	FIRST MADE FOR PM 1000		I C 3 6 2 2 G 1 N A 1
REV. 3	MADE BY KENNEY C. COX			I.C. NO. 3622		CONT. ON SH. 3.1 SH. NO. 3.0

APPLICATION NOTES

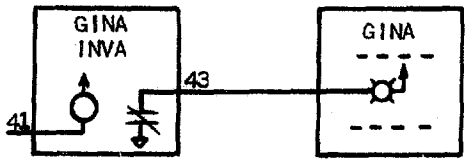
LIGHT IS ON WHEN INPUT IS 0 TO 4 VOLTS. LIGHT IS OFF WHEN INPUT IS 10 TO 15 VOLTS.
EACH INPUT IS ONE 18K LOAD, OR 0.82 MA TO COMMON.
INPUT POWER: +15V, 40 TO 55 MA, ALL LIGHTS OFF
+15V, 100 TO 130 MA, ALL LIGHTS ON.
INV A OR INV B MAY BE USED TO CONVERT ONE OR TWO LAMPS TO LIGHT ON AND OPEN INPUT INSTEAD OF A GROUNDED INPUT

SYMBOLS:

INPUT GROUNDED, LIGHT ON

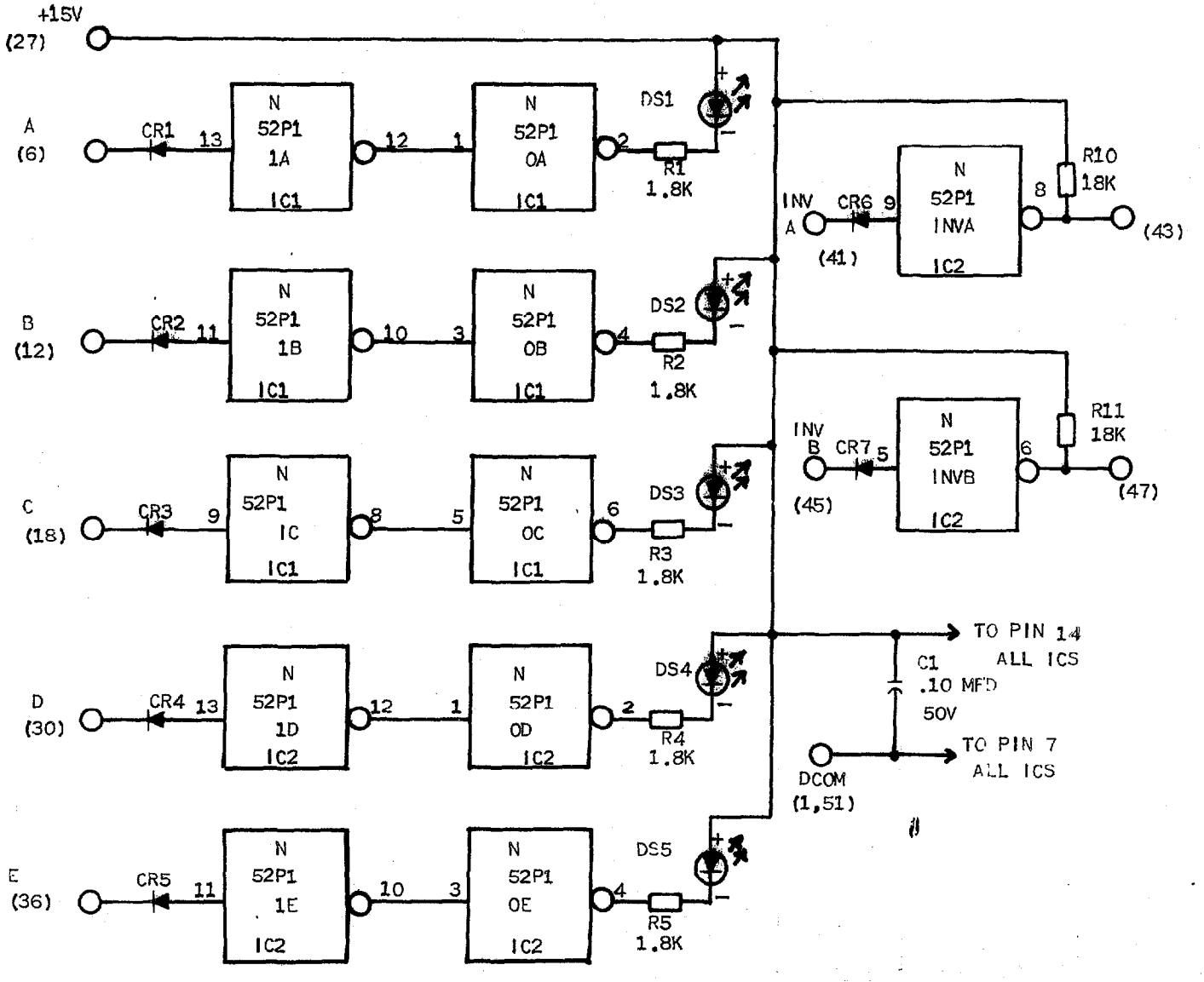


INPUT OPEN, LIGHT ON



PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO GV A6F 3T DL1	APPROVALS R E Flores	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM LED INDICATOR
REV. 2	ISSUED 4-17-72		P6A, A1F(4)	FIRST MADE FOR PM 1000		IC3622GINA1
REV. 3	MADE BY KENNEY C. COX WAY		CNVVN, 1338	I.C. NO. 3622		CONT. ON SH. FL. SH. NO. 3.1



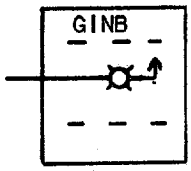
CIRCUIT	IN PIN	OUT PIN
A (TOP)	6	
B	12	
C	18	
D	30	
E (BOTTOM)	36	
INVA	41	43
INVB	45	47
+15V	27	
DCOM	1,51	
CARD GND	5	

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

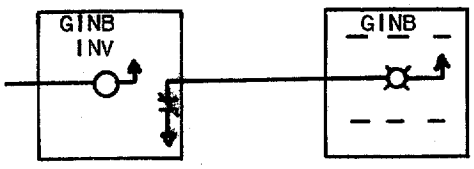
REV. 1	REV. 4	REV. 5	PRINTS TO A6F, P6A	APPROVALS LBB For REG	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 2	ISSUED 8-4-72	AIF(4),DL1,3T 1388,2530	FIRST MADE FOR PLAS. MOLD.		ELEMENTARY DIAGRAM LED INDICATOR
REV. 3	MADE BY W. M. HURD	CHVLVN, WBY	I.C. NO.		I C 3 6 2 2 G I N B CONT. ON SH. 3.1 SH. NO. 3.0

SYMBOLS:

LIGHT ON GROUNDED INPUT:



LIGHT ON AN OPEN INPUT:



APPLICATION NOTES

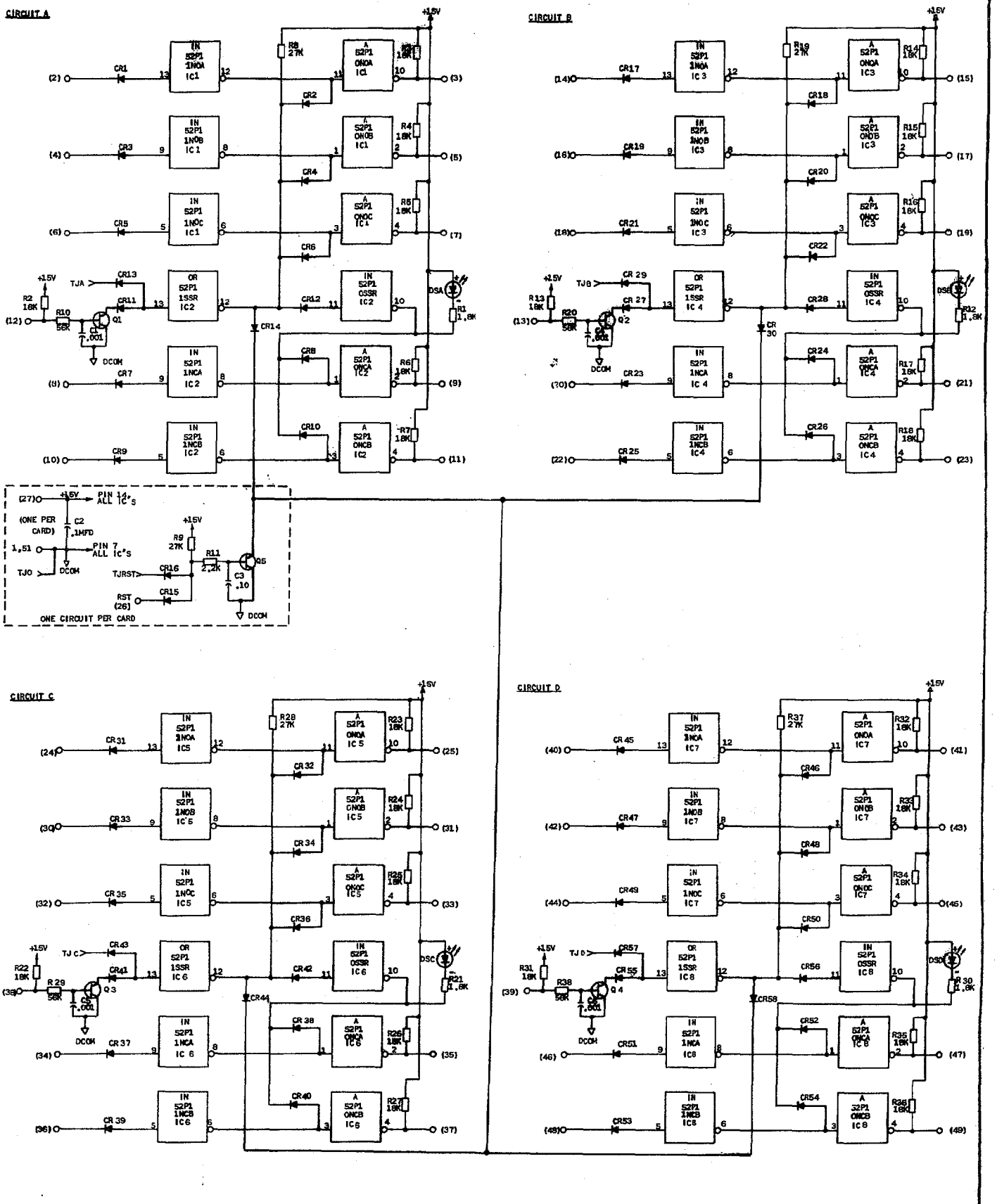
THERE ARE FIVE LIGHT EMITTING DIODE INDICATORS ON THIS CARD. THE INDICATORS ARE MOUNTED ON A METAL CARD FRONT AND MAY BE INDIVIDUALLY LABLED. THE CARD OCCUPIES ONE CARD ROW SLOT . THE INDICATORS TURN ON WHEN THE INPUT IS OV (GROUNDED TO DCOM.). TWO SPARE INVERTERS ARE PROVIDED SO THAT UP TO TWO INDICATORS CAN BE CONVERTED TO LIGHT "ON" ON AN OPEN (10 TO 15V) INPUT.

INPUT POWER:

+15V,	ALL LEDS OFF	16 MA TYPICAL, 42 MA MAX
+15V,	ALL LEDS ON	52 MA TYPICAL, 90 MA MAX

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO A6F, P6A, 3T	APPROVALS LGS For REG	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM LED INDICATOR
REV. 2	ISSUED 8-4-72		AIF(4) 2530 1338.005	FIRST MADE FOR PLAST. MOLD.		IC3622GINB
REV. 3	MADE BY W. M. HURD		CHVIVN, WGY	I.C. NO.		CONT. ON SH. FL. SH. NO. 3-1



REV.1	REV.4	REV.5	PRINTS TO WAY DL1,A6F	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM LOGIC RELAY 3N.O. 2N.C.
REV.2	ISSUED May 11, 1972		AIF(4),3T 6V,P6A	FIRST MADE FOR PLASTIC MOLDING	INDUSTRY CONTROL DEPT.	
REV.3	MADE BY S. B. HILL		CHVIVN 1338	LC. NO.	SALEM, VA., USA	IC3622GLEB1 CONT. ON SH. 3, SH. NO. 3

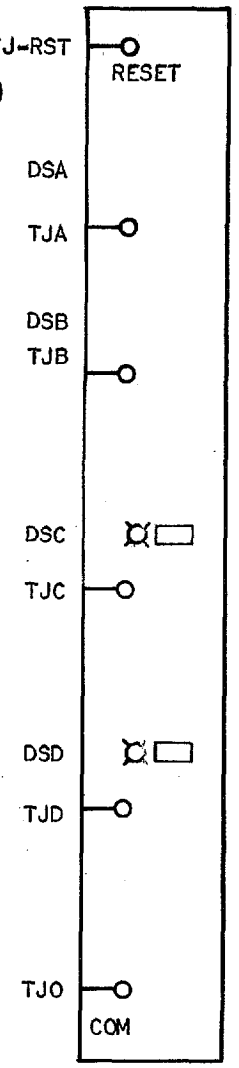
FORM IC 2349 (07-69)

A B C D E F G H J K L

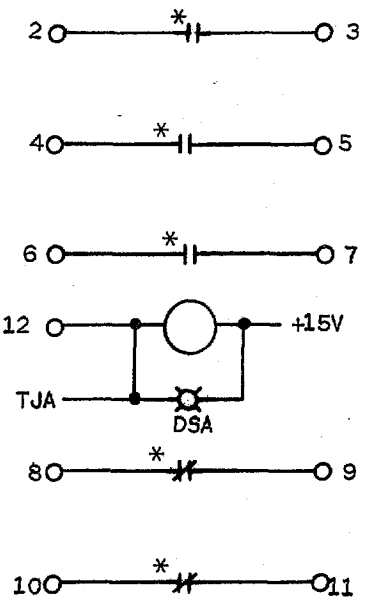
SYMBOL AND PIN ASSIGNMENT

LAMP AND TEST JACK ASSIGNMENT

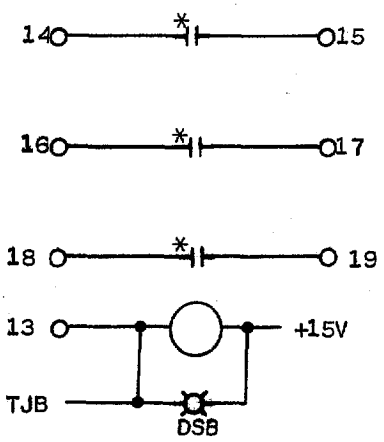
NOTE: SHOWS CUSTOMER NOMENCLATURE LABELS.



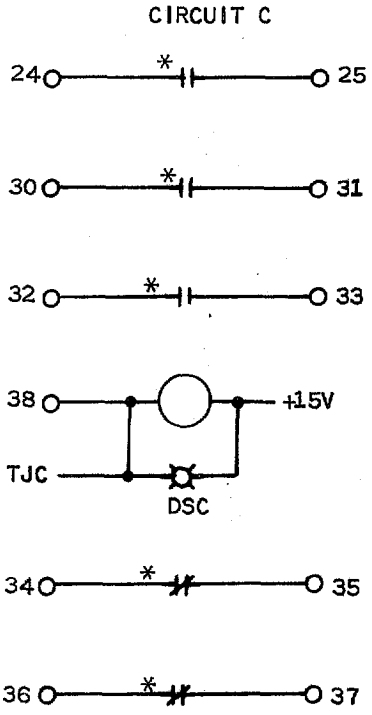
CIRCUIT A



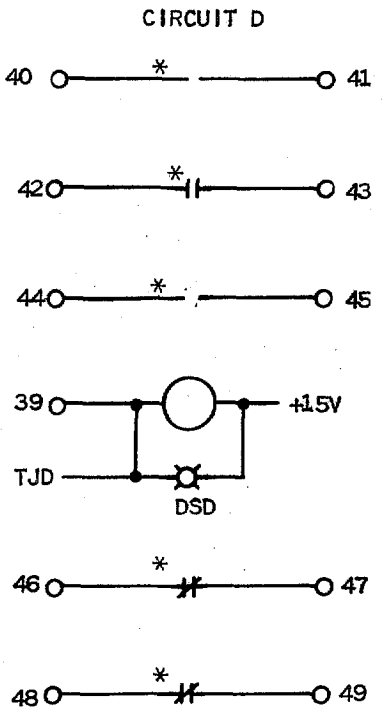
CIRCUIT B



CIRCUIT C



CIRCUIT D



NOTE:
TO PICK UP RELAY FOR TEST PURPOSES ONLY, JUMPER APPROPRIATE TEST JACK TO COMMON. LIGHT INDICATES PICKED UP STATE.

RESET IS (26)

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL1 6V.3T CHVIVN P6A.1338 WAY	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM RELAY 3NO-2NC
REV. 2	ISSUED May 1, 1972			FIRST MADE FOR PLASTIC MOLD		I C 3 6 2 2 G L E B 1
REV. 3	MADE BY S. B. HILL		AIF(4),A6F	I.C. NO.		CONT. ON SH. 3.2 SH. NO. 3.1

APPLICATION NOTES

IC3622GLEB1

THIS CARD CONTAINS FOUR FLOW-THROUGH LOGIC FUNCTIONS OF THE CONFIGURATION SHOWN ON SHEET 3.4. THE FLOW-THROUGH LOGIC CONCEPT ALLOWS SOLID STATE LOGIC TO BE DESIGNED USING RELAY NOMENCLATURE. "OR" FUNCTIONS ARE IMPLEMENTED BY PARALLEL CONTACTS, AND "AND" FUNCTIONS BY SERIES CONTACTS. THESE "COILS" AND "CONTACTS" ARE ACTUALLY SOLID STATE GATES ARRANGED AS FOLLOWS:

1) A NORMALLY OPEN CONTACT TRANSFERS ITS INPUT VOLTAGE TO ITS OUTPUT WHEN THE COIL IS PICKED UP. WHEN DROPPED OUT, THE OUTPUT IS A LOGIC "1" WHICH CORRESPONDS TO AN OPEN CIRCUIT.

2) A NORMALLY CLOSED CONTACT TRANSFERS ITS INPUT VOLTAGE TO THE OUTPUT WHEN THE COIL IS DROPPED OUT. WHEN PICKED UP, THE OUTPUT IS A LOGIC "1", OR OPEN.

3) THE COIL IS PICKED UP WHEN THE COIL INPUT IS AT 0 VOLTS. A LIGHT EMITTING DIODE INDICATES WHEN THE RELAY IS IN THE PICKED UP STATE.

4) ANY RELAY CAN BE OPERATED MANUALLY BY CONNECTING THE APPROPRIATE TEST POINT (ON CARD FRONT) TO COMMON, TP0

5) ALL RELAYS CAN BE FORCED TO DROP OUT BY CONNECTING RST TO COMMON. THIS IS ALSO AVAILABLE AS A TEST POINT ON THE CARD FRONT (RESET).

THE LOGIC TRUTH TABLE IS SHOWN BELOW:

COIL INPUT	N.O. INPUT	N.O. OUTPUT	N.C. INPUT	N.O. OUTPUT
1	1	1	1	1
1	0	1	0	0
0	1	1	1	1
0	0	0	0	1

THESE ARE HIGH LEVEL LOGIC, MINIMUM "1" IS 10V, MAXIMUM "0" IS 4V

THE FOLLOWING LOGIC RULES MUST BE APPLIED:

1) ALL CONTACT STRINGS START ON EITHER 0 VOLTS OR AT AN INPUT BUFFER OUTPUT WHICH IS 0 FOR A TRUE INPUT.

2) CONTACTS ARE NOT BI-DIRECTIONAL. LOGIC SIGNAL FLOW IS MAINTAINED IN THE DIRECTION FROM CONTACT INPUTS (SHOWN AS *, EVEN NUMBERED PINS) TO OUTPUTS (UNSTARRED, ODD NUMBERED PINS.)

IN GENERAL, RELAY TREES CAN BE DIRECTLY COPIED IN SOLID STATE. BRIDGE CIRCUITS, WHICH DEPENDS ON TWO WAY SIGNAL FLOW-THROUGH SOME CONTACTS CANNOT BE IMPLEMENTED DIRECTLY. THEY MUST FIRST BE CHANGED TO THE EQUIVALENT TREE CIRCUIT.

3) CONTACT STRINGS END ON EITHER A COIL INPUT OR A DRIVER CIRCUIT.

4) LOGIC LOADING RULES APPLY. EACH COIL, CONTACT INPUT, AND CONTACT OUTPUT IS ONE LOAD. EACH CONTACT OUTPUT CAN DRIVE SEVENTEEN EXTERNAL LOADS, PLUS THE ONE LOAD INTERNAL TO ITSELF. APPLY THIS RULE BY COUNTING THE TOTAL NUMBER OF CONTACTS AND COILS CONNECTED TOGETHER AT ANY CIRCUIT NODE. THE NUMBER MUST BE LESS THAN OR EQUAL TO EIGHTEEN.

5) TIMING CONSIDERATIONS. EACH COIL GATE CONTAINS A TIME DELAY WHICH SLOWS DOWN BOTH THE PICKUP AND DROP OUT RATES OF THE RELAYS. THIS SIMULATES THE RELAY TIMING SEQUENCE, ALTHOUGH THE ACTUAL PICKUP AND DROP OUT TIMES (APPROXIMATELY 40 MICROSECONDS) ARE MUCH FASTER THAN ELECTROMECHANICAL RELAYS.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL1, 6V, 3T	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM LOGIC RELAY, 3NO-2NC
REV. 2	ISSUED May 1, 1972		CHVIVN P6A, 1338	FIRST MADE FOR PLASTIC MOLD	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	IC3622GLEB1
REV. 3	MADE BY S. B. HILL		AIF(4), A6F	I.C. NO.		CONT. ON SH. 3.3 SH. NO. 3.2

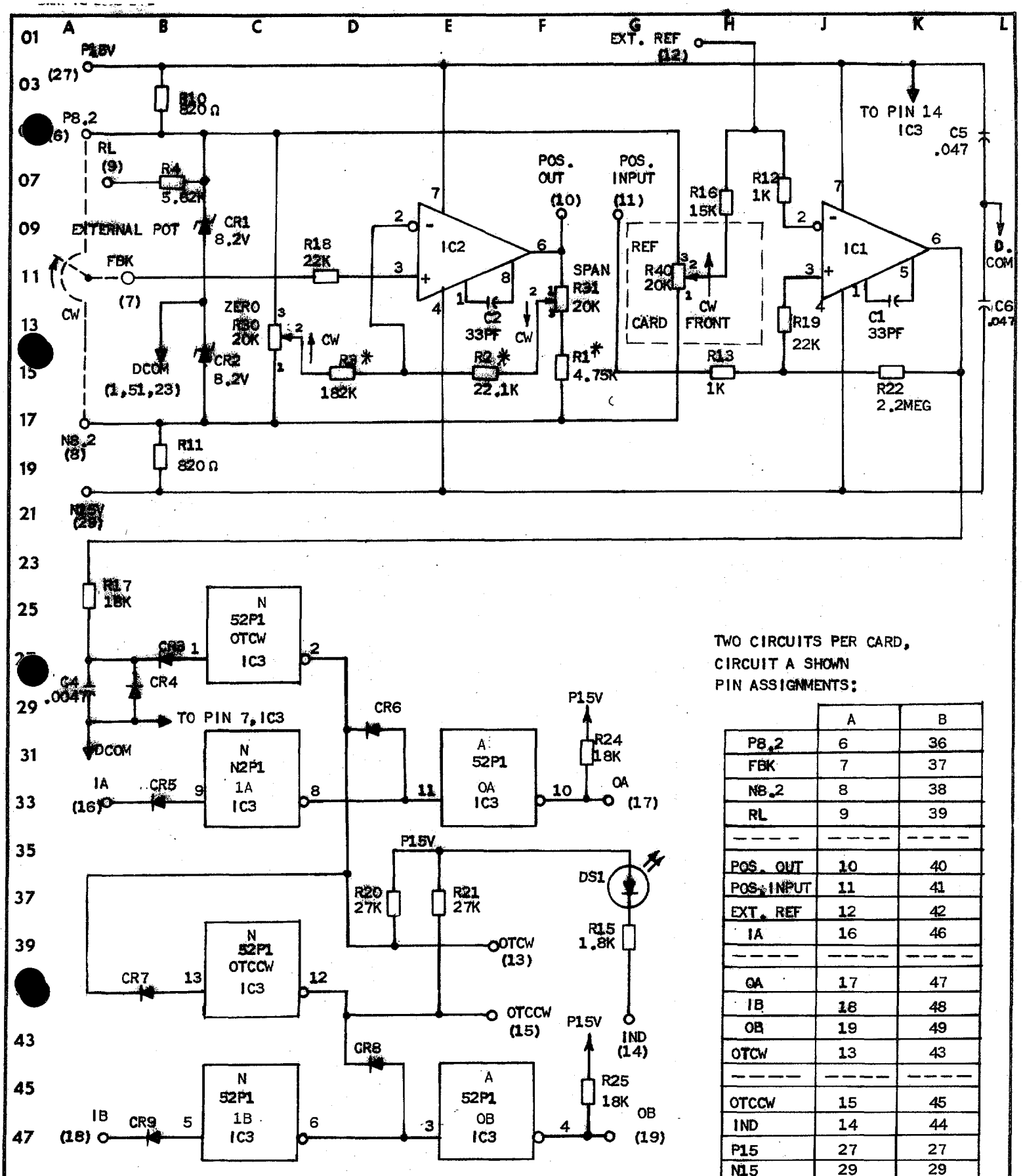
6) LATCH CIRCUITS. AN ADDITIONAL INPUT, RST, IS PROVIDED ON THE CARD. PULLING THIS TO ZERO FORCES ALL RELAYS ON THE CARD TO DROP OUT. THIS IS EQUIVALENT TO REMOVING ALL POWER FROM THE RELAY COILS. IF LATCH CIRCUITS ARE WIRED USING THE CARD, RST SHOULD BE USED AS A POWER APPLIED RESET TO PREVENT THE LATCH CIRCUITS FROM COMING UP SET. THE GTDB CARD CAN BE USED TO HOLD RST AT ZERO FOR APPROXIMATELY A HALF SECOND ON APPLICATION OF POWER. THIS MAKES CERTAIN THE CONTACT GATES ARE FUNCTIONING BEFORE ANY COILS CAN BE PICKED UP. RST IS ONE LOAD.

POWER SUPPLY: 15V, PIN 27
CURRENT: ALL COILS OFF -
65 MA TYPICAL, 110 MA MAXIMUM

ALL COILS PICKED UP -
100 MA TYPICAL, 155 MA MAXIMUM

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL1,6V,3T	APPROVALS R.E. Davis	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM LOGIC RELAY, 3NO-2NC
REV. 2	ISSUED May 1, 1972		CHVIVN P6A,1338	FIRST MADE FOR PLASTIC MOLD		IC3622GLEB1
REV. 3	MADE BY S. B. HILL		WAY A1F(4),A6F	I.C. NO.		CONT. ON SH. FL. SH. NO. 3.3



TWO CIRCUITS PER CARD,
CIRCUIT A SHOWN
PIN ASSIGNMENTS:

	A	B
P8.2	6	36
FBK	7	37
N8.2	8	38
RL	9	39
----	----	----
POS. OUT	10	40
POS. INPUT	11	41
EXT. REF	12	42
IA	16	46
----	----	----
OA	17	47
IB	18	48
OB	19	49
OTCW	13	43
----	----	----
OTCCW	15	45
IND	14	44
P15	27	27
N15	29	29
DCOM	1,51,23	1,51,23
GROUND	5	5

NOTE: ONLY ONE C5 AND C6 PER CARD.

* DENOTES PRECISION RESISTORS.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4 DL2,3T	REV. 5 2520, 1930, PCA,	PRINTS TO SF, A,F (3) 5V	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 2	ISSUED 8/9/72		CHVIVN WAY	FIRST MADE FOR PLAST. MOLD	
REV. 3	MADE BY GARY HARRIS			I.C. NO.	

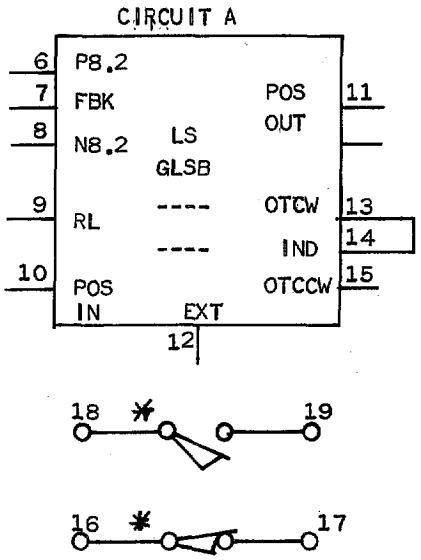
ELEMENTARY DIAGRAM
LIMIT SWITCH(PRECISION)

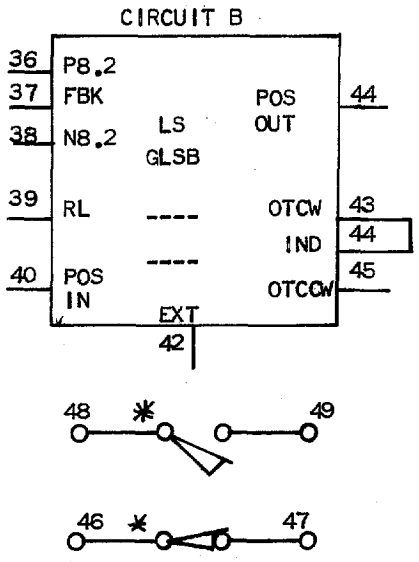
I C 3 6 2 2 G L S B 1

CONT. ON SH. 3,1 SH. NO. 3,0

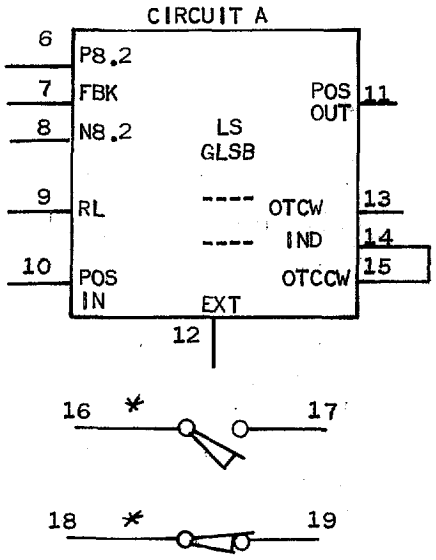
SYMBOLS:

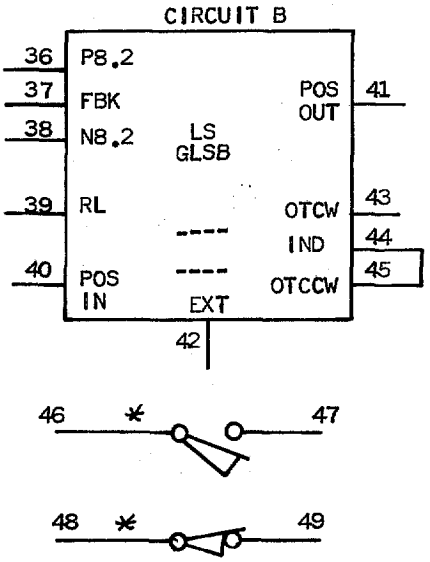
CLOCKWISE TRIP:





COUNTER CLOCKWISE TRIP:





NOTE: * DENOTES LOGIC LEVEL (0-15V) SOLID STATE CONTACTS

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4 P6A, REV. 5 A6F,3T A1F(4),DL1	PRINTS TO 1338 8530, 6V	APPROVALS R.L.Barnes	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM LIMIT SWITCH
REV. 2	ISSUED 8/9/72	CHVIVN WAY	FIRST MADE FOR PLAST. MOLD.		I C 3 6 2 2 G L S B 1
REV. 3	MADE BY GARY HARRIS		I.C. NO.		CONT. ON SH. 3.2 SH. NO. 3.1

APPLICATION DATA

THIS CARD OFFERS A SOLID STATE REPLACEMENT FOR MECHANICAL LIMIT SWITCHES. MACHINE POSITION IS SENSED BY MEASURING THE VOLTAGE FROM A MACHINE MOUNTED FEEDBACK POTENTIOMETER AND COMPARING IT TO PRESET REFERENCES ON THE CARD. WHEN THE TWO ARE EQUAL THE SOLID STATE LOGIC CONTACTS OPERATE AND THE CARD FRONT INDICATOR LIGHTS. THE CIRCUIT HAS THE FOLLOWING FEATURES:

1. FEEDBACK POT CAN BE ANY VALUE BETWEEN 5K AND 20K.
2. TWO POINT CALIBRATION OF REFERENCE DIALS, AT ZERO AND AT FULL TRAVEL.
3. SOLID STATE FLOW THROUGH LOGIC CONTACTS ON THE OUTPUT. ONE NORMALLY OPEN AND ONE NORMALLY CLOSED.
4. CAM OPERATION AVAILABLE BY USING TWO REFERENCES AND CONNECTING OUTPUT CONTACTS IN SERIES. ONE REFERENCE WILL SET PICK UP POINT, THE OTHER THE DROP OUT POINT.
5. INDICATOR CAN LIGHT EITHER ON CLOSING OR OPENING, SELECTABLE BY JUMPER.
6. ANALOG OUTPUT OF POSITION COMPATIBLE WITH PROCESS CONTROL.
7. EXTERNAL REFERENCE INPUT AVAILABLE FOR USE WITH PROCESS CONTROL.
8. UP TO 12 REFERENCES CAN BE ADDED USING THE ADD-ON CARDS GLSB2 OR GLSB3.

SPECIFICATIONS:

FEEDBACK POT

MANY SPECIFICATIONS DEPEND ON THE PROPERTIES OF THE EXTERNAL FEEDBACK POT USED. TO MEET THE LINEARITY AND CALIBRATION SPECS BELOW, THE POT SHOULD HAVE LINEARITY OF 0.2% OF FULL SCALE AND RESOLUTION BETTER THAN .05% OF FULL SCALE. ANY TOLERANCES ABOVE THIS OR ANY ERRORS IN MECHANICAL LINKAGE WILL ADD TO THE ERRORS LISTED DIRECTLY. THE POT SHOULD BE INSTALLED SO THAT WHEN THE MOTION IS AT THE DESIRED 000 ON THE REFERENCE, THE POT SHOULD HAVE LESS THAN 10% OF ITS ROTATION LEFT TO THE END STOP. THE EXACT ELECTRICAL ZERO CAN THEN BE SET WITH THE ZERO ADJUSTMENT ON THE CARD. FULL TRAVEL ON THE MACHINE SHOULD BE AT LEAST 20% OF THE RANGE OF THE POT, BUT OPTIMUM PERFORMANCE IS OBTAINED WHEN ABOUT 80% OF THE FULL ROTATION IS USED.

FEEDBACK POT WIRING RESTRICTIONS:

THE WIRES BETWEEN THE FEEDBACK POT AND THE CARD SHOULD BE CONSIDERED LOW LEVEL WIRING. IF THIS WIRING IS KEPT SEPARATED FROM HIGH LEVEL WIRING, THIS RUN MAY BE UP TO 150 FEET. HIGH LEVEL WIRING IS CONSIDERED TO BE ANY CIRCUIT WITH 110V OR MORE, OR CARRYING 20 AMPS OR MORE, OR HAVING ANY UNSUPPRESSED RELAY COILS IN THE CIRCUIT. IF THE FEEDBACK POT WIRING IS MIXED WITH HIGH LEVEL, THE THREE LEADS SHOULD BE TWISTED TOGETHER, OR SHOULD BE SHIELDED, AND THE LENGTH OF RUN SHOULD BE LESS THAN 25 FEET.

ZEROING

THE ZERO ADJUSTMENT COMPENSATES FOR THE END RESISTANCE OF THE FEEDBACK POT UP TO 10% OF THE VALUE OF THE POT. SET MACHINE TO THE DESIRED ZERO POSITION. WITH REFERENCE CLOSE TO ZERO (ABOUT .005) SET ZERO ADJUSTMENT UNTIL CIRCUIT TRIPS. THIS ALLOWS FOR SOME TOLERANCE TO INSURE THAT THE CIRCUIT WILL ALWAYS TRIP AT ZERO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4 P6A	REV. 5 A6F,3T A1F(4),DL1	PRINTS TO 6V 2530,1338	APPROVALS R Edwards	ELEMENTARY DIAGRAM LIMIT SWITCH (PRECISION)
REV. 2	ISSUED 8/9/72 CHVIVN WAY		FIRST MADE FOR PLAST. MOLD.	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	I C 3 6 2 2 G L S B 1
REV. 3	MADE BY G. HARRIS		I.C. NO.		CONT. ON SH. 3.3 SH. NO. 3.2

SPAN ADJUSTMENT

SET MACHINE TO MAXIMUM TRAVEL. SET REFERENCE TO DESIRED VALUE (CLOSE TO 999 PREFERRED) AND ADJUST SPAN TO TRIP.

LINEARITY OF SET POINTS TO MACHINE POSITION, MAXIMUM ERROR

TYPICALLY ±.05% OF FULL SCALE

MAXIMUM ± .2% OF FULL SCALE

LINEARITY OF FEEDBACK POT MUST BE ADDED TO THESE NUMBERS.

TRACKING BETWEEN REFERENCE POTS

ANY TWO DIGITAL TYPE REFERENCE DIALS SET TO THE SAME TRIP POINT WILL MATCH WITHIN ±0.15% TYPICALLY, ±0.5% MAXIMUM OF FULL SCALE.

STABILITY OF SET POINT WITH TEMPERATURE

THIS DEPENDS ON THE SPAN ADJUSTMENT USED. IF 80% OF THE ROTATION OF THE FEEDBACK POT IS USED, THE DRIFT IS LESS THAN ±0.05% OF FULL SCALE, 0°C TO 70°C. IF 20% SPAN IS USED, MAXIMUM DRIFT IS ±0.3% OF FULL SCALE.

HYSTERESIS

APPROXIMATELY 0.05% OF FULL SCALE. THIS ASSURES A SNAP ACTION ON THE OUTPUT AND ELIMINATES CONTACT BOUNCE EFFECTS.

SIGNALS: ALSO SEE WAVEFORMS SHOWN IN FIGURE 1.

POS OUT - THIS IS AN ANALOG VOLTAGE CORRESPONDING TO THE FEEDBACK POT POSITION. IT VARIES LINEARLY FROM -8.2V (AT 000 TRIP POSITION) TO +8.2V (AT 999 TRIP POSITION). THIS CAN DRIVE UP TO TWELVE POS INPUT LOADS PLUS A PROCESS CONTROL LOAD DRAWING 3 MILLIAMPERES OR LESS LOAD CURRENT. (EQUIVALENT TO A 2.7K LOAD TO COMMON).

POS INPUT - THIS IS THE COMPARATOR INPUT. A -8.2V SIGNAL TRIPS AT 000, WHILE A +8.2V SIGNAL TRIPS AT 999. APPROXIMATELY A 2 MEGOHM INPUT LOAD.

EXT. REF - EXTERNAL REFERENCE FOR PROCESS CONTROL USED IN EITHER OF TWO WAYS:

1. TO COMPLETELY OVERRIDE THE CARD FRONT SETTING, A VOLTAGE SOURCE BETWEEN +8.2 AND -8.2 VOLTS IS APPLIED FOR A REMOTE SETTING.
2. TO MODULATE THE CARD FRONT SETTING, A SUMMING RESISTOR CAN BE CONNECTED IN SERIES WITH THE INPUT. A 150K RESISTOR IN SERIES, WITH A ±8 VOLT SIGNAL APPLIED TO THE RESISTOR, WILL MODULATE THE CARD SETTING BY 10%.

OTCW INDICATOR DRIVER. JUMPER TO IND TO INDICATE CONTACT OPERATION ON CLOCKWISE ROTATION OF FEEDBACK POT. CLOCKWISE ROTATION IS AS DEFINED ON DIAGRAM SH 3.0, AS WIPER TRAVELING TOWARD POSITIVE LEAD OF POT.

OTCCW INDICATOR DRIVER. JUMPER TO IND TO INDICATE CONTACT OPERATION ON COUNTER CLOCK-WISE ROTATION OF FEEDBACK POT. THIS WOULD BE WIPER OF POT TRAVELING TOWARD NEGATIVE LEAD OF POT. OTCW AND OTCCW CAN SINK UP TO 14MA.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4 P6A,	REV. 5 A6F, 3T A1F(4), DL1	PRINTS TO 6V 2630.1388	APPROVALS R Stevens	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM LIMIT SWITCH(PRECISION)
REV. 2	ISSUED 8/9/72		CHVLVN WAY	FIRST MADE FOR PLAST. MOLD.		
REV. 3	MADE BY GARY HARRIS			I.C. NO.		I C 3 6 2 2 G L S B 1 CONT. ON SH. 3.4 SH. NO. 3.3

IND INDICATOR INPUT. OV LIGHTS. DRIVE MUST SINK 7.5 MA TO COMMON.

IA INPUT CONTACT A.ONE LOAD.

OA OUTPUT OF CONTACT A. CAN DRIVE 17 LOADS.

IB INPUT OF CONTACT B. ONE LOAD.

OB OUTPUT OF CONTACT B CAN DRIVE 17 LOADS.

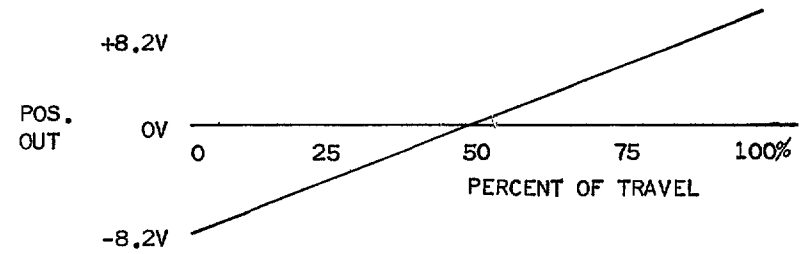
CONTACTS A AND B ARE LOGIC LEVEL FLOW THROUGH CONTACTS. THIS MEANS THAT AN INPUT GROUND (OV LEVEL) IS TRANSFERRED TO THE OUTPUT IF THE CONTACT IS CLOSED. ALL OTHER CONDITIONS ARE IDENTIFIED AS AN OPEN CIRCUIT CONDITION. THE APPLICATION OF THESE CONTACTS IN LOGIC CIRCUITS ARE AS EXPLAINED ON THE GLEB CARD. THE OPEN - CLOSED SEQUENCE OF THESE CONTACTS ARE AS SHOWN IN FIGURE 1.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

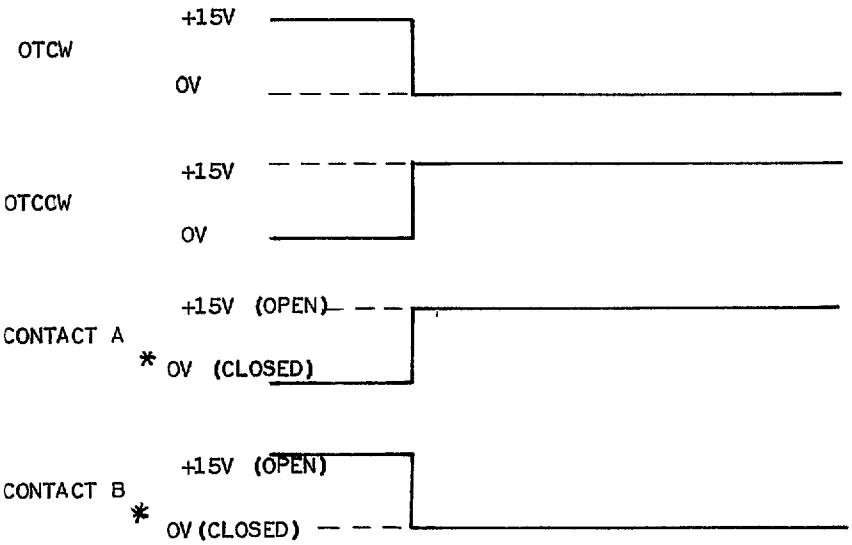
REV. 1	REV. 4 P6A.	REV. 5 A6F,3T A1F(4),DL1	PRINTS TO 6V 2530,1338	APPROVALS RC Harris	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM LIMIT SWITCH(PRECISION)
REV 2	ISSUED 8/9/72	CHVIVN WAY	FIRST MADE FOR PLAST. MOLD.			I C 3 6 2 2 G L S B 1
REV. 3	MADE BY GARY HARRIS		I.C. NO.			CONT ON SH. 3.5 SH. NO. 3.4

SIGNALS AVAILABLE

EXAMPLE TRIP POINT AT 25.0% OF TRAVEL (250 ON REFERENCE DIAL)



FEED BACK POT CLOCKWISE PER ELEMENTARY SH. 3.0



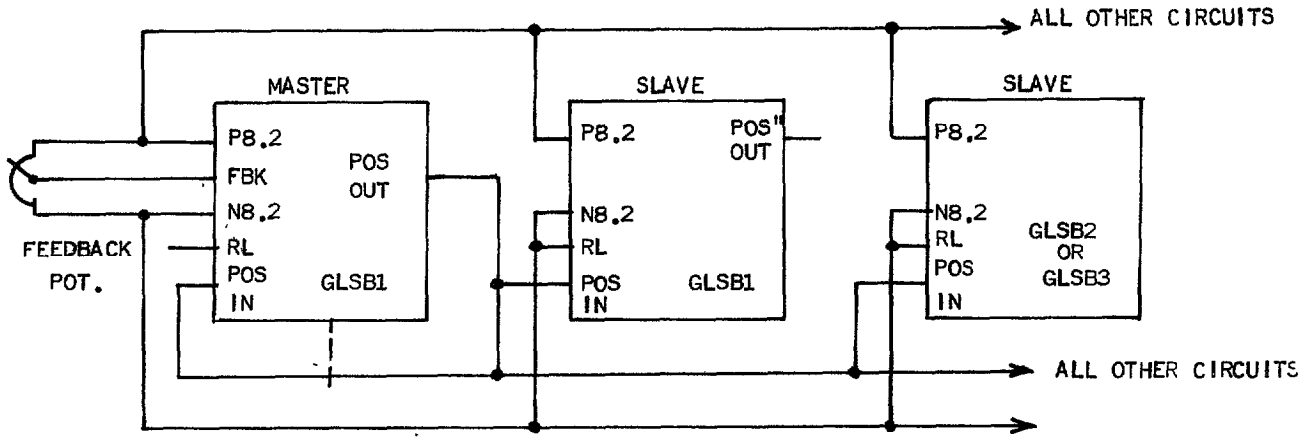
* CONTACT OUTPUT VOLTAGE IS ZERO WHEN CONTACT INPUT IS ZERO AND CONTACT IS CLOSED. SEE CONTACT SPECS PER IC3622GLEB CARD.

FIGURE 1

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV 4 P6A,	REV. 5 AGF, 3T AIF(4), DLI	PRINTS TO GV 2530, 1338	APPROVALS	ELEMENTARY DIAGRAM: LIMIT SWITCH (PRECISION)
REV. 2	ISSUED 8/9/72	CHVIVN WAY	FIRST MADE FOR PLAST. MOLD.	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	IC3622GLSB1
REV. 3	MADE BY GARY HARRIS		I.C. NO.		CONT. ON SH. 3.6 SH NO. 3.5

WIRING FOR ADDITIONAL REFERENCES, ALL SLAVED TO A SINGLE FEEDBACK POT:



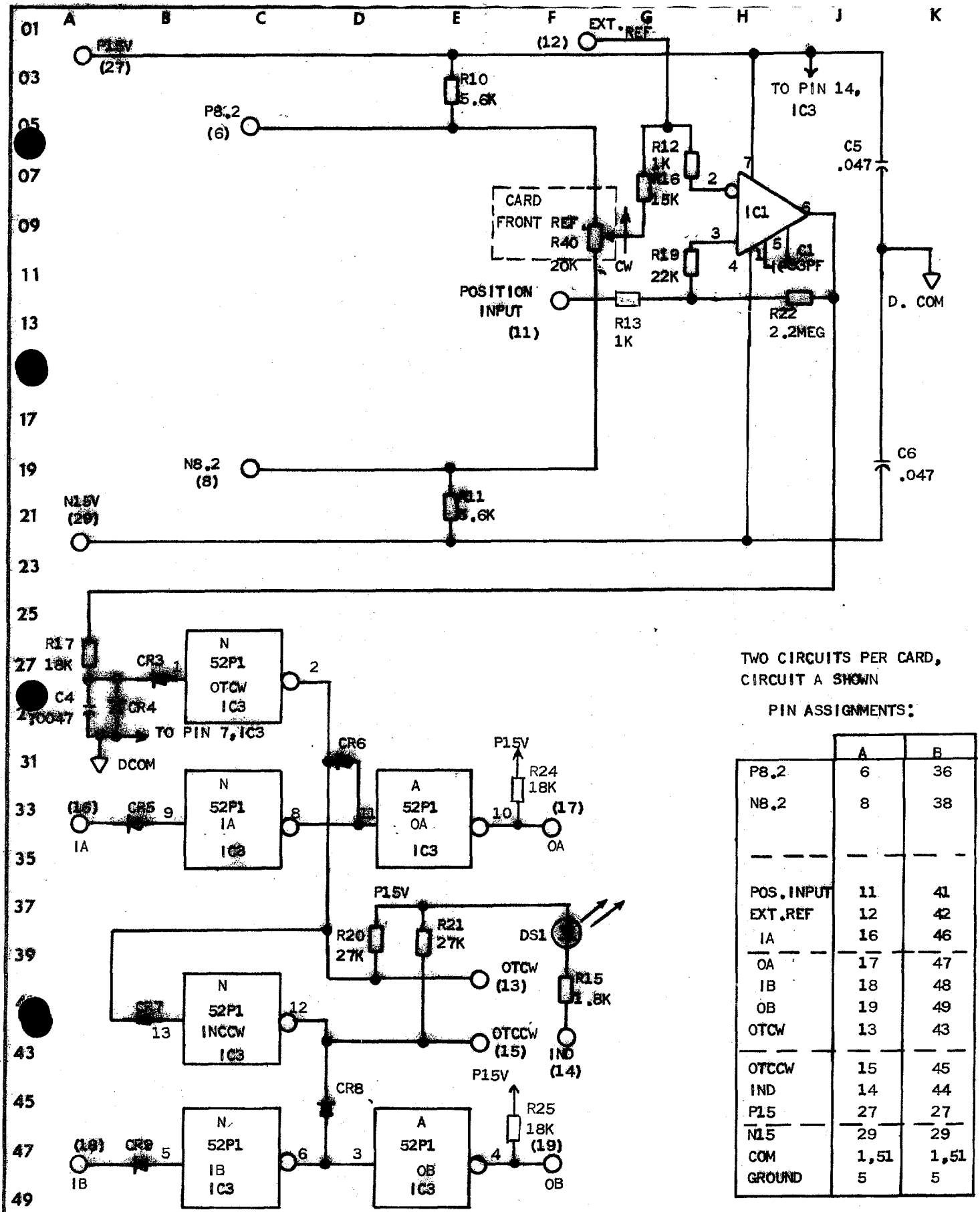
NOTE: WHEN TWO OR MORE GLSB1 CIRCUITS ARE USED AS SLAVE UNITS, THE LOAD RESISTOR RL SHOULD BE JUMPERED TO N8.2 TO PREVENT OVERLOADING THE 8.2V REGULATOR DIODES BUILT INTO THESE CARDS. THE CALIBRATION POTS ON THE SLAVE CIRCUITS HAVE NO EFFECT.

POWER: +15V (27), 62 MA TYP., 80 MA MAX.
-15V (29), 24 MA TYP., 32 MA MAX.

PACKAGING: STANDARD CARD WITH CARD FRONT, REQUIRES TWO CARD ROW SPACES. CARD FRONT (PIN 5) SHOULD BE GROUNDED.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4 A I F (4) 2830.3T	REV 5 1338 PSK, A6F	PRINTS TO 6V 2530.DL2	APPROVALS R.C. Garner	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM LIMIT SWITCH (PRECISION)
REV. 2	ISSUED 8/9/72	CHIVYN WAY	FIRST MADE FOR PLAST. MOLD.			IC3622GLSB1
REV. 3	MADE BY GARY HARRIS		I.C. NO			CONT. ON SH. FL. SH NO. 3.6



TWO CIRCUITS PER CARD,
CIRCUIT A SHOWN

PIN ASSIGNMENTS:

	A	B
P8.2	6	36
N8.2	8	38
POS. INPUT	11	41
EXT. REF	12	42
IA	16	46
OA	17	47
IB	18	48
OB	19	49
OTCW	13	43
OTCCW	15	45
IND	14	44
P15	27	27
N15	29	29
COM	1,51	1,51
GROUND	5	5

NOTE: ONLY ONE C5 AND C6 PER CARD.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5 /338	PRINTS TO	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 2	ISSUED	August 11, 1972	CANVAS WAY	FIRST MADE FOR PLAST. MOLD.	
REV. 3	MADE BY		I.C. NO.		

ELEMENTARY DIAGRAM LIMIT SWITCH (PRECISION) (SLAVE)

IC3622GLSB2

CONT. ON SH. 3.1 SH. NO. 3.0

THE GLSB2 IS USED TO ADD ADDITIONAL REFERENCES AND OUTPUTS TO THE GLSB1 MASTER UNIT. THIS CARD HAS TWO 10 TURN PRECISION DIALS WITH A THREE DIGIT READ OUT CALIBRATION, SPECS AND WIRING ARE AS EXPLAINED ON THE GLSB1 CARD. CARD OCCUPIES TWO CARD ROW SPACES.

POWER CONSUMPTION:

+15V, 45 MA TYPICAL, 60 MA MAX.

-15V, 6 MA TYPICAL, 9 MA MAX.

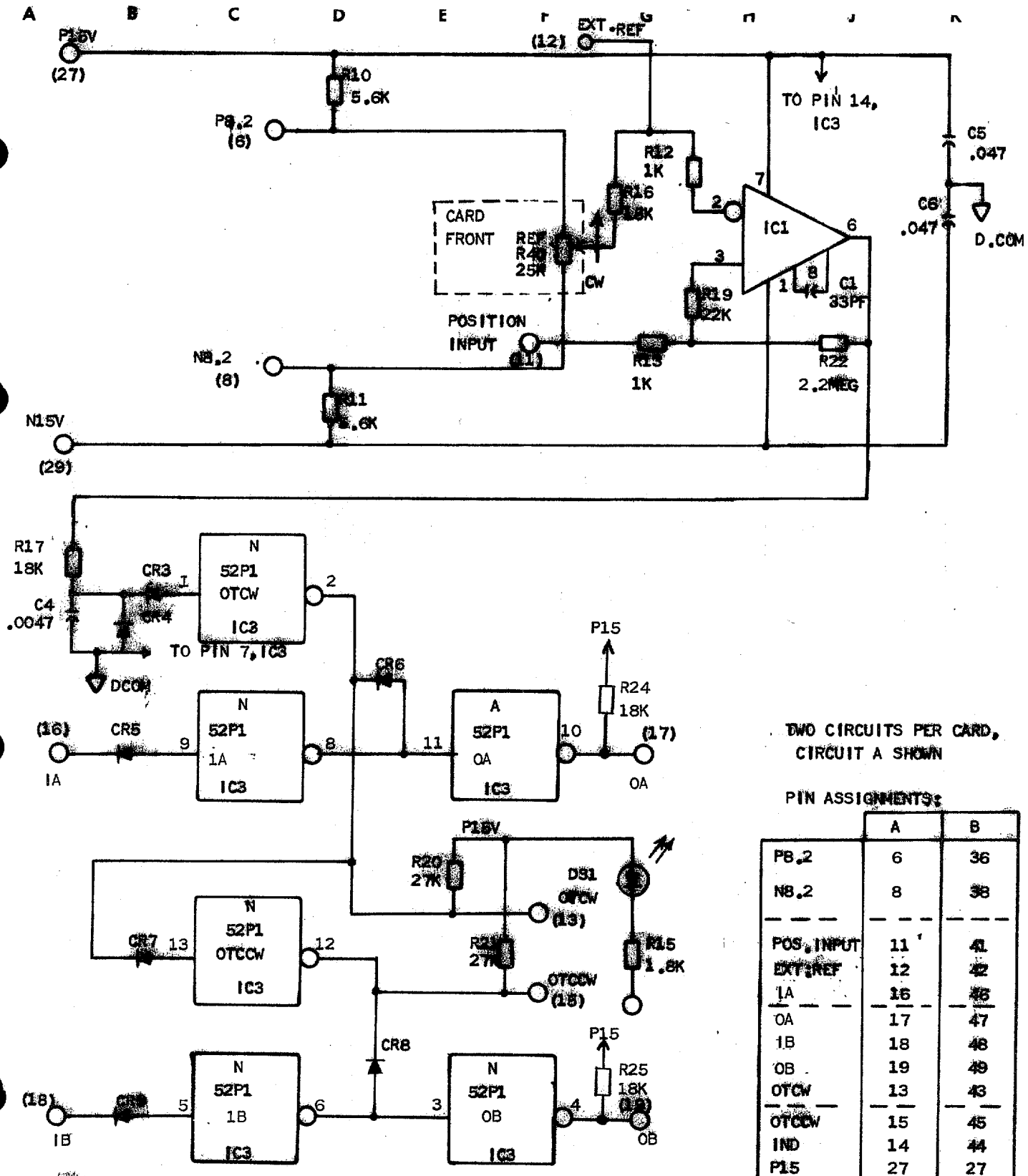
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4 2530	REV. 5 6V, AGF A1F(4), 3T	PRINTS TO P6A 1338, DL1	APPROVALS Rebarnes	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 2	ISSUED August 11, 1972		CHVIVN IWAY	FIRST MADE FOR PLAST MOLD	
REV. 3	MADE BY GARY. HARRIS			I.C. NO.	

ELEMENTARY DIAGRAM LIMIT SWITCH (PRECISION), SLAVE

IC3622GLSB2

CONT. ON SH FZ. SH NO 3.1



NOTE: ONLY ONE C5 AND C6 PER CARD.

TWO CIRCUITS PER CARD,
CIRCUIT A SHOWN

PIN ASSIGNMENTS:

	A	B
P8.2	6	36
N8.2	8	38
POS. INPUT	11	41
EXT. REF	12	42
1A	16	46
OA	17	47
1B	18	48
OB	19	49
OTCW	13	43
OTCCW	15	45
IND	14	44
P15	27	27
N15	29	29
COM	1,51	1,51
GROUND	5	5

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4 2539	REV. 5 6V, A6F A15(d), AT	PRINTS T P6A 333B D1 T	APPROVED	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 2	ISSUED August 11, 1972	ENVIRON WAY	FIRST MADE FOR RMS, HOLD,		ELEMENTARY DIAGRAM LIMIT SWITCH SLAVE NON-PRECISION
REV. 3	MADE BY GARY HARRIS		I.C. NO.		I C 3 6 2 2 G L S B 3 CONT. ON SH. 3.1 SH. NO. 3.0

THE GLSB3 IS USED TO ADD ADDITIONAL REFERENCES AND OUTPOTS TO THE GLSB1 MASTER UNIT. THIS CARD HAS TWO SINGLE TURN REFERENCE DIALS. SPECIFICATIONS AND WIRING ARE AS ON THE GLSB1 CARD WITH THE FOLLOWING EXCEPTIONS:

LINEARITY OF SET POINT TO MACHINE POSITION

MAXIMUM ERROR IS TYPICALLY ±5% OF FULL SCALE, MAXIMUM ±15% OF FULL SCALE.

TRACKING BETWEEN REFERENCE POTS

WHEN SET TO THE SAME TRIP POINT, DIALS WILL MATCH WITHIN ±5% OF FULL SCALE.

STABILITY WITH TEMPERATURE

MAXIMUM TEMPERATURE DRIFT IS ±1.0% OF FULL SCALE.

IN ADDITION TO USE AS A SLAVE UNIT, THE GLSB3 CAN BE USED AS AN INEXPENSIVE, UNCALIBRATED LIMIT SWITCH BY ITSELF. USE A 5K TO 10K FEEDBACK POT BETWEEN P8.2 AND N8.2. CONNECT FEEDBACK POT WIPER TO POSITION INPUT. RANGE OF THE REFERENCE POT IS EQUAL TO THE SPAN OF THE FEEDBACK POT, SO GENERALLY 75 TO 100% OF THE FEEDBACK POT ROTATION SHOULD BE USED. UP TO SIX REFERENCES CAN BE CONNECTED BY PARALLELING INPUTS. SPECS ARE AS ABOVE. NO PROVISION CAN BE MADE FOR PROCESS CONTROL IN THIS MODE.

PACKAGING

STANDARD CARD WITH METAL CARD FRONT OCCUPYING TWO CARD ROW SPACES.

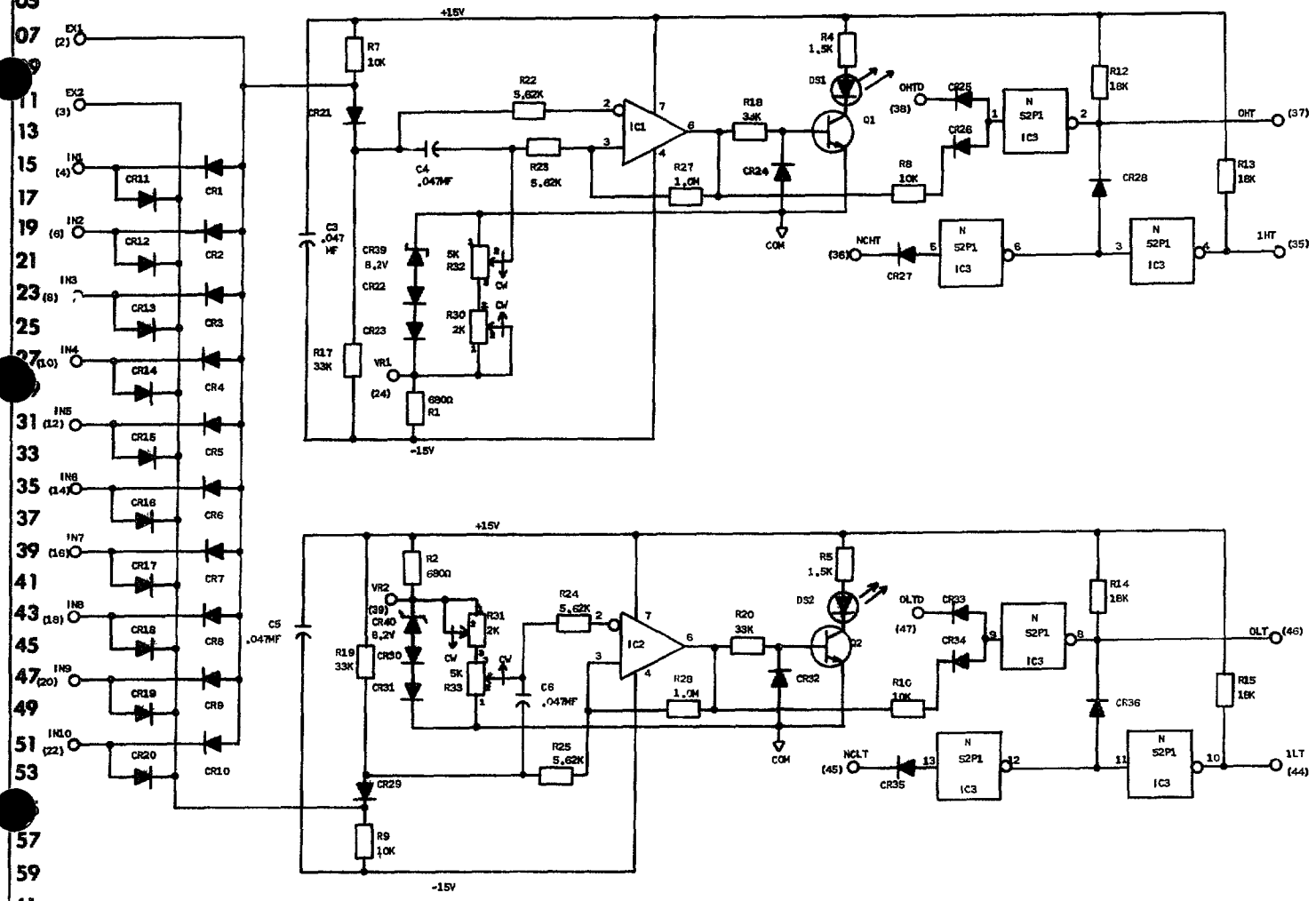
POWER SUPPLY

+15V, 45 MA TYPICAL, 60 MA MAX.

-15V, 6 MA TYPICAL, 9 MA MAX.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV 4 2530	REV 56V, A6F /A1F(4), 3T	PRINTS TO P6A 1338, DL1	APPROVALS RE Davis	ELEMENTARY DIAGRAM LIMIT SWITCH SLAVE (NON-PRECISION)
REV. 2	ISSUED August 11, 1972	CHVIVN WAY	FIRST MADE FOR PLAST. MOLD.	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	I C 3 6 2 2 G L S B 3
REV. 3	MADE BY GARY HARRIS		I.C. NO		CONT. ON SH FL SH. NO. 3.1



(27) → C1 1MF → +15V - TO PIN 14 OF IC3

(1) (51) → C2 1MF → COM - TO PIN 7 OF IC3

(29) → -15V

(5) → TO CARD FRONT

REV.1	REV.4	REV 5 2530 P6A, A1F-4,	PRINTS TO 1338, A6F 3T, 6V, DL1	APPROVALS K.B. Brown	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV.2	ISSUED August 11, 1972		CHVIVN WAV	FIRST MADE FOR PLAST. MOLD	INDUSTRY CONTROL DEPT. SALEM, VA., USA	TEMPERATURE MONITOR
REV.3	MADE BY RON MUCKENFUSS			T.C. NO.		IC3622GMFC1 C
						CONT. ON SH. 31 SH. NO. 30

FORM IC 6868 (01 69)

THEORY OF OPERATION

THIS CARD IS DESIGNED TO OPERATE WITH THE PLASTIC MOLDING TEMPERATURE REGULATOR , IC3622GTRD. THIS CARD MONITORS THE ERROR SIGNALS FROM UP TO TEN REGULATORS, AND PRODUCES AN OUTPUT SHOULD ANY ERROR EXCEED SOME PRE-SELECTED, ADJUSTABLE LIMIT.

EACH REGULATOR ERROR OUTPUT EI IS CONNECTED TO A MONTOR INPUT, IN1-10. THE ERROR SIGNAL IS SCALED SUCH THAT +7.50V REPRESENTS A 100°F UNDER TEMPERATURE ERROR AND -7.5V REPRESENTS A 100°F OVER TEMPERATURE ERROR.. THE INPUTS ARE MULTIPLEXED THROUGH DIODES CR1-20 TO TWO COMPARATORS IC1 AND IC2. IC2 IS THE POSITIVE COMPARATOR WHICH COMPARES POSITIVE ERROR WITH A PRE-SELECTED VOLTAGE SET BY R33. R33 WILL VARY THE COMPARATOR REFERENCE FROM OV TO +7.5V CORRESPONDING TO AN ERROR OF 0 - 100°F. SHOULD ANY ERROR SIGNAL EXCEED THE ERROR SET BY R33, THE OUTPUT OF IC2 WILL SWITCH FROM -14V TO +14V. THIS ACTION ALLOWS Q2 TO SATURATE, THUS TURNING ON DS2, AND PRODUCES LOGIC "0" AT THE OLT OUTPUT. INPUT OLTD AND CR33 PROVIDES A MEANS OF DISABLING THE OLT OUTPUT.

IC1 AND R32 CIRCUITRY FUNCTION IN A SIMILIAR MANNER, COMPARING NEGATIVE ERROR WITH A PRE-SELECTED VOLTAGE SET BY R32. R30 AND R31 COMPENSATE FOR CIRCUIT TOLERANCES AND ARE FACTORY CALIBRATED. IC3 COMPRISES ONE "NORMALLY CLOSED FLOW THROUGH CONTACT" AND ONE "NORMALLY OPEN TO COMMON FLOW THROUGH CONTACT" FOR EACH COMPARATOR.

SPECIFICATIONS

INPUTS

IN1-10 - ERROR INPUT. EACH INPUT WILL ACCEPT AN ANALOG ERROR VOLTAGE OF -15V TO +15V, HOWEVER, THE COMPARISON .. RANGE IS ADJUSTABLE ONLY BETWEEN -7.5V AND +7.5V. INPUT LOADING IS ±2.25MA MAXIMUM AT ±7.5V.

OLTD,OHTD - HIGH TEMPERATURE AND LOW TEMPERATURE DISABLE INPUTS. HTL LOGIC "0" ON OLTD WILL INHIBIT THE LOW TEMPERATURE OUTPUT CONTACTS, I.E. THE(NO) CONTACT REMAINS OPEN AND THE(NC) CONTACT REMAINS CLOSED. HTL LOGIC "0" ON OHTD SIMILARLY INHIBIT THE HIGH TEMPERATURE OUTPUT CONTACTS. EACH INPUT IS ONE HTL LOAD.

NCHT,NCLT - NORMALLY CLOSED FLOW THROUGH CONTACT INPUTS. EACH IS ONE HTL LOAD.

EX1,EX2 - EXPANDER INPUTS. A MAXIMUM OF TEN ADDITIONAL INPUTS MAY BE OBTAINED BY CONNECTING EXTERNAL DIODES TO EX1 AND EX2.

OUTPUTS

OHT,OLT - HIGH AND LOW TEMPERATURE ALARM OUTPUTS. EACH CAN DRIVE 17 HTL LOADS.

IHT,OLT - "NORMALLY CLOSED FLOW THROUGH CONTACTS" FOR HIGH TEMPERATURE AND LOW TEMPERATURE. FOR AN ALARM CONDITION THE NC CONTACT WILL OPEN AND IHT (OR ILT) WILL BECOME LOGIC "1". EACH OUTPUT CAN DRIVE 17 HTL LOADS.

ACCURACIES

DIAL ACCURACY - ±10% OF F.S. AT 25°C.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO 3T,A6F,	APPROVALS K.B. Brown	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM IC3622GMFC1
REV. 2	ISSUED 8-9-72	A1F(4),6V,DL1 P&A,1888 2530	FIRST MADE FOR PLASTIC MOLD			TEMP. MONITOR
REV 3	MADE BY RON MUCKENFUSS	CNVTVN WAY	I.C NO.			CONT. ON SH. 3.2 SH. NO. 3.1

REPEATIBILITY - ±1% OF F.S. AT STABLE AMBIENT TEMPERATURE

±2% OF F.S. OVER 0-70°C

POWER SUPPLY

+15V (+14V TO +16V)

21 MA TYPICAL

45 MA MAXIMUM

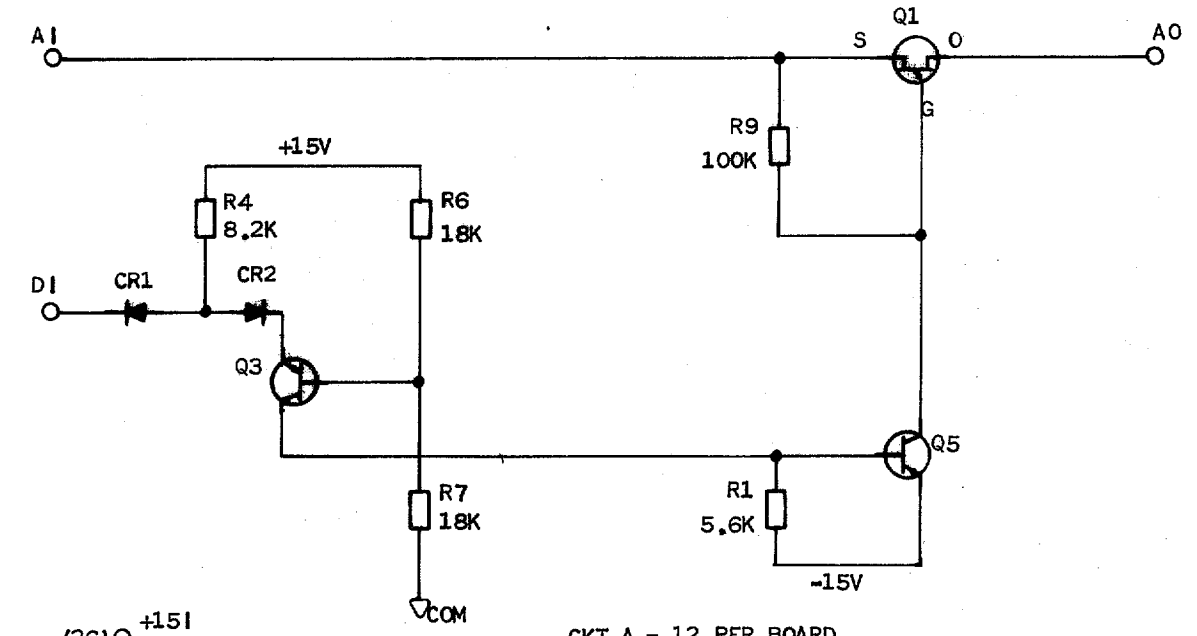
-15V (-14V TO -16V)

14 MA TYPICAL

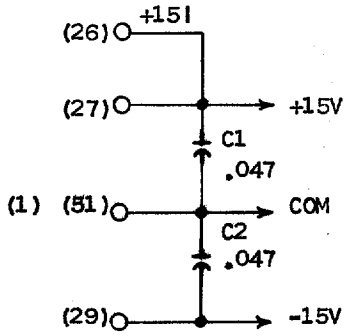
20 MA MAXIMUM

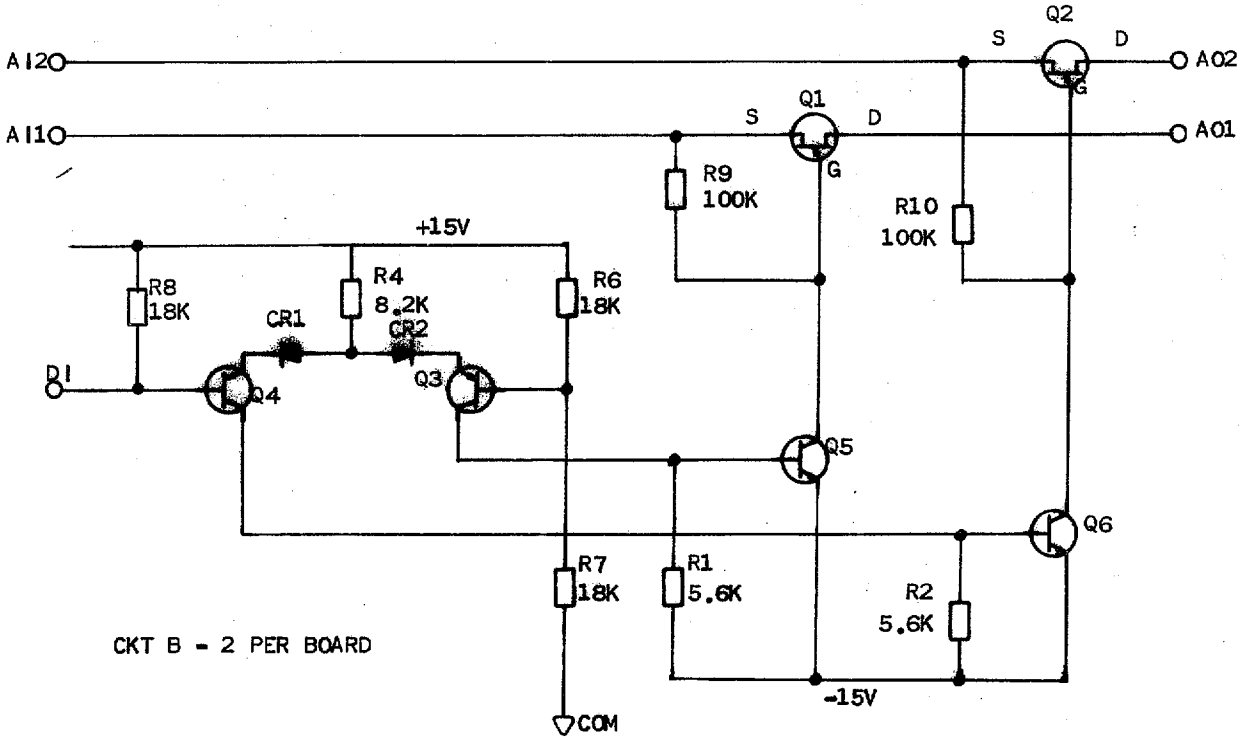
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO 3T,A6F,	APPROVALS L.B. Brown	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM I C 3 6 2 2 G M F C 1
REV. 2	ISSUED 8-4-72	AIF (4),6V,DL1, P&A 1338 2530		FIRST MADE FOR PLASTIC MOLD		
REV. 3	MADE BY RON MUCKENFUSS	CHVIVN WAY		I.C. NO.		TEMP. MONITOR CONT. ON SH. FL. SH. NO. 3,2



CKT A - 12 PER BOARD





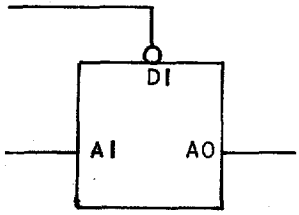
CKT B - 2 PER BOARD

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

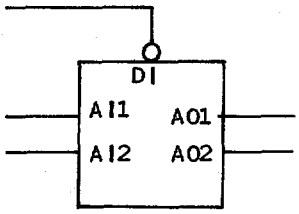
REV. 1	REV. 4	REV. 5	PRINTS TO A6F, 3T	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM MULTIPLEX INPUT
REV. 2	ISSUED 7-23-71.	A1F(4), DL1 P6A, 1338	FIRST MADE FOR PLASTIC MOLD		INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	I C 3 6 2 2 G M I A
REV. 3	MADE BY RON MUCKENFUSS		I.C. NO.			CONT. ON SH. 3.1 SH. NO. 3.0

	A CIRCUITS											
	1	2	3	4	5	6	7	8	9	10	11	12
AI	6	11	12	15	18	21	25	30	33	36	40	45
AO	7	10	13	16	19	22	24	31	34	37	39	46
DI	9	41	14	17	20	23	28	32	35	38	42	44

	B CIRCUITS	
	1	2
AI1	2	47
AI2	4	49
AO1	3	48
AO2	5	50
DI	8	43



A CIRCUIT SYMBOL



B CIRCUIT SYMBOL

THEORY OF OPERATION

THIS IS AN ANALOG MULTIPLEX CARD. IT CONSISTS OF 12 SINGLE-POLE JFET SWITCHES, 2 DOUBLE-POLE JFET SWITCHES, AND THEIR ASSOCIATED DRIVERS. ALL SWITCHES HAVE INDIVIDUAL OUTPUTS WHICH MAY BE JUMPERED TOGETHER TO FORM A DESIRED GROUPING. EACH DOUBLE-POLE SWITCH HAS ONE NORMALLY-OPEN AND ARE NORMALLY-CLOSED CONTACT WITH EACH POLE HAVING A SEPARATE OUTPUT.

THE DRIVERS ARE DESIGNED TO ACCEPT A HIGH THRESHOLD LOGIC (HTL) DIGITAL INPUT. WITH LOGIC INPUT DI AT LOGIC "1", Q3 WILL CONDUCT AND SATURATE Q5. Q5 SATURATING CONNECTS -15V TO THE GATE OF Q1, THUS TURNING OFF Q1. IN THE DOUBLE-POLE CONFIGURATION, Q4 AND Q6 WILL BE OFF AND Q2 WILL BE ON.

WITH DI AT LOGIC "0", Q3 AND Q5 WILL BE OFF AND Q1 WILL BE ON. IN THE DOUBLE POLE CONFIGURATION, Q4 AND Q6 WILL BE ON AND Q2 WILL BE OFF.

SPECIFICATIONS

DIGITAL INPUTS, DI - +4.6V MAX. "0" LEVEL, 10V MIN. "1" LEVEL. SINGLE-POLE INPUTS ARE 2.5 HTL LOADS, DOUBLE-POLE INPUTS ARE 1 HTL LOAD. ONE HTL LOAD IS 18K OHMS TO +15V.

ANALOG INPUTS, AI - +15V MAX. POSITIVE SIGNAL, -5V MIN. NEGATIVE SIGNAL.

ANALOG OUTPUTS, AO - WHEN THE SWITCH IS ON THERE WILL BE A MAXIMUM OF 100 OHMS BETWEEN AI AND AO IF SWITCH CURRENT IS LIMITED TO 4MA OR LESS. MAXIMUM SWITCH CURRENT IS 50MA.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO AGF, 3T	APPROVALS L.B. Brown	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 2	ISSUED 7-23-74	A1F(4),DL1	FIRST MADE FOR PLASTIC MOLD	I.C. NO.	ELEMENTARY DIAGRAM MULTIPLEX INPUT
REV. 3	MADE BY RON MUCKENFUSS	P6A,1338			I C 3 6 2 2 G M I A
					CONT. ON SH. 3.2 SH. NO. 3.1

WHEN THE SWITCH IS OFF, THERE WILL BE LESS THAN .2 MICRO-AMPS BETWEEN AI AND AO. WITH THE SWITCH OFF THE OUTPUT IS SUBJECT TO THE SAME VOLTAGE LIMITATIONS AS THE INPUT, I. E. +15V TO -8V.

POWER SUPPLY

+15V - 1.2 MA NOMINAL PER EACH OFF SWITCH (DI LOGIC "1"), 1.5 MA MAX.
2.2 MA NOMINAL PER EACH ON SINGLE POLE SWITCH, 2.7 MA MAX.
3.0 MA NOMINAL PER EACH ON DOUBLE POLE SWITCH, 3.6 MA MAX.
-15V - .9 MA NOMINAL PER EACH OFF SWITCH (DI LOGIC "1"), 1.1 MA MAX.
1.7 MA NOMINAL PER EACH ON DOUBLE POLE SWITCH (DI LOGIC "0"), 2.1 MA MAX.
0 MA PER EACH ON SINGLE POLE SWITCH (DI LOGIC "0")

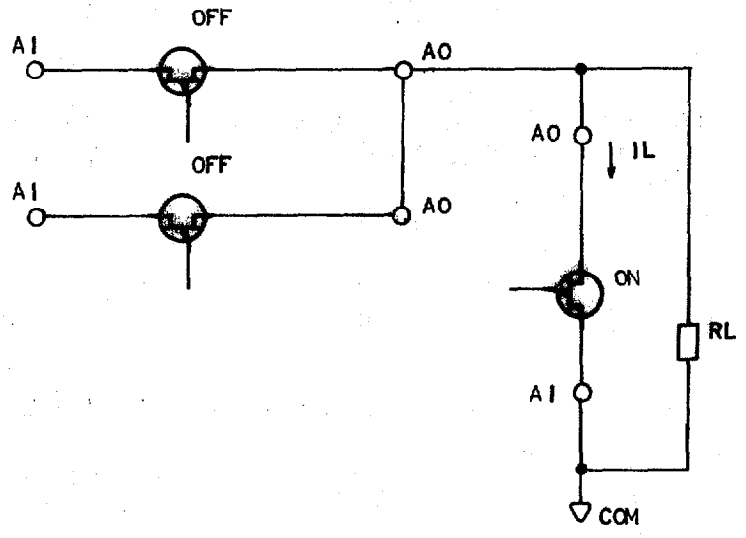
APPLICATION NOTES

AS PREVIOUSLY NOTED, MAXIMUM SWITCH CURRENT IS 50MA. THE SWITCH WILL BE PROTECTED AGAINST AN OUTPUT SHORT-CIRCUIT IF THE CURRENT IS EXTERNALLY LIMITED TO 50MA.

NOTE ALSO THAT WHEN A SWITCH IS OFF, THIS CIRCUIT PRESENTS A 100K OHM TO -15V LOAD ON THE INPUT SOURCE. IF THE SOURCE IS A POTENTIOMETER, THIS LOADING WILL INTRODUCE AN INPUT VOLTAGE ERROR. THIS IMPLIES THAT EACH SWITCH HAVE ITS OWN INPUT POTENTIOMETER. IF IT IS DESIRED TO DRIVE MORE THAN ONE SWITCH FROM A COMMON SOURCE, USE OF AN OP-AMP RATHER THAN A POT IS RECOMMENDED.

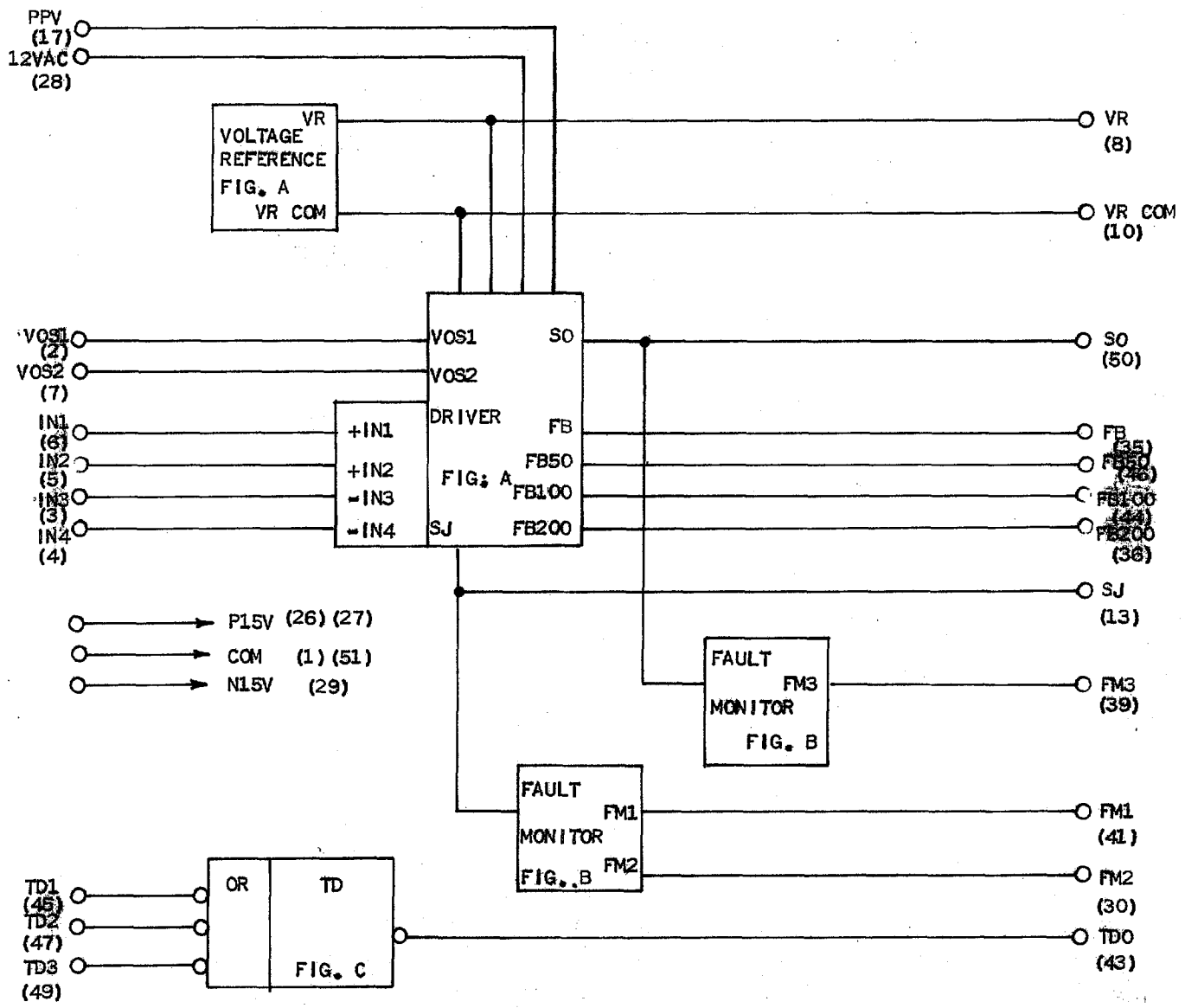
WHEN SEVERAL OUTPUTS ARE TIED TOGETHER AND ALL SWITCHES ARE OFF, THE LEAKAGE CURRENT OF EACH SWITCH WILL FLOW THROUGH THE LOAD RESISTOR. THIS WILL PRODUCE AN OFFSET ERROR WHICH MAY NOT BE NEGLIGIBLE.

ONE METHOD OF MINIMIZING THIS ERROR IS TO SHUNT THE OUTPUT OF A SWITCH GROUP WITH ANOTHER SWITCH AS SHOWN BELOW.



PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 ACW 11-15-71.	REV. 4	REV. 5	PRINTS TO A6F, 3T	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM MULTIPLEX INPUT
REV. 2	ISSUED 7-23-71.	A1F(4), DLE	FIRST MADE FOR PLASTIC MOLD			IC3622GMIA
REV. 3	MADE BY RON MUCKENFUSS	PGA, 1332	I.C. NO.			CONT. ON SH. FL SH. NO. 3.2



PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DLI, PGA	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM PROPORTIONAL VALVE DRIVER
REV. 2	ISSUED 7-26-91	1338, 3T A1F (4), A6F	FIRST MADE FOR PLASTIC MOLD			IC3622GPDA1
REV. 3	MADE BY KENNEY C. COX 6V		I.C. NO. 3622			CONT. ON SH. 3.1 SH. NO. 3.0

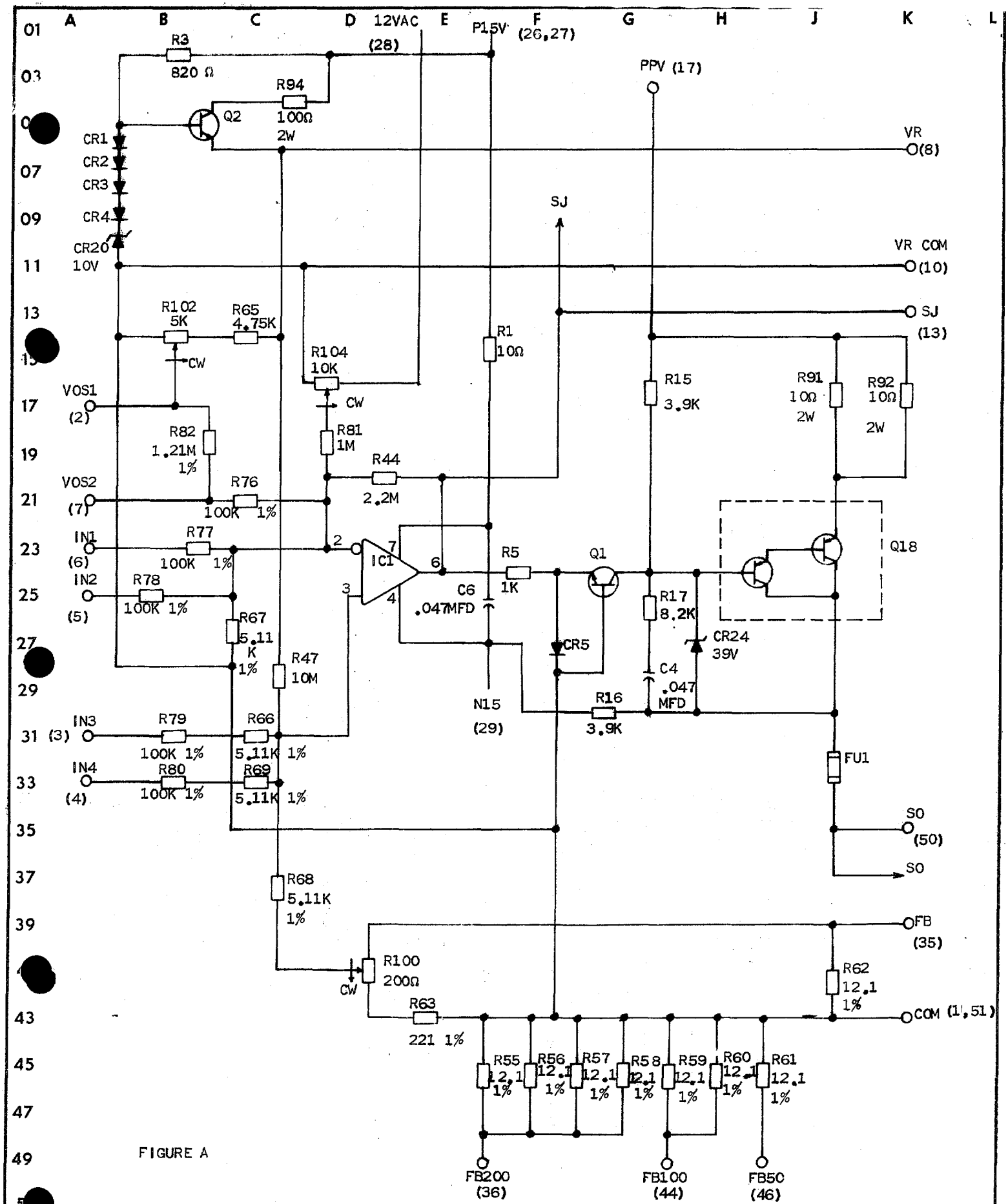


FIGURE A

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 KCC 2/20/71	REV. 4	REV. 5	PRINTS TO DL1, P6A	APPROVALS	ELEMENTARY DIAGRAM: PROPORTIONAL VALVE DRIVER
REV. 2	ISSUED 7-26-71 1338, 3T A1F (4), A6F			FIRST MADE FOR PLASTIC MOLD	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 3	MADE BY KENNEY C. COX 6V		I.C. NO. 3622		I C 3 6 2 2 G P D A 1
					CONT. ON SH. 3, 2 SH. NO. 3,1

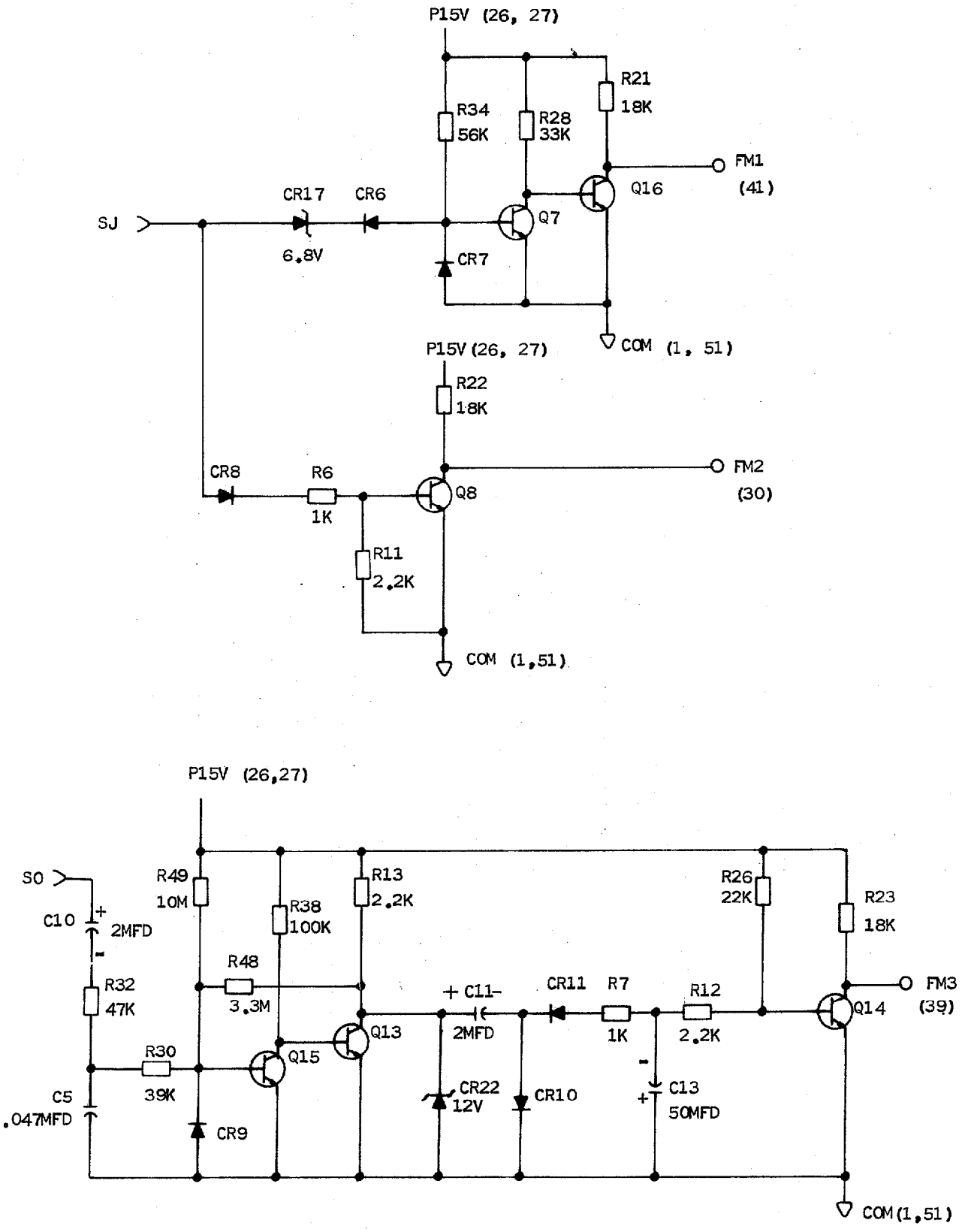


FIGURE B

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL1, P6A	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM PROPORTIONAL VALVE DRIVER
REV. 2	ISSUED 7-26-71.	1338, 3T A1F(4), A6F	FIRST MADE FOR PLASTIC MOLD			I C 3 6 2 2 G P D A 1
REV. 3	MADE BY KENNEY C. COX 6V		I.C. NO. 3622			CONT. ON SH. 3.3 SH. NO. 3.2

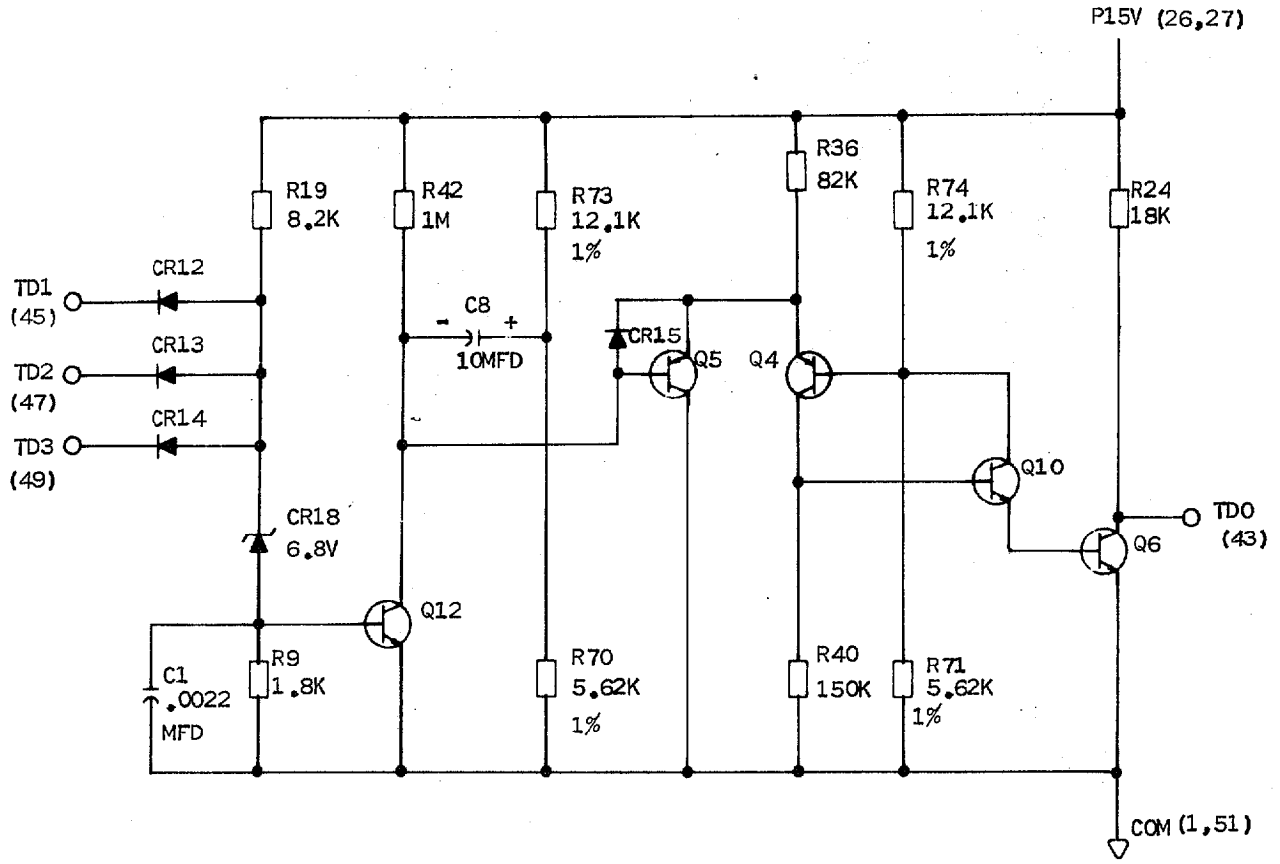


FIGURE C

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 KCC 9-30-71	REV. 4	REV. 5	PRINTS TO DL1, P6A	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM PROPORTIONAL VALVE DRIVER
REV. 2	ISSUED 7-26-71		1338, 3T A1F(4), A6F	FIRST MADE FOR PLASTIC MOLD		I C 3 6 2 2 G P D A 1
REV. 3	MADE BY KENNEY C. COX		6V	I.C. NO. 3622		CONT. ON SH. 3,4 SH. NO. 3,3

FUNCTION AND THEORY OF OPERATION

THIS CARD IS DESIGNED TO DRIVE ELECTROMAGNETICALLY CONTROLLED VALVES BY PROVIDING A CURRENT OUTPUT PROPORTIONAL TO VOLTAGE INPUT. THE BOARD CONSISTS OF THE AMPLIFIER, A VOLTAGE REFERENCE, THREE FAULT MONITORS, AND A TIME DELAY WITH THREE ORRED INPUTS. THE CURRENT OUTPUT MAY BE BIASED BY ADJUSTING THE OFFSET POTENTIOMETER. WITH INCREASE IN VOLTAGE AT EACH INPUT, THE INPUTS IN1 AND IN2, INCREASE THE OUTPUT CURRENT FROM THE BIAS POINT, WHILE IN3 AND IN4 DECREASE IT. THE MINIMUM CURRENT IS ZERO. IN ADDITION, SINCE MOST VALVES REQUIRE A DITHER A 60 HTZ SIGNAL INPUT TO THE AMPLIFIER IS PROVIDED. IT'S MAGNITUDE MAY BE CHANGED BY THE DITHER ADJUSTMENT POTENTIOMETER.

THE AMPLIFIER MAXIMUM OUTPUT CURRENT IS ADJUSTED BY STRAPPING COMBINATIONS OF FB50, FB100, AND FB200 TO FB. SMALLER ADJUSTMENTS ARE MADE BY ADJUSTING THE FEEDBACK POTENTIOMETER, R100 WITHOUT STRAPPING, THE NOMINAL MAXIMUM CURRENT IS 50MA. STRAPPING FB50, FB100, AND FB200 PROVIDE NOMINAL 50MA, 100MA, AND 200MA ADDITIONS. THEREFORE, BY STRAPPING, THE NOMINAL MAXIMUM CURRENT CAN BE ADJUSTED FROM 50MA TO 400MA IN 50MA STEPS. GAIN IS ADJUSTED FROM NOMINAL AFTER ZEROING.
IN ORDER TO DETECT FAULTS THAT MAY OCCUR IN THE AMPLIFIER OR THE LOAD, THREE FAULT MONITORS ARE PROVIDED. THESE HAVE A HIGH THRESHOLD LOGIC (HTL) AND LEVEL OUTPUT OF "1" DURING STEADY STATE NORMAL OPERATION AND AN HTL "0" WHEN A FAULT OCCURS. DURING NORMAL OPERATION TRANSIENT CONDITIONS MAY OCCUR WHICH CAN APPEAR TO BE FAULTS. IT IS RECOMMENDED THAT THE OUTPUTS OF THE FAULT MONITORS CONNECT TO THE TIME DELAY, AND THE OUTPUT OF THE TIME DELAY USED AS AN INDICATOR THAT A FAULT HAS OCCURED. THE FIRST MONITOR, OUTPUT FM1, DETECTS THE FOLLOWING: (1) OUTPUT SO SHORTED TO COMMON; (2) FEEDBACK, FB, SHORTED TO COMMON; (3) OUTPUT LOAD OPEN; AND (4) OPEN FEEDBACK. THE SECOND MONITOR, OUTPUT FM2, DETECTS A SHORTED OUTPUT TRANSISTOR AND THE OUTPUTS SHORTED TO PPV. THE THIRD MONITOR, OUTPUT FM3, DETECTS A SHORTED LOAD BY DETECTING THE ABSENSE OF A MINIMUM AMOUNT OF DITHER.

IN THE AMPLIFIER, IC1 ACTS AS A SUMMING JUNCTION FOR THE INPUTS, IN1, IN2, IN3, IN4, THE OFFSET VOLTAGE, DITHER VOLTAGE, AND THE FEEDBACK. TRANSISTOR Q1 THROUGH R5 CONVERTS THE VOLTAGE OUTPUT OF IC1 TO A PROPORTIONAL CURRENT. THIS CURRENT IS THEN AMPLIFIED BY A DARLINGTON PAIR CONTROLLING THE CURRENT FROM THE BUS PPV INTO THE LOAD ACROSS SO AND FB. THE CURRENT THROUGH THE FEEDBACK CREATES A VOLTAGE ACROSS R100 AND R63. THE VOLTAGE IS THEN FED TO THE SUMMING JUNCTION IC1 REGULATING THE AMOUNT OF CURRENT IN THE LOAD. R17 AND C4 PROVIDE STABILIZATION FOR THE LOOP. ZENER DIODE CR24 PROTECTS THE OUTPUT TRANSISTOR DURING INDUCTIVE KICKBACK, BY KEEPING THE VOLTAGE ACROSS IT AT A NOMINAL 35 VOLTS AND TURNING THE TRANSISTOR ON.

THE ZENER DIODE CR20 AND AN EMITTER FOLLOWER USING Q2 MAKE THE VOLTAGE REFERENCE. CR1-CR4 ARE USED FOR TEMPERATURE STABILIZATION. THE CIRCUIT SUPPLIES A REFERENCE VOLTAGE FOR DEVICES WHICH INPUT TO THE AMPLIFIER. IN ADDITION, IT PROVIDES AN OFFSET AND ZEROING VOLTAGE TO THE POTENTIOMETER R102, FOR THE AMPLIFIER. THE AMPLIFIER HAS TWO MODES OF OPERATION, WITH AND WITHOUT CURRENT BIAS. THE FIRST INPUT THROUGH R47 OFFSETS THE OUTPUT OF IC1 OVERCOMING THE OPERATIONAL AMPLIFIER MAXIMUM INHERENT OFFSET. THE SECOND INPUT IS THROUGH R82 AND R76 OR ONLY R76, WHEN R82 IS SHORTED BY STRAPPING FROM THE OFFSET ADJUST POTENTIOMETER R102. WHEN OPERATING WITHOUT BIAS, STRAPPING IS NOT USED. THE OUTPUT IS NULLED BY ADJUSTING R102 WITH IN1, IN2, IN3, IN4 CONNECTED TO COMMON, WITH NO DITHER, AND OBSERVING THE OUTPUT SJ. R102 IS ADJUSTED UNTIL THE OUTPUT CURRENT EXCEEDS 1.0MA AND THEN IS ADJUSTED BACK UNTIL IT IS JUST 0MA. WHEN BIAS IS USED VOS1 IS STRAPPED TO VOS2 AND THEN ZERO ADJUSTED AS BEFORE. BIAS IS ADJUSTED AFTER ZEROING AND GAIN ADJUSTMENTS BY R102.
THE FIRST AND SECOND MONITORS DETECT FAULT CONDITIONS BY MONITORING THE OUTPUT OF IC1. WHEN THE OUTPUT IS LESS THAN -6.8 VOLTS, THE FIRST MONITOR DETECTS IT, AND Q7 TURNS OFF TURNING Q16 ON BRINGING THE OUTPUT FM1 TO A LOGIC "0". IF THE OUTPUT GOES POSITIVE, THEN Q8 TURNS

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 KCC 9/30/71	REV. 2	APPROVED	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	
ISSUED 7-26-71 AGF	PRINTS TO A1F(4) DL1.3T	FIRST MADE FOR REQ. PLASTIC MOLDING	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	PROPORTIONAL VALVE DRIVER	
MADE BY KENNEY C. COX 6V	PGA 1338	I.C. NO.		IC3622GPDA1	SH. NO. 3.4
				CONT. ON SH. 3.5	

FORM IC 2219 (6)

ON TAKING THE OUTPUT OF THE SECOND MONITOR, FM2, TO A "0". THE THIRD MONITOR DETECTS THE ABSENCE OF A MINIMUM AMOUNT OF DITHER SIGNAL FROM THE OUTPUT SO TO COMMON. IF THE LOAD BECOMES SHORTED, THE DITHER VOLTAGE WILL BE REDUCED. IN THIS MONITOR C10, R32, C5, AND R30 ACT AS A FILTER FOR THE DITHER. DURING NORMAL OPERATION, THIS PROVIDES ENOUGH CURRENT TO TURN ON Q15 TURNING OFF Q13, GIVING POSITIVE FEEDBACK THROUGH R48. THIS HOLDS Q15 ON UNTIL THE DITHER GOES NEGATIVE ENOUGH TO TURN IT OFF. THE COLLECTOR OF Q13 IS CLAMPED TO A MAXIMUM OF 12 VOLTS BY CR22. WHEN IT IS AT 12 VOLTS, C11 IS CHARGED THROUGH CR10 TO COMMON. WHEN Q13 IS TURNED ON AND THE COLLECTOR GOES TO 0 VOLTS, C11 DISCHARGES THROUGH CR11 AND R7 CHARGING C13 NEGATIVE WITH RESPECT TO COMMON. THIS PULLS CURRENT THROUGH R12 AND R26 KEEPING Q14 OFF. IF THE DITHER VOLTAGE IS SMALL, AS WHEN THE LOAD IS SHORTED, Q15 WILL NOT TURN ON AND THE COLLECTOR OF Q13 WILL REMAIN AT 12 VOLTS. THIS ALLOWS THE VOLT ACROSS C13 TO GO POSITIVE VOLTAGE TURNING Q14 ON INDICATING A FAULT CONDITION.

THE TIME DELAY OPERATES WHEN ANY OF THE INPUTS TO1, TO2, OR TO3 GOES TO AN HTL "0". AT THE END OF THE DELAY, THE OUTPUT WILL CHANGE FROM AN HTL "1" TO "0" IF THE "0" INPUT REMAINS. THE TIME DELAY CIRCUIT IS SIMILAR TO THOSE ON THE IC3622GTDA.

SPECIFICATIONS

INPUTS

IN1, IN2, IN3, AND IN4 ARE THE SUMMING INPUTS TO THE AMPLIFIER; IN1, IN2 POSITIVELY, AND IN3, IN4 NEGATIVELY. EACH HAS AN INPUT IMPEDANCE OF 105K TO COMMON.

TD1, TD2, AND TD3 ARE HIGH LEVEL DIODE COUPLED INPUTS SUPPLYING CURRENT TO COMMON WHEN PULLED TO THE "0" STATE. INPUT LOADING IS 2.5 - 18K LOADS RETURNED TO THE P15V BUS. MINIMUM INPUT "1" - 10 VOLTS, MAXIMUM INPUT "0" - 4.6 VOLTS REFERRED TO BOARD COMMON.

VO1 AND VO2 ARE STRAPPED TOGETHER TO CHANGE THE OFFSET REFERENCE INPUT TO THE AMPLIFIER FROM A NULLING VOLTAGE TO AN OFFSET VOLTAGE. WHEN STRAPPED THE OFFSET REFERENCE PROVIDES A 0 TO 5.65 VOLT, ±5% INPUT TO THE AMPLIFIER BY ADJUSTMENT OF R102.

OUTPUTS

VR AND VR COM ARE THE VOLTAGE REFERENCE OUTPUT AND THE VOLTAGE REFERENCE COMMON RESPECTIVELY. THE OUTPUT VR IS 11.8 VOLTS ± 0.8V. IT CAN DRIVE UP TO 6 -10K LOADS BETWEEN VR AND VR COM.

FM1, FM2, FM3, TD4 ARE HIGH LEVEL LOGIC OUTPUTS. EACH ARE DESIGNED TO DRIVE 18-18K LOADS. THE MINIMUM OUTPUT "1" IS 2 VOLTS BELOW THE BUS, THE MAXIMUM OUTPUT "0" IS .4 VOLT.

SO, FB, FB50, FB100, AND FB200 CONSTITUTE THE OUTPUT, OUTPUT FEEDBACK AND FEEDBACK STRAPPING POINTS. THE FEEDBACK RESISTANCES, R16, AND RESPECTIVE NOMINAL MAXIMUM CURRENT OUTPUT, GAINED BY STRAPPING COMBINATIONS OF FB50, FB100, AND FB200 ARE SHOWN IN THE TABLE BELOW.

FORM IC 2219 (12

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 2	APPROVED	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
ISSUED 7-26-71 A1F(4) A6F	PRINTS TO DL1,3T	FIRST MADE FOR REQ. PLASTIC MOLDING	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	PROPORTIONAL VALVE DRIVER
MADE BY KENNEY C. COX 6U	P5A	I.C. NO.	IC3622GPDA1	SH. NO. 3 5

FB200	FB100	FB50	RFB	NOMINAL CURRENT
			12.1Ω	50MA
		X	6.05Ω	100MA
	X		4.04Ω	150MA
	X	X	3.03Ω	200MA
X			2.42Ω	250MA
X		X	2.02Ω	300MA
X	X		1.73Ω	350MA
X	X	X	1.53Ω	400MA

TABLE

NOTE: X INDICATE PINS STRAPPED

AMPLIFIER GAIN

THE GAIN IS DETERMINED BY THE FEEDBACK STRAPPING AND THE FEEDBACK ADJUST POTENTIOMETER, R100. THE FORMULA TO DETERMINE THE RANGE OF THE GAIN IS THE FOLLOWING:

$$\frac{(421 + RFB)10^3}{9.10\ RFB} \leq \frac{IC}{VM} \leq \frac{(421 + RFB)10^3}{4.78\ RFB} \quad \frac{MA.}{VOLT}$$

WHERE I_L IS THE OUTPUT CURRENT IN MILLIAMPERES IN THE LOAD AND VM IS THE SUMMATION OF THE INPUTS.

$$VM = (V_{IN1} + V_{IN2} + V_{OFFSET}) - (V_{IN3} + V_{IN4})$$

MAXIMUM STEADY STATE CURRENT

IL MAX IS DETERMINED BY THE FOLLOWING FORMULA:

$$I_{LMAX} = \frac{PPV-1.}{(5.5 + RFB + R_L)} \quad \text{BUT NOT} > .5 \text{ AMPERES}$$

WHERE RFB IS THE FEEDBACK RESISTANCE SHOWN IN THE TABLE AND R_L IS THE RESISTANCE IN OHMS OF THE LOAD.

DITHER

THE DITHER IS ADJUSTABLE AND HAS AN EFFECTIVE 0 TO 1.2 VOLT RMS 60HZ INPUT TO THE AMPLIFIER.

DRIFT

THE DRIFT OF THE AMPLIFIER IS .05% /°C OF THE SET MAXIMUM CURRENT.

LINEARITY

THE LINEARITY IS ± 1% OF THE SET MAXIMUM CURRENT

TIME DELAY

THE NOMINAL DELAY IS 3.7 SECOND

BUSSES

PPV, PERIPHERAL POSITIVE VOLTAGE. THE MAXIMUM AND MINIMUM LIMIT OF THIS BUS ARE DETERMINED BY THE FOLLOWING FORMULA:

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

FORM IC 2219 I12

REV. 1 KCC 9-30-71	REV. 2	APPROVED	GENERAL ELECTRIC	ELEMENTARY DIAGRAM PROPORTIONAL VALVE DRIVER
ISSUED 7-26-71 A6F A1F(4)	PRINTS TO DL1-3T	FIRST MADE FOR REQ. PLASTIC MOLDING	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	IC3622GPDA1 SH. NO. 3.6
MADE BY KENNEY C. COX	PCA	ECL NO.		

$$1 + I_{LMAX}\ (5.5 + R\ FB\ + R_L) \leq PPV \leq \sqrt{12\,(R_L + RFB\ + 5.5)}$$

BUT NOT > 32 VOLTS

$$\text{CURRENT REQUIRED} = \frac{PPV}{3.9}\ MA + I_{LMAX}$$

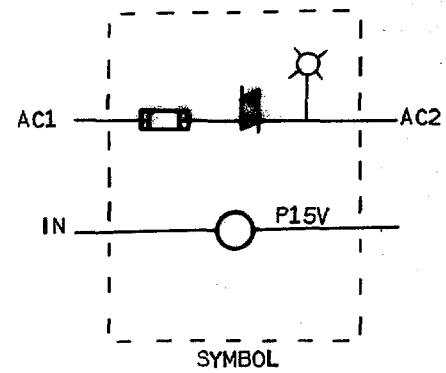
P15V - P14V TO P16V 25 MA TYPICAL, 30 MA MAX.

N15V - N14V TO N16V 12 MA TYPICAL, 16 MA MAX.

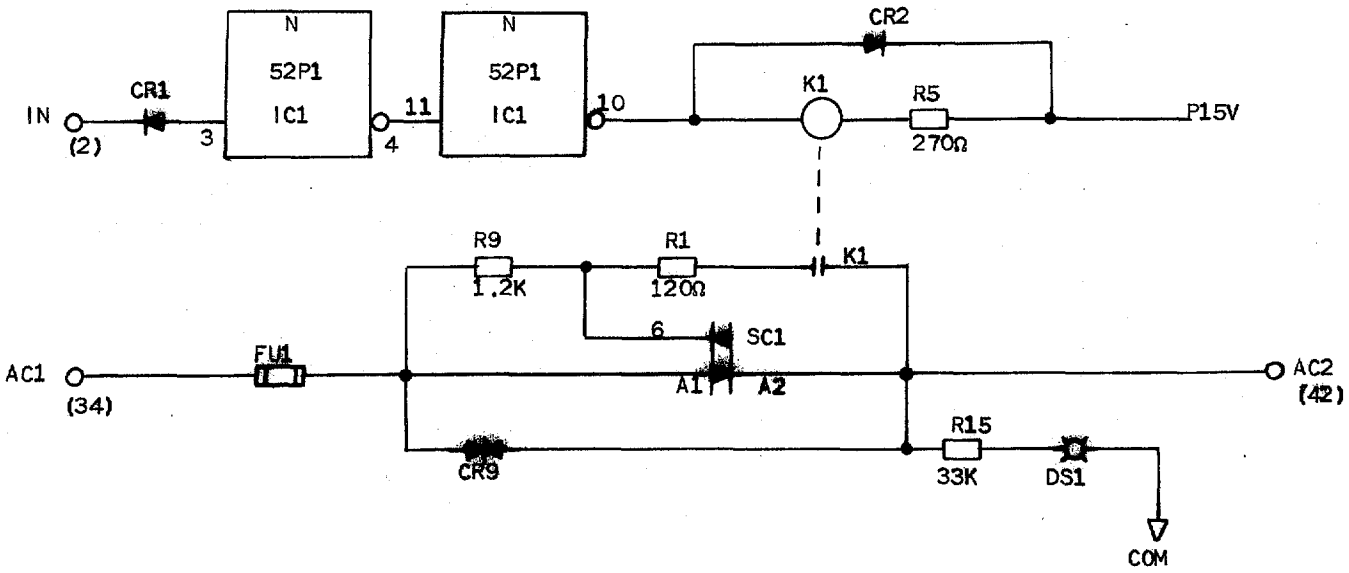
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

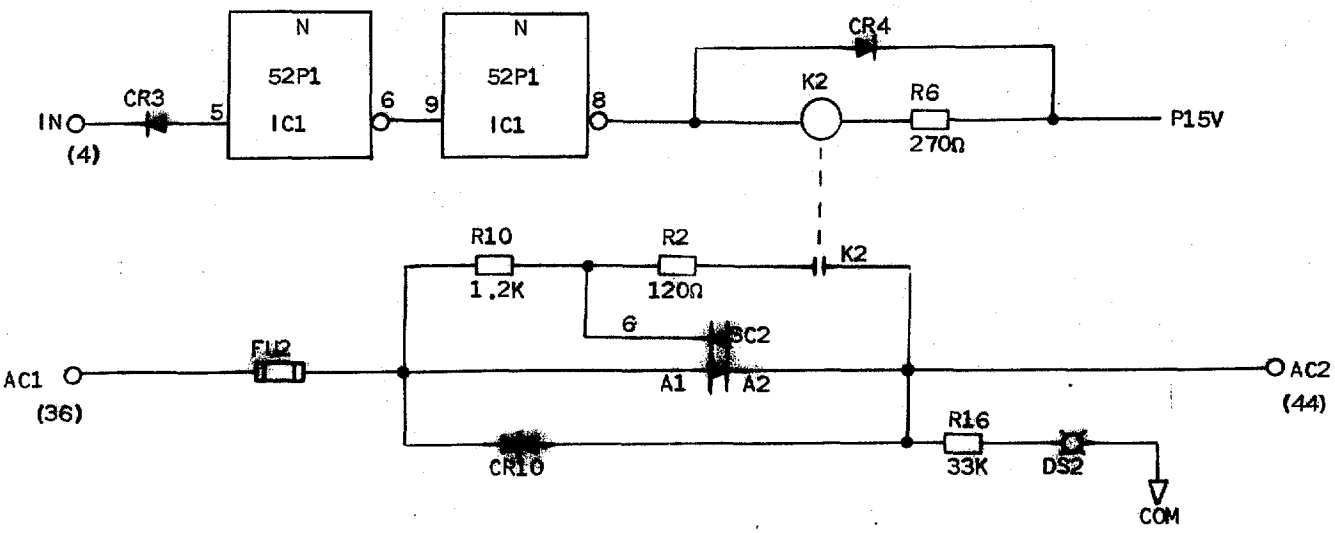
REV. 1 ACW 11-15-71.	REV. 2	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	PROPORTIONAL VALVE DRIVER
ISSUED 7-26-71. A6F A1F(4)	PRINTS TO DL1,3T	FIRST MADE FOR REQ. PLASTIC MOLDING	INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	IC3622GPDA1	SH. NO. 3 OF 7
MADE BY KENNEY C. COX 6V		ELC NO.			

FORM IC 2219 (12



CIRCUIT	IN	AC1	AC2
1	2	34	42
2	4	36	44
3	6	38	46
4	8	40	48





PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

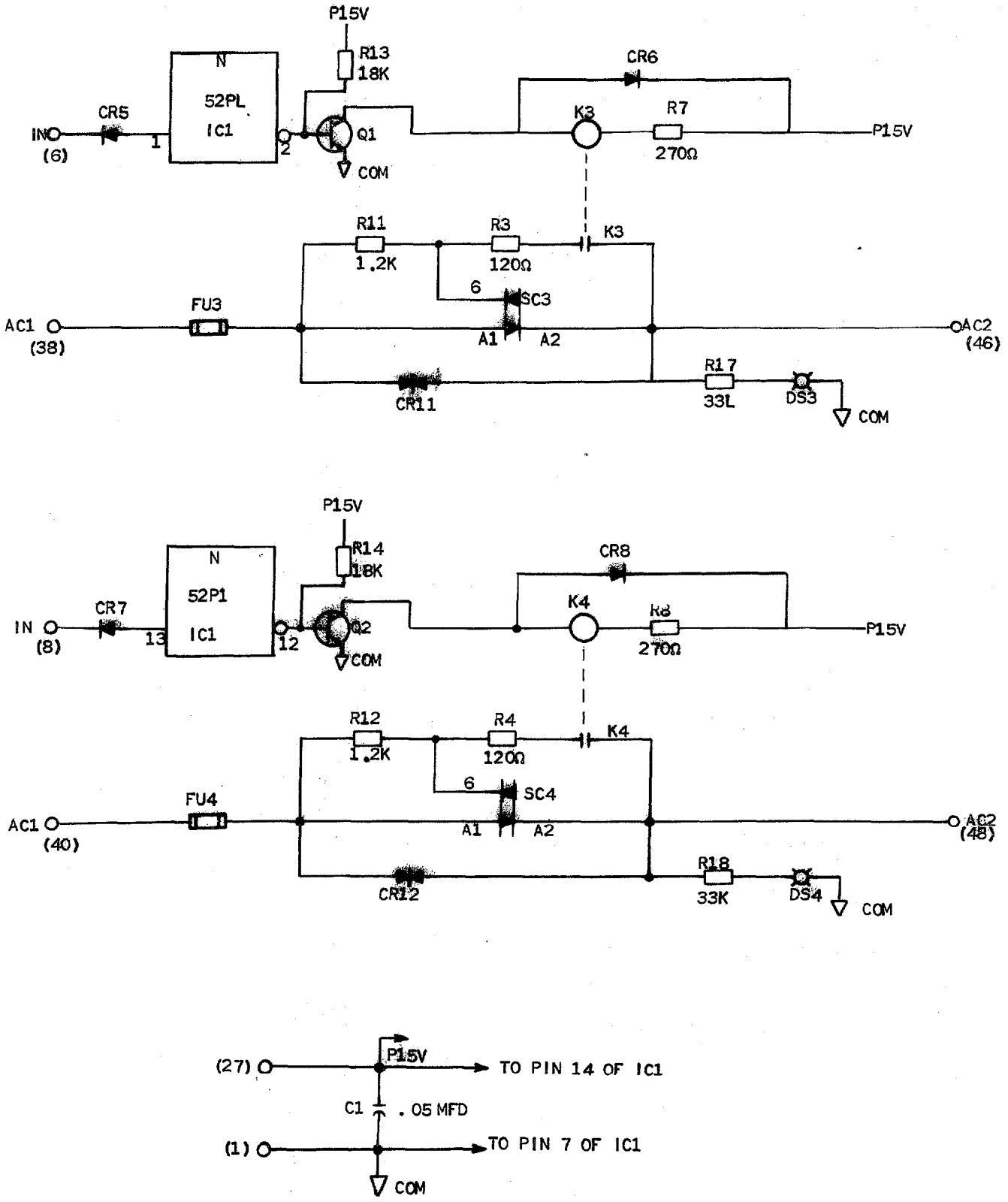
REV. 1	REV. 4	REV. 5	PRINTS TO DL1, PGA, GV	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 2	ISSUED May 12, 1972	2530 1838 AGF	FIRST MADE FOR PLASTIC MOLD		
REV. 3	MADE BY KENNEY C. COX	CHVTVN AIF (4) 3T	I.C. NO.		

ELEMENTARY DIAGRAM: SOLENOID DRIVER

I C 3 6 2 2 G S D E 1

CONT. ON SH. 3.1 SH. NO. 3.0

FORM 1C 2390 2-72



PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED May 12, 1972		FIRST MADE FOR PLASTIC MOLD		INDUSTRY CONTROL DEPT.	SOLENOID DRIVER
REV. 3	MADE BY KENNEY C. COX		I.C. NO.		SALEM, VA. U.S.A.	IC3622GSDE1
						CONT. ON SH. 3.2 SH. NO. 3.1

A B C D E F G H J K L

01

APPLICATION INFORMATION

03

EACH CIRCUIT ON THIS BOARD CONTAINS A STATIC A-C SWITCH (TRIAC SC1) WHICH PROVIDES A CONTROLLED CIRCUIT CLOSURE BETWEEN THE A-C SUPPLY INPUT AND THE LOAD. SINCE THE FOUR CIRCUITS ARE SIMILAR, THE FIRST WILL BE DISCUSSED. THE TRIAC CIRCUIT IS GATED ON BY THE CONTACT OF K1 CLOSING A CIRCUIT FROM ANODE 2 TO GATE OF SC1, SO THAT A POSITIVE OR NEGATIVE POTENTIAL ON ANODE 2 (WITH RESPECT TO THE GATE) PROVIDES A CURRENT THROUGH R1 AND THE GATE TURNING SC1 ON. WHEN SC1 IS THUS TURNED ON, IT WILL "LATCH" AND CONTINUE TO CONDUCT UNTIL THE LOAD CURRENT GOES THROUGH ZERO. IF K1 CONTACT IS CLOSED AT THE TIME THE CURRENT PASSES THROUGH ZERO, VOLTAGE ON A2 WILL AGAIN TURN SC1 ON FOR THE NEXT HALF CYCLE. IF K1 IS OPEN, THEN SC1 WILL OPEN AT THE CURRENT ZERO CROSSING AND REMAIN OPEN UNTIL K1 AGAIN CLOSES.

07

09

11

K1 COIL PROVIDES ISOLATION BETWEEN THE OUTPUT AND INPUT DRIVER. THE INPUT IS INTENDED TO BE DRIVEN FROM A HIGH THRESHOLD LOGIC ELEMENT.

15

EACH OUTPUT CIRCUIT IS FUSED WITH 2.0 AMPERE SLOWBLOW FUSE TO PROTECT OUTPUT WIRING.

17

SPECIFICATIONS - (EACH CIRCUIT)

INPUT IS A LOGIC INPUT. INPUT LOADING IS "ONE INPUT" 18K LOAD. INPUT 0 TO 4 VOLTS ACTUATES K1, INPUT 10-15 VOLTS RELEASES K1. MAXIMUM INPUT -- OUTPUT VOLTAGE DIFFERENTIAL -- ±200 VOLTS.

19

OUTPUT IS A SWITCHING CIRCUIT WHOSE CLOSURES FOLLOW K1 OPERATION.

21

MAXIMUM CLOSED CIRCUIT DROP ACROSS SWITCH IS ±3 VOLTS (EXCEPT FOR SHORT SPIKE ±20 VOLTS MAXIMUM PEAK AT EACH TRANSITION).

23

MAXIMUM OPEN CIRCUIT. LEAKAGE THROUGH SWITCH IS 4 MILLIAMPERES.
OUTPUT LOADING IS.2TO 1.6 AMPERES RMS STEADY STATE, 35 AMPERES 8 MILLISECONDS MAXIMUM INRUSH. SWITCH RATING IS BASED ON LINEAR BILATERAL LOADS, WITH NO RESTRICTIONS AS TO POWER FACTOR.

25

A-C SUPPLY INPUT AC1 IS 50-60 HZ SINE WAVE, 75-130 VOLTS RMS WITH RESPECT TO AC SUPPLY RETURN.
AC2 IS THE LOAD TERMINAL.

OPERATE TIME (INPUT CLOSURE TO OUTPUT CLOSURE) - 2 MILLISECONDS MAXIMUM.

29

RELEASE TIME (INPUT OPEN TO OUTPUT OPEN) - 2 MILLISECONDS PLUS 1/2 CYCLE OF SUPPLY PERIOD MAXIMUM.

BUS

31

P15V - 14V - 16V
20 MILLIAMPERES MAXIMUM PER ACTUATED CIRCUIT.

33

.24 WATTS MAXIMUM PER ACTUATED CIRCUIT.

35

REPETITION RATE - ANY DRIVER (OR ALL DRIVERS) MAY BE OPERATED CONTINUOUSLY AT AN AVERAGE RATE OF UP TO 2 CYCLES PER SECOND PROVIDED MINIMUM CYCLE PERIODS IS 1/6 SECOND, AND AVERAGE IS TAKEN OVER A PERIOD NOT EXCEEDING 12 SECONDS. ATTENTION IS CALLED TO THE FACT THAT WHEN RUN CONTINUOUSLY AT THIS RATE THE REED SWITCH MAY BE EXPECTED TO WEAR OUT IN A YEAR.

37

39

APPLICATION NOTES:

ON INDICATOR OPERATION. WHEN UNIT IS USED FOR DRIVING AN INDICATOR LAMP, AND NOT FOR CONTROL PURPOSES, MINIMUM LOAD RESTRICTION DOES NOT APPLY, EXCEPT WHERE NEON LAMP IS DRIVEN, AT LEAST 15,000 OHM PARALLEL RESISTIVE LOAD REQUIRED. THIS CARD IS DESIGNED TO PLUG IN TO A MODIFIED DOM II CARD ROW. THE AVERAGE NUMBER OF DRIVERS CONTINUOUSLY ENERGIZED SHOULD BE NO MORE THAN 4 PER 2 ADJACENT CARD SLOTS AND THESE SHOULD NOT ALL BE ON THE SAME CARD.

43

45

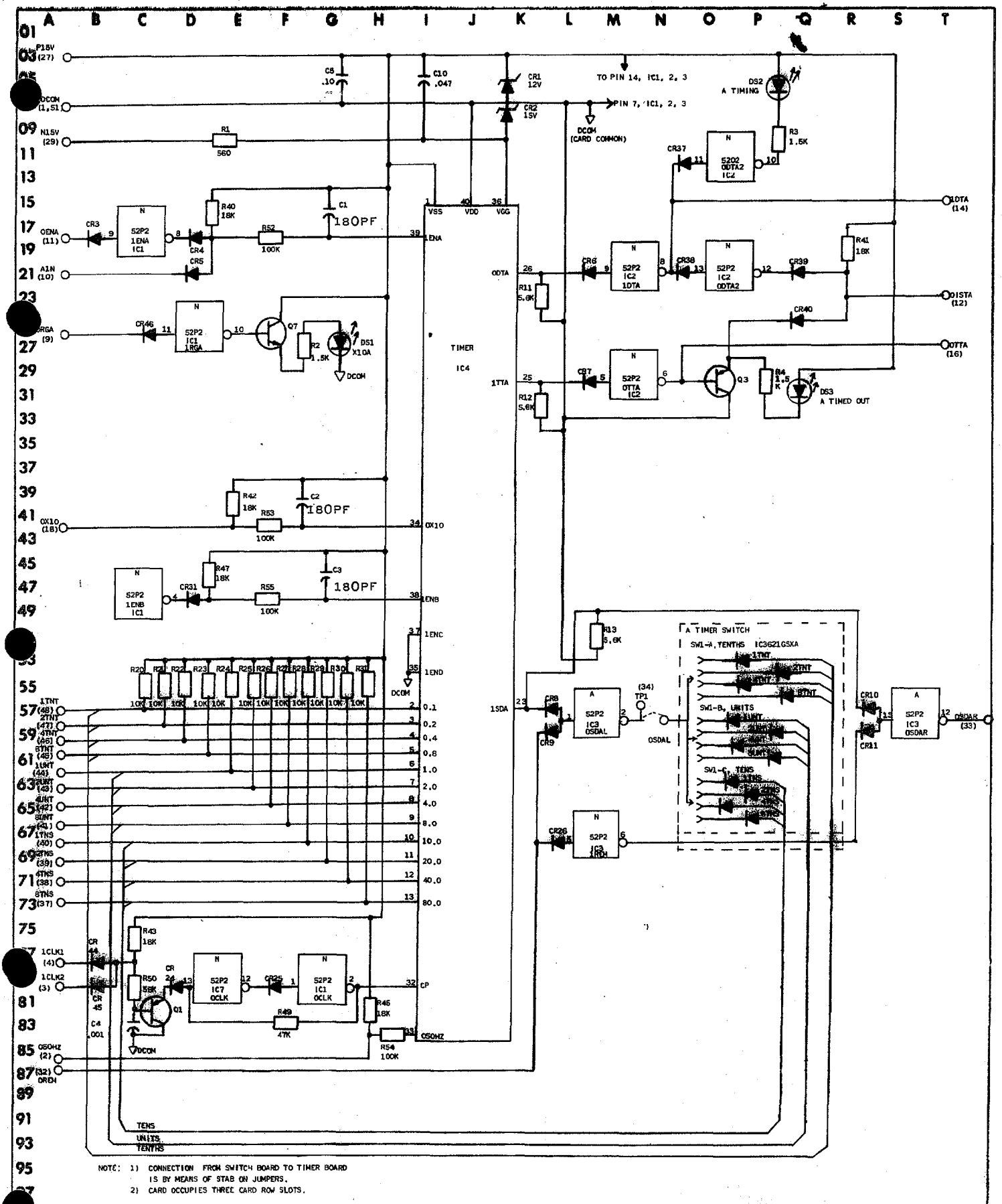
47

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51

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL1, PGA	APPROVALS -	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM SOLENOID DRIVER
REV. 2	ISSUED May 12, 1972		CHVIVN 1338.6V.3T	FIRST MADE FOR PLASTIC MOLD		IC3622GSDE1
REV. 3	MADE BY KENNEY C. COX		2530 TWAY AIF (4)AGF	I.C. NO.		CONT. ON SH. FL. SH. NO. 3.2



NOTE: 1) CONNECTION FROM SWITCH BOARD TO TIMER BOARD IS BY MEANS OF STAB ON JUMPERS.
2) CARD OCCUPIES THREE CARD ROW SLOTS.

REV.1 wmy 4-5-73	REV.4	REV.5	PRINTS TO DL.D1E5 2550	APPROVALS RC Reeves	GENERAL ELECTRIC	ELEMENTARY DIAGRAM TIMER CARD, 3DIGIT ONE CIRCUIT
REV.2	ISSUED September 1, 1972 CHVIVN		FIRST MADE FOR PLASTIC MOLD	INDUSTRY CONTROL DEPT. SALEM, VA., U.S.A.		IC3622GTCB1
REV.3	MADE BY GARY HARRIS	RIC	I.C.NO.			CONT. ON SH. 3, 1SH NO. 3.0

FORM IC 2349 (07-69)

APPLICATION NOTES

DIGITAL TIMER

THIS CARD HAS ONE DIGITAL TIMER ON THE CARD. THE TIMER USES A CLOCK SIGNAL DERIVED FROM THE AC POWER LINE FREQUENCY FOR HIGH TIMER STABILITY. FOUR RANGES ARE AVAILABLE DEPENDING ON THE INPUT CLOCK FREQUENCY AND THE CONDITION OF THE OX10 INPUT. RANGE AND ACCURACY SPECIFICATIONS ARE SHOWN IN TABLE 1.

1CLK FREQUENCY	OX10 VOLTS	RANGE SECONDS	ACCURACY SECONDS	REPEATABILITY SECONDS
120	15	0-99.9	.01 TO .05	.01
120	0	0-999	.01 TO .05	.01
1200	15	0-9.99	.001 TO .005	.001
1200	0	0-99.9	.001 TO .005	.001

TABLE 1

A TIMER CIRCUIT CAN BE SHIFTED BY A FACTOR OF 10 BY SWITCHING TH OX10 INPUT.

THE 1CLK INPUT IS NORMALLY WIRED TO EITHER 120 OR 1200 HZ, RESTRICTING THE TIMER TO TWO OF THE FOUR AVAILABLE RANGES, BY SWITCHING OX10. BOTH 1CLK AND OX10 ARE LOGIC LEVEL SO THEY CAN BE SWITCHED BY EXTERNAL LOGIC IF REQUIRED.

TIMER INPUTS AND OUTPUTS ARE ARRANGED SO IT CAN BE USED WITH EITHER STANDARD LOGIC OR FLOW THROUGH LOGIC DESIGNS. WHEN USED WITH FLOW THROUGH LOGIC, OISTA AND OTTA OUTPUTS ARE USED TO PICK UP RELAYS ON A GLEB CARD. THESE RELAYS GIVE THE INSTANTANEOUS AND TIMED OUT CONTACTS NORMALLY ASSOCIATED WITH A TIMER. FIGURE 1 GIVES AN EXAMPLE CIRCUIT. FIGURE 2 GIVES CIRCUIT INPUT AND OUTPUT WAVEFORMS. IT SHOULD BE NOTED THAT THERE IS A SHORT DELAY OF 4 CLOCK PULSES IN PICKING UP THE TIMER. WHEN THE TIMER IS SET TO 000, 1DTA DOES NOT SWITCH, AND/OISTA AND OTTA SWITCH AT THE SAME TIME.

THE CLOCK FREQUENCY IS AVAILABLE FROM TWO SOURCES. IF THE 0.00-9.99 SECOND RANGE IS USED, THE IC3622GPGA CARD IS USED TO SUPPLY THE CLOCK SIGNALS AT EITHER 120 OR 1200 HZ. IF ONLY THE TWO LONGER RANGES ARE REQUIRED, THE SCHEME SHOWN IN FIGURE 3 CAN BE USED. THE TWO INPUT DIODES ACT AS A FULL WAVE RECTIFIER TO GENERATE THE 120 HZ REFERENCE FREQUENCY.

THE TIMER CAN BE SET REMOTELY AND INDEPENDENTLY OF THUMBWHEEL SWITCH SETTINGS. THE DIAGRAM OF FIGURE 4 SHOWS THE ARRANGEMENT. WHEN OREM IS MADE 0 VOLTS, THE INTERNAL STROBE LINES WILL NOT OPERATE. INSTEAD, WHEN THE TIMER IS INITIATED, STROBE PULSES (PULSES TO ZERO VOLTS) WILL APPEAR AT OSDAR OR OSDBR. THESE PULSES LAST FOR ONE CLOCK CYCLE AND APPEAR AT TIMES CORRESPONDING TO THE LEAST SIGNIFICANT DIGIT OF THE RANGED SELECTED (.01,0.1, OR 1 SECOND). WHEN THE STROBE PULSE APPEARS THE EXTERNAL DATA IS SUPPLIED TO THE TWELVE DATA LINES 1TNT THROUGH 8TNS. THESE CORRESPOND TO THE 1, 2, 4, 8 BITS OF THE BCD CODED THUMBWHEEL SWITCHES WITH THREE DECADES, (TENTHS, UNITS, TENS). THE TIMER STARTS COUNTING WHEN INITIATED AND STOPS (TIMES OUT) WHEN ITS INTERNAL COUNTER MATCHES THE APPLIED DATA. CAUTION: ALL SWITCH MULTIPLEX CIRCUITS CONNECTED TO THE DATA LINES MUST BE CAPABLE OF WIRE-OR (PARALLEL CONTACT) OPERATION TO ALLOW THE INTERNAL TIMER SWITCHES TO FUNCTION.

INPUTS:

OENA (11) A ZERO VOLT LEVEL INITIATES TIMING — AIN (12) SHOULD BE OPEN

AIN (12) A 15 VOLT LEVEL INITIATES TIMING IF THIS INPUT IS USED. ONE LOAD. — OENA (11) SHOULD BE JUMPERED TO OV,

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL 50	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM TIMER CARD
REV. 2	ISSUED August 23, 1972	2530	FIRST MADE FOR CALVINE Plastic Mold		IC3622GTCB1
REV. 3	MADE BY GARY HARRIS	RIC	I.C. NO.		CONT. ON SH. 3.2 SH. NO. 3.1

APPLICATION INFORMATION

INPUTS (CONTINUED)

ORGA (9) A ZERO VOLT LEVEL LIGHTS THE X10A INDICATION. THIS IS NORMALLY USED TO INDICATE A TIMES 10 SHIFT ON THE SWITCH RANGE. ONE LOAD.

OX10 (18) OPEN CIRCUIT OR 15 VOLTS GIVES NORMAL TIMING RANGE (0-99.9 SEC. FOR 120 HZ CLOCK). ZERO VOLTS MULTIPLIES TIME RANGE BY A FACTOR OF 10.

1CLK1 (4) CLOCK INPUT, EITHER 120 OR 1200 HZ FOR 60 HZ POWER LINE, OR 100 OR 1 KHZ FOR 50 HZ
1CLK2 (3) POWER LINE FREQUENCY. ONE LOGIC LOAD.

050HZ (2) OPEN CIRCUIT FOR 60 HZ APPLICATIONS. CONNECT TO COMMON FOR 50 HZ OPERATION.

OREM (32) OPEN CIRCUIT OR +15V FOR NORMAL OPERATION. OV FOR REMOTE OPERATION. ONE LOGIC LOAD.

1TNT THROUGH 9TNS (PINS 37 THROUGH 48). THESE ARE INPUT DATA LINES FOR REMOTE TIME SETTING. EACH INPUT IS TWO LOADS. CONTACT CLOSURES TO OV SETS DATA WHEN OSDAR IS ZERO.

GND (5) PIN 5 IS USED TO GROUND THE CARD FRONT.

OUTPUTS

1DTA (14) GOES TO 15V WHILE TIMER IS TIMING. GOES TO OV WHEN RESET OR TIMED OUT. CAN DRIVE 17 LOADS

OISTA (12) 15 VOLTS WHEN RESET. GOES TO O VOLTS WHEN TIMING AND REMAINS O WHEN TIMED OUT. AS SHOWN IN FIGURE 2, THERE IS A DELAY OF THREE OR FOUR CLOCK PULSES BETWEEN SWITCHING THE INPUT AND THE PICK UP OF OISTA. CAN DRIVE 17 LOADS.

OTTA (16) GOES TO ZERO WHEN TIMED OUT. +15 VOLTS WHEN RESET OR TIMING. CAN DRIVE 17 LOADS.

OSDAR (33) STROBE SIGNAL FOR REMOTE SET POINT. GOES TO ZERO TO MULTIPLEX INPUT DATA. CAN DRIVE 17 LOADS.

POWER SUPPLY

+15V (27). TYPICAL CURRENTS: 45 MA. WHEN RESET. ADD 5 MA FOR EACH TIMER TIMING OR TIMED OUT. ADD ADDITIONAL 5 MA FOR EACH X10 INDICATOR LIT, TOTAL 55MA, TYPICAL, 70 MA MAX.

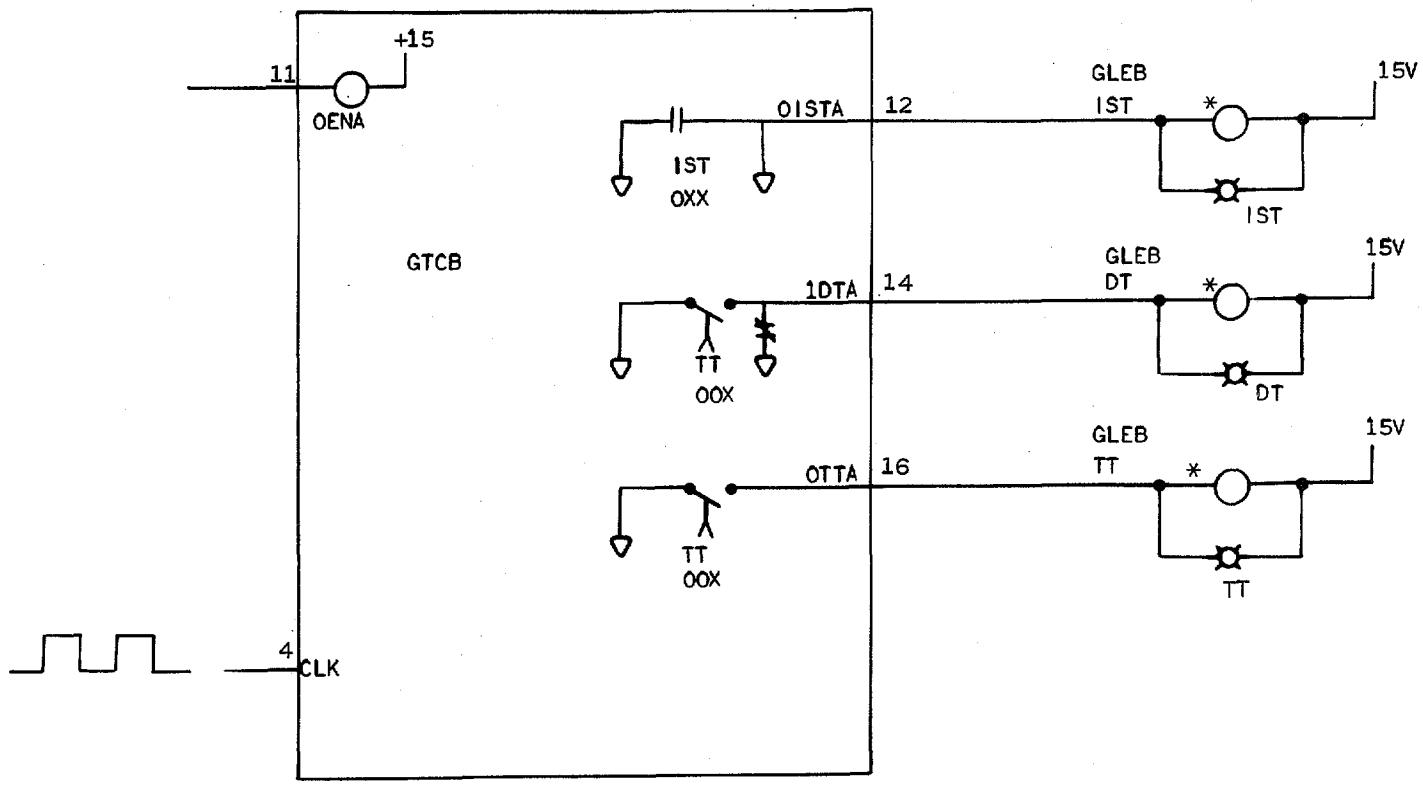
-15V (29) - 6.5 MA TYPICAL, 10 MA MAX

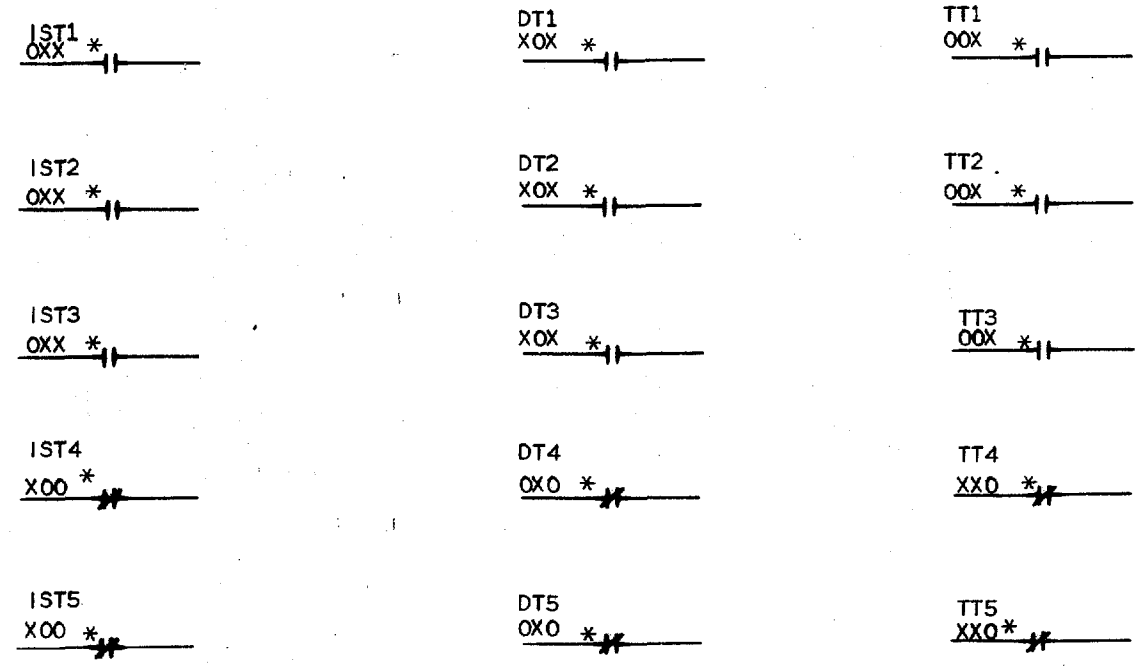
CARD SIZE:

STANDARD DOM II WITH CARD FRONT, OCCUPIES THREE CARD ROW SPACES.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL1, DL58	APPROVED R.E. Banks	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM TIMER CARD
REV. 2	ISSUED August 23, 1972	2530 2HVIVN	FIRST MADE FOR Electric Mill			I C 3 6 2 2 G T C B 1
REV. 3	MADE BY GARY R. HARRIS	RIC	I.C. NO.			CONT. ON SH. 3 3 SH. NO. 3 2





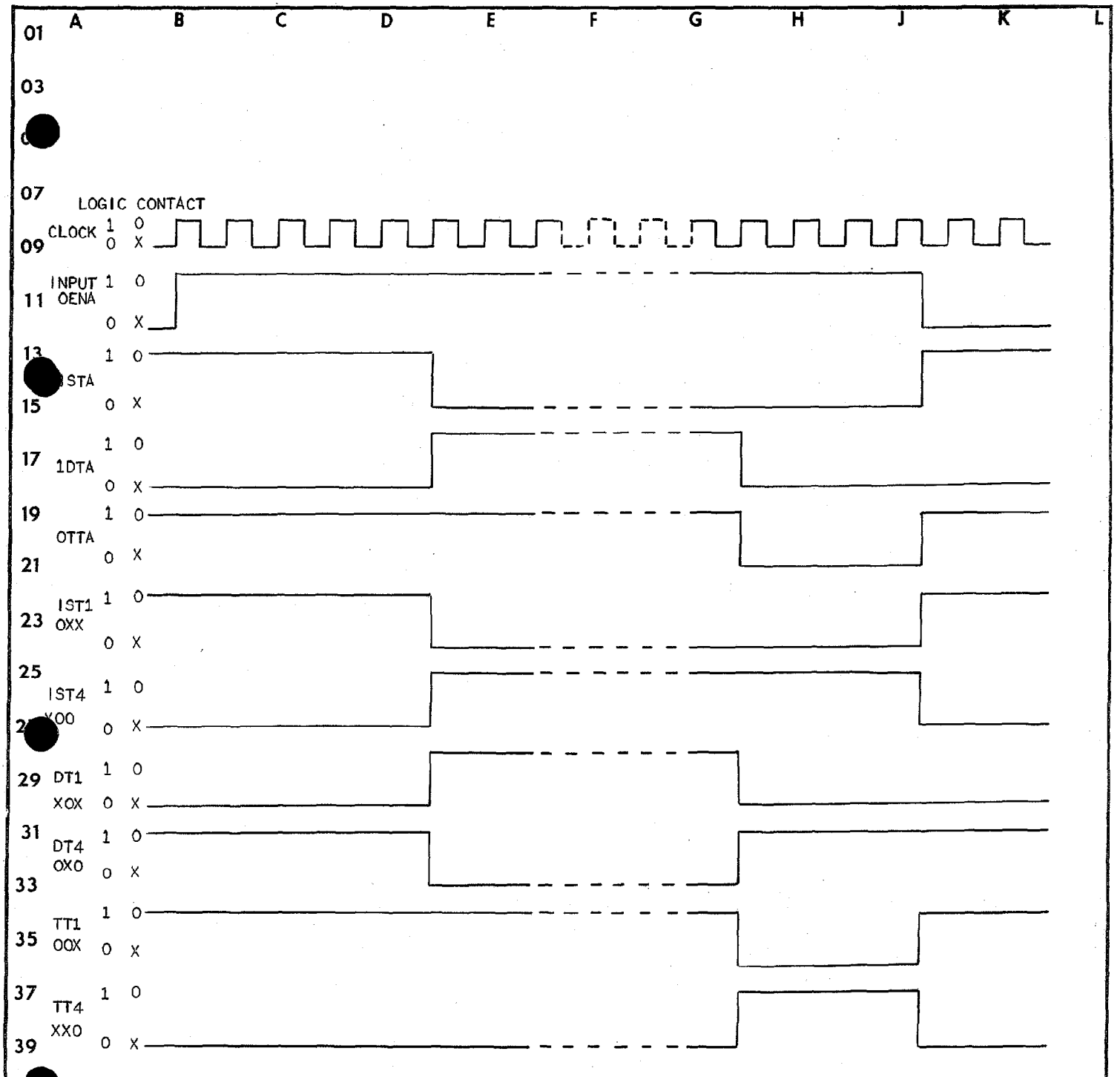
TYPICAL WIRING OF TIMER FOR FLO-THROUGH LOGIC

FIGURE 1

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DUPLEX 2530	APPROVED Pete Ferris	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM TIMER CARD
REV. 2	ISSUED August 23, 1972			TEST MADE FOR Post Mold		IC3622GTCB1
REV. 3	MADE BY GARY HARRIS		RIC	I.C. NO.		CONT. ON SH. 3,4 SH. NO. 3,3

FORM 1C 2390 2-72



DEFINITIONS:

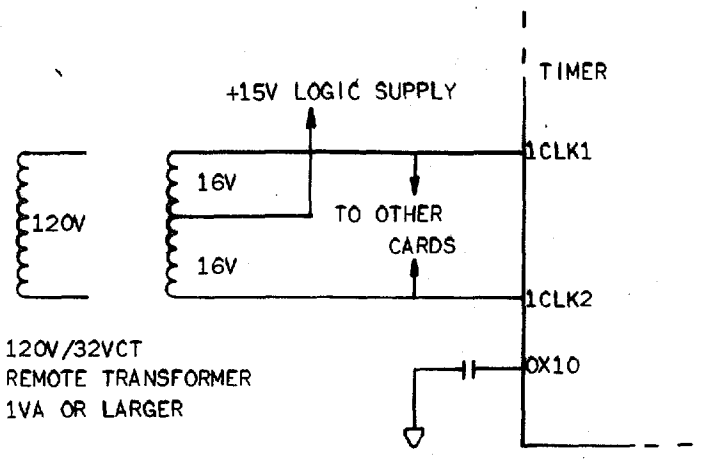
LOGIC 1 10 TO 15 VOLTS
LOGIC 0 0 TO 4 VOLTS
CONTACT 0 OPEN
CONTACT X CLOSED

TIMER SIGNALS AVAILABLE

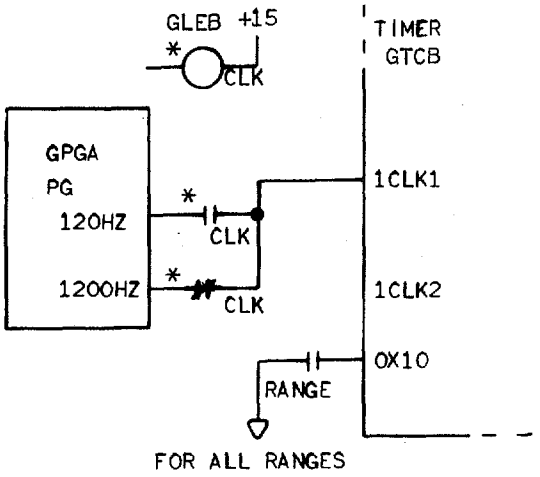
FIGURE 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO IDL1, DL58 2530	APPROVED	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 2	ISSUED August 23, 1972 CHVIVN		FIRST MADE FOR PLAST Mdd		ELEMENTARY DIAGRAM TIMER CARD
REV. 3	MADE BY GARY HARRIS RIC		I.C. NO.		1C3622GTCB1 CONT. ON SH. 3.5 SH. NO. 3.4

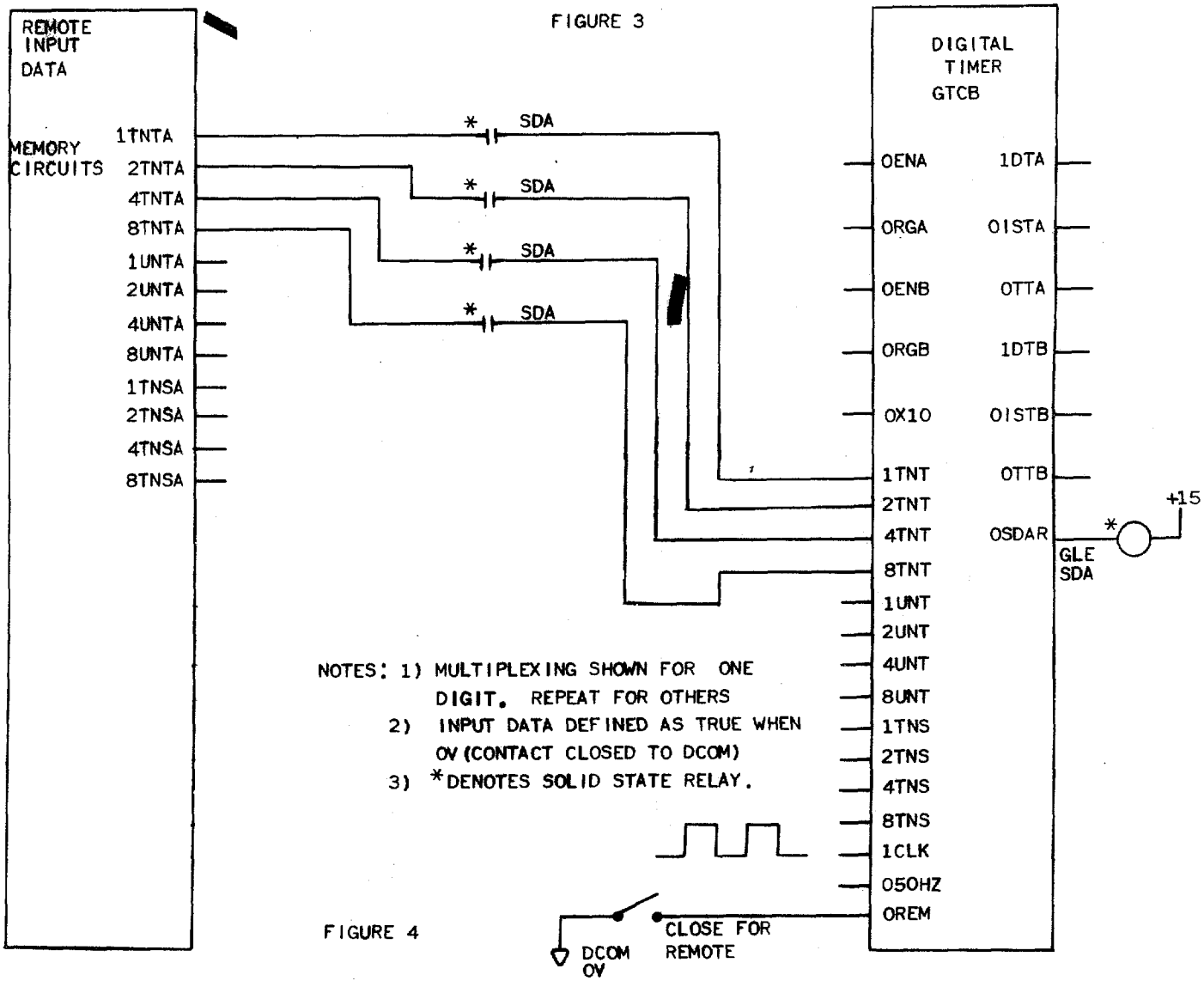


FOR 120HZ ONLY
2 RANGES
FIGURE 3A



FOR ALL RANGES
FIGURE 3B

ALTERNATE CLOCK INPUTS
FIGURE 3

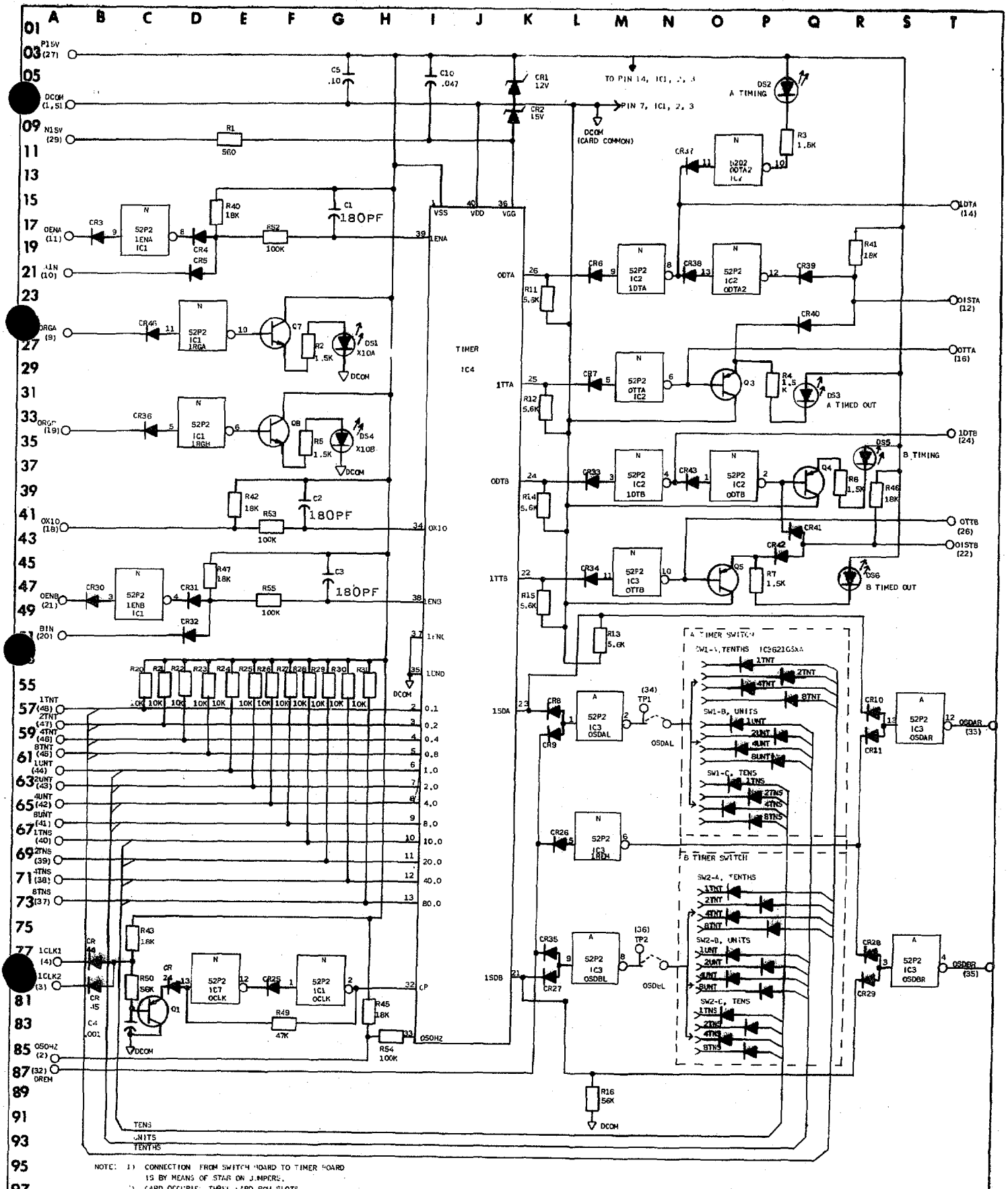


NOTES: 1) MULTIPLEXING SHOWN FOR ONE DIGIT. REPEAT FOR OTHERS
2) INPUT DATA DEFINED AS TRUE WHEN 0V (CONTACT CLOSED TO DCOM)
3) *DENOTES SOLID STATE RELAY.

FIGURE 4

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL58	R.C. Jones	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 2	ISSUED August 23, 1982	CHVIVN 253DL1	FIRST MADE FOR PLAST. Mold		ELEMENTARY DIAGRAM TIMER CARD
REV. 3	MADE BY GARY HARRIS	RIC	I.C. NO.		IC3622GTCB1
					CONT. ON SH. FL. SH. NO. 3.5



NOTE: 1) CONNECTION FROM SWITCH BOARD TO TIMER BOARD IS BY MEANS OF STAR ON JUMPERS.
2) CARD OCCUPIES THREE CARD ROM SLOTS.

1 WMH 4-5-73	REV.4	REV.5	PRINTS TO DL1, DL58	APPROVALS RC Davis	GENERAL ELECTRIC
REV.2	ISSUED May 17, 1972		CHV1VN 2530, R(c)	FIRST MADE FOR	INDUSTRY CONTROL DEPT. SALEM, VA., USA
REV.3	MADE BY KENNEY C. COX			I.C. NO.	

ELEMENTARY DIAGRAM
TIMER CARD

IC3622GTCB2

CONT. ON SH. 3.1 SH. NO. 3.0

APPLICATION NOTES

DIGITAL TIMER

THIS CARD IS AVAILABLE WITH EITHER ONE OF TWO DIGITAL TIMERS ON THE CARD. THE TIMER USES A CLOCK SIGNAL DERIVED FROM THE AC POWER LINE FREQUENCY FOR HIGH TIMER STABILITY. FOUR RANGES ARE AVAILABLE DEPENDING ON THE INPUT CLOCK FREQUENCY AND THE CONDITION OF THE OX10 INPUT. RANGE AND ACCURACY SPECIFICATIONS ARE SHOWN IN TABLE 1.

1CLK FREQUENCY	OX10 VOLTS	RANGE SECONDS	ACCURACY SECONDS	REPEATABILITY SECONDS
120	15	0-99.9	.01 TO .05	.01
120	0	0-999	.01 TO .05	.01
1200	15	0-9.99	.001 TO .005	.001
1200	0	0-99.9	.001 TO .005	.001

TABLE 1

A TIMER CIRCUIT CAN BE SHIFTED BY A FACTOR OF 10 BY SWITCHING TH OX10 INPUT.

THE 1CLK INPUT IS NORMALLY WIRED TO EITHER 120 OR 1200 HZ, RESTRICTING THE TIMER TO TWO OF THE FOUR AVAILABLE RANGES, BY SWITCHING OX10. BOTH 1CLK AND OX10 ARE LOGIC LEVEL SO THEY CAN BE SWITCHED BY EXTERNAL LOGIC IF REQUIRED.

BECAUSE OF SHARING COMMON CIRCUITRY, THERE ARE TWO RESTRICTIONS ON APPLICATION OF CARDS WITH TWO TIMERS. THE FIRST IS THAT ONE TIMER SHOULD NOT BE INITIATED WHILE THE OTHER IS ALREADY TIMING. THIS WOULD RESULT IN THE SECOND TIMER PERIOD BEING SHORTENED BY THE AMOUNT OF DELAY FROM STARTING THE FIRST TIMER TO STARTING THE SECOND TIMER. THE OTHER RESTRICTION IS THAT IF TWO TIMERS ARE INITIATED AT THE SAME TIME, BOTH TIMERS SHOULD BE CONNECTED FOR THE SAME TIME RANGE, SINCE THE CLOCK AND OX10 INPUTS ARE COMMON FOR THE TWO TIMERS.

TIMER INPUTS AND OUTPUTS ARE ARRANGED SO IT CAN BE USED WITH EITHER STANDARD LOGIC OR FLOW THROUGH LOGIC DESIGNS. WHEN USED WITH FLOW THROUGH LOGIC, OISTA (OR OISTB) AND OTTA (OR OTTB) OUTPUTS ARE USED TO PICK UP RELAYS ON THE GLEB.CARD. THESE RELAYS GIVE THE INSTANTANEOUS AND TIMED OUT CONTACTS NORMALLY ASSOCIATED WITH A TIMER. FIGURE 1 GIVES AN EXAMPLE CIRCUIT. FIGURE 2 GIVES CIRCUIT INPUT AND OUTPUT WAVEFORMS. IT SHOULD BE NOTED THAT THERE IS A SHORT DELAY OF 4 CLOCK PULSES IN PICKING UP THE TIMER. WHEN THE TIMER IS SET TO 000, 1DTA DOES NOT SWITCH, AN OISTA AND OTTA SWITCH AT THE SAME TIME.

THE CLOCK FREQUENCY IS AVAILABLE FROM TWO SOURCES. IF THE 0.00-9.99 SECOND RANGE IS USED, THE IC3622GPGA CARD IS USED TO SUPPLY THE CLOCK SIGNALS AT EITHER 120 OR 1200 HZ. IF ONLY THE TWO LONGER RANGES ARE REQUIRED, THE SCHEME SHOWN IN FIGURE 3 CAN BE USED. THE TWO INPUT DIODES ACT AS A FULL WAVE RECTIFIER TO GENERATE THE 120 HZ REFERENCE FREQUENCY.

THE TIMER CAN BE SET REMOTELY AND INDEPENDENTLY OF THUMBWHEEL SWITCH SETTINGS. THE DIAGRAM OF FIGURE 4 SHOWS THE ARRANGEMENT. WHEN OREM IS MADE 0 VOLTS, THE INTERNAL STROBE LINES WILL NOT OPERATE. INSTEAD, WHEN THE TIMER IS INITIATED, STROBE PULSES (PULSES TO ZERO VOLTS) WILL APPEAR AT OSDAR OR OSDBR. THESE PULSES LAST FOR ONE CLOCK CYCLE AND APPEAR AT TIMES CORRESPONDING TO THE LEAST SIGNIFICANT DIGIT OF THE RANGED SELECTED (.01, 01 OR 1 SECOND). WHEN THE STROBE PULSE APPEARS THE EXTERNAL DATA IS SUPPLIED TO THE TWELVE DATA LINES 1TNT THROUGH 8TNS. THESE CORRESPOND TO THE 1, 2, 4, 8 BITS OF THE BCD CODED THUMBWHEEL SWITCHES WITH THREE DECADES, (TENTHS, UNITS, TENS). THE TIMER STARTS COUNTING WHEN INITIATED AND STOPS (TIMES OUT) WHEN ITS INTERNAL COUNTER MATCHES THE APPLIED DATA. CAUTION: ALL SWITCH MULTIPLEX CIRCUITS CONNECTED TO THE DATA LINES MUST BE CAPABLE OF WIRE-OR (PARALLEL CONTACT) OPERATION TO ALLOW THE INTERNAL TIMER SWITCHES TO FUNCTION.

INPUTS:

OENA (11) A ZERO VOLT LEVEL INITIATES TIMING ON THE TOP TIMER. AIN (12) SHOULD BE OPEN

AIN (12) A 15 VOLT LEVEL INITIATES TIMING ON THE TOP TIMER. OENA (11) SHOULD BE JUMPERED TO OV. ONE INPUT LOAD.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL58, RIC, 2530, DL1	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM TIMER CARD
REV. 2	ISSUED May 12, 1972			FIRST MADE FOR		IC3622GTCB2
REV. 3	MADE BY KENNEY C. COX		CHV1VN	I.C. NO.		CONT. ON SH. 3.2 SH. NO. 3.1

APPLICATION INFORMATION

INPUTS (CONTINUED)

ORGA (9) A ZERO VOLT LEVEL LIGHTS THE X10A INDICATION. THIS IS NORMALLY USED TO INDICATE A TIMES 10 SHIFT ON THE SWITCH RANGE. ONE LOAD.

OENB (21) BIN (20), ORGB (19), THESE ARE INPUTS FOR THE BOTTOM TIMER CORRESPONDING TO THE ABOVE.

OX10 (18) OPEN CIRCUIT OR 15 VOLTS GIVES NORMAL TIMING RANGE (0-00.9 SEC. FOR 120 HZ CLOCK). ZERO VOLTS MULTIPLIES TIME RANGE BY A FACTOR OF 10.

1CLK1 (4) CLOCK INPUT, EITHER 120 OR 1200 HZ FOR 60 HZ POWER LINE, OR 100 OR 1 IHZ FOR 50 HZ
1CLK2 (3) POWER LINE FREQUENCY. ONE LOGIC LOAD.

050HZ (2) OPEN CIRCUIT FOR 60 HZ APPLICATIONS. CONNECT TO COMMON FOR 50 HZ OPERATION.

OREM (32) OPEN CIRCUIT OR +15V FOR NORMAL OPERATION. OV FOR REMOTE OPERATION. ONE LOGIC LOAD.

1TNT THROUGH 9TNS (PINS 37 THROUGH 48). THESE ARE INPUT DATA LINES FOR REMOTE TIME SETTING. EACH INPUT IS TWO LOADS. CONTACT CLOSURES TO OV SETS DATA.

GND (5) PIN 5 IS USED TO GROUND THE CARD FRONT.

OUTPUTS

1DTA (14) GOES TO 15V WHILE TIMER IS TIMING. GOES TO OV WHEN RESET OR TIMED OUT. CAN DRIVE 17 LOADS

OISTA (12) 15 VOLTS WHEN RESET. GOES TO 0 VOLTS WHEN TIMING AND REMAINS 0 WHEN TIMED OUT. AS SHOWN IN FIGURE 2, THERE IS A DELAY OF THREE OR FOUR CLOCK PULSES BETWEEN SWITCHING THE INPUT AND THE PICK UP OF OISTA. CAN DRIVE 17 LOADS.

OTTA (16) GOES TO ZERO WHEN TIMED OUT. +15 VOLTS WHEN RESET OR TIMING. CAN DRIVE 17 LOADS.

OSDAR (33) STROBE SIGNAL FOR REMOTE SET POINT. GOES TO ZERO TO MULTIPLEX INPUT DATA. CAN DRIVE 17 LOADS.

1DTB (24) OISTB (22), OTTB (26), AND OSDBR (35) ARE THE CORRESPONDING OUTPUTS OF THE BOTTOM TIMER.

POWER SUPPLY

+15V (27). TYPICAL CURRENTS: 45 MA. WHEN RESET. ADD 5 MA FOR EACH TIMER TIMING OR TIMED OUT. ADD ADDITIONAL 5 MA FOR EACH X10 INDICATOR LIT, TOTAL 55MA, TYPICAL, 70 MA MAX.

-15V (29) - 6.5 MA TYPICAL, 10 MA MAX

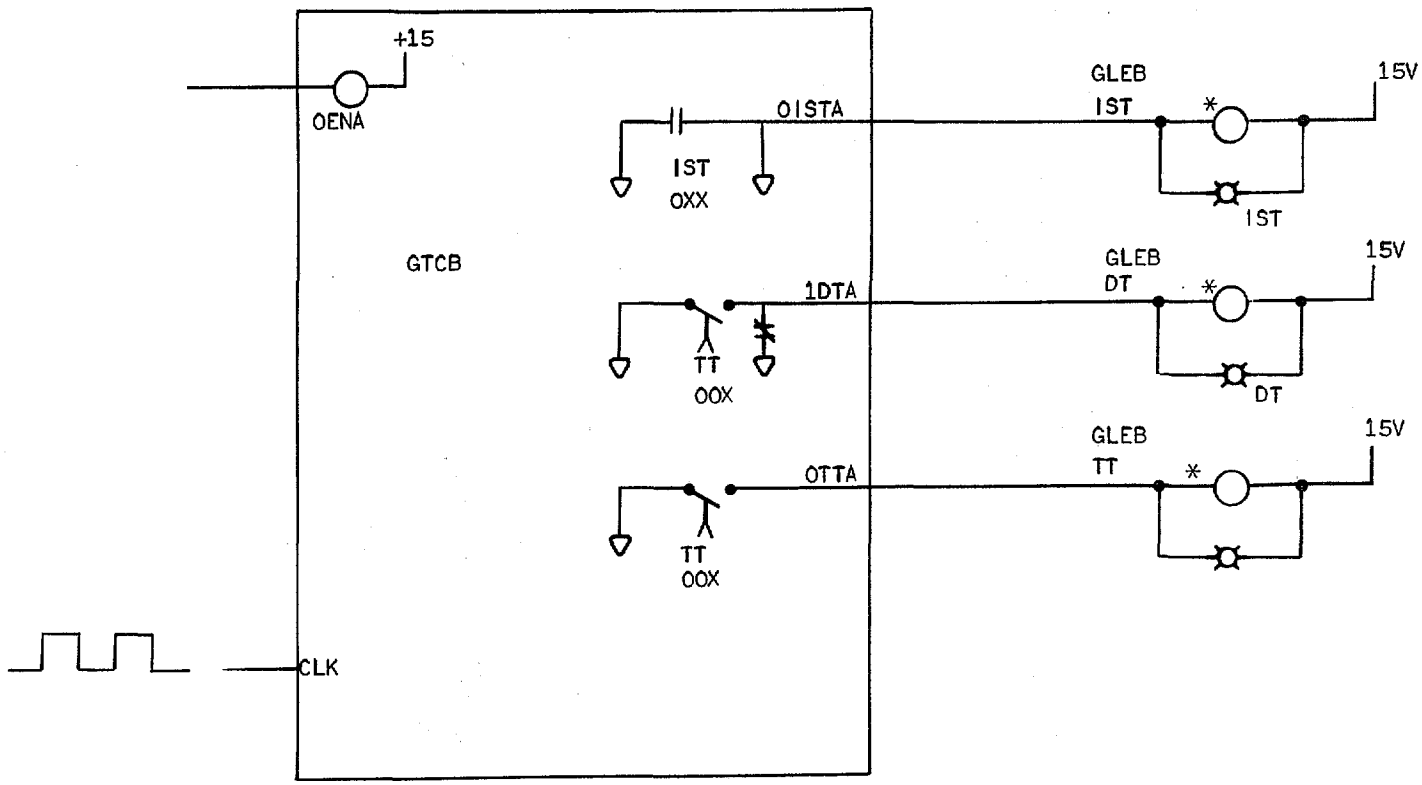
CARD SIZE:

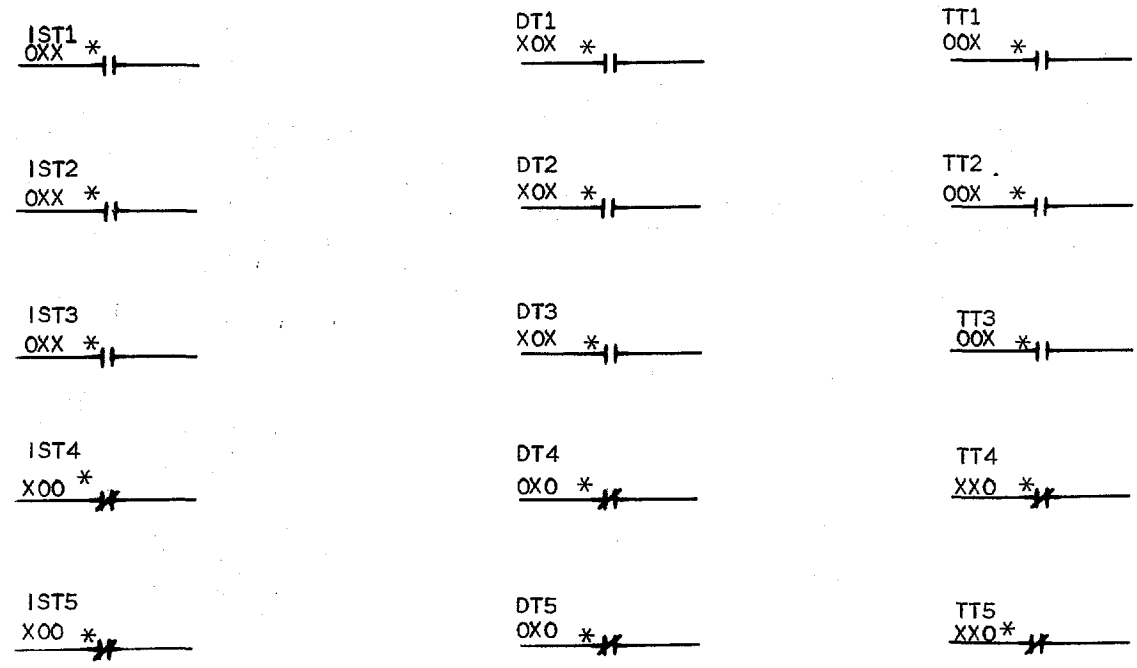
STANDARD DOM II WITH CARD FRONT, OCCUPIES THREE CARD ROW SPACES.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRI DL58, RIC,	APPROVAL AC Davis	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM TIMER CARD
REV. 2	ISSUED May 12, 1972	CHV1VN		FIRST MADE FOR		I C 3 6 2 2 G T C B 2
REV. 3	MADE BY KENNEY C. COX	DL1, 2530		I.C. NO.		CONT. ON SH. 3.3 SH. NO. 3.2

FORM 1C 2390 2-72





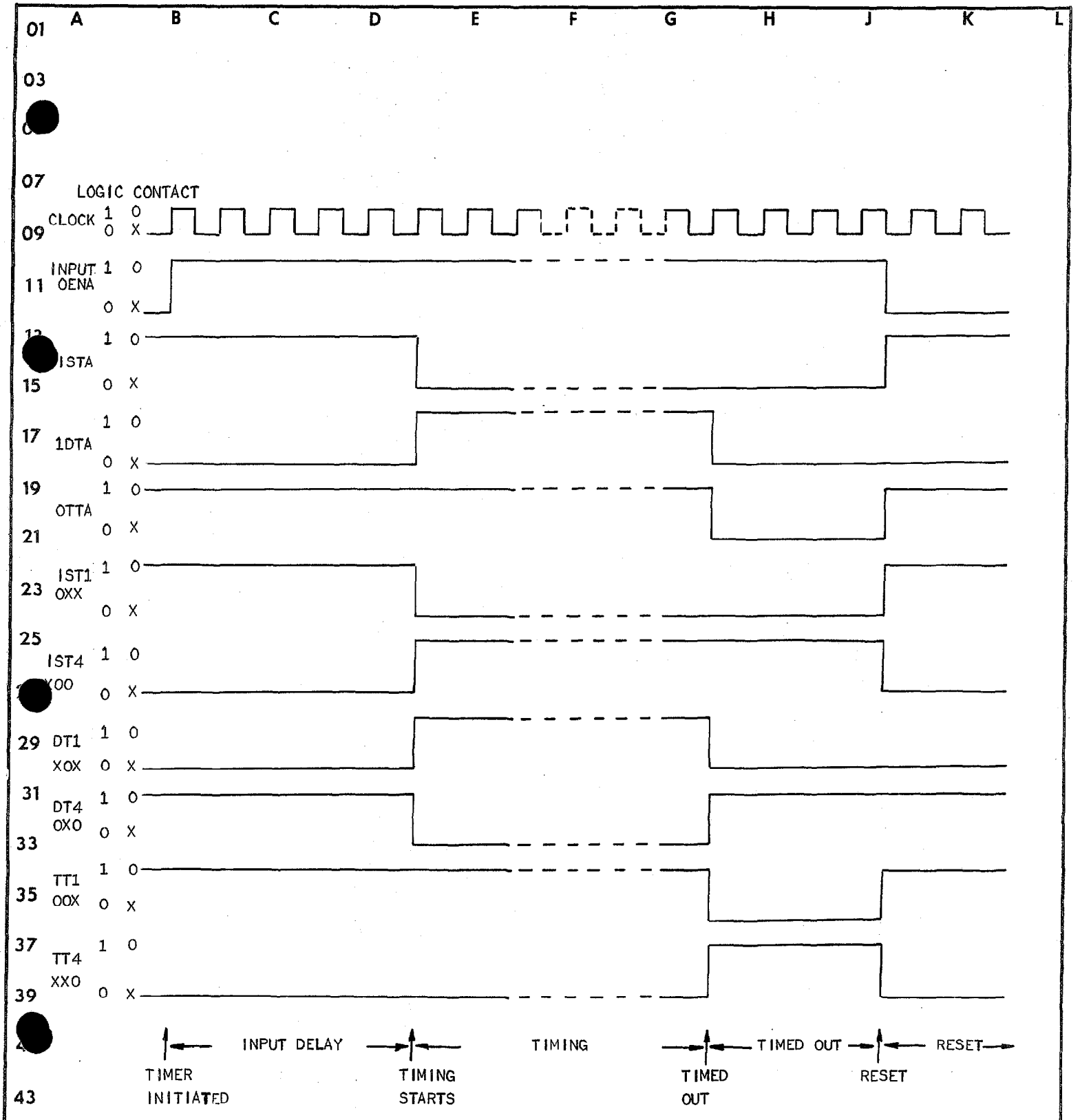
TYPICAL WIRING OF TIMER FOR FLO-THROUGH LOGIC

FIGURE 1

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL58, R I C.	APPROVED	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM TIMER CARD
REV. 2	ISSUED May 12, 1972		DL1, 2530	FIRST MADE FOR		I C 3 6 2 2 G T C B 2
REV. 3	MADE BY KENNEY C. COX		CHV1VN	I.C. NO.		CONT. ON SH. 3.4 SH. NO. 3.3

FORM 1C 2390 2-72



DEFINITIONS:

LOGIC 1 10 TO 15 VOLTS
LOGIC 0 0 TO 4 VOLTS
CONTACT 0 OPEN
CONTACT X CLOSED

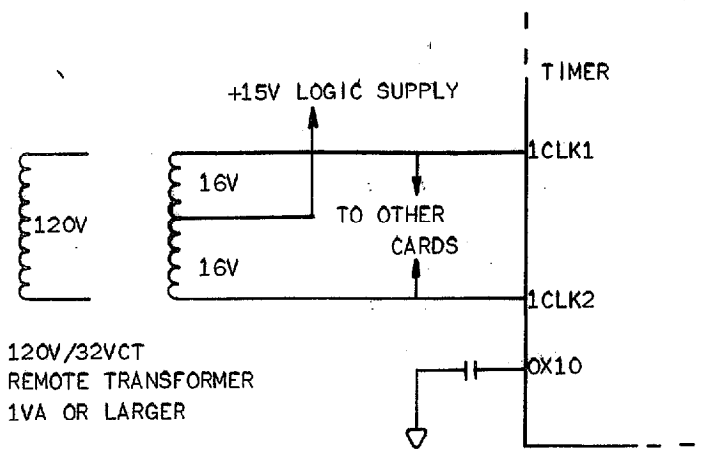
TIMER SIGNALS AVAILABLE

FIGURE 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL58, DL1,	APPROVALS	ELEMENTARY DIAGRAM TIMER CARD
REV. 2	ISSUED May 12, 1972		CHV1VN,	FIRST MADE FOR	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 3	MADE BY		2530, RIC	I.C. NO.	IC3622GTCB2

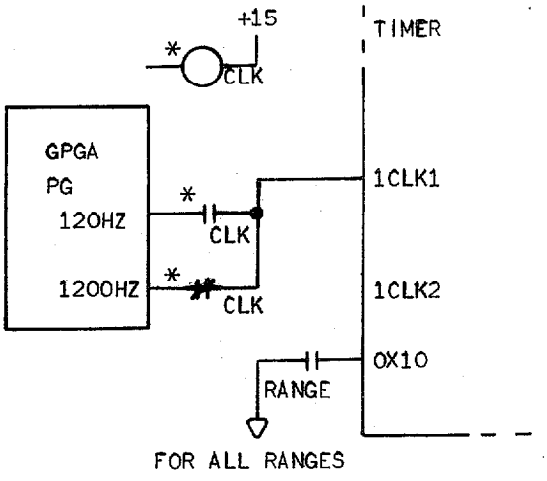
FORM IC 2990F2-72



120V/32VCT
REMOTE TRANSFORMER
1VA OR LARGER

FOR 120HZ ONLY
2 RANGES

FIGURE 3A

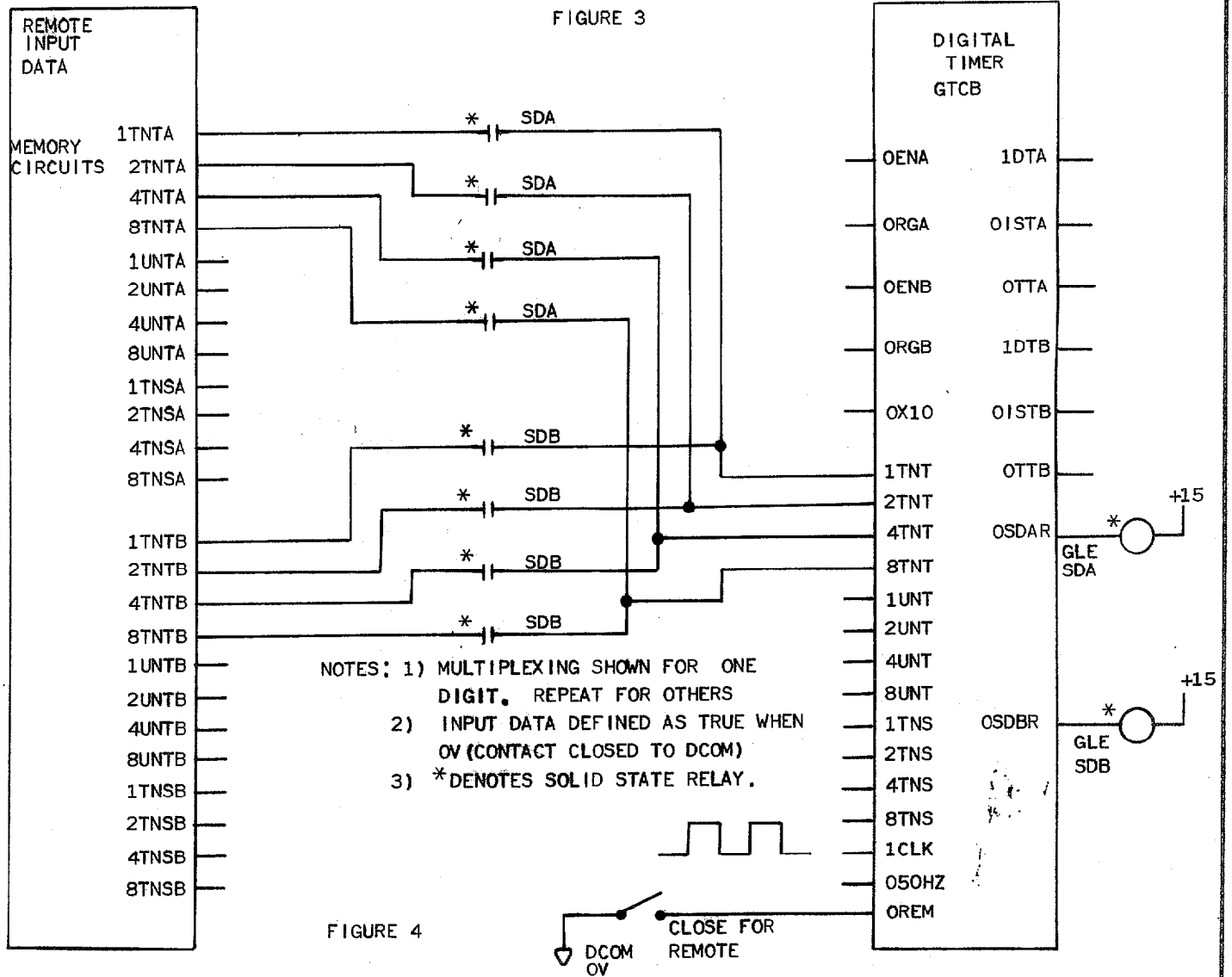


FOR ALL RANGES

FIGURE 3B

ALTERNATE CLOCK INPUTS

FIGURE 3

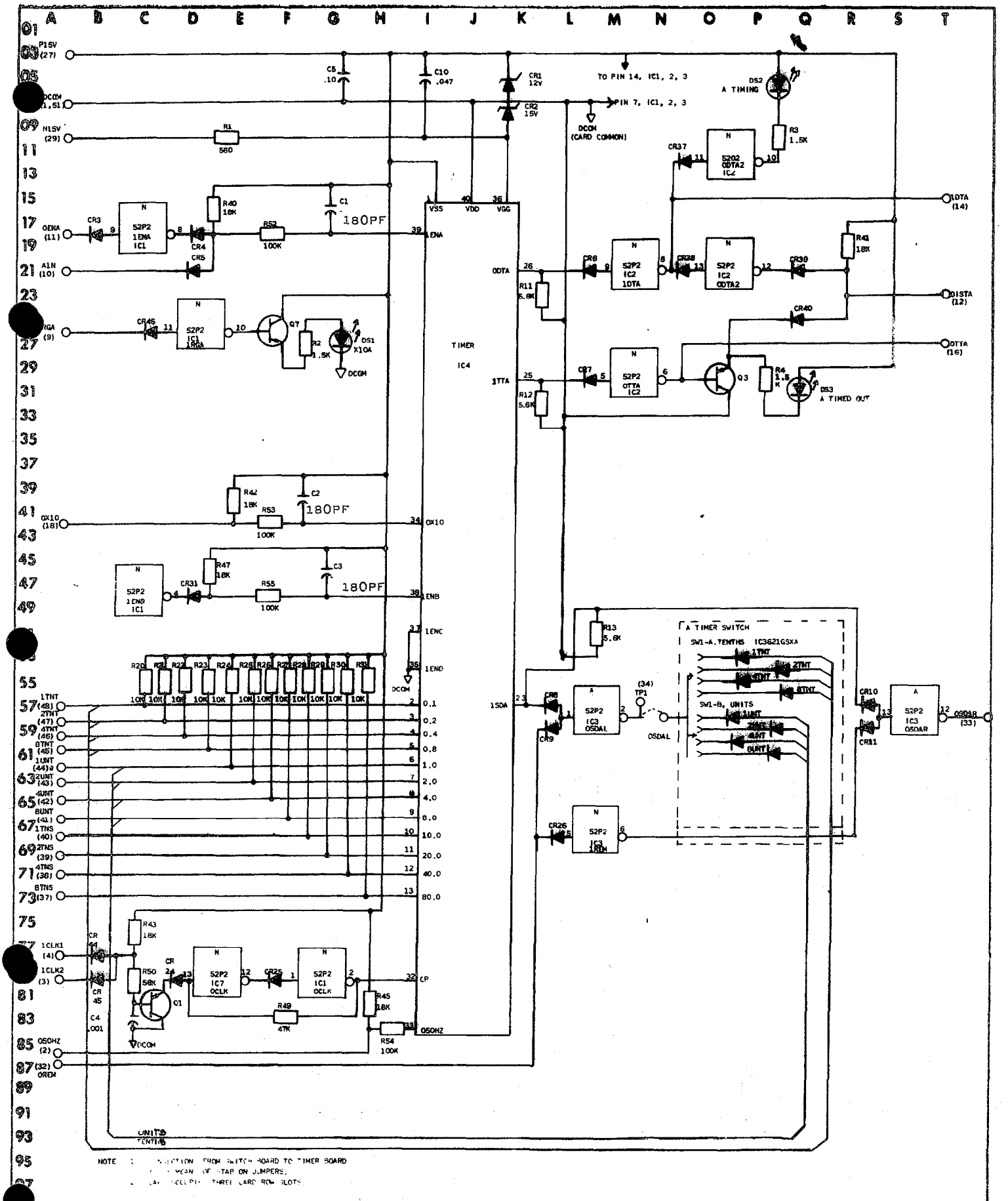


NOTES: 1) MULTIPLEXING SHOWN FOR ONE DIGIT. REPEAT FOR OTHERS
2) INPUT DATA DEFINED AS TRUE WHEN 0V (CONTACT CLOSED TO DCOM)
3) *DENOTES SOLID STATE RELAY.

FIGURE 4

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL58, 2530-T	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM TIMER CARD
REV. 2	ISSUED May 12, 1972		CHV1VN,	FIRST MADE FOR	IC3622GTCB2
REV. 3	MADE BY KENNEY C. COX		DL1, RIC	I.C. NO.	CONT. ON SH. FL. SH. NO. 3.5



V1 WMH 4-5-73	REV. 4	REV. 5	PRINTS TO DL1, DL58	APPROVALS R C Jones	GENERAL ELECTRIC	ELEMENTARY DIAGRAM TIMER CARD, 2-DIGIT ONE CIRCUIT
REV. 2	ISSUED January 9, 1973		CHV1VN 2530	FIRST MADE FOR PLASTIC MOLD	INDUSTRY CONTROL DEPT. SALEM, VA., U.S.A.	IC3622GTCB3
REV. 3	MADE BY MIKE OTEY			I.C. NO.		CONT. ON SH. 3, 1SH. NO. 3.0

APPLICATION NOTES

DIGITAL TIMER

THIS CARD HAS ONE DIGITAL TIMER ON THE CARD. THE TIMER USES A CLOCK SIGNAL DERIVED FROM THE AC POWER LINE FREQUENCY FOR HIGH TIMER STABILITY. FOUR RANGES ARE AVAILABLE DEPENDING ON THE INPUT CLOCK FREQUENCY AND THE CONDITION OF THE OX1O INPUT. RANGE AND ACCURACY SPECIFICATIONS ARE SHOWN IN TABLE 1.

1CLK FREQUENCY	OX1O VOLTS	RANGE SECONDS	ACCURACY SECONDS	REPEATABILITY SECONDS
120	15	0-- 9.9	.01 TO .05	.01
120	0	0-- 99	.01 TO .05	.01
1200	15	0-- .99	.001 TO .005	.001
1200	0	0-- 9.9	.001 TO .005	.001

TABLE 1

A TIMER CIRCUIT CAN BE SHIFTED BY A FACTOR OF 10 BY SWITCHING TH OX1O INPUT.

THE 1CLK INPUT IS NORMALLY WIRED TO EITHER 120 OR 1200 HZ, RESTRICTING THE TIMER TO TWO OF THE FOUR AVAILABLE RANGES, BY SWITCHING OX1O. BOTH 1CLK AND OX1O ARE LOGIC LEVEL SO THEY CAN BE SWITCHED BY EXTERNAL LOGIC IF REQUIRED.

TIMER INPUTS AND OUTPUTS ARE ARRANGED SO IT CAN BE USED WITH EITHER STANDARD LOGIC OR FLOW THROUGH LOGIC DESIGNS. WHEN USED WITH FLOW THROUGH LOGIC, OISTA AND OTTA OUTPUTS ARE USED TO PICK UP RELAYS ON A GLEB CARD. THESE RELAYS GIVE THE INSTANTANEOUS AND TIMED OUT CONTACTS NORMALLY ASSOCIATED WITH A TIMER. FIGURE 1 GIVES AN EXAMPLE CIRCUIT. FIGURE 2 GIVES CIRCUIT INPUT AND OUTPUT WAVEFORMS. IT SHOULD BE NOTED THAT THERE IS A SHORT DELAY OF 4 CLOCK PULSES IN PICKING UP THE TIMER. WHEN THE TIMER IS SET TO 000, 1DTA DOES NOT SWITCH,AND/OISTA AND OTTA SWITCH AT THE SAME TIME.

THE CLOCK FREQUENCY IS AVAILABLE FROM TWO SOURCES. IF THE 0.00 - .99 SECOND RANGE IS USED, THE IC3622GPGA CARD IS USED TO SUPPLY THE CLOCK SIGNALS AT EITHER 120 OR 1200 HZ. IF ONLY THE TWO LONGER RANGES ARE REQUIRED, THE SCHEME SHOWN IN FIGURE 3 CAN BE USED. THE TWO INPUT DIODES ACT AS A FULL WAVE RECTIFIER TO GENERATE THE 120 HZ REFERENCE FREQUENCY.

THE TIMER CAN BE SET REMOTELY AND INDEPENDENTLY OF THUMBWHEEL SWITCH SETTINGS. THE DIAGRAM OF FIGURE 4 SHOWS THE ARRANGEMENT. WHEN OREM IS MADE O VOLTS, THE INTERNAL STROBE LINES WILL NOT OPERATE. INSTEAD, WHEN THE TIMER IS INITIATED, STROBE PULSES (PULSES TO ZERO VOLTS) WILL APPEAR AT OSDAR OR OSDBR. THESE PULSES LAST FOR ONE CLOCK CYCLE AND APPEAR AT TIMES CORRESPONDING TO THE LEAST SIGNIFICANT DIGIT OF THE RANGED SELECTED (.01,0.1, OR 1 SECOND). WHEN THE STROBE PULSE APPEARS THE EXTERNAL DATA IS SUPPLIED TO THE TWELVE DATA LINES 1TNT THROUGH 8TNS. THESE CORRESPOND TO THE 1, 2, 4, 8 BITS OF THE BCD CODED THUMBWHEEL SWITCHES WITH THREE DECADES, (TENTHS, UNITS, TENS). THE TIMER STARTS COUNTING WHEN INITIATED AND STOPS (TIMES OUT) WHEN ITS INTERNAL COUNTER MATCHES THE APPLIED DATA. CAUTION: ALL SWITCH MULTIPLEX CIRCUITS CONNECTED TO THE DATA LINES MUST BE CAPABLE OF WIRE-OR (PARALLEL CONTACT) OPERATION TO ALLOW THE INTERNAL TIMER SWITCHES TO FUNCTION.

INPUTS:

OENA (11) A ZERO VOLT LEVEL INITIATES TIMING — AIN (12) SHOULD BE OPEN

AIN (12) A 15 VOLT LEVEL INITIATES TIMING IF THIS INPUT IS USED. ONE LOAD. — OENA (11) SHOULD BE JUMPERED TO OV,

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

APPLICATION INFORMATION

INPUTS (CONTINUED)

ORGA (9) A ZERO VOLT LEVEL LIGHTS THE X10A INDICATION. THIS IS NORMALLY USED TO INDICATE A TIMES 10 SHIFT ON THE SWITCH RANGE. ONE LOAD.

OX10 (18) OPEN CIRCUIT OR 15 VOLTS GIVES NORMAL TIMING RANGE (0- 9.9 SEC. FOR 120 HZ CLOCK). ZERO VOLTS MULTIPLIES TIME RANGE BY A FACTOR OF 10.

1CLK1 (4) CLOCK INPUT, EITHER 120 OR 1200 HZ FOR 60 HZ POWER LINE, OR 100 OR 1 KHZ FOR 50 HZ
1CLK2 (3) POWER LINE FREQUENCY. ONE LOGIC LOAD.

O50HZ (2) OPEN CIRCUIT FOR 60 HZ APPLICATIONS. CONNECT TO COMMON FOR 50 HZ OPERATION.

OREM (32) OPEN CIRCUIT OR +15V FOR NORMAL OPERATION. OV FOR REMOTE OPERATION. ONE LOGIC LOAD.

1TNT THROUGH 9TNS (PINS 37 THROUGH 48). THESE ARE INPUT DATA LINES FOR REMOTE TIME SETTING. EACH INPUT IS TWO LOADS. CONTACT CLOSURES TO OV SETS DATA WHEN OSDAR IS ZERO.

GND (5) PIN 5 IS USED TO GROUND THE CARD FRONT.

OUTPUTS

1DTA (14) GOES TO 15V WHILE TIMER IS TIMING. GOES TO OV WHEN RESET OR TIMED OUT. CAN DRIVE 17 LOADS

OISTA (12) 15 VOLTS WHEN RESET. GOES TO 0 VOLTS WHEN TIMING AND REMAINS 0 WHEN TIMED OUT. AS SHOWN IN FIGURE 2, THERE IS A DELAY OF THREE OR FOUR CLOCK PULSES BETWEEN SWITCHING THE INPUT AND THE PICK UP OF OISTA. CAN DRIVE 17 LOADS.

OTTA (16) GOES TO ZERO WHEN TIMED OUT. +15 VOLTS WHEN RESET OR TIMING. CAN DRIVE 17 LOADS.

OSDAR (33) STROBE SIGNAL FOR REMOTE SET POINT. GOES TO ZERO TO MULTIPLEX INPUT DATA. CAN DRIVE 17 LOADS.

POWER SUPPLY

+15V (27). TYPICAL CURRENTS: 45 MA. WHEN RESET. ADD 5 MA FOR EACH TIMER TIMING OR TIMED OUT. ADD ADDITIONAL 5 MA FOR EACH X10 INDICATOR LIT, TOTAL 55MA, TYPICAL, 70 MA MAX.

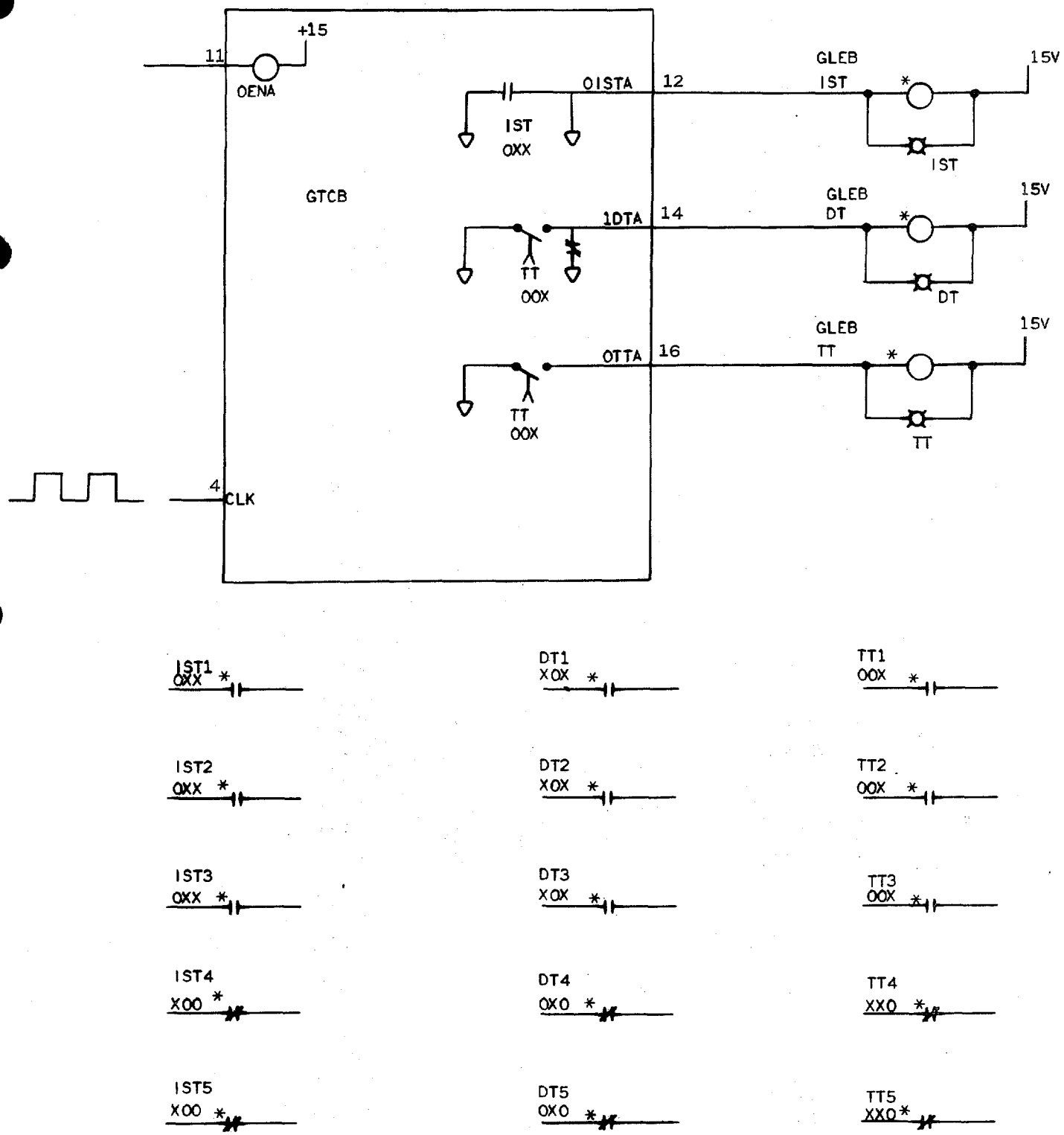
-15V (29) - 6.5 MA TYPICAL, 10 MA MAX

CARD SIZE:

STANDARD DOM II WITH CARD FRONT, OCCUPIES THREE CARD ROW SPACES.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL1, DL58	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM TIMER CARD, 2 DIGIT
REV. 2	ISSUED January 9, 1973	2530, CHV1 VN	FIRST MADE FOR PLASTIC MOLD		I C 3 6 2 2 G T C B3
REV. 3	MADE BY MIKE OTEY	RIC	I.C. NO.		CONT. ON SH. 3.3 SH. NO. 3.2

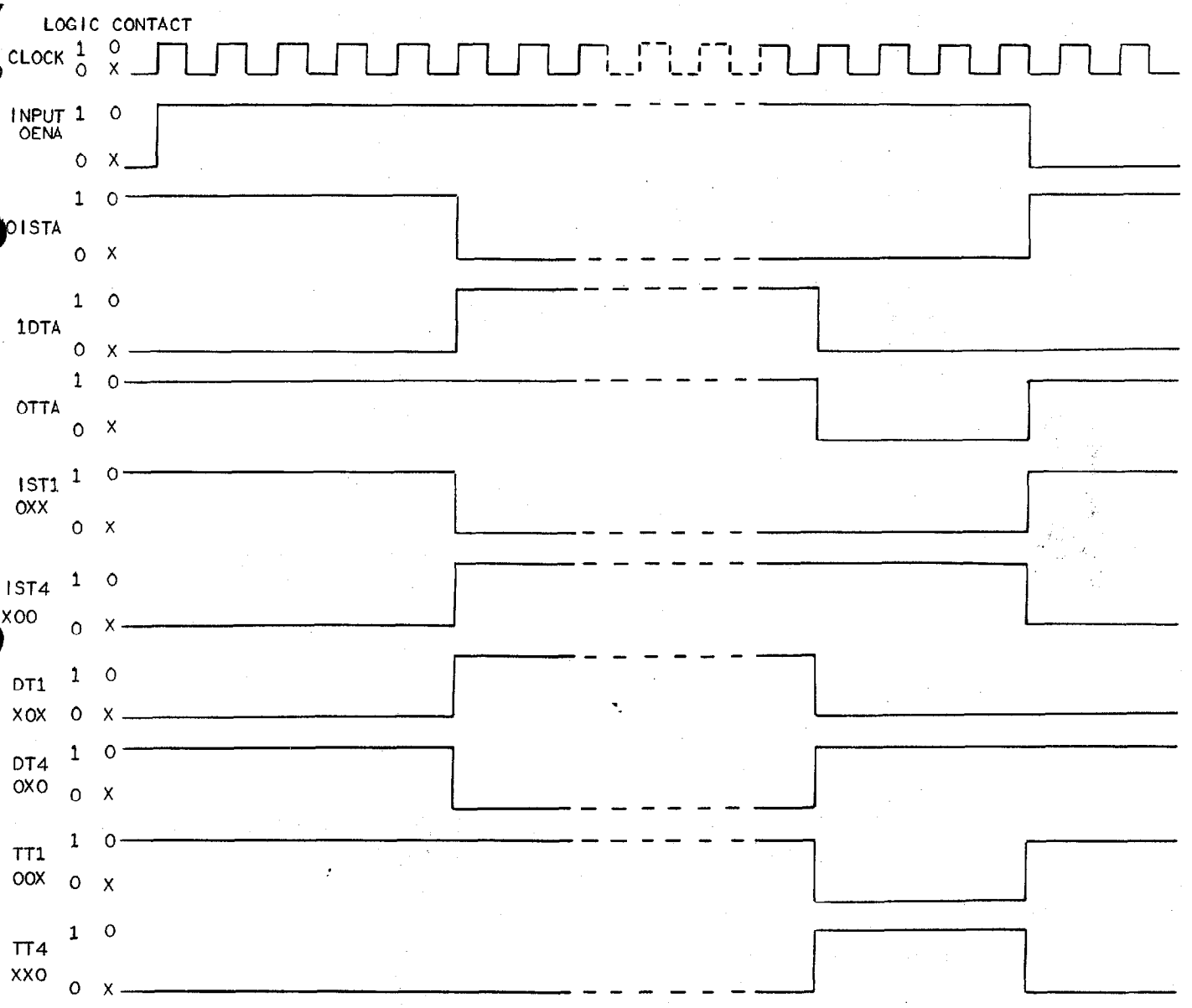


TYPICAL WIRING OF TIMER FOR FLO-THROUGH LOGIC

FIGURE 1

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL1, DL58	APPROVED RC Bayes	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM TIMER CARD, 2 DIGIT
REV. 2	ISSUED January 9, 1973	2530, CHV1 VN	FIRST MADE FOR PLASTIC MOLD			IC3622GTCB3
REV. 3	MADE BY MIKE OTEY	RIC	I.C. NO.			CONT. ON SH. 3,4 SH. NO 3,3



DEFINITIONS:

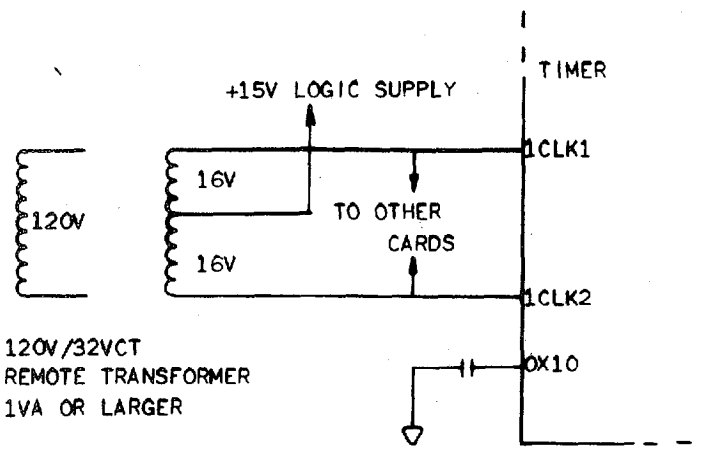
LOGIC 1 10 TO 15 VOLTS
LOGIC 0 0 TO 4 VOLTS
CONTACT 0 OPEN
CONTACT X CLOSED

TIMER SIGNALS AVAILABLE

FIGURE 2

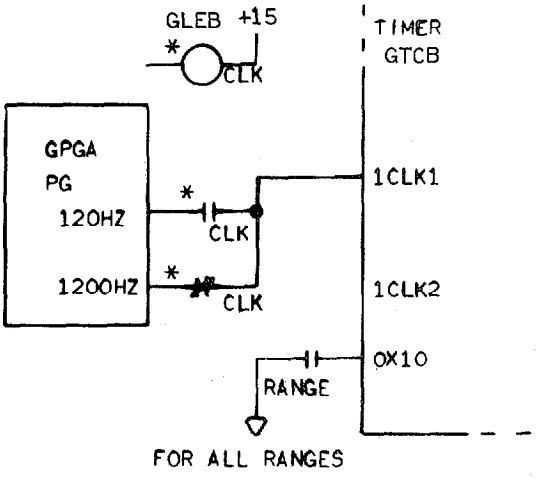
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL1, DL58	APP'D	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM TIMER CARD, 2 DIGIT
REV. 2	ISSUED January 9, 1973 2530, CHG1 VN		FIRST MADE FOR PLASTIC MOLD			IC3622GTCB3
REV. 3	MADE BY MIKE OTEY	RIC	I.C. NO.			CONT. ON SH. 3,5 SH. NO 3,4



FOR 120HZ ONLY
2 RANGES

FIGURE 3A

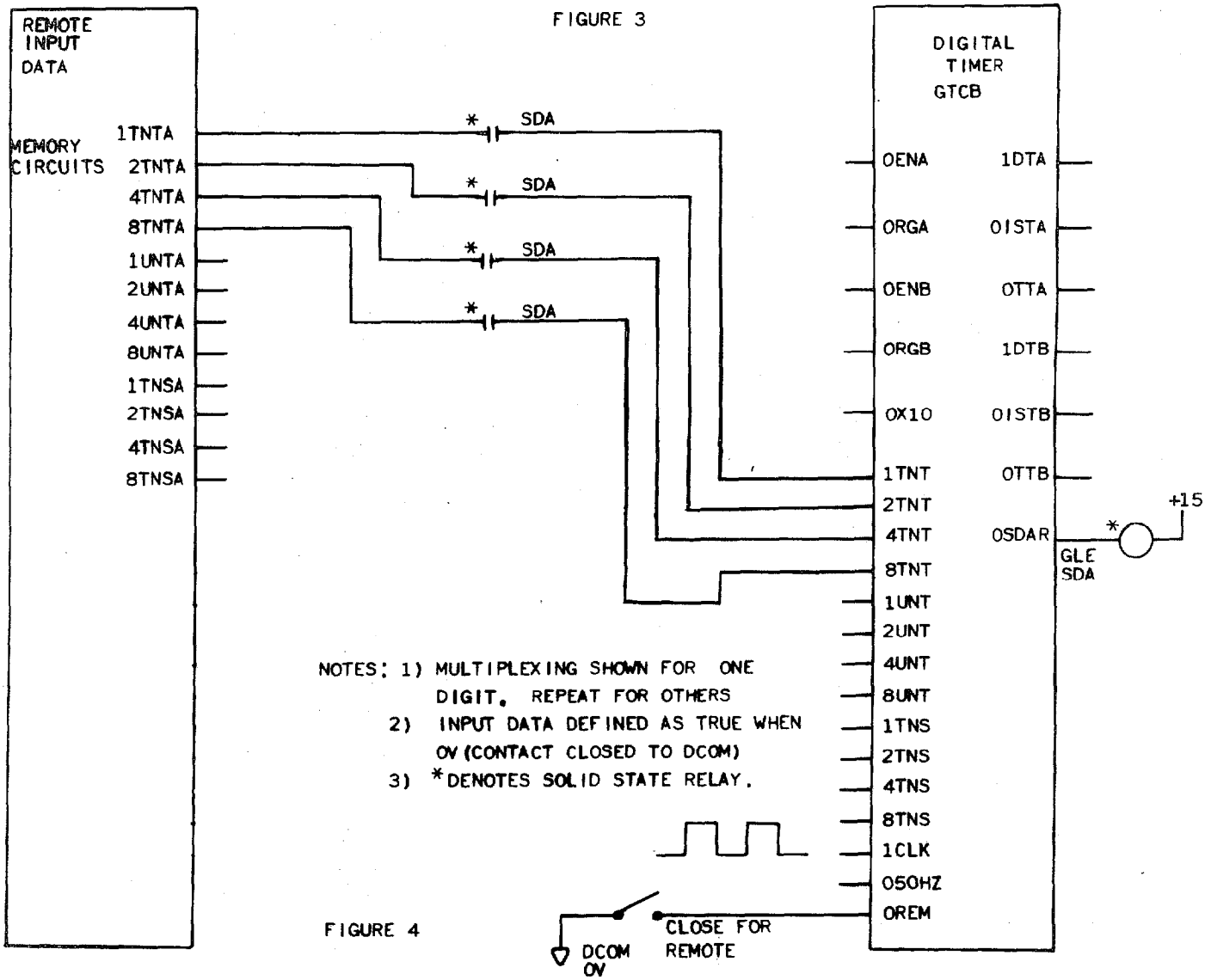


FOR ALL RANGES

FIGURE 3B

ALTERNATE CLOCK INPUTS

FIGURE 3

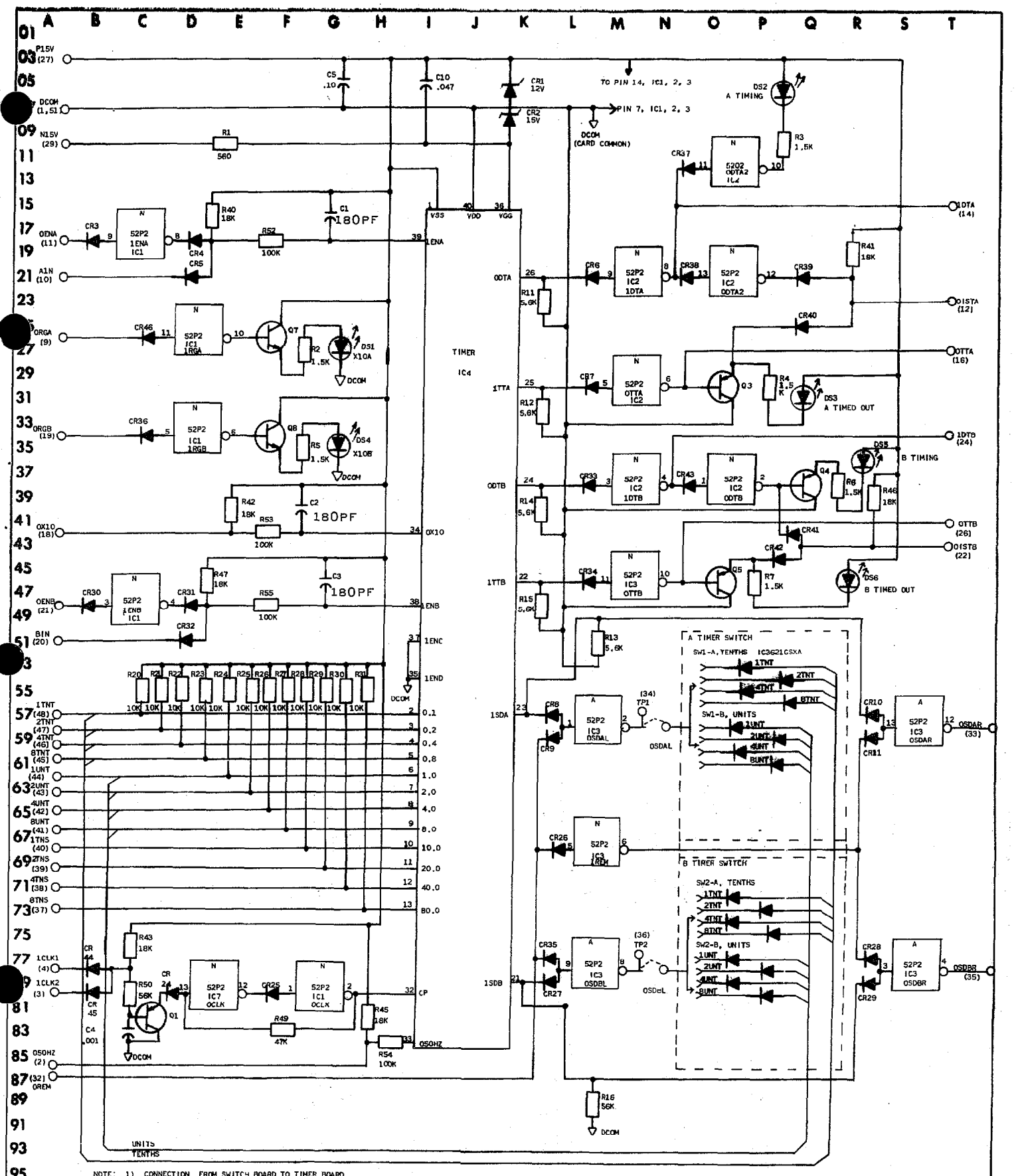


NOTES: 1) MULTIPLEXING SHOWN FOR ONE DIGIT. REPEAT FOR OTHERS
2) INPUT DATA DEFINED AS TRUE WHEN 0V (CONTACT CLOSED TO DCOM)
3) *DENOTES SOLID STATE RELAY.

FIGURE 4

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL1, DL58	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED January 9, 1973 2530, CHV1 VN		FIRST MADE FOR PLASTIC MOLD	INDUSTRY CONTROL DEPT.	TIMER CARD, 2 DIGIT
REV. 3	MADE BY MIKE OTEY	RIC	C NO.	SALEM, VA. U.S.A.	IC3622GTCB3
					CONT. ON SH. FL. SH. NO. 3.5



NOTE: 1) CONNECTION FROM SWITCH BOARD TO TIMER BOARD IS BY MEANS OF STAB ON JUMPERS.
2) CARD OCCUPIES THREE CARD ROW SLOTS.

REV. 1 WMH 4-5-73	REV. 4	REV. 5	PRINTS TO DL58, DL1, R1C, CHV1VN	APPROVALS	GENERAL ELECTRIC
REV. 2	ISSUED September 1, 1972	2530	FIRST MADE FOR PLASTIC MOLD		INDUSTRY CONTROL DEPT. SALEM, VA., U.S.A.
REV. 3	MADE BY GARY HARRIS		I.C. NO.		

ELEMENTARY DIAGRAM
TIMER CARD
TWO DIGIT, TWO CIRCUIT

IC3622GTCB4

CONT. ON SH. 3.1 SH. NO. 3.0

FORM IC 2348 (01 69)

APPLICATION NOTES

DIGITAL TIMER

THIS CARD HAS TWO DIGITAL TIMERS. THE TIMER USES A CLOCK SIGNAL DERIVED FROM THE AC POWER LINE FREQUENCY FOR HIGH TIMER STABILITY. FOUR RANGES ARE AVAILABLE DEPENDING ON THE INPUT CLOCK FREQUENCY AND THE CONDITION OF THE OX10 INPUT. RANGE AND ACCURACY SPECIFICATIONS ARE SHOWN IN TABLE 1.

1CLK FREQUENCY	CX10 VOLTS	RANGE SECONDS	ACCURACY SECONDS	REPEATABILITY SECONDS
120	15	0- 9.9	.01 TO .05	.01
120	0	0- 99	.01 TO .05	.01
1200	15	0- .99	.001 TO .005	.001
1200	0	0- 9.9	.001 TO .005	.001

TABLE 1

A TIMER CIRCUIT CAN BE SHIFTED BY A FACTOR OF 10 BY SWITCHING TH OX10 INPUT.

THE 1CLK INPUT IS NORMALLY WIRED TO EITHER 120 OR 1200 HZ, RESTRICTING THE TIMER TO TWO OF THE FOUR AVAILABLE RANGES, BY SWITCHING OX10. BOTH 1CLK AND OX10 ARE LOGIC LEVEL SO THEY CAN BE SWITCHED BY EXTERNAL LOGIC IF REQUIRED.

BECAUSE OF SHARING COMMON CIRCUITRY, THERE ARE TWO RESTRICTIONS ON APPLICATION OF CARDS WITH TWO TIMERS. THE FIRST IS THAT ONE TIMER SHOULD NOT BE INITIATED WHILE THE OTHER IS ALREADY TIMING. THIS WOULD RESULT IN THE SECOND TIMER PERIOD BEING SHORTENED BY THE AMOUNT OF DELAY FROM STARTING THE FIRST TIMER TO STARTING THE SECOND TIMER. THE OTHER RESTRICTION IS THAT IF TWO TIMERS ARE INITIATED AT THE SAME TIME, BOTH TIMERS SHOULD BE CONNECTED FOR THE SAME TIME RANGE, SINCE THE CLOCK AND OX10 INPUTS ARE COMMON FOR THE TWO TIMERS.

TIMER INPUTS AND OUTPUTS ARE ARRANGED SO IT CAN BE USED WITH EITHER STANDARD LOGIC OR FLOW THROUGH LOGIC DESIGNS. WHEN USED WITH FLOW THROUGH LOGIC, OISTA (OR OISTB) AND OTTA (OR OTTB) OUTPUTS ARE USED TO PICK UP RELAYS ON THE GLEB CARD. THESE RELAYS GIVE THE INSTANTANEOUS AND TIMED OUT CONTACTS NORMALLY ASSOCIATED WITH A TIMER. FIGURE 1 GIVES AN EXAMPLE CIRCUIT. FIGURE 2 GIVES CIRCUIT INPUT AND OUTPUT WAVEFORMS. IT SHOULD BE NOTED THAT THERE IS A SHORT DELAY OF 4 CLOCK PULSES IN PICKING UP THE TIMER. WHEN THE TIMER IS SET TO 000, 1DTA DOES NOT SWITCH, AN OISTA AND OTTA SWITCH AT THE SAME TIME.

THE CLOCK FREQUENCY IS AVAILABLE FROM TWO SOURCES. IF THE 0.00-0.99 SECOND RANGE IS USED, THE IC3622GPGA CARD IS USED TO SUPPLY THE CLOCK SIGNALS AT EITHER 120 OR 1200 HZ. IF ONLY THE TWO LONGER RANGES ARE REQUIRED, THE SCHEME SHOWN IN FIGURE 3A CAN BE USED. THE TWO INPUT DIODES ACT AS A FULL WAVE RECTIFIER TO GENERATE THE 120 HZ REFERENCE FREQUENCY.

THE TIMER CAN BE SET REMOTELY AND INDEPENDENTLY OF THUMBWHEEL SWITCH SETTINGS. THE DIAGRAM OF FIGURE 4 SHOWS THE ARRANGEMENT. WHEN OREM IS MADE 0 VOLTS, THE INTERNAL STROBE LINES WILL NOT OPERATE. INSTEAD, WHEN THE TIMER IS INITIATED, STROBE PULSES (PULSES TO ZERO VOLTS) WILL APPEAR AT OSDAR OR OSDBR. THESE PULSES LAST FOR ONE CLOCK CYCLE AND APPEAR AT TIMES CORRESPONDING TO THE LEAST SIGNIFICANT DIGIT OF THE RANGED SELECTED (.01,0.1,OR 1 SECOND). WHEN THE STROBE PULSE APPEARS THE EXTERNAL DATA IS SUPPLIED TO THE TWELVE DATA LINES 1TNT THROUGH 8TNS. THESE CORRESPOND TO THE 1, 2, 4, 8 BITS OF THE BCD CODED THUMBWHEEL SWITCHES WITH THREE DECADES, (TENTHS, UNITS, TENS). THE TIMER STARTS COUNTING WHEN INITIATED AND STOPS (TIMES OUT) WHEN ITS INTERNAL COUNTER MATCHES THE APPLIED DATA. CAUTION: ALL SWITCH MULTIPLEX CIRCUITS CONNECTED TO THE DATA LINES MUST BE CAPABLE OF WIRE-OR (PARALLEL CONTACT) OPERATION TO ALLOW THE INTERNAL TIMER SWITCHES TO FUNCTION.

INPUTS:

OENA (11) A ZERO VOLT LEVEL INITIATES TIMING ON THE TOP TIMER. AIN (12) SHOULD BE OPEN

AIN (12) A 15 VOLT LEVEL INITIATES TIMING ON THE TOP TIMER. OENA (11) SHOULD BE JUMPERED TO OV, IF THIS INPUT IS USED. ONE LOAD.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL58, 2530	R C Davis	GENERAL ELECTRIC	ELEMENTARY DIAGRAM TIMER CARD
REV. 2	ISSUED Aug. 29, 1972	CHV1VN,	FIRST MADE FOR		INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	I C 3 6 2 2 G T C B 4
REV. 3	MADE BY GARY HARRIS	DL1, R I C	I.C. NO.			CONT. ON SH. 3.2 SH. NO. 3.1

APPLICATION INFORMATION

INPUTS (CONTINUED)

ORGA (9) A ZERO VOLT LEVEL LIGHTS THE X10A INDICATION. THIS IS NORMALLY USED TO INDICATE A TIMES 10 SHIFT ON THE SWITCH RANGE. ONE LOAD.

OENB (21) BIN (20), ORGB (19), THESE ARE INPUTS FOR THE BOTTOM TIMER CORRESPONDING TO THE ABOVE.

OX10 (18) OPEN CIRCUIT OR 15 VOLTS GIVES NORMAL TIMING RANGE (0-00.9 SEC. FOR 120 HZ CLOCK). ZERO VOLTS MULTIPLIES TIME RANGE BY A FACTOR OF 10.

1CLK1 (4) 1CLK2 (3) CLOCK INPUT, EITHER 120 OR 1200 HZ FOR 60 HZ POWER LINE, OR 100 OR 1 IHZ FOR 50 HZ POWER LINE FREQUENCY. ONE LOGIC LOAD.

050HZ (2) OPEN CIRCUIT FOR 60 HZ APPLICATIONS. CONNECT TO COMMON FOR 50 HZ OPERATION.

OREM (32) OPEN CIRCUIT OR +15V FOR NORMAL OPERATION. OV FOR REMOTE OPERATION. ONE LOGIC LOAD.

1TNT THROUGH 9TNS (PINS 37 THROUGH 48). THESE ARE INPUT DATA LINES FOR REMOTE TIME SETTING. EACH INPUT IS TWO LOADS. CONTACT CLOSURES TO OV SETS DATA.

GND (5) PIN 5 IS USED TO GROUND THE CARD FRONT.

OUTPUTS

1DTA (14) GOES TO 15V WHILE TIMER IS TIMING. GOES TO OV WHEN RESET OR TIMED OUT. CAN DRIVE 17 LOADS

OISTA (12) 15 VOLTS WHEN RESET. GOES TO 0 VOLTS WHEN TIMING AND REMAINS 0 WHEN TIMED OUT. AS SHOWN IN FIGURE 2, THERE IS A DELAY OF THREE OR FOUR CLOCK PULSES BETWEEN SWITCHING THE INPUT AND THE PICK UP OF OISTA. CAN DRIVE 17 LOADS.

OTTA (16) GOES TO ZERO WHEN TIMED OUT. +15 VOLTS WHEN RESET OR TIMING. CAN DRIVE 17 LOADS.

OSDAR (33) STROBE SIGNAL FOR REMOTE SET POINT. GOES TO ZERO TO MULTIPLEX INPUT DATA. CAN DRIVE 17 LOADS.

1DTB (24) OISTB (22), OTTB (26), AND OSDBR (35) ARE THE CORRESPONDING OUTPUTS OF THE BOTTOM TIMER.

POWER SUPPLY

+15V (27). TYPICAL CURRENTS: 45 MA. WHEN RESET. ADD 5 MA FOR EACH TIMER TIMING OR TIMED OUT. ADD ADDITIONAL 5 MA FOR EACH X10 INDICATOR LIT, TOTAL 55MA, TYPICAL, 70 MA MAX.

-15V (29) - 6.5 MA TYPICAL, 10 MA MAX

CARD SIZE:

STANDARD DOM II WITH CARD FRONT, OCCUPIES THREE CARD ROW SPACES.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO RIC, DL58,	APPROVALS RE Davis	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 2	ISSUED July 21, 1972		CHV1VN,	FIRST MADE FOR	
REV. 3	MADE BY D. M. MILLER		DL1, 2530	I.C. NO.	

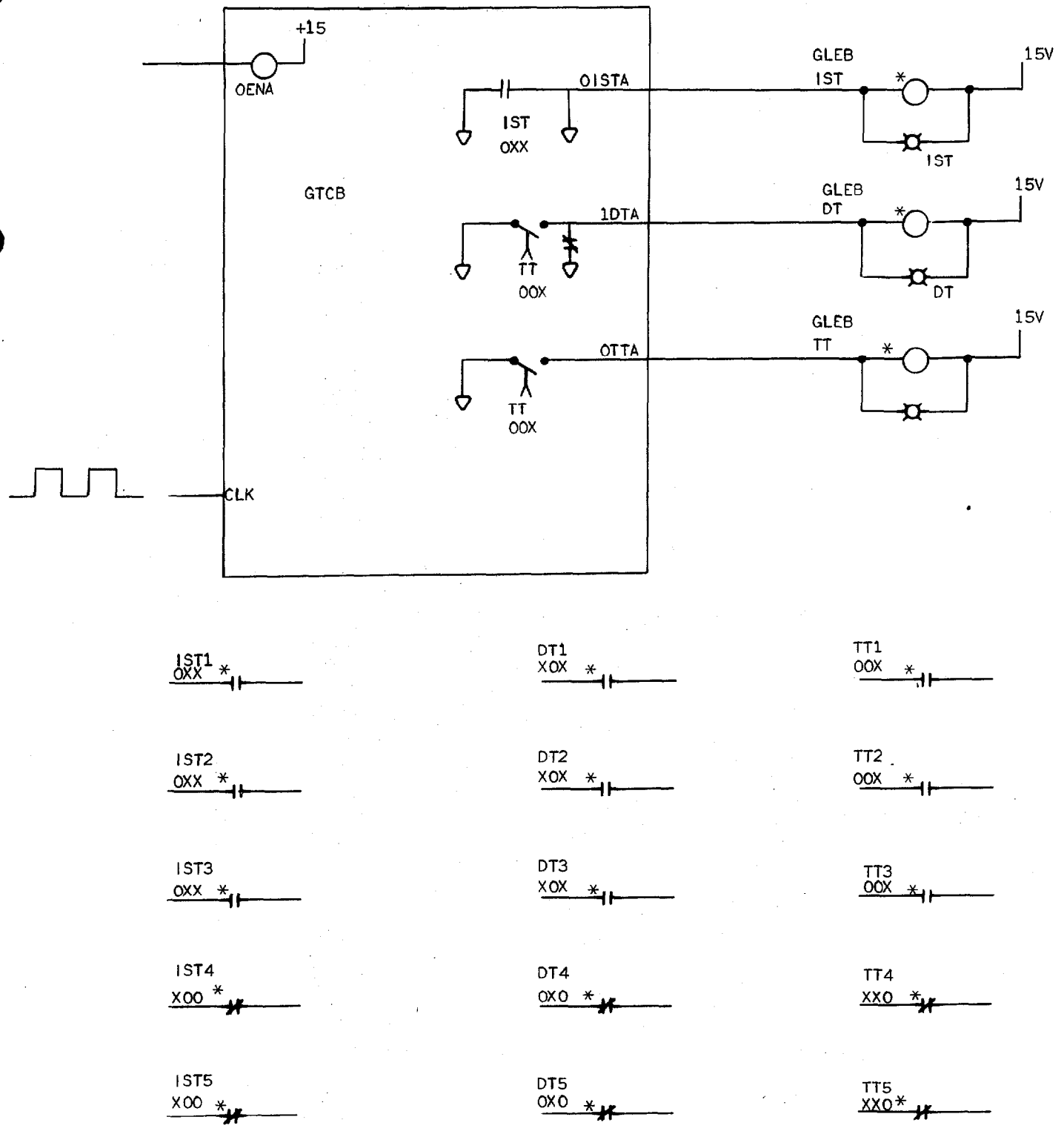
ELEMENTARY DIAGRAM

TIMER CARD

IC3622GTCB4

CONT. ON SH. 3.3 SH. NO. 3.2

FORM 1C 2390 2-72



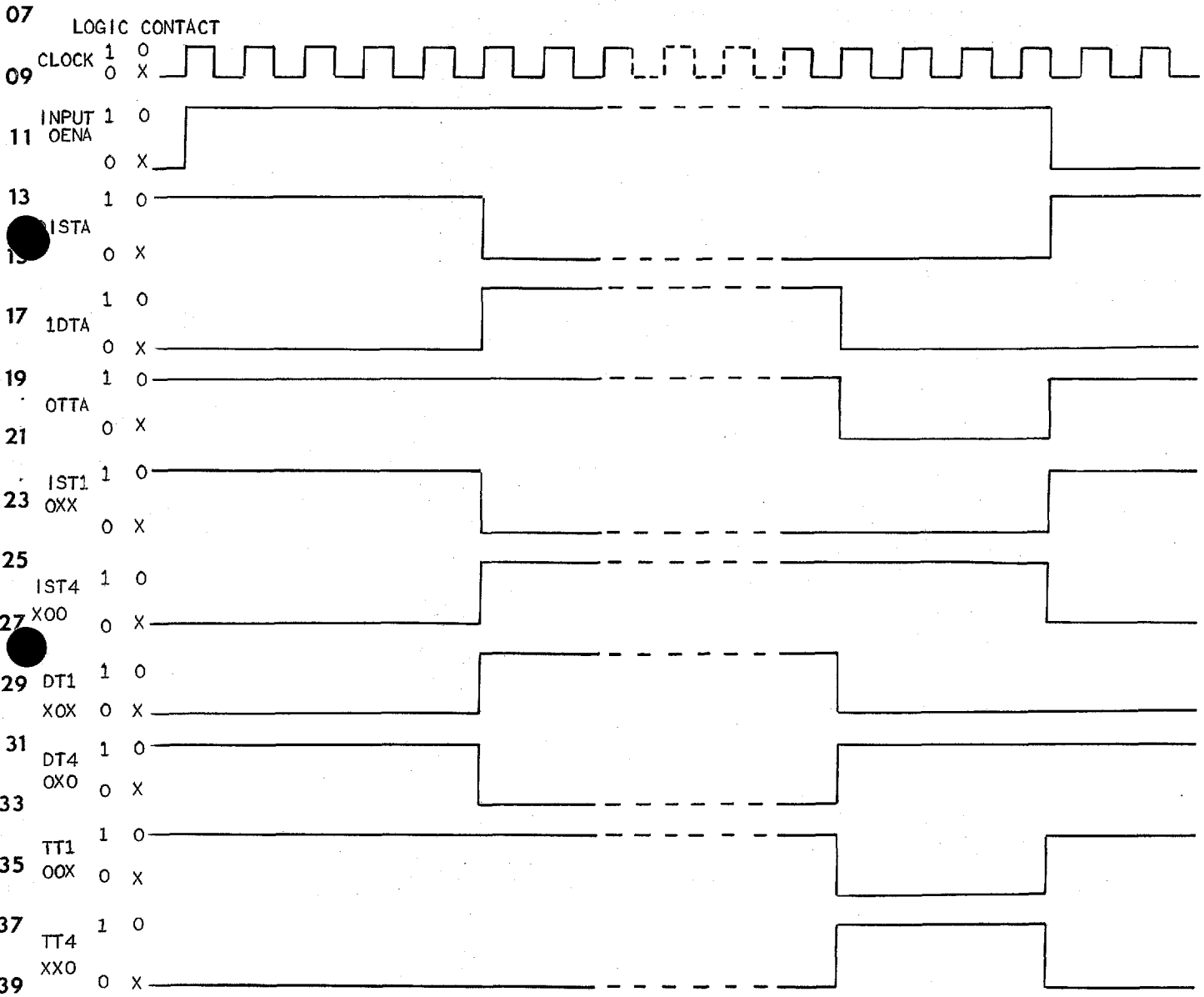
TYPICAL WIRING OF TIMER FOR FLO-THROUGH LOGIC

FIGURE 1

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO CHVIVN,	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 2	ISSUED July 21, 1972		DL1, 2530, DL58, RIC	FIRST MADE FOR	
REV. 3	MADE BY D. M. MILLER			I.C. NO.	

ELEMENTARY DIAGRAM
TIMER CARD
IC3622GTCB4
CONT. ON SH. 3,4 SH. NO. 3,3



DEFINITIONS:

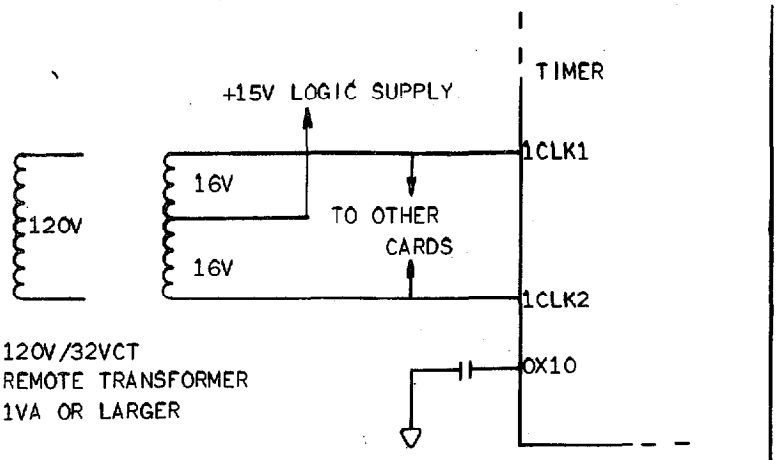
LOGIC 1 10 TO 15 VOLTS
LOGIC 0 0 TO 4 VOLTS
CONTACT 0 OPEN
CONTACT X CLOSED

TIMER SIGNALS AVAILABLE

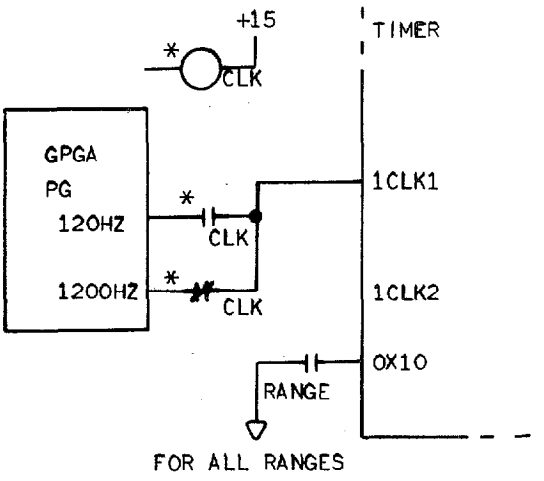
FIGURE 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO CHV1VN	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM TIMER CARD
REV. 2	ISSUED July 21, 1972		DL1, 2530, DL58, RIC	FIRST MADE FOR		IC3622GTCB4
REV. 3	MADE BY D. M. MILLER			I.C. NO.		CONT. ON SH. 3.5 SH. NO. 3.4

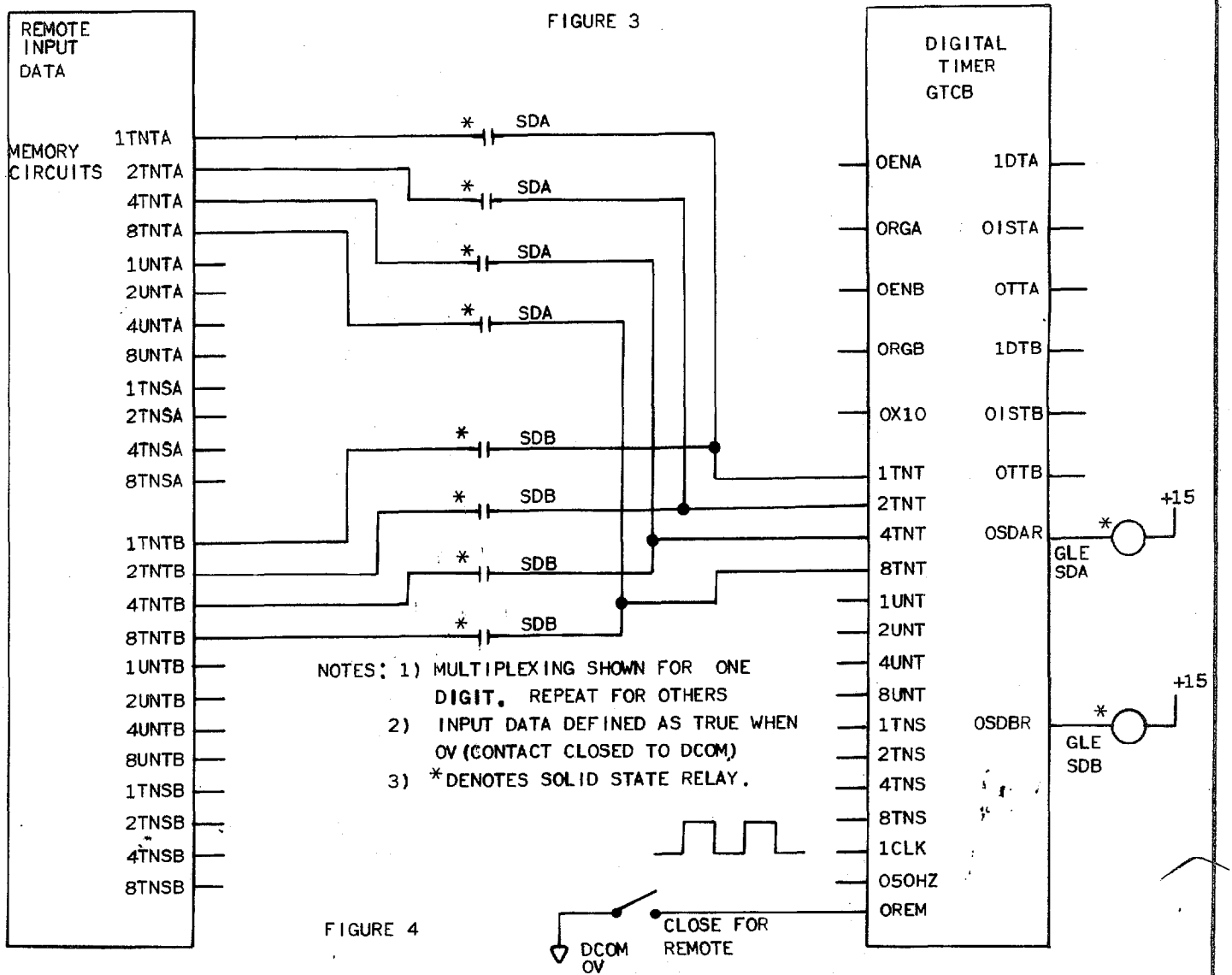


FOR 120HZ ONLY
2 RANGES
FIGURE 3A



FOR ALL RANGES
FIGURE 3B

ALTERNATE CLOCK INPUTS
FIGURE 3



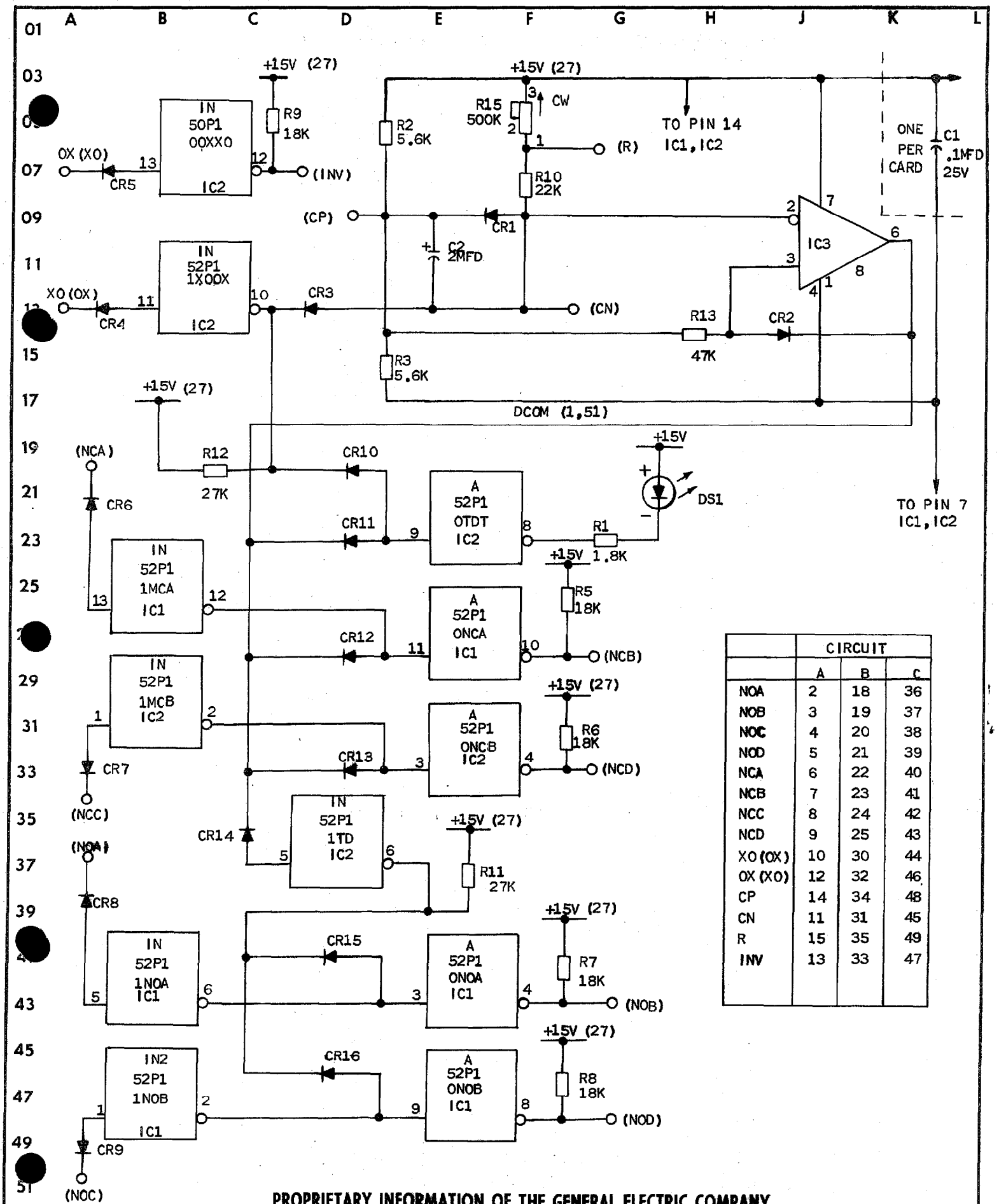
NOTES: 1) MULTIPLEXING SHOWN FOR ONE DIGIT. REPEAT FOR OTHERS
2) INPUT DATA DEFINED AS TRUE WHEN OV (CONTACT CLOSED TO DCOM)
3) *DENOTES SOLID STATE RELAY.

FIGURE 4

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO RIC, DL58, CHVIVN DL1, 2530		ELEMENTARY DIAGRAM: TIMER CARD
REV. 2	ISSUED July 21, 1972			FIRST MADE FOR	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 3	MADE BY D. M. MILLER			I.C. NO.	I C 3 6 2 2 G T C B 4
					CONT. ON SH. FL. SH. NO. 3.5

FORM IC 2390 2-72



	CIRCUIT		
	A	B	C
NOA	2	18	36
NOB	3	19	37
NOC	4	20	38
NOD	5	21	39
NCA	6	22	40
NCB	7	23	41
NCC	8	24	42
NCD	9	25	43
XO (OX)	10	30	44
OX (XO)	12	32	46
CP	14	34	48
CN	11	31	45
R	15	35	49
INV	13	33	47

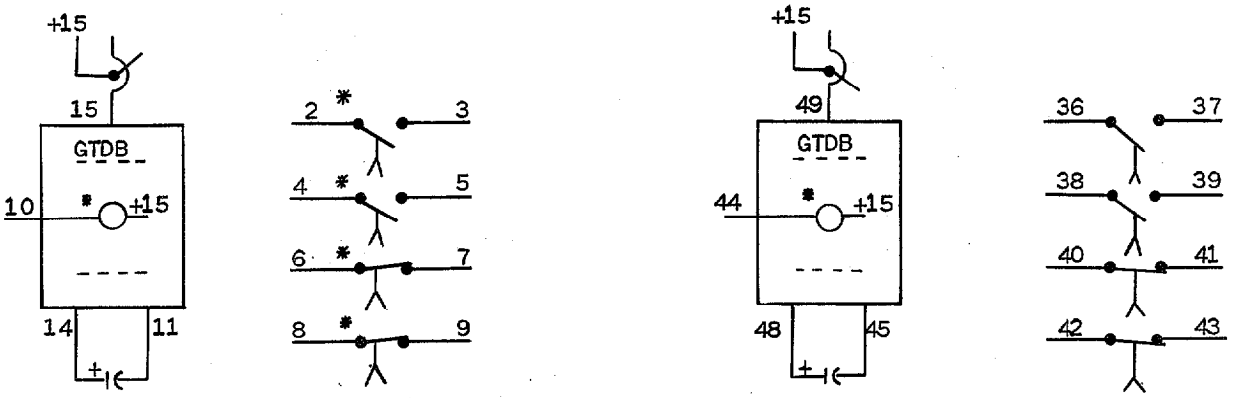
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

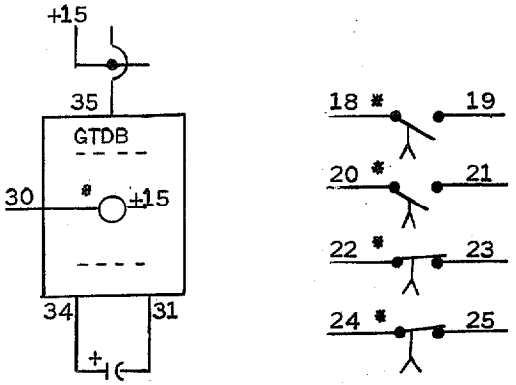
REV. 1	REV. 4	REV. 5	PRINTS TO DL1, P6A	APPROVALS W.A.M.	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 2	ISSUED May 9, 1972	1338 A6F CHVLVN	FIRST MADE FOR		ELEMENTARY DIAGRAM TIME DELAY
REV. 3	MADE BY KENNEY C. COX	A1F(4),3T	I.C. NO.		I C 3 6 2 2 G T D B 1
					CONT. ON SH. 3.1 SH. NO. 3.0

FORM 1C 2390 2-72

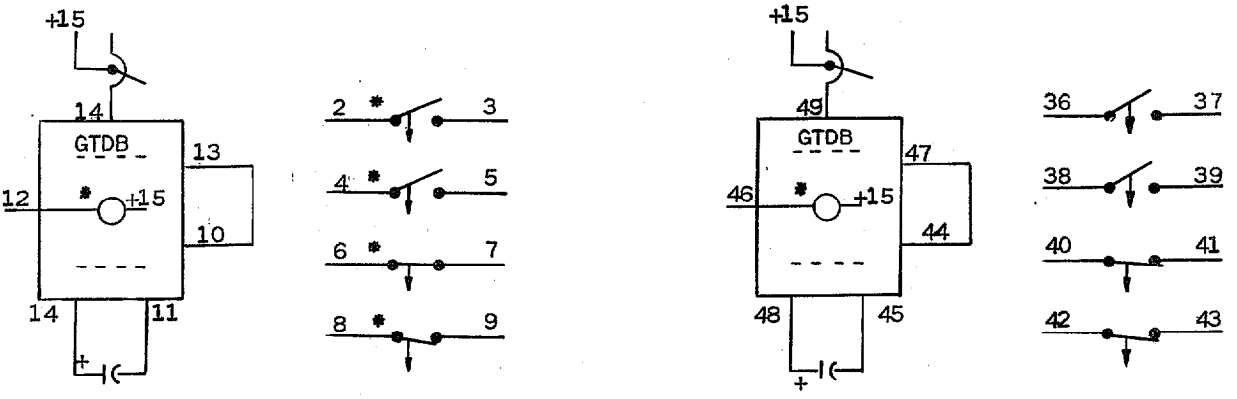
SYMBOLS

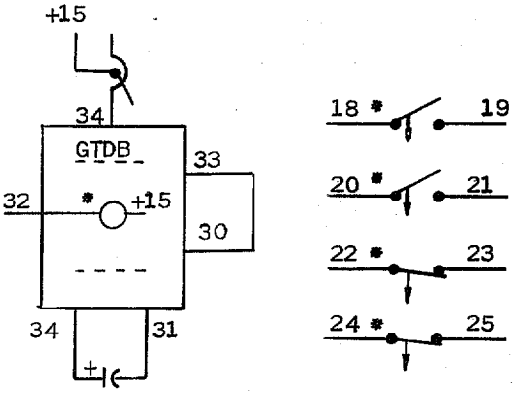
TIME ON PICK UP:





TIME ON DROP OUT:





PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL1, P6A, 3T CAVIVN 1338, A6F, GV	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM TIME DELAY
REV. 2	ISSUED May 9, 1972			FIRST MADE FOR		I C 3 6 2 2 G T D B 1
REV. 3	MADE BY KENNEY C. COX		A1F (4) WAY	I.C. NO.		CONT. ON SH. 3.2 SH. NO. 3.1

APPLICATION NOTES
TIME DELAY

THIS CARD CONTAINS THREE IDENTICAL TIME DELAYS WITH OUTPUTS COMPATIBLE WITH FLOW THROUGH LOGIC. FLOW THROUGH LOGIC IS SOLID STATE LOGIC ARRANGED SO THAT ITS APPLICATION, SYMBOLS AND NOMENCLATURE IS LIKE RELAY LOGIC. THE CIRCUIT CAN BE USED AS A TIME DELAY RELAY WHICH TIMES ON PICK UP, OR BY CHANGING THE PIN CONFIGURATION, TIMES ON DROP OUT. THE CIRCUIT SYMBOLS SHOW BOTH CONFIGURATIONS. AN INDICATOR SHOWS WHEN THE CIRCUIT IS TIMING.

TIME DELAY RANGE:

THE DELAY ON THE CARD CAN BE ADJUSTED WITH THE INTERNAL RHEOSTAT R15. THE TABLE BELOW SHOWS THE TIME RANGE THAT CAN BE SET. THE TOLERANCE SHOWN ARE DUE TO COMPONENT TOLERANCES.

R15	TD MIN.	TD. TYP.	TD MAX.
FULL CCW	30 MS	40 MS	60 MS
FULL CW	580 MS	700 MS	1100 MS

LONGER TIMES MAY BE OBTAINED BY CONNECTING EXTERNAL CAPACITANCE ACROSS THE CP AND CN TERMINALS. IF ELECTROLYTICS ARE USED, THEY SHOULD BE CONNECTED WITH THE POSITIVE LEAD TO CP. IF +80%, -20% TOLERANCE CAPACITORS ARE USED, THE NEW TIME RANGE IS OBTAINED BY MULTIPLYING THE TIMES GIVEN ABOVE BY $\frac{C+2}{2}$ WITH EXTERNAL CAPACITORS, TEMPERATURE STABILITY OF TIMING IS ±10% WITH ALUMINUM ELECTROLYTICS (68A7053), ±5% WITH TANTALYTIC (68A7054), ±3% WITH METALLIZED MYLAR (68A7052) OVER A 0°C TO 70°C CARD TEMPERATURE.

REMOTE SETTING:

THE DELAY TIME MAY BE SET BY AN EXTERNAL RHEOSTAT CONNECTED BETWEEN TERMINAL R AND +15V. R15 SHOULD BE SET FULL CLOCKWISE. THE DELAY TIME IS GIVEN BY

$$TD = (.74)\left(\frac{500R}{R+500} + 22\right)(C+2)(10^{-3}) \text{ SECONDS}$$

R IN K OHMS

C IN MICROFARADS

A 500K RHEOSTAT IS RECOMMENDED BUT SMALLER VALUES CAN BE USED IF LESS RANGE IS DESIRED. THE TOLERANCE ON THIS TIME IS THE SAME AS THE TOLERANCE ON CAPACITOR VALUE.

RESET TIME:

THERE IS A 20 MICROSECOND DELAY IN CONTACT OPERATION WHEN THE TIMER IS RETURNED TO NORMAL AFTER A TIMING CYCLE. IN ADDITION SOME TIME MUST BE ALLOWED TO RECHARGE THE TIMING CAPACITOR BEFORE A NEW TIMING PERIOD IS STARTED. THE RESET TIME SHOULD BE GREATER THAN .0084 (C+2) SECONDS. (C IN MICROFARADS). SHORTER TIMES WILL SHORTEN THE DELAY TIME.

LOGIC DESCRIPTION

A LOGIC "0" (0 TO 4.6 VOLTS) IS CONSIDERED A PICKED UP CONDITION, WHILE A LOGIC "1" (10 TO 15 VOLTS) IS CONSIDERED OFF OR OPEN. THE INPUT IS THEREFORE CONSIDERED PICKED UP WHEN DRIVEN BY A "0" LEVEL. THE "CONTACTS" ARE ARRANGED AS FOLLOWS:

CONTACT STATE	INPUT	OUTPUT
OPEN	1	1
OPEN	0	1
CLOSED	1	1
CLOSED	0	0

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 MLO 3-15-78	REV. 4	REV. 5	PRINTS TO DL58, DL [illegible]	APPROVALS [illegible]	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM TIME DELAY
REV. 2	ISSUED May 9, 1972		QHV1VN 2530 RIC	FIRST MADE FOR PLASTIC MOLD		I C 3 6 2 2 G T D B 1
REV. 3	MADE BY KENNEY C. COX			I.C. NO.		CONT. ON SH. 3.3 SH. NO. 3.2

APPLICATION NOTES (CONTINUED)

A CLOSED CONTACT HAS AN OUTPUT LEVEL THE SAME AS THE INPUT.

AN OPEN CONTACT HAS AN OUTPUT "1" LEVEL CORRESPONDING TO AN OPEN CONDITION. LOGIC IS ARRANGED WITH PARALLEL AND SERIES CONTACT ARRANGEMENTS SUBJECT TO THE FOLLOWING RESTRICTIONS:

1) LOGIC SIGNAL FLOW STARTS AT 0 VOLTS OR ON AN INPUT BUFFER WHICH HAS A 0 VOLT OUTPUT FOR A TRUE INPUT.

2) CONTACTS ARE NOT BI-DIRECTIONAL. SIGNAL FLOW IS IN THE DIRECTION FROM CONTACT INPUT (STARRED) TO CONTACT OUTPUT.

3) THE SIGNAL FLOW THROUGH A CONTACT STRING ENDS ON A SOLID STATE RELAY COIL OR DRIVER CIRCUIT.

4) LOADING RESTRICTIONS APPLY. EACH INPUT IS ONE LOAD AND EACH OUTPUT CAN DRIVE 17 LOADS. A PARRALLEL CONTACT OUTPUT REPRESENTS ONE LOAD TO THE CONTACT THAT IS CLOSED. THE NUMBER OF INPUTS AND OUTPUTS CONNECTED TO ANY CIRCUIT NODE MUST NOT EXCEED 18.

CIRCUIT OPERATION:

TIMING CAPACITOR IS NORMALLY CHARGED TO 7.5 VOLTS BY THE R2-R3 DIVIDER AND THE INPUT INVERTER 1XOOX. DURING TIMING THE OUTPUT OF 1XOOX GOES TO VOLTS REVERSE BIASING CR3. THIS ALLOWS C2 TO DISCHARGE SO CN MOVES TOWARD 15V. WHEN CN REACHES 7.5 VOLTS THE COMPARATOR IC3 CHANGES STATE. CR1 PREVENTS REVERSE BIASING THE CAPACITOR. THE COMPARATOR DRIVES THE HTL GATES FORMING THE CONTACT LOGIC.

POWER APPLIED RESET:

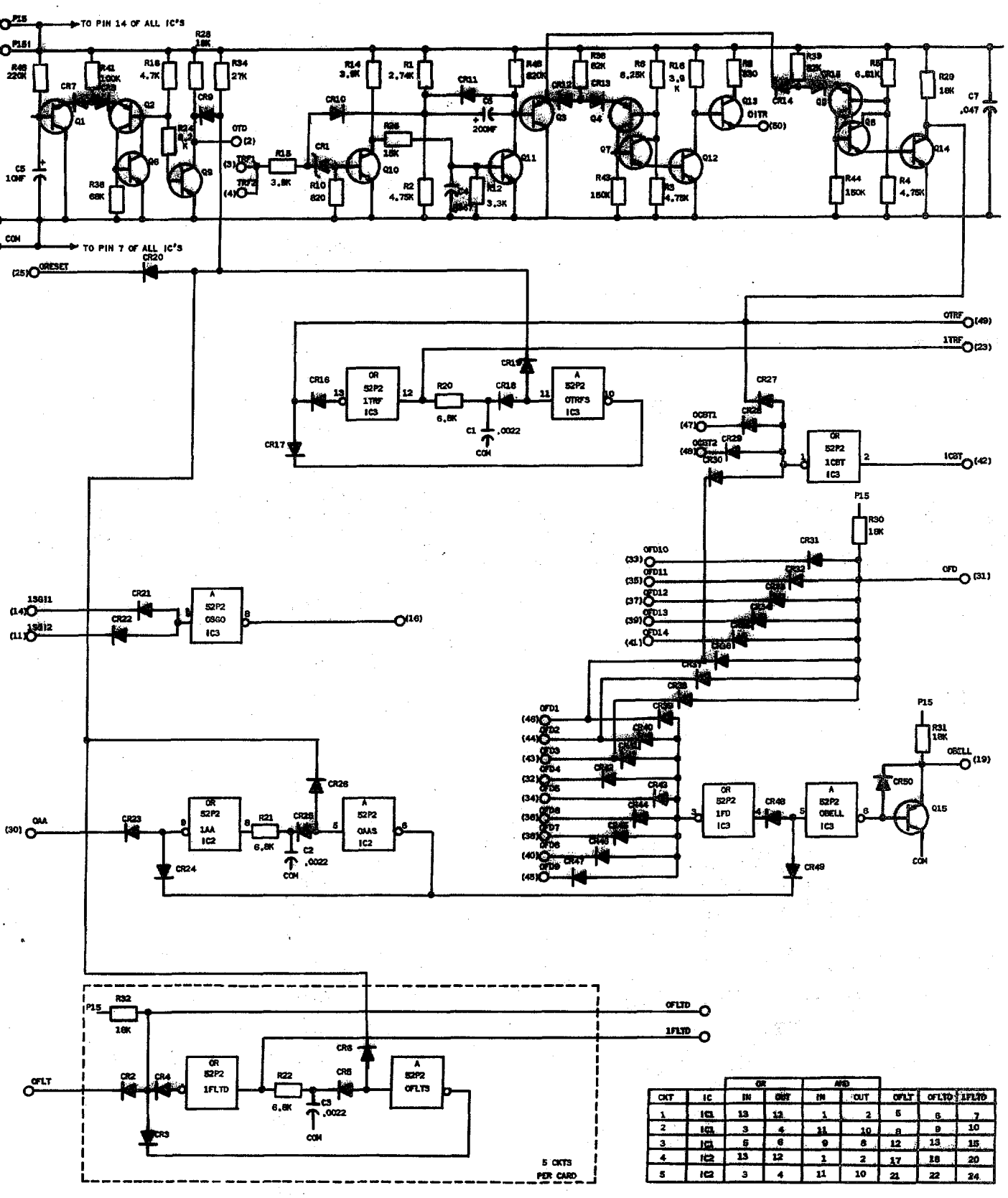
THE CIRCUIT MAY BE MADE TO TIME OUT ON POWER APPLICATION, AS IN A POWER APPLIED RESET, BY CONNECTING A 5 MFD CAPACITOR FROM CN TO D COM, WITH CN POSITIVE. CONNECT ALL INPUTS TO COMMON. SET R15 FOR ONE SECOND.

POWER SUPPLY

+ 15 VOLTS, 60 MA TYPICAL WHEN NOT TIMING;
75 MA TYPICAL WHEN ALL CIRCUITS TIMING SIMULTANEOUSLY.
100 MA MAXIMUM CURRENT.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL1, P6A, 3T	APPROVED	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM TIME DELAY
REV. 2	ISSUED May 9, 1972		C#VIVN 1338, A6F, 6W	FIRST MADE FOR PLASTIC MOLD		I C 3 6 2 2 G T D B 1
REV. 3	MADE BY KENNEY C. COX		WAY, A1F (4)	I.C. NO.		CONT. ON SH. FL. NO. 3.3



CKT	IC	OR IN	OR OUT	AND IN	AND OUT	OFLT	OFLTD	1FLTD
1	IC1	13	12	1	2	5	6	7
2	IC1	3	4	11	10	8	9	10
3	IC1	5	6	9	8	12	13	15
4	IC2	13	12	1	2	17	18	20
5	IC2	3	4	11	10	21	22	24

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

A WRIGHT JULY 16, 71	REV.4	REV.5	PRINTS TO DL1, A6F 1336, 3T	APPROVALS	GENERAL ELECTRIC
REV.2	ISSUED 5-26-71		A1F(4), P6A CHVI	FIRST MADE FOR PLASTIC MOLDING	INDUSTRY CONTROL DEPT. SALEM, VA., USA
REV.3	MADE BY SANDY HILL			TC. NO.	

ELEMENTARY DIAGRAM
TRIAC FAILURE AND ALARM

IC3622GTFA1

CONT. ON SH. 3.1 SH. FOR 3.0

A B C D E F G H J K L

01

03

THEORY OF OPERATION

07 THE TRIAC FAILURE PORTION OF THIS CARD IS DESIGNED TO OPERATE WITH THE PLASTIC MOLDING TEMPERATURE REGULATOR IC3628R AND HEATER DRIVER IC3625D. THIS CIRCUIT PERIODICALLY CHECKS THAT AC POWER CAN BE REMOVED FROM EACH HEATER ZONE, THUS 09 VERIFYING THAT THE TRIACS IN THE HEATER DRIVER ARE FUNCTIONING PROPERLY.

11 TRF1 AND TRF2 FORM THE INITIATE AND RESET TERMINAL OF A RC TIMER CONSISTING OF Q11, C6, AND R48. THERE WILL BE FOUR REED RELAY CONTACTS (ONE REED ON EACH HEATER DRIVER) 13 CONNECTED IN PARALLEL BETWEEN THE INITIATE TERMINALS AND +15V. IF NO ZONE IS RECEIVING HEAT ALL CONTACTS WILL BE OPEN, Q10 WILL BE OFF, Q11 WILL BE SATURATED, 15 AND C6 WILL BE CHARGED. THIS IS THE RESET STATE, WITH OUTPUTS OITR AND OTRF BEING A LOGIC "1".

17 IF ONE OR MORE ZONES ARE RECIEVING HEAT, THE ASSOCIATED CONTACTS WILL BE CLOSED, Q10 WILL BE ON, Q11 OFF, AND C6 WILL DISCHARGE THROUGH R48. AS C6 DISCHARGES, Q3 19 BASE VOLTAGE RISES. IF THE TIMER IS NOT RESET WITHIN APPROXIMATELY 60 SECONDS, Q3 BASE VOLTAGE EXCEEDS Q4 BASE VOLTAGE. AT THIS POINT Q3 TURNS OFF AND Q4, Q7, AND Q12 21 SATURATE. Q13 IS NOW OFF, THUS PROVIDING A SIGNAL TO THE TEMPERATURE REGULATOR TO GATE ALL TRIACS OFF. IF HEATER POWER IS REMOVED FROM ALL ZONES, THE TIMER WILL BE RESET. 23 HOWEVER IF ANY ZONE CONTINUES TO RECEIVE HEAT THE TIMER WILL CONTINUE TO TIME AND APPROXIMATLEY 15 SECONDS LATER, Q5, Q8, AND Q14 WILL SATURATE. OUTPUT 1CBT WILL NOW 25 BECOME LOGIC "1" WHICH WILL INITIATE THE TRIPPING OF THE HEATER CIRCUIT BREAKER. Q14 SATURATING WILL ALSO LATCH THE SEAL CIRCUIT, THUS GIVING A "1" AND "0" OUTPUT ON 1TRF AND OTRF.

HEATER POWER CANNOT BE RE-APPLIED UNTIL THE SEAL CIRCUIT HAS BEEN RESET. RESET IS 29 ACCOMPLISHED BY APPLYING A LOGIC "0" TO THE ORESET TERMINAL.

31 FIVE ADDITIONAL SEAL CIRCUITS ARE PROVIDED TO DETECT AND REMEMBER THAT A FAULT CONDITION HAS OCCURRED. A LOGIC "0" ON THE OFLT INPUT WILL LATCH THE SEAL AND PROVIDE A FAULT 33 INDICATION AT THE OUTPUTS. WHEN THE FAULT HAS BEEN CLEARED, THE SEAL MUST BE RESET BY THE ORESET INPUT. NOTE THAT IF THE FAULT STILL EXISTS, THE SEAL CANNOT BE CLEARED BY THE RESET SIGNAL.

35

37 THE OFLTD OUTPUT OF ANY SEAL CIRCUIT MAY BE CONNECTED TO ANY INPUT OFD1 - OFD14 TO BRING UP AN ALARM AND/OR GIVE A LOGIC INDICATION AT OUTPUT OFD. THE OFLTD OUTPUT MAY ALSO BE USED TO TRIP THE HEATER CIRCUIT BREAKER BY CONNECTING IT TO OCBT1 OR OCBT2.

39 INPUT OAA PROVIDES A MEANS OF SILENCING THE ALARM ONCE A FAULT HAS BEEN DETECTED. LOGIC "0" ON OAA WILL LATCH THE ALARM ACKNOWLEDGE SEAL AND CAUSE OBELL OUTPUT TO BECOME LOGIC "1". THIS SEAL IS CLEARED BY THE COMMON RESET SIGNAL.

43 Q1, Q2, Q6, Q9 AND ASSOCIATED CIRCUITRY FORM A TIME DELAY TO INSURE ALL SEAL CIRCUITS ARE CLEARED WHEN POWER IS FIRST APPLIED. THE R46, C5 TIME CONSTANT IS APPROXIMATELY 45 TEN TIMES THAT OF THE +15V SUPPLY. WHEN POWER IS FIRST APPLIED Q9 WILL BE ON, THUS HOLDING THE RESET LINE TO ALL SEAL CIRCUITS AT LOGIC "0". AFTER APPROXIMATELY 2.5 47 SECONDS, Q9 WILL TURN OFF AND REMOVE THE RESET SIGNAL.

49

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO DL1, P6A	APPROVALS E.B. Brown	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM TRIAC FAILURE AND ALARM
REV. 2	ISSUED 5-20-71		3T 1338 ALF 14, 16F	FIRST MADE FOR PLASTIC MOLD		IC3622GTFA1
REV. 3	MADE BY SANDY HILL			I.C. NO.		CONT. ON SH. 3.2 SH. NO. 3.1

INPUT LOADING IS TABULATED BELOW. ONE LOAD IS DEFINED AS 18K OHMS TO +15V.

INPUT	LOADS
OAA	1
OCBT1	1
OCBT2	1
OFD1	3 + OFD
OFD2	2 + OFD
OFD3	2 + OFD
OFD4 - OFD9	1
OFD10-OFD14	1 + OFD
ORESET	8
OFLT	2 + OFLTD
1SGI1	1
1SGI2	1

TRF1, TRF2 - CONNECTING TRF1 OR TRF2 TO +15V WILL INITIATE THE TRIAC FAILURE TIMER. RESET REQUIRES THAT TRF1 AND TRF2 BE OPEN-CIRCUITED FOR .5 MS MINIMUM.

OUTPUT DRIVE CAPABILITY IS TABULATED BELOW. IN SOME CASES THE DRIVE CAPABILITY OF AN OUTPUT IS LIMITED BY CERTAIN INPUTS. IN THESE CASES THE LIMITING INPUTS ARE LISTED.

OUTPUT	MAX. LOADING	LIMITED BY
OTD	18	
OTRF	15	
1TRF	17	
OFLTD	16	OFLT
1FLTD	17	
1CBT	18	
OBELL	30	
OFD	17	OFD1-3, OFD10-14
OSGO	18	

OITR - WILL DRIVE 2K OHMS TO -15V WITH A LOGIC "1" GREATER THAN 10V.

POWER REQUIREMENTS

+15 V, 30 MA NOMINAL, 60 MA MAX.

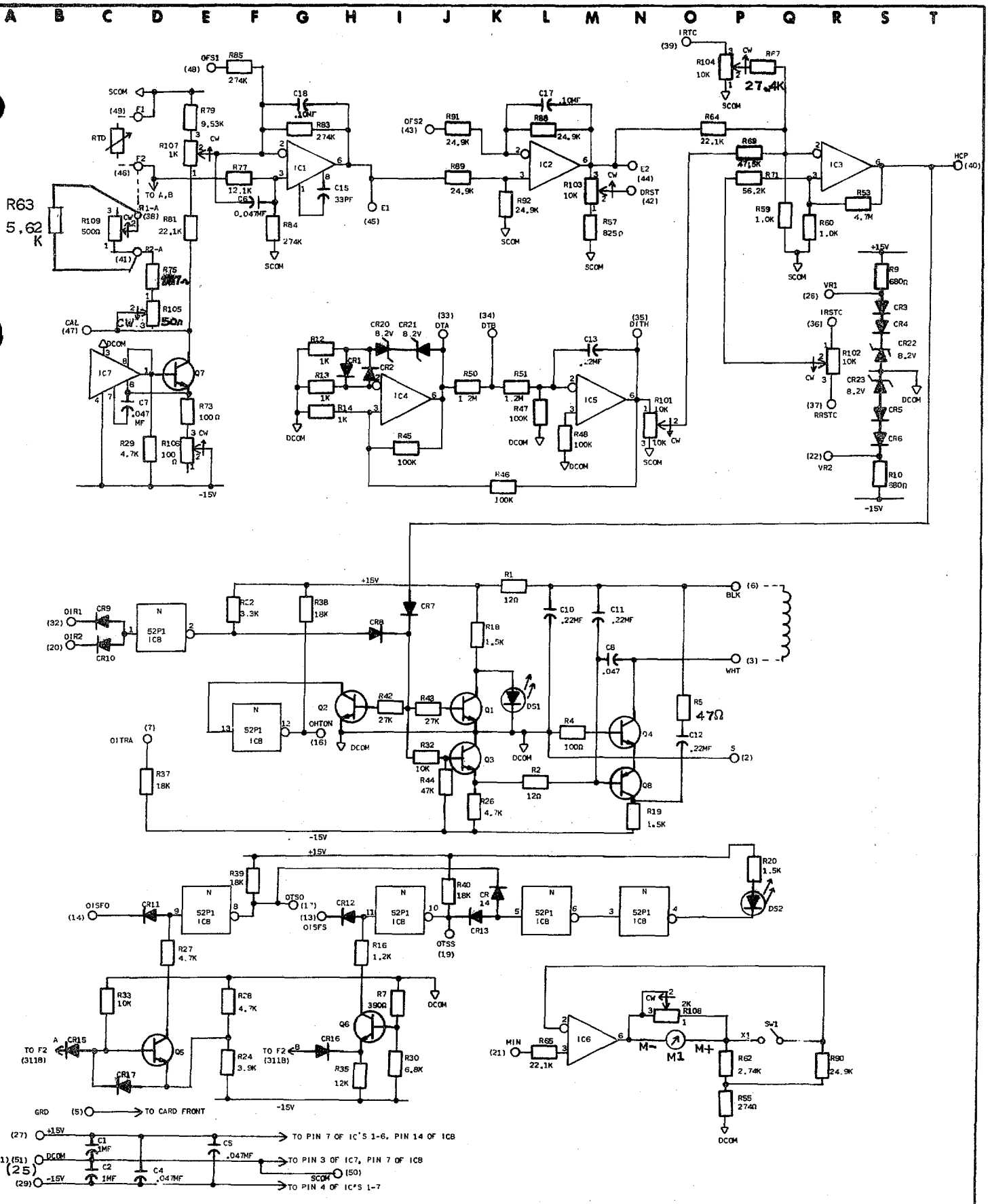
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 ACW 11-15-71.	REV. 4	REV. 5	PRINTS TO DLI, P6A	APPROVALS C.B. Brown	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 2	ISSUED 5-20-71	3T, 1338 A1F (4), A6F	FIRST MADE FOR PLASTIC MOLD		
REV. 3	MADE BY SANDY HILL WAY		I.C. NO.		

ELEMENTARY DIAGRAM: TRIAC FAILURE AND ALARM

I C 3 6 2 2 G T F A 1

CONT. ON SH. FL SH. NO. 3.2



DMM 9/6/72	REV.4	REV. 5	PRINTS TO DL58 NEW CHV1VN	APPROVALS C.B. Brown	ELEMENTARY DIAGRAM TEMPERATURE REGULATOR
REV.2 MLO 2-21-73	ISSUED May 12, 1972		2530 MOLDING	FIRST MADE FOR PLASTIC	
REV.3	MADE BY D. M. MILLER			I.C. NO.	I C 3 6 2 2 G T R D 1
			GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA., USA		CONT. ON SH. 3.1 SH. NO. 3.0

FORM IC 8849 (09 67)

APPLICATION INFORMATION

FUNCTION AND THEORY OF OPERATION

THIS BOARD CONTAINS A BASIC TIME PROPORTIONING TEMPERATURE REGULATOR WITH MANUAL RESET. THE BASIC REGULATOR INCLUDES A DIGITAL SETPOINT REFERENCE, A DUAL RANGE DEVIATION METER, AND INDICATOR LAMPS FOR HEAT ON AND RTD MONITOR. AUTOMATIC RESET AND RATE FUNCTIONS CAN BE OBTAINED BY ADDITION OF AUXILIARY CARD, IC3622GTRE. THIS BOARD IS DESIGNED TO DRIVE IC3625D HEATER DRIVER.

THE REGULATOR FRONT END IS DESIGNED TO ACCEPT MILLIVOLT SIGNALS FROM A PLATINUM RTD TEMP. SENSOR WHOSE RESISTANCE IS 100Ω AT 0°C AND 0C±.00392 OHMS/OHM/°C. THE RTD AND REFERENCE POT R109, ALONG WITH R75 AND R105, COMPRISE ONE LEG OF A WHEATSTONE BRIDGE. R79, R81, AND R107 COMPRISE THE OTHER LEG. IC7 AND Q7 PROVIDE A CONSTANT CURRENT SOURCE FOR THE BRIDGE. R106 IS A CURRENT ADJUSTMENT AND R107 IS A LINEARITY ADJUSTMENT, BOTH FACTORY CALIBRATED AND SEALED. R105 IS A FRONT PANEL ADJUSTMENT, CAL, AND MAY BE ADJUSTED TO COMPENSATE FOR RTD LEAD RESISTANCE. IC1 AMPLIFIES THE BRIDGE OUTPUT AND PROVIDES FOR AN EXTERNAL OFFSET. IC2 IS A UNITY-GAIN AMPLIFIER AND ALSO PROVIDES FOR AN EXTERNAL OFFSET. IC4 AND IC5 COMPRISE A RAMP GENERATOR WHICH PROVIDES A NOMINAL CYCLE TIME OF 2 SECONDS. THE RAMP OUTPUT IS FED THROUGH GAIN ADJUSTMENT R101 TO A COMPARATOR, IC3. R102 IS THE RESET ADJUSTMENT AND FEEDS THE RESET TERM TO COMPARATOR IC3. IC3 COMPARES THE ERROR, RAMP, RESET AND RATE TERMS AND DETERMINES THE ON-OFF RATIO FOR THE OSCILLATOR.

Q4, Q8, C8, C11, C12, R[illegible], R1[illegible] AND THE PRIMARY OF THE PULSE TRANSFORMER ON THE HEATER DRIVER FORM A SINE WAVE OSCILLATOR TO PROVIDE FIRING PULSES TO THE HEATER DRIVER. THE OSCILLATOR MAY BE GATED OFF BY LOGIC INPUTS OIR1 AND OIR2.

Q5, Q6, IC8 AND ASSOCIATED CIRCUITRY FORM THE RTD MONITOR. SHOULD THE RTD FAIL EITHER SHORTED OR OPEN, D52 WILL LIGHT AND AN APPROPRIATE LOGIC SIGNAL WILL APPEAR AT PINS 17 OR 19.

IC6 IS THE DEVIATION METER AMPLIFIER. R108 IS A METER ADJUSTMENT AND SW1 CHANGES THE METER RANGE. THE AMPLIFIER INPUT MAY BE DRIVEN FROM EITHER OUTPUT E1 OR E2.

SPECIFICATIONS

INPUTS

F1, F2 - FEEDBACK CONNECTIONS FROM RTD.

R1A, R2A - REFERENCE POT CONNECTIONS. NOTE THAT AN EXTERNAL CONNECTION MUST BE MADE BETWEEN F2 AND R1.

OFS1 - EXTERNAL OFFSET INPUT, MUST BE DRIVEN FROM A LOW IMPEDANCE VOLTAGE SOURCE. INPUT PRESENTS A MAXIMUM LOADING OF .04MA AT +7.5V. +7.50V IS EQUIVALENT TO SUBTRACTING 100°F ± 2.5°F, FROM THE REFERENCE SETTING. -7.50 V IS EQUIVALENT TO ADDING 100°F ± 2.5°F TO THE REFERENCE SETTING. WHEN UNUSED, INPUT MUST BE TIED TO S COM.

OFS2 - EXTERNAL OFFSET INPUT, MUST BE DRIVEN FROM A LOW IMPEDANCE VOLTAGE SOURCE. INPUT PRESENTS A MAXIMUM LOADING OF .60MA AT + 7.5V. +7.50V IS EQUIVALENT TO SUBTRACTING 100°F ± 1.25°F FROM REFERENCE SETTING. -7.50V IS EQUIVALENT TO ADDING 100°F ± 1.25°F TO REFERENCE SETTING. WHEN UNUSED, INPUT MUST BE TIED TO S COM.

OIR1, OIR2 - REGULATOR INHIBIT INPUTS. HTL LOGIC "0" WILL INHIBIT THE OSCILLATOR. EACH INPUT IS ONE HTL LOAD.

OITRA - CONNECTING OITRA TO OIR1 (OR OIR2) PROVIDES A PULL DOWN RESISTOR WHICH MAKES THE OIR1 INPUT COMPATIBLE WITH THE IC3622GTFA OITR OUTPUT.

OISFO - HTL LOGIC "0" WILL INHIBIT THE SENSOR FAILED OPEN CIRCUIT. INPUT IS ONE HTL LOAD.

OISFS - HTL LOGIC "0" WILL INHIBIT THE SENSOR FAILED SHORTED CIRCUIT. INPUT IS ONE HTL LOAD.

MIN - DEVIATION METER AMPLIFIER INPUT. AN INPUT OF ±7.50V WILL RESULT IN FULL SCALE DEFLECTION WHEN SW1 IS CLOSED. AN INPUT OF±.750V WILL RESULT IN FULL SCALE DEFLECTION WHEN SW1 IS OPEN. LOADING TS 5 MICROAMPS MAXIMUM AT 7.5V.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 DMM 9/6/72	REV. 4	REV. 5	PRINTS TO 6V,DL1,3T	APPROVALS	ELEMENTARY DIAGRAM: TEMPERATURE REGULATOR
REV. 2	ISSUED May 12, 1972 2530, P6A,1338		FIRST MADE FOR PLASTIC MOLD	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	IC3622GTRD1
REV. 3	MADE BY D. M. MILLER	A6F,A1F (4)	I.C. NO.		CONT. ON SH. 3.2 SH. NO. 3.1

APPLICATION INFORMATION

SPECIFICATIONS

DTA,DTB - CYCLE TIME INPUTS, CONNECTING DTA TO DTB DECREASES CYCLE TIME TO APPROXIMATELY 1 SECOND. CONNECTING EXTERNAL RESISTANCE FROM DTB TO COMMON INCREASES CYCLE TIME. 330K TO COMMON GIVES APPROXIMATELY 5 SEC. AND 120K TO COMMON GIVES APPROXIMATELY 10 SECONDS.

IRTC - RATE INPUT WHEN GTRE CARD IS SUPPLIED. A 10.0V INPUT WILL OFFSET THE REGULATOR A MINIMUM OF 100° WITH R104 FULLY CW, AND A MAXIMUM OF 5° WITH R104 FULLY CCW. A POSITIVE INPUT VOLTAGE CALLS FOR A GREATER ON TIME, A NEGATIVE INPUT VOLTAGE LESS ON TIME. INPUT PRESENTS A MAXIMUM LOADING OF 1.4 MA AT 10V. WHEN UNUSED, INPUT SHOULD BE TIED TO S COM.

IRSTC, RRSTC - RESET INPUTS. FOR MANUAL RESET IRSTC IS JUMPERED TO VR1 AND RRSTC IS JUMPERED TO VR2. WITH R102 CENTERED, THE RESET TERM IS ZERO. R102 TO EITHER EXTREME WILL OFFSET THE REGULATOR A MINIMUM OF 40°F. CW INCREASES ON TIME, CCW DECREASES ON TIME.

FOR AUTOMATIC RESET WHEN GTRE CARD IS SUPPLIED, RRSTC IS CONNECTED TO S COM AND IRSTC IS CONNECTED TO RSTC OUTPUT OF GTRE CARD. R102 NOW BECOMES THE INTEGRATOR SPAN ADJUSTMENT AND SHOULD BE ADJUSTED COMPATIBLE WITH THE PROPORTIONAL BAND ADJUSTMENT, R101. FOR A PROPORTIONAL BAND OF ±40°F, R101 AND R102 WOULD BOTH BE FULL CCW. AN INTEGRATOR OUTPUT OF 10V (40°) WOULD NOW OFFSET THE REGULATOR BY 40°, OR 50% OF THE TOTAL PROPORTIONAL BAND. FOR A PROPORTIONAL BAND OF ±20F, R101 AND R102 WOULD BOTH BE MID-RANGE. NOW AN INTEGRATOR OUTPUT OF 10V WOULD OFFSET THE REGULATOR BY 20°, AGAIN 50% OF THE TOTAL BAND.

OUTPUTS

E1 - ERROR AMPLIFIER OUTPUT. SCALING IS A NOMINAL 75 MV PER DEGREE OF ERROR AT 500°F. MAXIMUM LOADING IS 2.5K OHMS TO COMMON. FOR STABLE 25°C AMBIENT TEMPERATURE, ACCURACIES ARE AS FOLLOWS:
ZEROING ACCURACY (BRIDGE BALANCED AT ANY REFERENCE SETTING BETWEEN 70°F AND 800°F)
± .187% OF FULL SCALE
ERROR AMPLIFICATION ACCURACY (BRIDGE UNBALANCED A MAXIMUM OF ±100°F AT ANY REFERENCE SETTING BETWEEN 250°F AND 800°F) ± .875% OF FULL SCALE.
EXTERNAL OFFSET ACCURACY - ±.25% OF FULL SCALE.
TEMPERATURE DRIFT WILL BE LESS THAN .250% OF FULL SCALE

E2 - ERROR AMPLIFIER OUTPUT. SAME AS OUTPUT E1, EXCEPT EXTERNAL OFFSET ACCURACY IS ±.05% OF FULL SCALE, AND MAXIMUM LOADING IS 3.0K OHMS TO COMMON.

DRST - AUTOMATIC RESET OUTPUT, CONNECTS TO RSTF INPUT OF GTRE CARD WHEN SUPPLIED. R103 IS THE INTEGRATE TIME ADJUSTMENT. R103 FULL CW GIVES A NOMINAL 75MV PER DEGREE OF ERROR WHICH CORRESPONDS TO A NOMINAL RESET TIME OF 1 MINUTE. R101 FULL CCW GIVES A NOMINAL OF 6MV PER DEGREE OF ERROR WHICH CORRESPONDS TO A NOMINAL RESET TIME OF 10 MINUTES.

HCP - COMPARATOR OUTPUT MAXIMUM LOADING IS 5K OHMS TO COMMON. OUTPUT WILL BE +10.0V MINIMUM INDICATING NO HEAT IS REQUIRED, OR -10.0V MAXIMUM, INDICATING HEAT IS REQUIRED.

DITH - PROPORTIONAL BAND OSCILLATOR OUTPUT. OUTPUT IS A RAMP WAVEFORM OF 16.0V MINIMUM PEAK TO PEAK. FREQUENCY IS A NOMINAL .50 HZ PER SECOND WHICH PROVIDES A 2 SECOND CYCLE TIME. R101 IS THE GAIN ADJUSTMENT. FULL CW PROVIDES A PROPORTIONAL BAND OF 5°F MAXIMUM. FULL CCW PRODUCES A PROPORTIONAL BAND OF 80°F MINIMUM. MAXIMUM LOADING IS 3.5K OHMS TO COMMON.

VR1,VR2 - REFERENCE VOLTAGE OUTPUTS TO SUPPLY A STABLE VOLTAGE FOR R102 WHEN MANUAL RESET IS PROVIDED. VR1 WILL BE +8.8V TO +9.9V. VR2 WILL BE -8.8V TO -9.9V. MAXIMUM LOADING IS 5K OHMS BETWEEN VR1 AND VR2.

OHTON - LOGIC OUTPUT GIVING LOGIC "0" WHEN HEAT IS REQUIRED AND REGULATOR IS NOT INHIBITED. OUTPUT DRIVE CAPABILITY IS 17 FLOW-THRU LOGIC LOADS.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 DMM 9/6/72	REV. 4	REV. 5	PRINTS TO 6V,DL1,3T	APPROVALS P.B. Brown	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM TEMPERATURE REGULATOR
REV. 2	ISSUED May 12, 1972 2530, WAY P6A,1338 CHVTVN A6F,A1F(4)		FIRST MADE FOR PLASTIC MOLD			I C 3 6 2 2 G T R D 1
REV. 3	MADE BY D. M. MILLER		I.C. NO.			CONT ON SH 3,3 SH NO. 3,2

APPLICATION INFORMATION

SPECIFICATIONS

OTSO, OTSS - LOGIC OUTPUTS GIVING LOGIC "0" FOR SENSOR FAILED OPEN AND SENSOR FAILED SHORTED. OUTPUT DRIVE CAPABILITY IS 17 FLOW-THRU LOGIC LOADING.

OSCILLATOR OUTPUTS BLK,WHT, S - THE BLK, WHT AND S (PINS 6,3,2) OUTPUTS WILL CONNECT TO THE BLK, WHT AND S LEADS FROM HEATER DRIVER IC3625D. NOTE THAT THE SHIELD MUST BE LEFT OPEN AT THE DRIVER END, THAT IS, IT MUST NOT CONNECT TO THE TRANSFORMER PRIMARY. THE OSCILLATOR OUTPUT WILL BE 60 KHZ SINE WAVE BURSTS OF 30V PEAK TO PEAK. THE BURSTS WILL OCCUR AT APPROXIMATELY 400 MICRO-SECOND INTERVALS AND THEIR DURATION WILL VARY WITH LOADING.

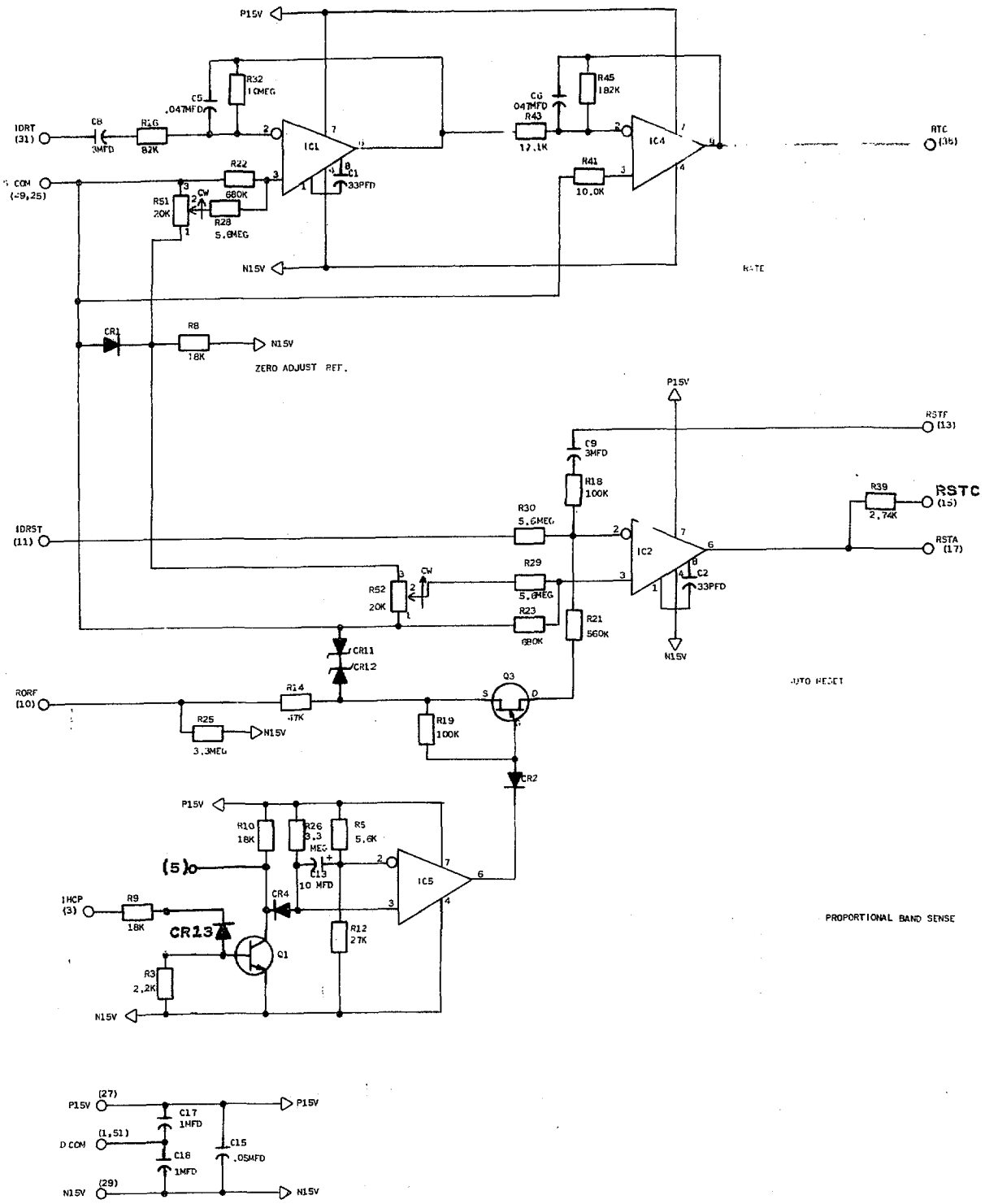
POWER SUPPLY

+15V (+14V TO +16V)
52 MA TYPICAL
60 MA MAXIMUM

-15V (-14V TO -16V)
52 MA TYPICAL
58 MA MAXIMUM.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO 6V,DL1,3T	APPROVALS F.B. Brown	ELEMENTARY DIAGRAM TEMPERATURE REGULATOR
REV. 2	ISSUED May 12, 1972	2530, P6A, 1338 CHIVVY WAY	FIRST MADE FOR PLASTIC MOLD	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	IC3622GTRD1
REV. 3	MADE BY D. M. MILLER	A6F,A1F(4)	I.C. NO.		CONT. ON SH. FL. SH. NO. 3.3



PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV.1	REV.4	REV.5	PRINTS TO 6V,2530	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM TEMPERATURE REGULATOR RATE/RESET AUXILIARY
REV.2	ISSUED MAY 12, 1972	CHV|VW, DL1, P6A, WAY	FIRST MADE FOR PLASTIC MOLD.		INDUSTRY CONTROL DEPT. SALEM, VA., U.S.A.	I C 3 6 2 2 G T R E 1
REV.3	MADE BY RON MUCKENFUSS		I.C. NO.			CONT. ON SH. 3, 1 SH. NO. 3, 0

APPLICATION INFORMATION

THIS BOARD PROVIDES A RATE (DIFFERENTIAL) STABILIZING FUNCTION PLUS AN AUTOMATIC RESET (INTEGRATE) LONG TERM ERROR ZEROING FUNCTION. INCLUDED WITH THE RESET CIRCUIT IS A PROPORTIONAL BAND SENSE SWITCH WHICH CAN BE USED TO SWITCH AN OVERRIDE INPUT INTO THE AUTOMATIC RESET CIRCUIT# SUCH THAT THE INTEGRATOR IS HELD TO DEMAND BELOW SET TEMPERATURE (OR OPTIONALLY, TO NEAR ZERO RESET ACTION) WHEN THE REGULATOR IS OUTSIDE PROPORTIONAL BAND. THUS, WHEN OUTSIDE THE PROPORTIONAL BAND (E.G. DURING STARTUP), THE INTEGRATOR WILL NOT SET UP TO DEMAND A LARGE TEMPERATURE OVERSHOOT DURING SETTLING.

THIS BOARD IS DESIGNED TO PROVIDE THE AUXILIARY RATE/RESET FUNCTIONS FOR TEMPERATURE REGULATOR IC3622GTRD. IT IS MOUNTED ON A BOARD WITH A CUT-OUT IN THE BOARD WHICH PERMITS IT TO BE MOUNTED BENEATH THE REGULATOR FRONT, IN A BOARD SOCKET ADJACENT TO THE BASIC REGULATOR. SET-UP POTENTIOMETERS ARE PROVIDED ON THE BASIC REGULATOR.

SPECIFICATIONS

RATE CIRCUIT CONSISTS OF IC1 AND IC4 CIRCUITRY, CAPACITOR C8 DIFFERENTIATES THE ERROR SIGNAL INPUT FROM THE TEMPERATURE REGULATOR THEN IC1 AND IC4 AMPLIFY THE DIFFERENTIATED SIGNAL TO PROVIDE THE RATE STABILIZATION OUTPUT.

THE ERROR OUTPUT OF THE REGULATOR IS 75 MILLIVOLTS PER DEGREE FAHRENHEIT, AND THE RATE STABILIZATION OUTPUT WILL BE 0.56 VOLTS NOMINAL PER DEGREE FAHRENHEIT PER MINUTE. THE MAXIMUM RATE OF CHANGE OF ERROR WHICH CAN BE ACCEPTED WITHOUT SATURATION IS A NOMINAL 20°F PER MINUTE. THIS MAXIMUM OUTPUT IS SCALED ON THE REGULATOR TO REPRESENT 100°F, THE RATE CIRCUIT CAN ACCOMODATE UP TO 5 MINUTES OF TRANSPORT DELAY TIME. INPUT TO THE REGULATOR BOARD IS ADJUSTABLE TO LOWER RATIOS OF THE OUTPUT, SO THAT TRANSPORT DELAYS OF LESS THAN 5 MINUTES CAN BE ACCOMODATED.

RATE INPUT IS A MINIMUM OF 75,000 OHMS.

RATE OUTPUT CAN DRIVE 3500 OHMS MINIMUM RETURNED TO COMMON.

RESET CIRCUIT CONSISTS OF IC2 CIRCUITRY. CAPACITOR C9 PROVIDES INTEGRATOR FEEDBACK FOR THE CIRCUIT. TO SECURE OPERATION, RSTF (RESET FEEDBACK INPUT) MUST BE STRAPPED TO RSTA (RESET AMPLIFIER OUTPUT). CAPACITOR C9 AND RESISTOR INPUT R30 PROVIDE A BASIC INTEGRATOR RATE OF 16.8 SECONDS, HOWEVER, THE INPUT (FROM THE ERROR OUTPUT OF THE REGULATOR) IS SCALED AT 75 MILLIVOLTS PER DEGREE F, AND THE MAXIMUM INTEGRATOR OUTPUT IS SCALED FOR 250 MILLIVOLTS/DEGREE F SO A MINIMUM INTEGRATE TIME IS 250/75 (16.8) OR 56 SECONDS. THE REGULATOR ERROR OUTPUT FEEDING THE RESET INPUT CAN BE DECREASED

(BY POTENTIOMETER ON THE REGULATOR) SO THAT INPUT MILLIVOLTS/DEGREE F CAN BE REDUCED, THUS INCREASING RESET TIME (UP TO A MAXIMUM OF 10 MINUTES. THE OUTPUT OF THE RESET CIRCUIT IS ALSO SCALED BY A POTENTIOMETER ON THE REGULATOR ADJUSTING RESET SPAN COMPATIBLE WITH PROPORTIONAL BAND ACTION.

A SWITCHED INPUT TO THE INTEGRATOR, RORF (RESET OUT OF RANGE FEEDBACK) IS INTERNALLY WIRED SUCH THAT, WHEN NO INPUT IS CONNECTED TO RORF, WHEN Q3 CLOSES THE RORF CIRCUIT TO INTEGRATOR INPUT, R25 FROM INPUT TO N15V DRIVES THE OUTPUT POSITIVE, CONSTITUTING A CONTROL SIGNAL DEMANDING REQUIRED HEAT. THUS, WHEN THE SYSTEM IS WARMING UP WITH NO INPUT TO RORF THE INTEGRATOR WILL CAUSE PROPORTIONING ACTION AT A LOWER TEMPERATURE, Q3 SWITCH OPENS AND THE INTEGRATOR SLOWLY RESETTING WILL SERVE TO SLOW DOWN THE HEAT RATE EARLY AND MINIMIZE OVERSHOOT. IF ON THE OTHER HAND, RORF IS CONNECTED TO THE RSTA OUTPUT, THE INTEGRATOR BECOMES (WITH Q3 SWITCH CLOSED) A 1/9 GAIN AMPLIFIER, SUCH THAT WHEN Q3 OPENS IN THE PROPORTIONAL BAND, IC2 OUTPUT IS APPROXIMATELY 5 DEGREES OFF ZERO OUTPUT., WHEN THE ERROR INPUT IS MAXIMUM VALUE ON ENTERING PROPORTIONAL BAND.

RESET INPUT IS NEGLIGIBLE LOAD.

RORF INPUT MAY BE REGARDED AS 100K MAXIMUM LOAD.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 6-27-72 Rem	REV. 4	REV. 5	PRINTS TO WAY 6V,2530	APPROVALS 5/12/72	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 2	ISSUED May 12, 1972 3T,A6F/D-1		FIRST MADE FOR PLASTIC MOLD		ELEMENTARY DIAGRAM TEMPERATURE REGULATOR RATE/RESET AUXILIARY
REV. 3	MADE BY RON MUCKENFUSS AIF(4)		I.C. NO.		I C 3 6 2 2 G T R E 1
					CONT. ON SH. 3.2 SH. NO. 3.1

APPLICATION INFORMATION

RSTF INPUT (FEEDBACK) MUST BE STRAPPED TO RSTA OUTPUT.

RSTC OUTPUT SPECIFICALLY DRIVES RESET (SPAN) POTENTIOMETER ON REGULATOR.

RORF OUTPUT CAN DRIVE 7000 OHMS MINIMUM RETURNED TO COMMON.

RANGE SENSE CIRCUIT SENSES THE REGULATOR OPERATION IN THE TIME PROPORTIONAL BAND AND CUTS OFF Q3 SWITCH IN THE INTEGRATOR RORF INPUT. THE SENSE INPUT (IHCP) IS DRIVEN BY THE OUTPUT OF THE REGULATOR COMPARATOR. IF THE COMPARATOR OUTPUT IS CONTINUOUSLY HIGH (HEAT FULL OFF) OR GOES THROUGH A SWITCHING CYCLE AT LEAST ONCE EVERY 24 SECONDS, IC5 OUTPUT IS DRIVEN NEGATIVE, HOLDING Q3 OFF.

IF THE COMPARATOR OUTPUT REMAINS LOW FOR 130 SECONDS OR LONGER, IC5 OUTPUT IS DRIVEN POSITIVE, TURNING Q3 ON.

IHCP INPUT REPRESENTS A MAXIMUM LOADING OF 1.4 MN AT +15V.

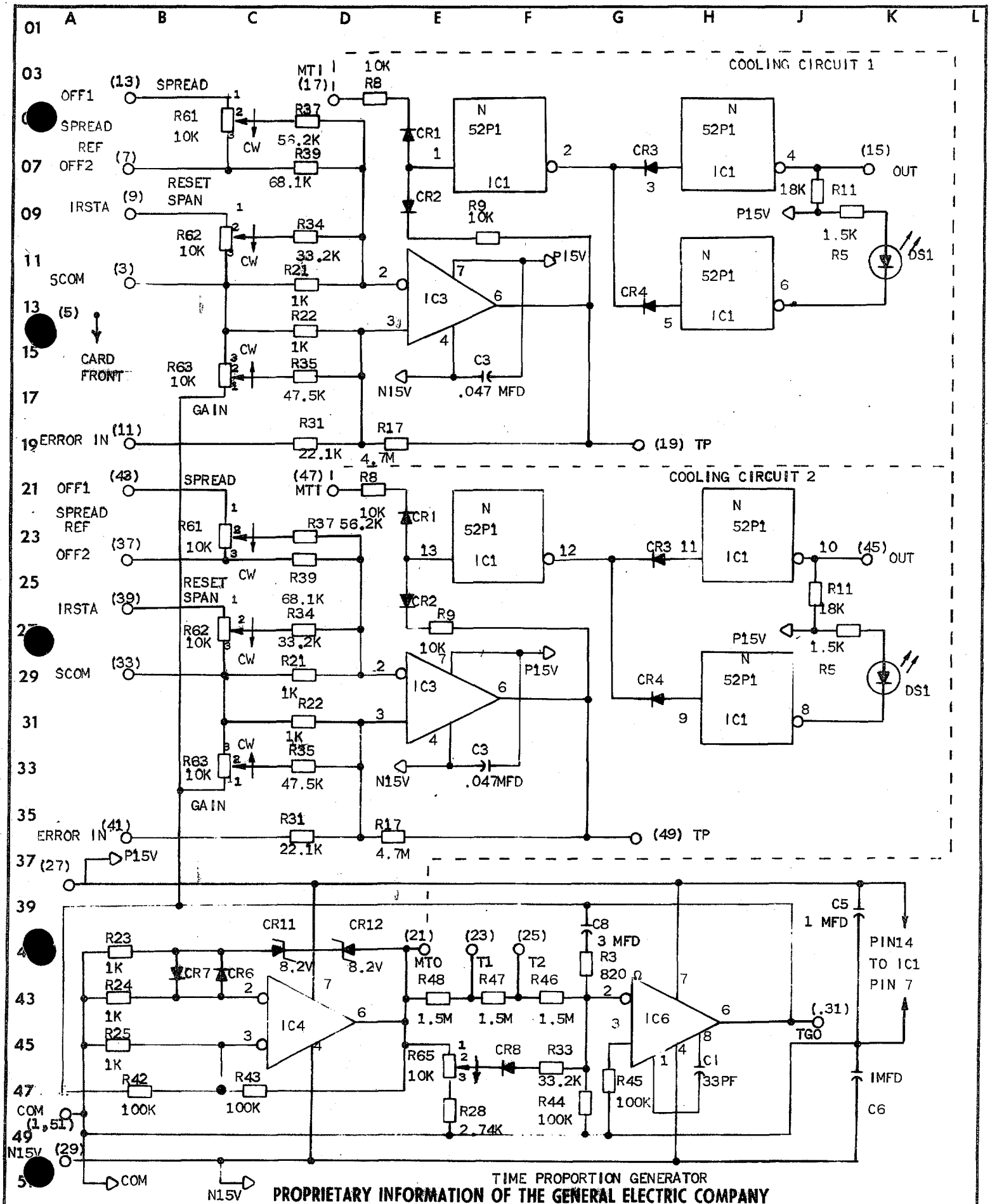
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P15V - 13.8 TO 16 VOLTS
10.5 MILLIAMPERES MAXIMUM, 4.6 MILLIAMPERE TYPICAL

N15V - 13.9 TO 16 VOLTS
11.5 MILLIAMPERE MAXIMUM, 5.4 MILLIAMPERES TYPICAL

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1 6-27-72 REV. 4	REV. 5	PRINTS TO 6V.2530 WAY	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM TEMPERATURE REGULATOR RATE/RESET AUXILIARY
REV. 2	ISSUED May 12, 1972	CHVIVN 3T.A6F /D21	FIRST MADE FOR PLASTIC MOLD		IC 3622GTRE1
REV. 3	MADE BY RON MUCKENFUSS	AIF(4)	I.C. NO.		CONT. ON SH. FL. SH. NO. 3.2



TIME PROPORTION GENERATOR

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO 6V, DL1, P6A	APPROVALS R.C. CLARK	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.
REV. 2	ISSUED July 25, 1972	1338, AIF (4), 3T AGF	FIRST MADE FOR PLASTIC M.		ELEMENTARY DIAGRAM TEMP. REGULATOR (COOLING AUX.)
REV. 3	MADE BY JOHN TURNER	CHVIVN, WAY	I.C. NO. IC3622GTRF1		I C 3 6 2 2 G T R F 1
					CONT. ON SH. 3.1 SH. NO. 3

FUNCTION AND THEORY OF OPERATION

THIS BOARD PROVIDES CIRCUITRY WHICH CAN BE USED IN CONJUNCTION WITH TEMPERATURE REGULATOR IC3622GTRD AND RATE RESET AUXILIARY IC3622GTRE TO PROVIDE A CONTROLLED COOLING FUNCTION AS A PART OF THE TEMPERATURE REGULATOR SYSTEM. EACH COOLING REGULATOR AUXILIARY BOARD CONTAINS TWO COOLING CONTROL CIRCUITS PLUS ONE TIME PROPORTIONING GENERATOR USED INTERNALLY ON THE TWO CIRCUITS. THE TIME PROPORTIONING GENERATOR PROVIDES A TIME VARIABLE PULSE (VERY SHORT COMPARED TO THE TIME PROPORTIONING PERIOD) WHICH MAY BE USED AS AN INPUT TO EITHER OR BOTH COOLING CONTROL CIRCUITS FOR MINUMUM COOLING OUTPUT CONTROL.

IN EACH COOLING CONTROL CIRCUIT, IC3 PROVIDES A BASIC COMPARATOR WITH ITS OUTPUT DRIVING IC1 AND WITH IC1 PROVIDING A LOGIC LEVEL (CONTACT CLOSURE) OUTPUT AS WELL AS A LIGHT INDICATOR DRIVE SHOWING WHEN THE COOL FUNCTION IS ON. A NEGATIVE GOING IC3 OUTPUT PROVIDES A FLOW THROUGH LOGIC CONTACT CLOSURE TO COMMON AND A LIGHT INDICATION ON DS1. THE S COM CONNECTION AND THE ERROR INPUT CONNECTION FOR EACH COOLING CIRCUIT COMES FROM THE TEMPERATURE REGULATOR WITH WHICH THAT CIRCUIT IS ASSOCIATED. THE ERROR OUTPUT OF THE BASIC TEMPERATURE REGULATOR GOES NEGATIVE FOR INCREASING TEMPERATURE ERROR AND THIS SIGNAL APPLIED TO THE DIRECT INPUT OF IC3 WILL MAKE IC3 OUTPUT NEGATIVE FOR OVERTEMPERATURE IN TURN OPERATING THE CONTACT OUTPUT AND THE LIGHT. THE TIME PROPORTIONING GENERATOR OUTPUT IS A SAW TOOTH WITH A SLOW NEGATIVE GOING RAMP WHICH IS APPLIED THROUGH THE GAIN CONTROL TO ADD TO THE ERROR SIGNAL ON THE DIRECT INPUT OF IC3. THUS AN INCREASING AMOUNT OF ERROR INPUT WILL CAUSE AN INCREASING PROPORTION OF GENERATOR CYCLE TIME DURING WHICH THE OUTPUT OF IC3 IS NEGATIVE CORRESPONDING TO AN INCREASING PROPORTION OF COOLING TIME. WHEN THE GAIN CONTROL IS TURNED MAXIMUM CLOCKWISE, FOR A MAXIMUM GAIN CONDITION, THEN THE TIME PROPORTIONING GENERATOR IS EFFECTIVELY CUTOFF AND IC3 SWITCHES BACK AND FORTH FULL ON AND OFF WITH A CHANGE IN ERROR SIGNAL OF APPROXIMATELY $1^{\circ}F$. AN INTEGRATOR CONTROL INPUT SIGNAL FROM THE TEMPERATURE REGULATOR AUXILIARY OUTPUT RSTA IS APPLIED THROUGH A RESET SPAN CONTROL POTENTIOMETER TO THE INVERSE INPUT OF IC3. THE INTEGRATOR OUTPUT GOES INCREASINGLY POSITIVE THE LONGER A NEGATIVE (OVERTEMPERATURE) OUTPUT EXISTS FROM THE REGULATOR SO THAT THE INTEGRATOR INPUT WILL TEND TO INCREASE COOLING TIME PROPORTION FOR INCREASED OVERTEMPERATURE ERROR TIME ON THE REGULATOR. RESET SPAN ON A COOLING REGULATOR AUXILIARY WILL BE SET OUT OF CONSIDERATION FOR THE PROPORTIONAL BAND (GAIN) PLUS THE SPREAD SETTING. NORMALLY A SPAN SETTING AS FAR COUNTER-CLOCKWISE AS THE GAIN SETTING WILL BE SATISFACTORY. NOTE THAT THE SPAN SETTING FOR THE REGULATOR PROPER SHOULD BE ALTERED TO BE ABOUT TWICE AS FAR COUNTER-CLOCKWISE AS THE COOLING REGULATOR AUXILIARY. WITH THIS SETTING RESET REPEATS PER MINUTE ARE TWICE THAT UNDER OPERATION WITHOUT THE AUXILIARY (I.E. MINIMUM RESETS/MINUTE IS NOW 0.2). SPREAD CONTROL REFERENCE POTENTIOMETER ON THIS BOARD IS CONNECTED TO THE MANUAL RESET SUPPLY FOR THE BASIC TEMPERATURE REGULATOR BOARD. THE INPUT ON THIS POTENTIOMETER CONNECTS TO THE INVERTING INPUT TO IC3 AND FUNCTIONS TO OFFSET THE OPERATION OF IC3 FROM THAT OF THE BASIC REGULATOR COMPARATOR (I.E. - SPREADS THE OPERATING CENTERPOINTS OF THE TWO COMPARATORS).

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO 6V, DL1, WAY	APPROVALS R.C. CLARK/kms	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM TEMP REGULATOR (COOLING AUX)
REV. 2	ISSUED Jul 25, 1972		P6A, 1338	FIRST MADE FOR PLASTIC M.		IC3622GTRF1
REV. 3	MADE BY JOHN M. TURNER		CHVIVN 3T, AIF(4)	I.C. NO. IC3622GTRF1		CONT. ON S 3.2 SH. NO. 3.1

THE TIME PROPORTION GENERATOR PROVIDES A SAW TOOTH VOLTAGE INPUT TO THE GAIN CONTROL POTS ON EACH COOLING CIRCUIT BY FUNCTIONS OF IC4 AND IC6 CIRCUITRY. IC6 SLOWLY INTEGRATES NEGATIVE DUE TO A POSITIVE INPUT FROM IC4 UNTIL ITS OUTPUT THROUGH R42 CAUSES IC4 TO SWITCH NEGATIVE. THIS WILL OCCUR APPROXIMATELY AT THE POINT WHERE IC6 OUTPUT BECOMES EQUAL IN MAGNITUDE NEGATIVE TO THE POSITIVE MAGNITUDE OF IC4 OUTPUT. WHEN THIS SWITCH POINT IS REACHED, IC4 GOES NEGATIVE CAUSING IC6 TO INTEGRATE POSITIVE MORE RAPIDLY THAN IT DID NEGATIVE DUE TO THE ADDED GAIN INPUT OF R33 THROUGH CR8. THIS ADDED GAIN IS ADJUSTABLE BY USE OF R65. BECAUSE OF THE FASTER INTEGRATING RATE, IC6 WILL BECOME POSITIVE IN A MUCH SHORTER TIME CAUSING IC4 TO SWITCH BACK TO POSITIVE SO THAT THE OUTPUT OF IC6 IS A SAW TOOTH WITH A SLOW NEGATIVE SLOPE AND A RAPID POSITIVE SLOPE. THE SHORT NEGATIVE OUTPUT OF IC4 CAN BE USED AS A MINUMUM COOLING TIME PULSE WHEN CONNECTED TO MINIMUM TIME INPUT (MTI) TO IC1 ON EACH COOLING CIRCUIT.

SPECIFICATIONS

INPUTS

ERROR INPUT IS DRIVEN BY ERROR OUTPUT E2 FROM BASIC TEMPERATURE REGULATOR AND IS A MAXIMUM OF 22K LOAD TO COMMON ON THAT OUTPUT.

INTEGRATOR INPUT IRSTA IS DRIVEN BY RSTA OUTPUT OF RATE RESET AUXILIARY BOARD USED WITH THE BASIC REGULATOR AND REPRESENTS A MAXIMUM LOAD OF 8.5K TO COMMON.

SPREAD REFERENCE INPUTS OFF1, OFF2,

DERIVE THEIR REFERENCE SUPPLY VOLTAGE FROM OUTPUTS VR1 AND VR2 CORRESPONDINGLY ON THE BASIC REGULATOR BOARD.

MINIMUM TIME INPUT (MTI) IS SPECIFICALLY DESIGNED TO BE DRIVEN BY IC4 OUTPUT (MTO) TO PROVIDE A MINIMUM TIME PERIOD PER PROPORTIONING CYCLE ON THE COOLING CIRCUIT OUTPUT WHEN SO DESIRED.

SCOM IS A COMMON REFERENCE CONNECTED TO THE BASIC REGULATOR ASSOCIATED WITH THE COOLING CIRCUIT.

OUTPUTS

COOLING CIRCUIT OUTPUT (OUT) IS DESIGNED TO BE USED AS A FLOW THROUGH CONTACT CONNECTED TO COMMON AND WILL DRIVE 17 LOADS.

MTO OUTPUT (MTO) IS THE OUTPUT FOR THE TIME PROPORTIONING GENERATOR IC4 USED TO DRIVE THE MTI INPUTS OF THE BASIC COOLING CIRCUITS. THIS OUTPUT GOES NEGATIVE FOR A SHORT CONTROLLABLE PERIOD EACH CYCLE ADJUSTABLE BY R65 FROM .3 TO 1.0 SECONDS. IT IS ALSO USED WITH

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO 6V, DLI	APPROVALS R.C. CLARK/CWS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM TEMP REGULATOR (COOLING AUX)
REV. 2	ISSUED JULY 26, 1972	CHVIVN P6A.1338, WAY	FIRST MADE FOR PLASTIC M.			
REV. 3	MADE BY JOHN M. TURNER	3T.AIF(4)	I.C. NO. IC3622GTRF1			IC3622GTRF1
						CONT. ON SH. 3.3 SH. NO. 3.2

STRAPPING TO VARY THE TIME OF THE LONG RAMP TO BE DESCRIBED LATER FOR STRAPPING CIRCUITS T1 AND T2.

TIME GENERATOR OUTPUT (TGO) IS A SAW TOOTH OF A MINIMUM 16.0 VOLT PEAK TO PEAK AMPLITUDE. THIS IS BASICALLY A TEST POINT BUT CAN BE USED WITH AN EXTERNAL LOAD AS LOW AS 5.0K TO COMMON.

INDIVIDUAL COOLING CIRCUIT ADJUSTMENTS ARE DESCRIBED BELOW AS THEY SPECIFICALLY RELATE TO PREVIOUSLY DESCRIBED STRAPPING TO THE BASIC REGULATOR IC3622GTRD AND THE RATE RESET AUXILIARY IC3622 GTRE.

GAIN (ON BOARD-R63) CAN BE ADJUSTED SUCH THAT COOLING OPERATION PROPORTIONS OVER A MINIMUM OF 100°F WITH THE GAIN POTENTIOMETER FULLY COUNTER-CLOCKWISE OR THE COOLING BAND IS A SWITCH WITH A HYSTERESIS OF ABOUT 1°F WITH THE POTENTIOMETER SET FOR MAXIMUM GAIN (FULLY CLOCKWISE).

RESET SPAN (ON BOARD-R62) CUTS OFF INTEGRATOR ACTION COMPLETELY WHEN SET FULLY CLOCKWISE BUT MOVES THE COOLING ACTION BY A MINIMUM OF ±100°F IN RESPONSE TO ERROR WHEN SET FULLY COUNTER-CLOCKWISE.

SPREAD (FRONT PANEL) - WHEN SET FULLY COUNTER-CLOCKWISE CAUSES THE CENTER OF THE COOLING BAND PROPORTIONING ZONE TO BE A MINIMUM OF 5° BELOW THE CENTER OF THE HEATING BAND PROPORTIONING ZONE. WHEN SET FULLY CLOCKWISE, CAUSES THE CENTER OF THE COOLING BAND PROPORTIONING ZONE TO BE A MINIMUM OF 95° ABOVE THE CENTER OF THE HEATING BAND PROPORTIONING ZONE.

STRAPPING ADJUSTMENTS

TIME PROPORTIONING GENERATOR STRAPPING IS USED TO VARY THE SLOW RAMP PORTION OF THE TIME PROPORTIONING GENERATOR SAWTOOTH. WITH NO STRAPPING, THE SLOW RAMP PORTION OF THE TIME PROPORTIONING GENERATOR TAKES 24 TO 30 SECONDS OF EACH TIME PROPORTIONING PERIOD. WITH MTO STRAPPED TO T1 THIS SAME TIME OF 16 TO 20 SECONDS. WITH MTO STRAPPED TO T2 THE TIME IS 8 TO 10 SECONDS.

BUS

P15V + 13.8 TO 16.0 VOLTS WITH RESPECT TO COMMON
39 MILLIAMPERES MAXIMUM, 37 MILLIAMPERES TYPICAL

N15V -13.8 TO 16.0 VOLTS WITH RESPECT TO COMMON
15 MILLIAMPERES MAXIMUM, 9 MILLIAMPERES TYPICAL

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REV. 1	REV. 4	REV. 5	PRINTS TO WAY 6V.DLI, CHVIVN, P6A.1338	APPROVALS R.C. Clarklus	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	ELEMENTARY DIAGRAM TEMP REGULATOR (COOLING AUX)
REV. 2	ISSUED JULY 26, 1972			FIRST MADE FOR PLASTIC M.		IC3622GTRF1
REV. 3	MADE BY JOHN M. TURNER 3T.AIF(4)			I.C. NO. IC3622GTRF1		CONT. ON SH. FL SH. NO 3.3

GENERAL ELECTRIC

PWC245A1920

CONT ON SHEET 2 SH NO. 1

REV. NO.

PWC245A1920

CONT ON SHEET 2 SH NO. 1

TITLE

CARD

FIRST MADE FOR

PART	DESCRIPTION	ORDERING INFORMATION	FORM AND QUANTITY 1	2	3
1	CARD	IC3600 3CBN1	1	1	1
2	CARD FRONT	245A3260P1	1	1	1
3	NAME PLATE	245A1921P1		1	1
4	NAME PLATE	245A1923P1	1		
5	DIAL	218A4186P1	1	2	2
6	POT (10K) RH1, RH2	68A7110P100F1	1	2	
7	SLEEVE	68A7806P3	1	2	2
8	SCREW	N51P13008B6	2	2	2
9	NUT	N210P13B6	2	2	2
10	WASHER	N402P7B6	2	2	2
11	L' WASHER	N403P13C	2	2	2
12	SPACER	68A994281P1	2	2	2
13	NP	167A4671P2	1	1	1
14	NP	STAMP PART 13 WITH CAT.NO.	X	X	X
15	JUMPER	68A9922 99P2	1	1	1
16	NOTE	OMIT RH2	X		
17	CR1 MOUNT IN POS C,L	68A7806P1	1	2	2
18	NOTE	OMIT POS L	X		
19	RES MOUNT IN POS D,M	68A7000P182	1	2	2
20	NOTE	OMIT POS M	X		
21	POT (5K) RH1	68A7110P500E1			1
22	POT (500Ω) RH2	68A7110P500D1			1
23					

REVISION

1 ECA 8-21-72 BU23500C/4

2 ECA 11-3-72 BU23500CR

3 ECA 11-22-72 BU23500CV

4 ECA 12-12-72 BU23500CX

REVISION

G1 ONE POT

G2 2 POTS

PRINTS TO

IME1 (3)

IEH1

4AA1

RIC1 VM

068

MADE BY Earl Abbott

ISSUED 7 27 72

APPROVALS

ECA

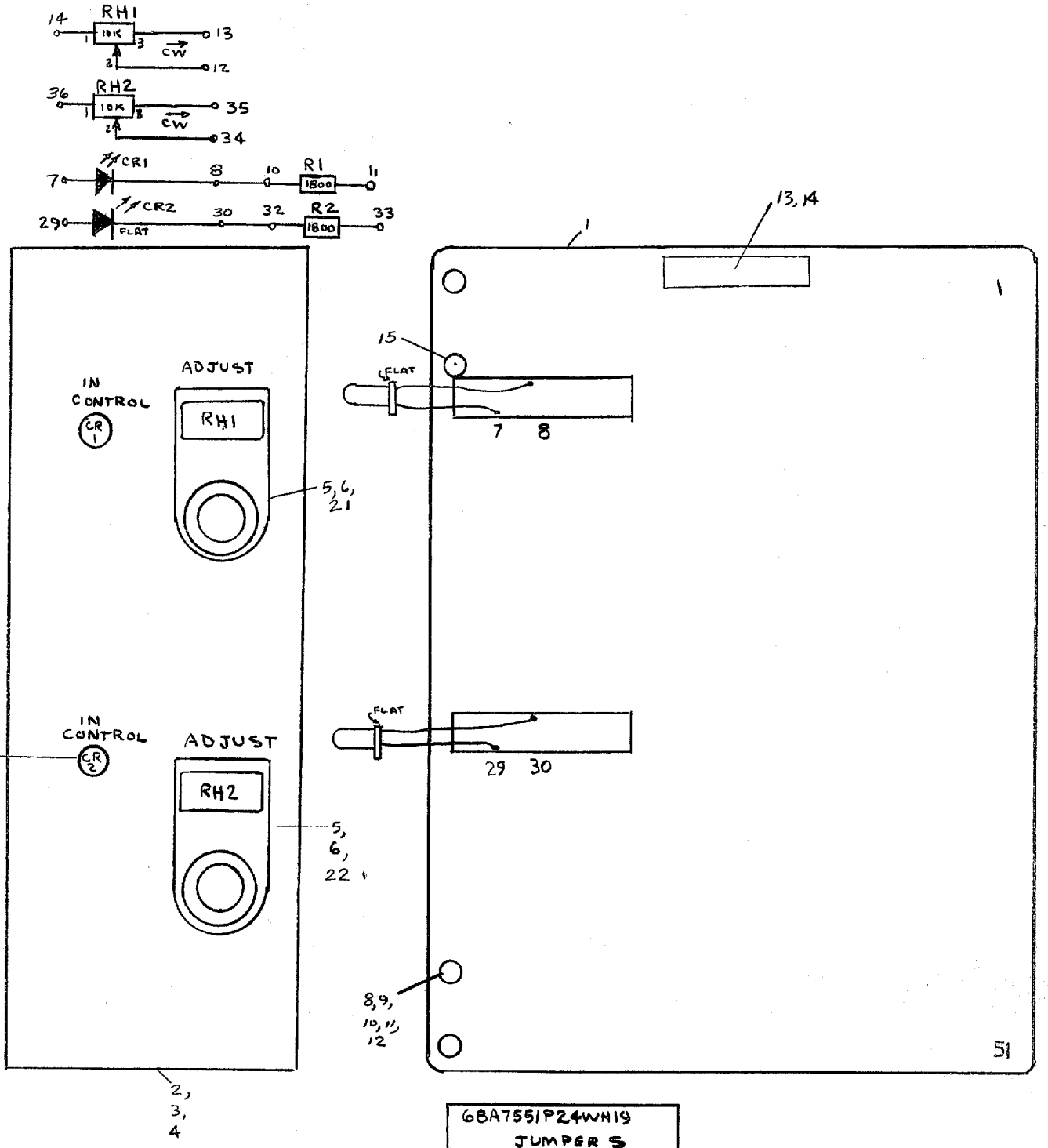
INDUSTRY CONTROL DIV OR DEPT

SALEM, VIRGINIA LOCATION

PWC245A1920

CONT ON SHEET 2 SH NO. 1

FORM 1C 2143E (9-64)



GBA755IP24WH19 JUMPER S	
FROM	TO
RH1(1)	PIN 14
RH1(2)	PIN 12
RH1(3)	PIN 13
RH2(1)	PIN 36
RH2(2)	PIN 34
RH2(3)	PIN 35
PIN 8	PIN 10
PIN 30	PIN 32

PART 15 →	CARD FT	PIN 5

068

REV.	REV. (1) E79-8-21-72 BU 23500CH	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE SET UP CARD		
REV.	REV. 2 E79 11-22-72 ADD. 1 PT NO. FOR G3				
MADE BY Earl Abbott	APPROVALS E79	PRINTS TO	IME1(3) RICHMOND		PWC 245A1920
ISSUED 7-27-72			IEW 4AA1	CONT ON SHEET 51	SH. NO. 2

FORM IC 2143B (9-64)

GENERAL ELECTRIC

245A1928

CONT ON SHEET FL SH NO. 1

REV NO.

245A1928

CONT ON SHEET FL SH NO. 1

TITLE

TEMP INDICATOR ASSEMBLY

FIRST MADE FOR IC7966

G2	G1	PT	NAME	DWG NO
X	X	1	ASM	FACTORY TO ASSEMBLE
1	1	2	METER	218A4111P1
1	1	3	CARD	IC3621GMSA1
2	2	4	SPACER	68A999859P3
1	1	5	CONNECTOR	AMP# 582382-9
3	3	6	CONN PINS	AMP# 66150-2
	1	7	CABLE	68A7527P24GN19B LENGTH= 10 FEET
	4	8	TERMINALS	577A499P200
	1	9	WIRE	68A7551P24WH19 LENGTH=10 FEET
	1	10	SPAGHETTI	FOR #24 WIRE, STAMP F2
	1	11	SPAGHETTI	FOR #24 WIRE, STAMP F1
	1	12	SPAGHETTI	FOR #24 WIRE, STAMP N15V
X	X	13	DRAWING	IC3621GMSA1 SH 3.0
X	X	14	DRAWING	IC3621GMSA1 SH 3.1
	1	15	SPAGHETTI	FOR #24 WIRE, STAMP SHLD
2	2	16	NUT	N211P20
X		17	NOTE	FACTORY TO PUT CONN PINS, PART 6, IN
1		18	ENVELOPE	ENVELOPE FF581A
X		19	NOTE	AND ATTACH PER ASM DWG
X	X	20	DRAWING	MAIL THIS SHEET TO CUSTOMER

REVISIONS

1	ECN BU23500 C10 10-10-72	
2	ECN 11-21-72 NOTE CHANGES, ADDED WIRING	

CABLE: TO METER

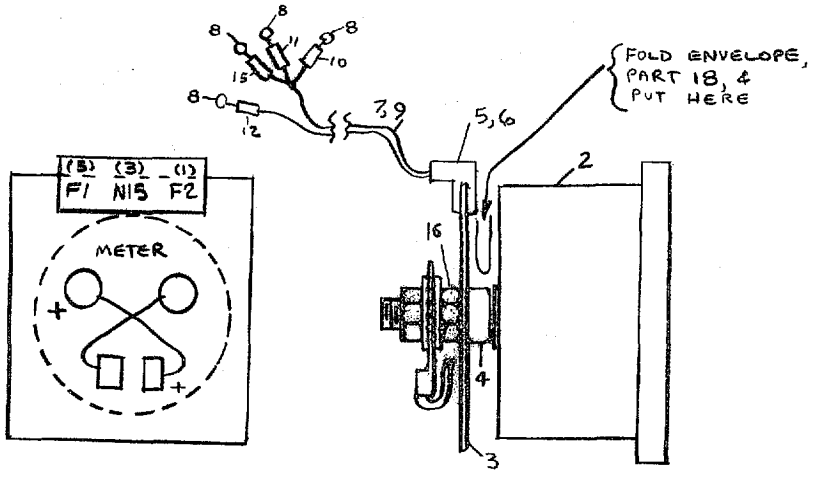
F2 WHT TERM (1)

F1 BLK TERM (5)

SHLD, DO NOT CONNECT TO METER

WIRE: TO METER

N15V WHT TERM (3)



1

NOTE: TERM F1 OF METER SHOULD BE WIRED TO PIN 49 OF TEMP REG IC3622GTRD1

TERM F2 SHOULD GO TO PIN 46

THESE CONNECTIONS MAY BE MADE AT RTD TERM BOARD.

AIF(4)

AGT

3T

PRINTS TO

MADE BY Earl Abbott

ISSUED July 13, 1972

APPROVALS ECA

INDUSTRY CONTROL DIV OR DEPT.

SALEM, VA LOCATION

245A1928

CONT ON SHEET FL SH NO. 1

CODE IDENT NO.

FF-803 WF (1-72) PRINTED IN U.S.A.

GENERAL ELECTRIC

PWC245A1943

CONT ON SHEET 2 SH NO. 1

REV. NO.

PWC245A1943

CONT ON SHEET 2 SH NO. 1

TITLE ALARM MODULE CARD FRONT

FIRST MADE FOR STD LINE

PART	DESCRIPTION	ORDERING INFORMATION	FORM AND QUANTITY
			1
1	ASM	THIS DWG. SH. 2	X
2	CARD	IC3600SCBN1	1
3	RESISTOR MOUNT IN POS B,D,K,T,V	68A7000P182	5
4	RESISTOR MOUNT IN POS H,M	68A7000P183	2
5	NAMEPLATE	245A3273P1	1
6	CARD FRONT	245A4555P1	1
7	LED	68A7806P1	5
8	SLEEVE (FOR PART 7)	68A7806P3	5
9	JUMPER	68A992299P2	1
10	SCREW	N51P13008B6	2
11	NUT	N210P13B6	2
12	WASHER	N402P7B6	2
13	L'WASH	N405P37C6	2
14	SPACER	68A994281P1	2
15	NP	167A4671P2	1
16	NP NOTE.	STAMP PT 15 WITH CAT. NO.	X
17	PUSHBUTTON	68A7332P2	2
18	TEST SPEC	68A989079	X

REVISION	REVISION	PRINTS TO
2 S74 9-13-72 BU2350CL JEEa 9-6-72 BU2350CI		PID AIF(3) PIC AGF CHVI VN 3T RICI VN

068

MADE BY Earl Abbott 8-15-72

ISSUED Aug 18, 1972

APPROVALS EJGhrat

INDUSTRY CONTROL — DIV OR DEPT

SALEM, VIRGINIA — LOCATION

PWC245A1943

CONT ON SHEET 2 SH NO. 1

FORM 1C 2143E (9-64)

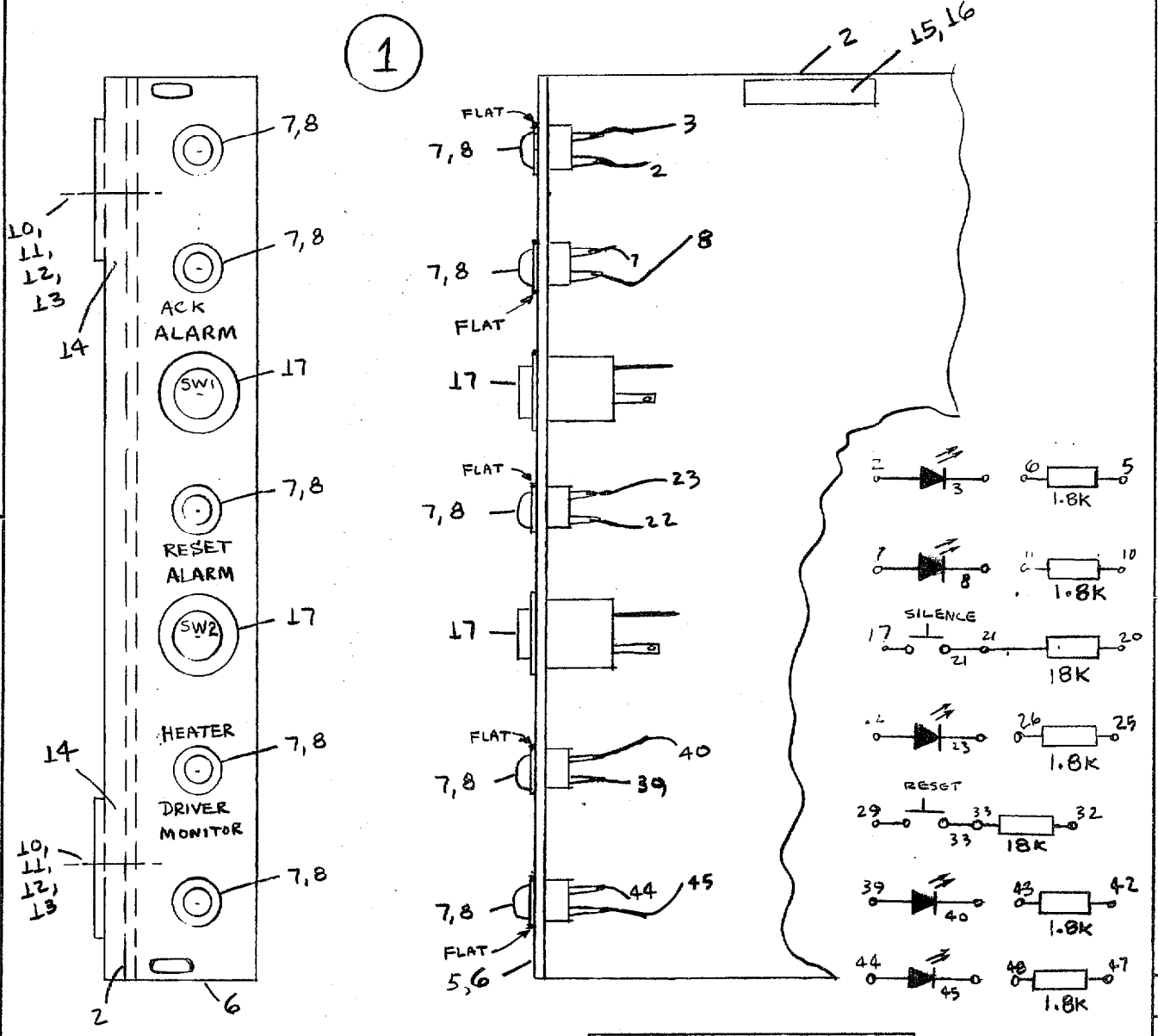
GENERAL ELECTRIC

PWC245A1943
CONT ON SHEET FL SH NO. 2

REV NO.	TITLE
PWC245A1943 CONT ON SHEET FL SH NO. 2	CARD FRONT FIRST MADE FOR STD LINE

IC7966

REVISIONS



68A7551P24WH19	
FROM	TO
SW1(C)	CARD (17)
SW2(C)	CARD (29)
SW1(A)	CARD (21)
SW2(A)	CARD (33)

1 IEIA 9-13-72
2 EDA 11-17-72 Part 5 was changed to BU. "ACK ALARM"

IMEI (2)
IQAI
IEHI
4AAI

RICI VM CHVI VM
068 PRINTS TO

MADE BY	APPROVALS	DIV OR DEPT.	
Earl Abbott		IND. CONT.	PWC245A1943
ISSUED Aug 18, 1972		LOCATION SALEM, VA.	CONT ON SHEET FL SH NO. 2

FF-803 WF (1-72)
PRINTED IN U.S.A.

CODE IDENT NO.

GENERAL ELECTRIC

PWC245A1944
CONT ON SHEET 1-1 SH NO. 1

REV NO: PWC245A1944
CONT ON SHEET 1-1 SH NO. 1

TITLE: ROTARY SWITCH CARD FRONT

FIRST MADE FOR

PART	DESCRIPTION	ORDERING INFORMATION	FORM AND QUANTITY 1	2	3	4	5
1	TEST SPEC	68A989079	X	X	X	X	X
2	CARD	IC3600SCBN1	1	1	1	1	1
3	SWITCH	68A7607P22	2				
4	NAMEPLATE	245A3275P1	1				
5	CARD FRONT	245A4552P1	1	1	1	1	1
6	SLEEVE	68A7806P3	2	2	1	1	1
7	JUMPER	68A992299P2	1	1	1	1	1
8	SCREW	N5IP13008B6	2	2	2	2	2
9	NUT	N210P13B6	2	2	2	2	2
10	WASHER	N402P7B6	2	2	2	2	2
11	L'WASH	N405P37C6	2	2	2	2	2
12	SPACER	68A994281P1	2	2	2	2	2
13	NP	167A4671P2	1	1	1	1	1
14	NP NOTE	STAMP PT 13 WITH CAT NO	X	X	X	X	X
15	KNOB	68A7605P1	2	2	1	1	1
16	TEST SPEC	68A989079	X	X	X		
17	NAMEPLATE			1			
18	WIRE TABLE & ASM	THIS DWG SH 2.0	X				
19	WIRE TABLE & ASM	THIS DWG SH 2.1		X			
20	WIRE TABLE & ASM	THIS DWG SH. 2.2			X		
21	NAMEPLATE	245A3276P1			1		
22	LED, CR1,2, MOUNT IN POS C,L	68A7806P1	2	2	1	1	2
23	SWITCH	68A7607P26		2	1		
24	LED NOTE	OMIT CR2 IN POS L			X	X	
25	RES POS T,V	68A7000P182	2	2	1	1	2
26	RES NOTE	OMIT RES IN POS V			X	X	
27	LED & RES ASM	THIS DWG SH 2.3	X	X	X	X	X

REVISION		REVISION							PRINTS TO	
1	ECA 9-8-72 BU23500CK		G1 2 SWITCHES 16 POSITION	G2 2 SWITCHES 8 POSITION	G3 1 SWITCH 8 POSITION	G4 1 SW 5 POS	G5 2 SW 5 POS	C68	4AA1(3)	1ME(3)
2	ECA 2-6-73 BU23500DF									1QA9
										1EH1

MADE BY: Earl Abbott 8-15-72
ISSUED: 8/15/72
APPROVALS: E Abbott

INDUSTRY CONTROL — DIV OR DEPT
SALEM, VIRGINIA — LOCATION

PWC245A1944
CONT ON SHEET 1-1 SH NO. 1

FORM IC 2143E (9-64)

GENERAL ELECTRIC

PWC245A1944
CONT ON SHEET 2.0 SH NO. 1.1

REV. NO. PWC245A1944
CONT ON SHEET 2.0 SH NO. 1.1

TITLE ROTARY SWITCH CARD FRONT

FIRST MADE FOR

PART	DESCRIPTION	ORDERING INFORMATION	FORM AND QUANTITY 4	5
28	SWITCH	68A7607P11	1	2
29	WIRE TABLE & ASM	THIS DWG SH. 2.4	X	
30	WIRE TABLE & ASM	THIS DWG SH. 2.5		X
31	NAME PLATE	245A7637P1		1
32	NAMEPLATE	245A7638P1	1	

REVISION	REVISION	PRINTS TO
		4AA(3) 1MEA(3)
		1QA9
		1EH1
	068	

MADE BY O.C. GILBERT
ISSUED FEB. 7, 1973
APPROVALS E/A
INDUSTRY CONTROL — DIV OR DEPT
SALEM, VIRGINIA — LOCATION
PWC245A1944
CONT ON SHEET 2.0 SH NO. 1.1

FORM 1C 2143E (9-64)

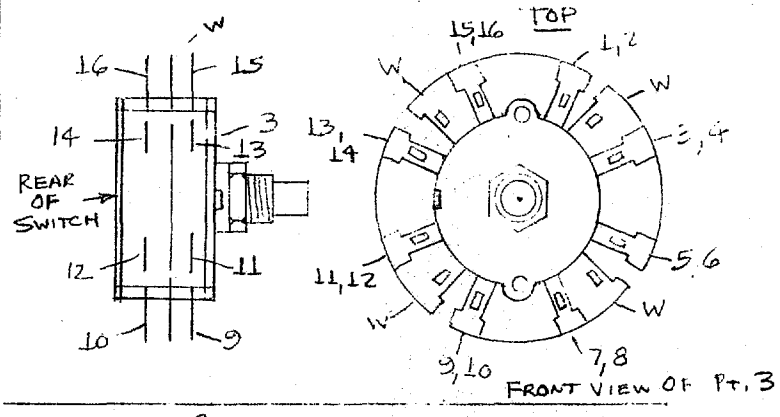
GENERAL ELECTRIC

PWC245A1944

CONT ON SHEET 2.1 SH NO. 2.0

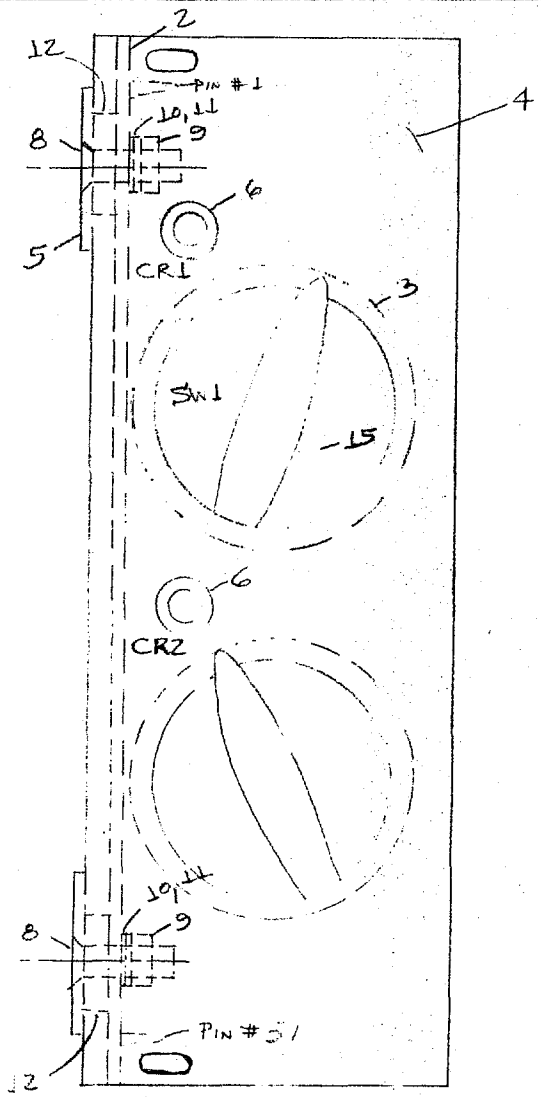
REV NO.	TITLE
PWC245A1944	ROTARY SWITCH & CARD FRONT
CONT ON SHEET 2.1 SH NO. 2.0	FIRST MADE FOR LC7966

REVISIONS



FRONT VIEW OF PT. 3

SEQUENTIAL ASSEMBLY IS RECOMMENDED.
CUT WIRES TO 4 1/2 INCH LENGTH,
SOLDER TO SWITCH & TAG LOOSE END.
THEN TERMI POINT LOOSE WIRE END
TO CARD PINS.
CORD TIE (TIE-WRAP) WIRES TOGETHER.



68A7551P24WH19		68A7551P24WH19	
FROM	TO	FROM	TO
SW2 (1)	CARD (2)	SW1 (W)	CARD (2)
(1)	22	(1)	(3)
2	23	2	4
3	24	3	5
4	25	4	6
5	28	5	9
6	31	6	10
7	32	7	11
8	33	8	12
9	34	9	13
10	35	10	14
11	36	11	15
12	37	12	16
13	38	13	17
14	39	14	18
15	40	15	19
16	41	16	20
CD FRONT	CARD (1)		

1EH1
1QA9
4AA16
1MEK3

PRINTS TO

MADE BY	APPROVALS	DIV OR DEPT.	
ISSUED		INDUSTRY CONTROLS LOCATION SALEM, VA.	PWC245A1944 CONT ON SHEET 2.1 SH NO. 2.0

GENERAL ELECTRIC

PWC245A1944

CONT ON SHEET 2.2 SH NO. 2.1

REV NO.

PWC245A1944

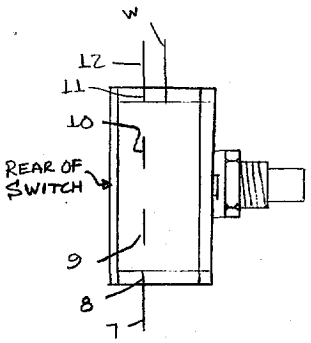
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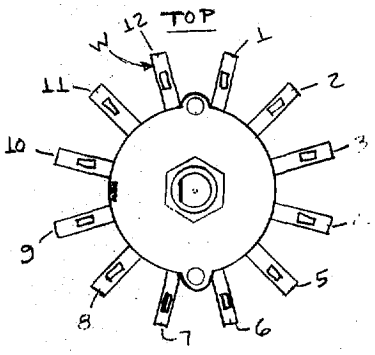
TITLE ROTARY SWITCH & CARD FRONT

FIRST MADE FOR IC7966

REVISIONS

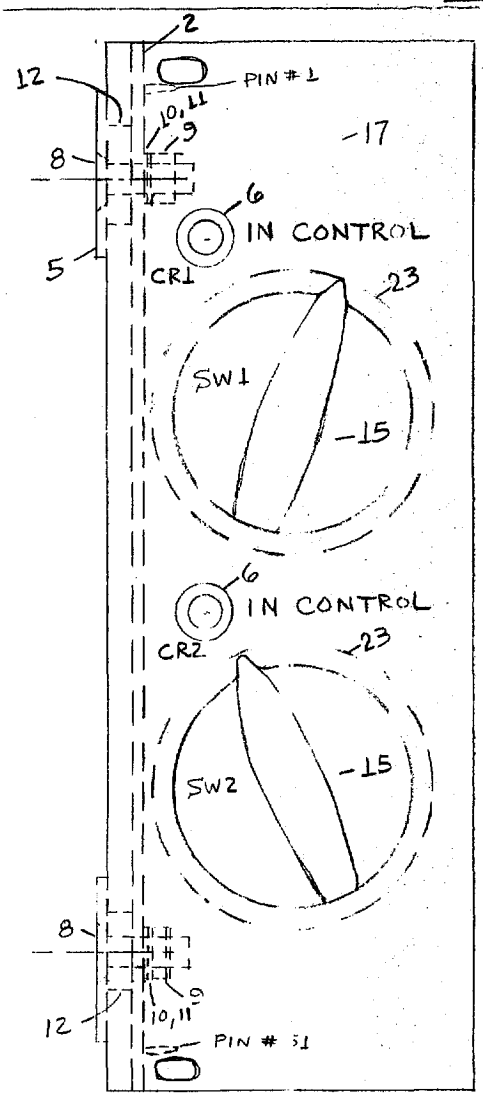
1	ECR 2-6-73 BU235-0D





FRONT VIEW OF PT. 23

SEQUENTIAL ASSEMBLY IS RECOMMENDED.
CUT WIRES TO 4 1/2 INCH LENGTH,
SOLDER TO SWITCH & TAG LOOSE END.
THEN TERMI POINT LOOSE WIRE END
TO CARD PINS.
CORD TIE (TIE-WRAP) WIRES TOGETHER.



NOTE: FACTORY TO SET STOP ON SW1 & SW2 FOR FIRST 8 POSITIONS.

68A7551P24WH19

FROM SW1 (W)	TO CARD (2)
(1)	(3)
2	4
3	5
4	6
5	9
6	10
7	11
8	12

SW2 (W)	CARD (21)
1	13
2	14
3	15
4	16
5	17
6	18
7	19
8	20
CD FRONT	CARD (1)

IEH 1
10A9
4AA1
1ME1 3

068

PRINTS TO

MADE BY DKHowe 8/17/72
ISSUED

APPROVALS EJOHall

INDUSTRY CONT. DIV OR DEPT.
SALEM, VA. LOCATION

PWC245A1944
CONT ON SHEET 2.2 SH NO. 2.1

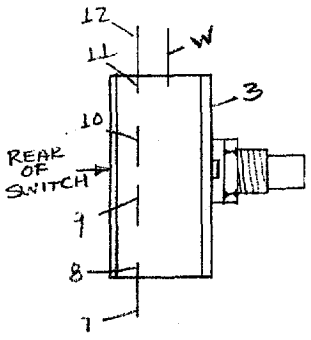
FE 803-WE (8-71)

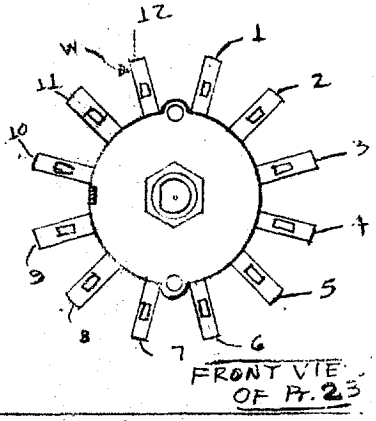
CODE IDENT NO.

GENERAL ELECTRIC

PWC245A1944
CONT ON SHEET 2.3 SH NO. 2.2

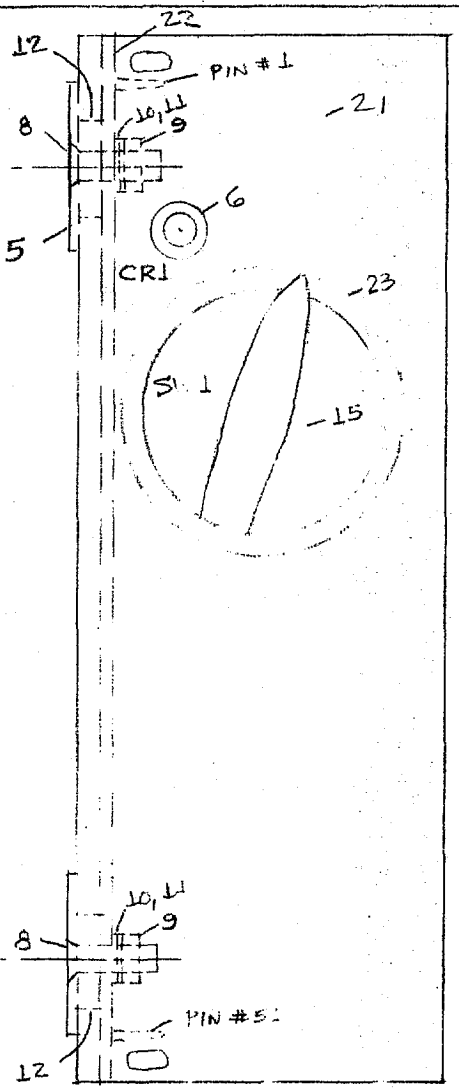
REV NO.	TITLE
PWC245A1944	ROTARY SWITCH & CARD FRONT
CONT ON SHEET 2.3 SH NO. 2.2	FIRST MADE FOR IC7966





FRONT VIEW OF PT. 23

SEQUENTIAL ASSEMBLY IS RECOMMENDED.
CUT WIRES TO 4 1/2 INCH LENGTH, SOLDER TO SWITCH & TAG LOOSE END. THEN TERM| POINT LOOSE WIRE END TO CARD PINS.
CORD TIE (TIE-WRAP) WIRES TOGETHER.



68A7551P24WH19	
FROM	TO
CD FRONT	CARD (1)
SW1 (W)	(2)
(1)	(3)
(2)	4
(3)	5
(4)	6
(5)	9
(6)	10
(7)	11
(8)	12

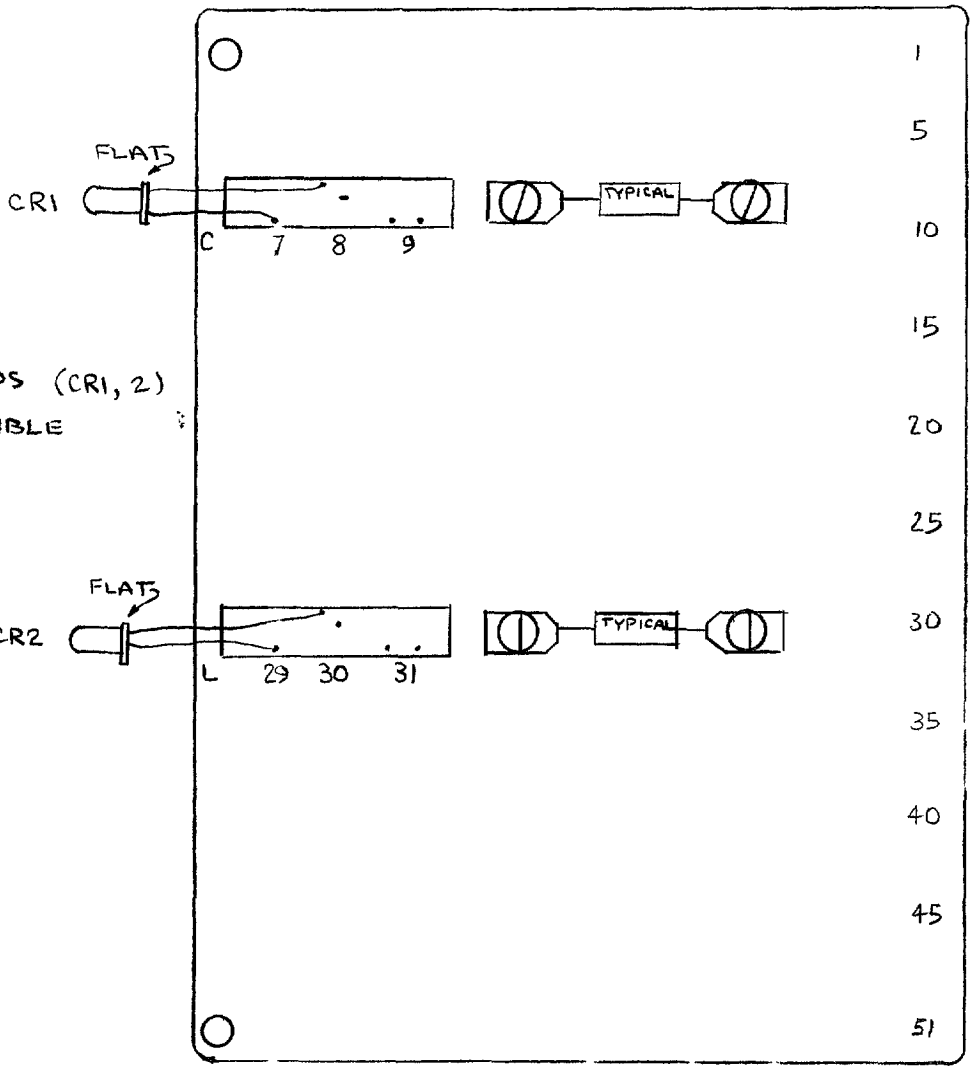
NOTES: FACTORY TO SET STOP ON SW1 FOR FIRST 8 POSITIONS.

REVISIONS

IEM1
IQA9
4AA13

068 PRINTS TO

MADE BY	APPROVALS	DIV OR DEPT.	
8/17/77		IND. CONT.	PWC245A1944 C
ISSUED		LOCATION SALEM, VA.	CONT ON SHEET 2.3 SH NO. 2.2



FACTORY NOTE:

LEAVE DEVICE LEADS (CR1, 2) AS LONG AS POSSIBLE PER SKETCH.

068

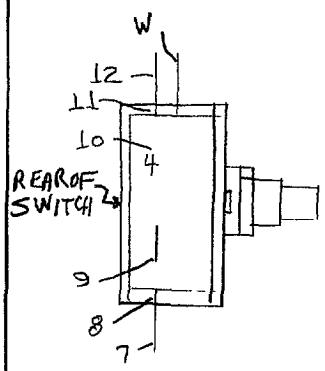
REV.	REV.	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA. U.S.A.	TITLE ROTARY SWITCH & CARD FRONT			
REV.	REV.					
MADE BY Earl Abbott	APPROVALS	PRINTS TO	1EH1	4AA13		PWC245A1947
ISSUED						

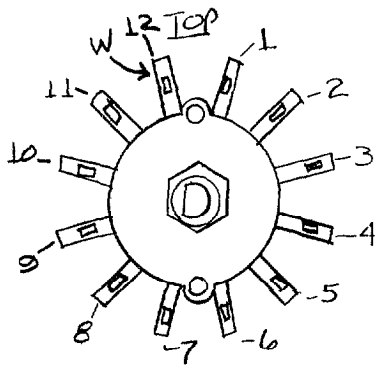
IC 2143B (9-64)

GENERAL ELECTRIC

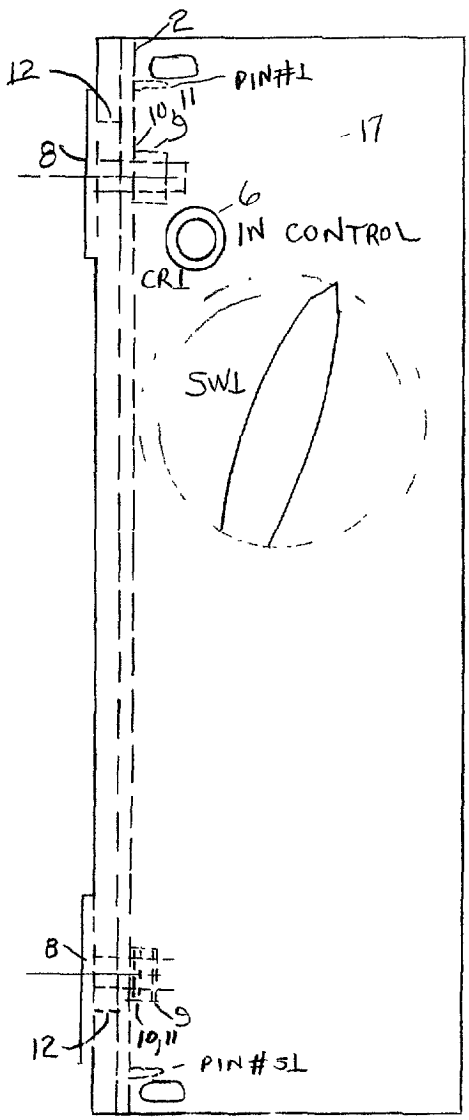
PWC245A1944
CONT ON SHEET 2.5 SH NO. 2.4

REV NO.	TITLE
PWC245A1944	
CONT ON SHEET 2.5 SH NO. 2.4	FIRST MADE FOR





SEQUENTIAL ASSEMBLY IS RECOMMENDED. CUT WIRES TO 4 1/2 INCH LENGTH, SOLDER TO SWITCH & TAG LOOSE END, THEN TERMINATE LOOSE END TO CARD PINS. CORD TIE (TIE-WRAP) WIRES TOGETHER.



NOTE: FACTORY TO SET STOP ON SW1 FOR FIRST 5 POSITIONS.

68A7551P24WH19	
FROM	TO
SW1 (W)	CARD (2)
(1)	(3)
(2)	(4)
(3)	(5)
(4)	(6)
(5)	(9)
CD FRONT	CARD (1)

68A7551P24WH19	
FROM	TO

REVISIONS

1EH1
1ME1(B)
4AA13
1QA9

B68 PRINTS TO

MADE BY	APPROVALS		
O.C GILBERT		ICPD (DIV OR DEPT.)	PWC245A1944
ISSUED FEB. 3, 1973		SALEM, VA. (LOCATION)	CONT ON SHEET 2.5 SH NO. 2.4

FF-803-WF (2-71)

CODE IDENT NO.

GENERAL ELECTRIC

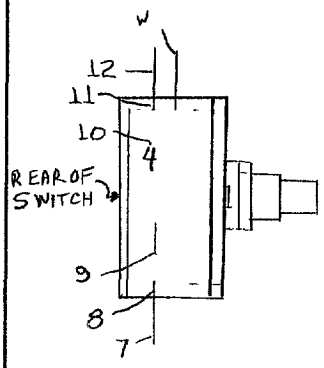
PWC245A1944
CONT ON SHEET FL SH NO. 2.5

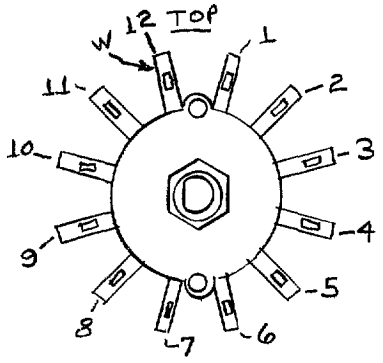
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CONT ON SHEET FL SH NO. 2.5

TITLE ROTARY SWITCH & CARD FRONT

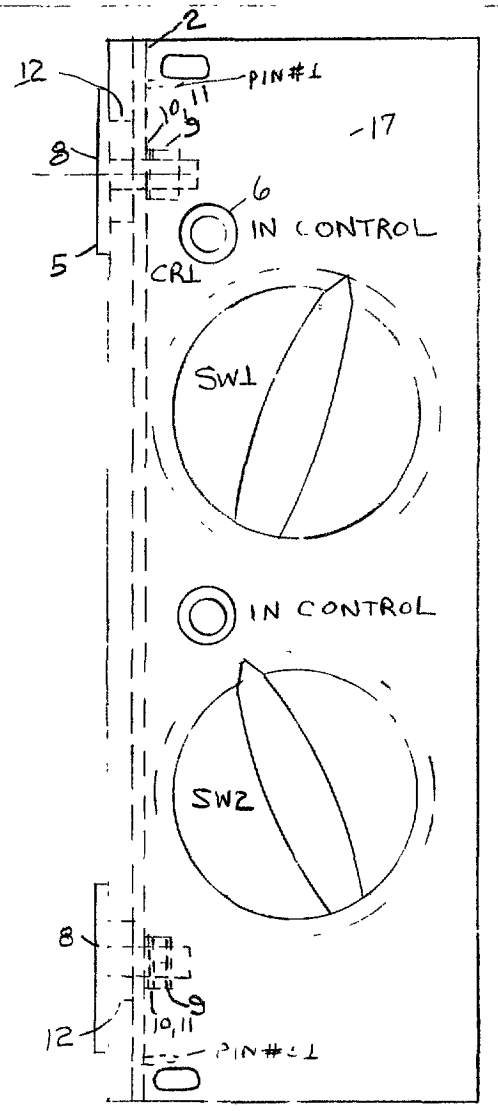
FIRST MADE FOR

REVISIONS





SEQUENTIAL ASSEMBLY IS RECOMMENDED. CUT WIRES TO 4 1/2 INCH LENGTH, SOLDER TO SWITCH & TAG LOOSE END. THEN TERM I POINT LOOSE WIRE END TO CARD PINS. CORD TIE (TIE-WRAP) WIRES TOGETHER.



NOTE: FACTORY TO SET STOP ON SW1 & SW2 FOR FIRST 5 POSITIONS.

68A7551P24WH19	
FROM	TO
SW1 (W)	CARD 2
(1)	3
2	4
3	5
4	6
5	9
CD FRONT	CARD (1)

68A7551P24WH19	
FROM	TO
SW2 (W)	CARD 21
1	22
2	23
3	24
4	25
5	28

1EHL
1ME1(3)
4AA13
1QA9

068

PRINTS TO

MADE BY O C GILBERT
ISSUED FEB 3, 1973
APPROVALS
ICPD DIV OR DEPT.
SALEM, VA. LOCATION
PWC245A1944
CONT ON SHEET FL SH NO. 2.5

FF-803-WF (9-69)
CODE IDENT NO.

DRAWING NO IC3625D CONT ON SHEET 1.1 SH NO 1.0

GENERAL ELECTRIC

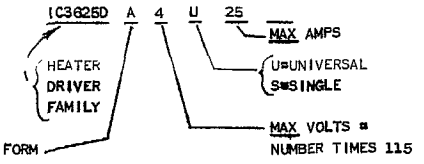
IC3625D CONT ON SHEET 1.1 SH NO 1.0

REV NO. IC3625D CONT ON SHEET 1.1 SH NO 1.0

TITLE HEATER DRIVER

FIRST MADE FOR REQ. PLASTIC MOLDING CONTROL S.O

ORDERING EXAMPLE



AVAILABLE FORMS

A4U25 (440V1Ø, OR 220V 3Ø VEE)
A4U05 (440V1Ø, OR 220V 3Ø VEE)
A2S25 (220V1Ø OR 110V1Ø)
A2S05 (220V1Ø OR 110V1Ø)
C4U25 (440V1Ø OR 220V 3ØNFE)
C2S25 (220V1Ø OR 110V 1 Ø)

B4S50 (440V 1Ø, OR 220V 1Ø) HEAT SINK ASM. ONLY - NO BASE OR COVER

B4S50A (440V 1Ø, OR 220V 1Ø) COMPLETE UNIT

THE ABOVE EXAMPLE IS FOR A UNIVERSAL HEATER DRIVER WITH 25 AMPS PER LINE MAX CURRENT. THE DRIVER MAY BE CONNECTED EITHER 440 VOLTS SINGLE PHASE OR 220 VOLT THREE PHASE VEE. THE FUSE IN THE LINE TO THE VEE APEX IS NOT INCLUDED IN THE DRIVER PACKAGE. FOR MAX VOLTS ÷ 2 (220 VOLTS) OPERATION THE USER WILL JUMPER TOGETHER J1 AND J2 (#24 WIRE IS OK)

COMPUTER ORDERING LOGIC

IC3625D	A	4	U	25	SPACE
FAMILY 01	02	03	04	05	06

ARG NOS.

IC3625DB4S50A
01 02 04 06
03 05

FORM DEFINITION

A HEAT IS REMOVED BY CONDUCTION THRU MODULE BASE.
THIS MODULE CAN BE MOUNTED IN NEMA 12 ENCLOSURE

B HEAT IS REMOVED BY AIR PASSING OVER FINNED HEAT SINK
THIS MODULE MAY OR MAY NOT HAVE A VENTILATED COVER

C. HEAT IS REMOVED BY CONDUCTION THRU MODULE BASE.
THIS MODULE CAN BE MOUNTED IN NEMA 12 ENCLOSURE.
ALSO THIS MODULE HAS THE NEW RCA POTTED TRIAC.

IEHI
RICHMI

IMEI
IEHI
PRINTS TO

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REVISIONS			
4	2-14-72 2/17/72 ADDED C FORM	3	1-21-71
4	2-23-72 CORRECTED FORM	2	JULY 16, 71 A WRIGHT

MADE BY E. T. ABBOTT
ISSUED 6-17-70
APPROVALS
INDUSTRY CONTROL DIV OR DEPT
SALEM, VIRGINIA LOCATION
IC3625D CONT ON SHEET 1.1 SH NO 1.0

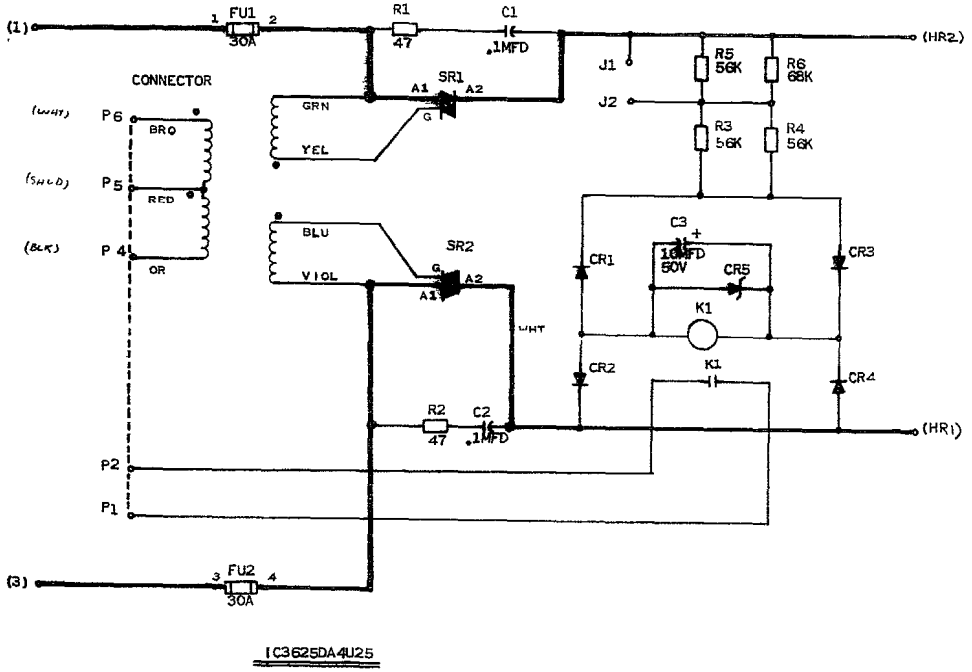
FF 806 WF (6-69) PRINTED IN U S A

DRAWING NO 1C3625D CONT ON SHEET 3.1 SH NO 3.0

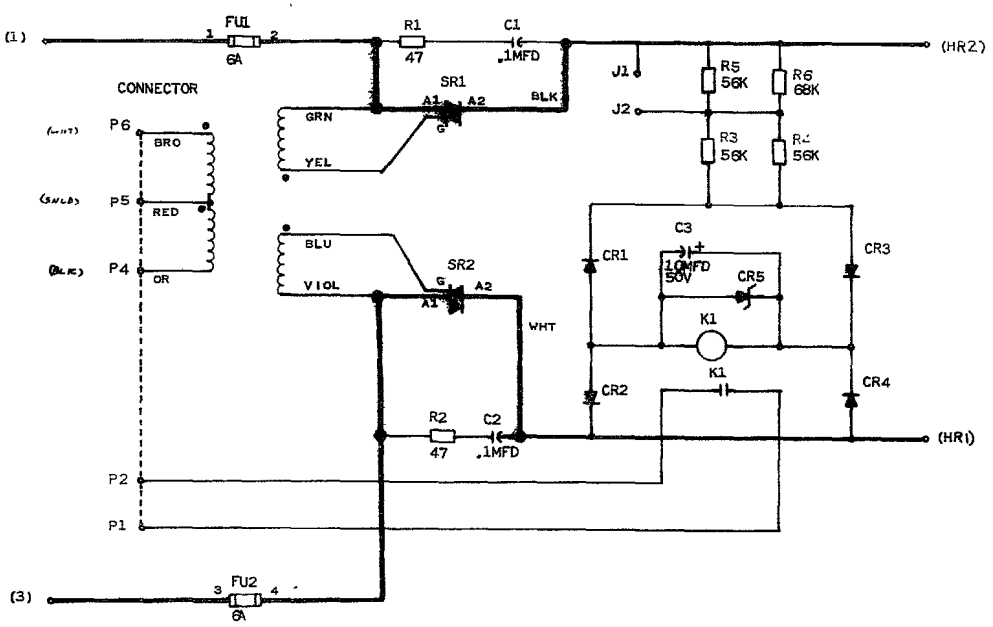
GENERAL ELECTRIC

1C3625D CONT ON SHEET 3.1 SH NO 3.0

REV NO	TITLE
1C3625D CONT ON SHEET 3.1 SH NO 3.0	HEATER DRIVER ELEMENTARY SINGLE OR 3 PHASE VEE
	FIRST MADE FOR REQ PLASTIC MOLDING S.0



1C3625DA4U25



1C3625DA4U05

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

RIC VNI IQAK IMS 3 IEHI IELI PRINTS TO

REVISIONS		
1	L7A 1-21-71 Film copy made so that outputs work full 8V	
2	L7A 4-12-71 Added Pin #'s	

MADE BY	APPROVALS	DIV OR DEPT	
ET ABBOTT		INDUSTRY CONTROL	1C3625D
ISSUED 6-19-70.	6/1/70	LOCATION SALEM, VIRGINIA	CONT ON SHEET 3.1 SH NO 3.0

FF 805 WF (8 68)

DRAWING NO IC3625D SH NO 3.1 CONT ON SHEET

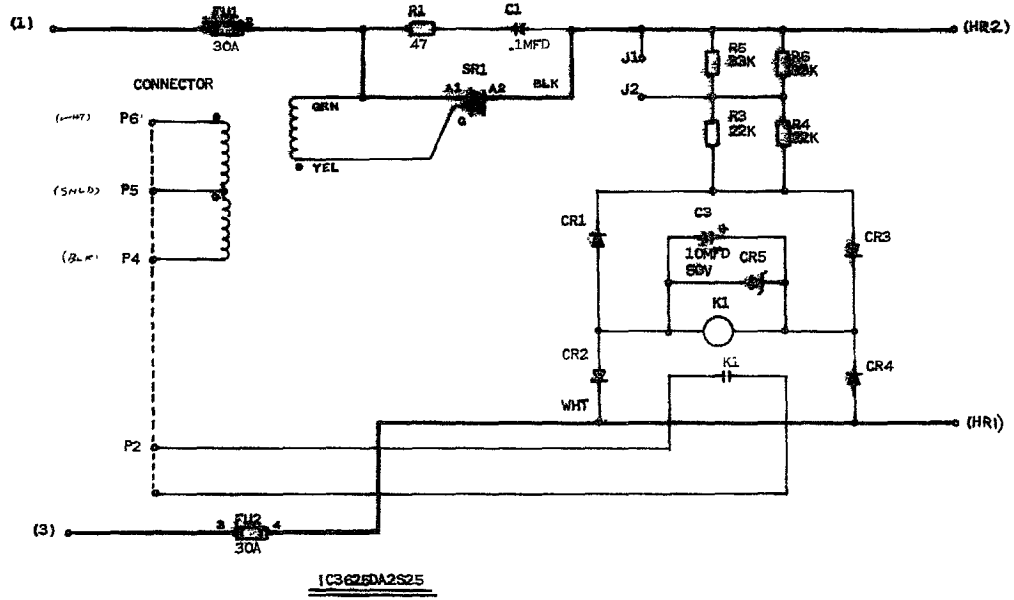
GENERAL ELECTRIC

IC3625D

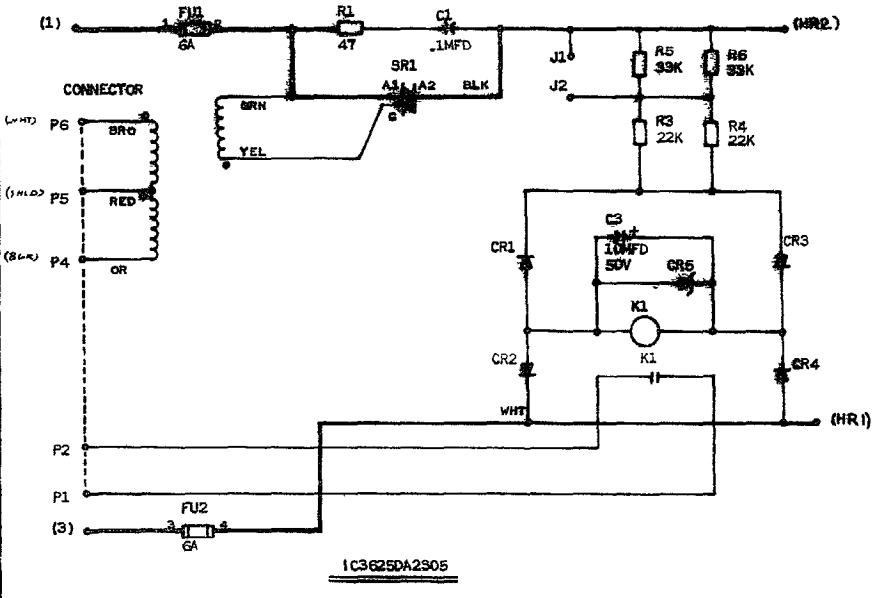
CONT ON SHEET SH NO 3.1

TITLE HEATER DRIVER ELEMENTARY
SINGLE PHASE

FIRST MADE FOR REQ. S.O.



IC3625DA2S25



IC3625DA2S05

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

RIC VNI
SAI
IMGH2
IEH1
IEL1
PRINTS TO

REVISIONS		
1	2 Dec 1-21-71 film copy made to outboard won't roll off	
2	ECR #-13-71 updated mark drawing	

MADE BY ET ABBOTT
ISSUED 6-17-70.

APPROVED R.S. Eh 6/17/70

INDUSTRY CONTROL — DIV OR DEPT
SALEM, VIRGINIA — LOCATION

IC3625D
CONT ON SHEET SH NO 3.1

FF-506 WF (9-68)
PRINTED IN U.S.A.

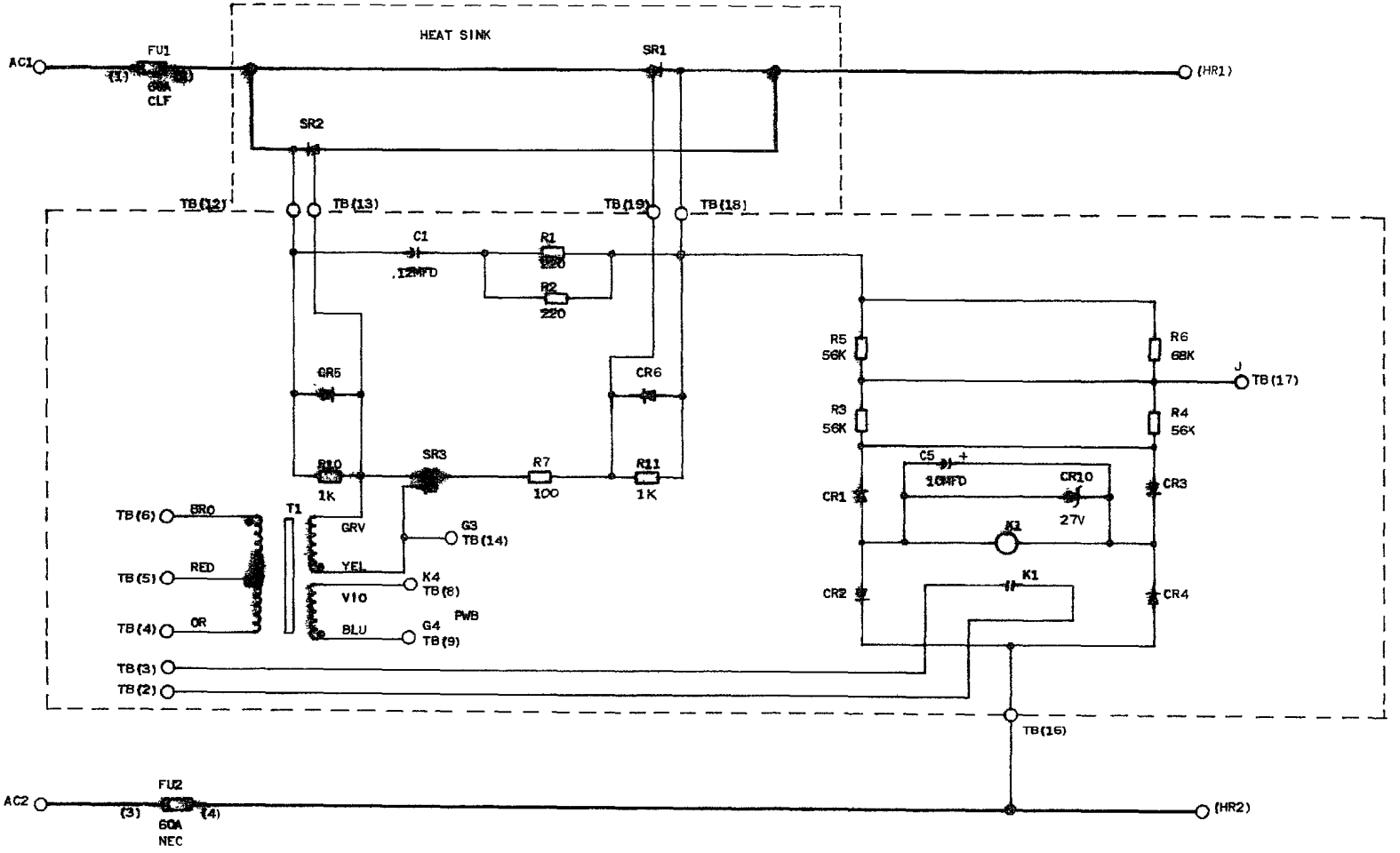
GENERAL ELECTRIC

IC3625D
CONT ON SHEET SH NO 3.2

REV BD

TITLE 50 AMP DRIVER

FIRST MADE FOR REQ S.O.



IC3625DB4S50

IC3625DB4S50A

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

		1			
		2			

MADE BY E. T. [illegible] ISSUED 2-8-71

APPROVALS

INDUSTRY CONTROL DIV OR DEPT

SALEM, VIRGINIA LOCATION

IC3625D CONT ON SHEET SH NO 3.2

RIC VNI IMEK (3) CA

1 EA

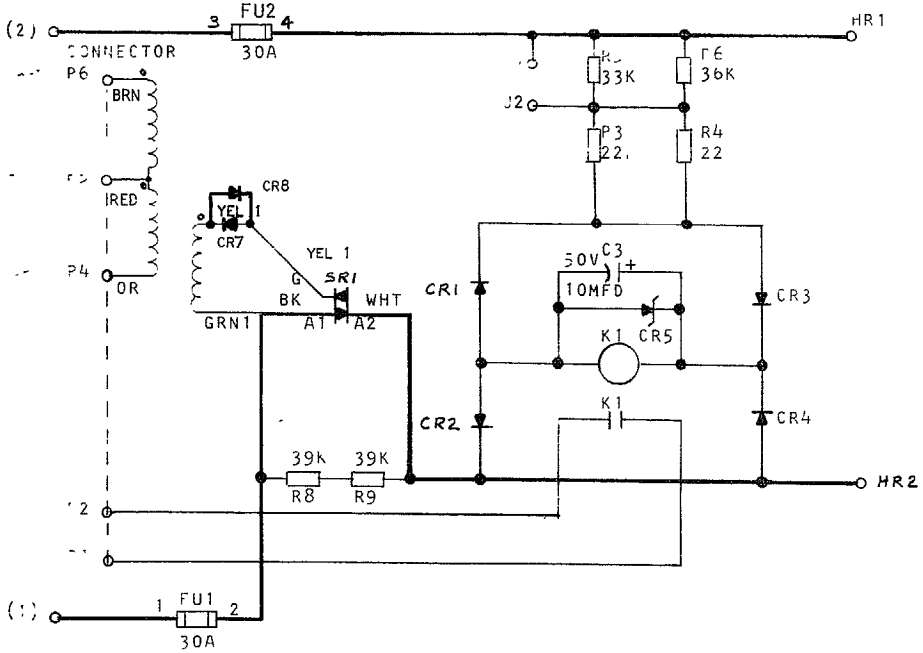
1 EA

PRINTS TO

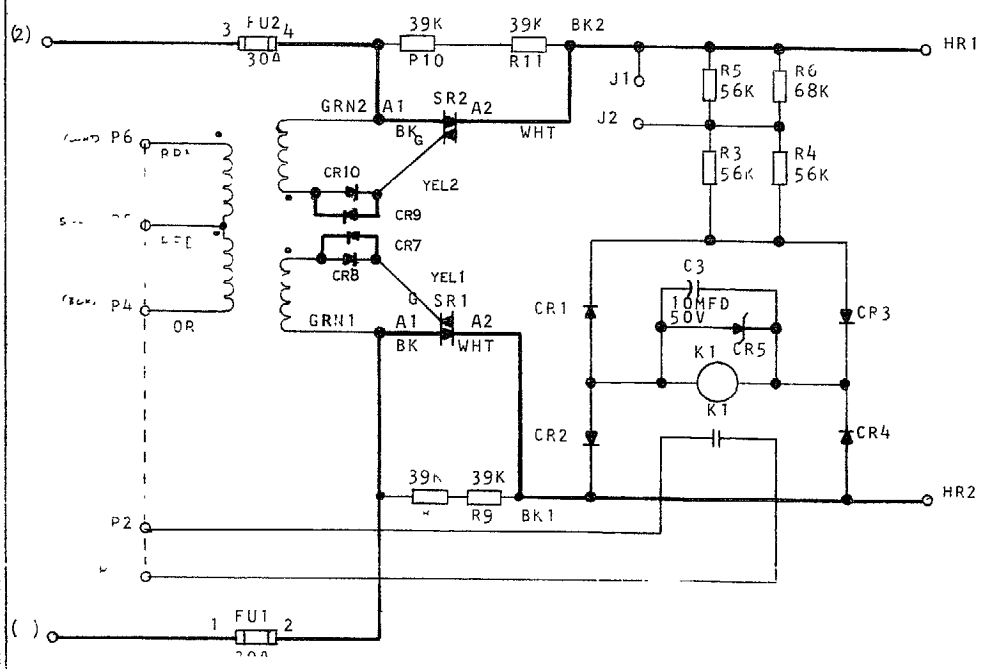
FOLD

GENERAL ELECTRIC

REV NO	TITLE	IC3625D
IC3625D	HEATER DRIVER ELEMENTARY	CONT ON SHEET SH NO 3.3
CONT OF SHEET SH NO 3.3	FIRST MADE FOR REQ PLASTIC MOLDING SO	



IC3625DC2S25



IC3625DC4U25

RIC VN1
IQAI
IEHI
IELI
PRINTS TO

REVISIONS			
2	4-3-72 ORL 2-29-72 CR10 ADDED		
1			

MADE BY Benjamin 2-14-72
ISSUED February 14, 1972
APPROVALS

INDUSTRIAL CONTROL DIV OF PT
SALEM, VIRGINIA LOCATION

IC3625D
CONT ON SHEET SH NO 3.3

IC36250 | TITLE [illegible]
CONT ON SHEET 4.1 SH NO. 4.0 | FIRST MADE FOR REG. S.O.

THESE UNITS PROVIDE SWITCHING OF A-C POWER IN RESPONSE TO A HIGH FREQUENCY TRANSFORMER COUPLED INPUT SIGNAL. THE TRANSFORMER COUPLING PROVIDES INPUT ISOLATION COMPATIBLE WITH UP TO 480 VOLT A-C SUPPLY CIRCUITRY.

THE INPUT TRANSFORMER IS ALSO A PART OF THE OSCILLATOR CIRCUIT PROVIDING THE HIGH FREQUENCY INPUT. THE REMAINDER OF THE OSCILLATOR IS LOCATED ON REGULATOR MODULE IC36[illegible]. THERE ARE 2 TO 4 SUCH CIRCUIT PORTIONS ON THE REGULATOR MODULE, EACH CONSTITUTING AN OUTPUT TO A DRIVER, AND EACH IN TURN BEING CONTROLLED BY ONE OF THE REGULATOR CIRCUITS ON THE REGULATOR MODULE.

FIGURE A SHOWS THE DRIVER TRANSFORMER CONNECTED TO THE REMAINDER OF THE OSCILLATOR.

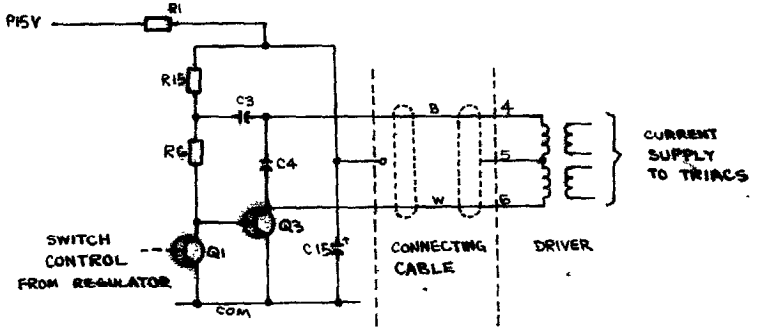


FIGURE A - CONNECTION BETWEEN REGULATOR AND DRIVER

THIS CONNECTION IS MADE WITH AN INSULATED SHIELDED CABLE, THE SHIELD SERVING AS POSITIVE SUPPLY CONNECTION TO THE OSCILLATOR, AND THE OTHER TWO WIRES (W AND B) CONNECTING THE TRANSISTOR COLLECTOR AND REGENERATIVE BASE DRIVE CIRCUIT RESPECTIVELY. THE CAPACITOR C4 TUNES THE TRANSFORMER INDUCTANCE TO MAKE THE CIRCUIT OSCILLATE AT APPROXIMATELY 60 KILLOHERTZ. THE VOLTAGE ON THE COLLECTOR LEAD WILL BE A DISTORTED SINE WAVE (FIGURE B) WITH APPROXIMATELY TWICE THE PEAK AMPLITUDE OF THE SUPPLY.



FIGURE B
COLLECTOR WAVE FORM ON DRIVER.

THE BASE DRIVE RESISTORS IN THE REGULATOR ARE SELECTED SUCH THAT THE OSCILLATOR MAY "SKEG" OR OPERATE IN INTERMITENT BURSTS (FIGURE C) WHEN THERE IS NO A-C CURRENT PASSING THROUGH THE DRIVER (AC - POWER OR LOAD NOT CONNECTED); WHEN CURRENT IS PASSING THROUGH THE SWITCH, THIS CONDITION DISAPPEARS, BUT THE OUTPUT IS MODULATED SLIGHTLY BY LINE FREQUENCY

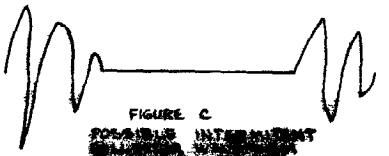


FIGURE C
POSSIBLE INTERMITTENT [illegible]



FIGURE D
[illegible]

THE TRANSISTOR DRIVE ON THE PRIMARY IS COUPLED THROUGH THE TRANSFORMER TO THE GATE CIRCUITS ON THE SECONDARY SIDE OF THE TRANSFORMER. HERE, THE EFFECT OF SWITCH CURRENT IS EVEN MORE PROMINENT. WITH NO CURRENT PASSING THROUGH THE DRIVER THE VOLTAGE ACROSS A TRANSFORMER SECONDARY (GATE TO COMMON VOLTAGE ON A TRIAC) WILL BE SIMILAR TO FIGURE C EXCEPT THAT THE PEAKS WILL BE FLATTENED. WITH LOAD THE WAVEFORM WILL BE AS SHOWN IN FIGURE E.

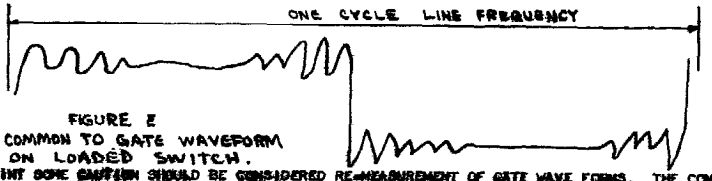


FIGURE E
COMMON TO GATE WAVEFORM ON LOADED SWITCH.

AT THIS POINT SOME CAUTION SHOULD BE CONSIDERED RE-MEASUREMENT OF GATE WAVE FORMS. THE COMMON AND GATE TERMINALS ARE AT THE POTENTIAL OF THE INPUT POWER SOURCE. MEASUREMENTS SHOULD BE MADE USING EXTREME CAUTION CONSIDERING THIS, AND THE OSCILLOSCOPE SHOULD BE [illegible]

THE OUTPUT IS MONITORED FOR TRIAC FAILURE (CONTINUOUSLY ON TRIAC) BY REED SWITCH K1. THE VOLTAGE AT THE OUTPUT TERMINALS IS DROPPED BY RESISTOR NETWORK R3-R6, RECTIFIED BY CR1-CR4 DIODE BRIDGE AND AND FILTERED BY CAPACITOR C3 TO DRIVE K1 COIL. THUS, WHEN LINE VOLTAGE IS APPLIED TO LOAD TERMINALS K1 COIL WILL PICK UP. EACH DRIVER HAS TWO BASIC OPERATING VOLTAGES: 220/440 OR 110/220. FOR THE LOWER VOLTAGE J1 AND J2 ARE JUMPERED TOGETHER IN THE RELAY CIRCUIT.

FOR DRIVERS LABELLED SINGLE PHASE OR 3 PHASE VEE (UNIVERSAL) THE DESIGN INTENT IS TO PROVIDE A DRIVER CAPABLE OF HANDLING UP TO 11 KILLOWATTS OF HEATER LOAD ON A 25 AMP DRIVER OR 2200 WATTS ON A 5 AMP DRIVER ON EITHER 440 OR 220 VOLT 3 PHASE LINE. FOR THE 440 VOLT APPLICATION, THE LOAD IS CONNECTED WITH ONE TRIAC IN EACH LINE. FOR THE 220 VOLT APPLICATION, ONE TRIAC DRIVES 1/2 THE LOAD BETWEEN THE FIRST PHASE AND A FUSED CONNECTION TO THE THIRD PHASE (EXTERNAL TO THE DRIVER). WHERE THE OTHER TRIAC DRIVES THE OTHER HALF OF THE LOAD BETWEEN THE SECOND PHASE AND THE FUSED CONNECTION TO THE THIRD PHASE. NOTE THAT THE EXTERNAL FUSE SHOULD BE 1.7 TIMES THE FUSE IN THE PACKAGE, BUT NEED NOT BE AN ELECTRONIC FUSE AS ARE THE FUSES ON THE PACKAGE.

RICHMOND (3)
IQAI
IEMI
IGLI
PRINTS TO

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REVISIONS		
1	BJA 12-21-71 Also copy made to sheet 4 [illegible]	
2	BJA 5-18-71 add'l note appropriate info ...	

MADE BY E. T. ABBOTT | APPROVALS | INDUSTRY CONTROL | DIV OR DEPT | IC36250
ISSUED 6-15-70. | 6/17/70 | SALEM, VIRGINIA | LOCATION | CONT ON SHEET 4.1 | SH NO 4.0

GENERAL ELECTRIC

IC3625D CONT ON SHEET FL. SH NO 4.1

REV NO

IC3625D CONT ON SHEET FL. SH NO 4.1

TITLE FUNCTION AND THEORY OF OPERATION

FIRST MADE FOR REQ. S O.

SPECIFIED RATINGS

IC3625DA	2S05	2S25	4U05	4U25	4S50, 4S50A
AC INPUT					
VOLTAGE (50/60 HZ)					
A-J1 AND J2 STRAPPED VOLTS	100-120	100-120	200-240	210-240	210-240
B-J1 AND J2 OPEN VOLTS	200-240	200-240	400-480	400-480	400-480
MAX LOAD CURRENT EACH LEG AMPS.	5	25	5	25	50

NOTES

1. CURRENT RATING IS DEPENDENT ON CASE HEAT SINKING, RATING SHOWS MAXIMUM ALLOWABLE, BUT PARTICULARLY FOR THE 25 AMPERE UNIT, MAY BE LESS IN THE MOUNTING PACKAGE.

2. MINIMUM VOLTAGE IS ESTABLISHED FOR OPERATION OF MONITOR RELAY.

SPECIFIED RATINGS

IC3625DC	2S25	4U25
AC INPUT		
VOLTAGE (50/60 HZ)		
A-J1 AND J2 STRAPPED VOLTS	100-120	210-240
B-J1 AND J2 OPEN VOLTS	200-240	400-480
MAX LOAD CURRENT EACH LEG AMPS.	25	25

RICVNI (MG 3) IQA IEHI IELI PRINTS TO

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

REVISIONS		
1	ECO 1-21-74 added note for 4550+4S50A	
2	ECO 5-10-71 added note proprietary info	
3	2/14/72 2-14-72 ADDED DC FORM	

MADE BY E. T. ABBOTT ISSUED 6-19-70.

APPROVALS 6/17/71

INDUSTRY CONTROL DIV OR DEPT

SALEM, VIRGINIA LOCATION

IC3625D CONT ON SHEET FL. SH NO 4.1

GENERAL ELECTRIC

PARTS, DRAWING OR MODEL LIST IC3622GMIC3
CONT. ON SHEET 1.1 SH NO. 1.0

EV NO

CATALOG OR MODEL NUMBER: IC3622GMIC3

TITLE: MULTIPLEX INPUT GMIC3

FIRST MADE FOR PLASTIC MOLDING

PART NO.	IDENTIFICATION OF PART		REFERENCE	CLASS	SOURCE	QUANTITY
001	44B294138-001	BLANK BOARD		DZ		001
006	68A997520P2	CONNECTOR PIN		SPC		025
007	68A997520P3	CONNECTOR PIN		SPC		025
029	IC3622GMIC3	ASMB DWG SH 2		IN	L01	001
030	44B294138	TAPE MANUSCRIPT		IN	L99	001
031	IC3622GMIC3	ELEM SH 3.03.13.2		IN	L01	001
032	44B294227-002	TOP ARTWORK		IN	L99	001
033	44B294227-003	BOT ARTWORK		IN	L99	001
036	44B294227-006	MRK DWG		IN	L99	001
039	68B997530P2	CARD BLANK		IN	L99	001
040	44A288940	TEST INSTR		IN	L02	001
150	259A9010P562	CC 5600 OHM 5% .5W	R35-R46	RR		012
154	259A9010P822	CC 8200 OHM 5% .5W	R1-R11	RR		011
155	259A9010P183	CC 18K 5% .5W	R12-R34	RR		023
156	259A9010P104	CC 100K 5% .5W	R71-R82	RR		012
157	259A9010P681	CC 680 OHM 5% .5W	R47-R70	RR		024
351	68A7068P0470J	CAPACITOR .047MF	C1,C2	EVC		002
400	259A9240P1	DIODE SIL-SIGNAL	CR1-CR22	QQ		022
401	68A8203P093	ZENER DIODE 15V .4W 5%	CR23-CR32, CR43, CR45	PM1		012
402	68A8203P013	ZENER DIODE 6.8V .4W 5%	CR33-CR42 CR44, CR46	PM1		012
450	68A7727P1	TRANSISTOR	Q1-Q12	PM1		012

REVISIONS →

PRINTS TO

1338	1EL1
53	2590
1RA1	30VN
29VN	

MADE BY C.Coffey August 30,1979
ISSUED SEP 04 1979

APPROVALS D Hypes

DATA COMMUNICATION PRODUCTS DEPT.
WAYNESBORO, VIRGINIA

PARTS, DRAWING OR MODEL LIST IC3622GMIC3
CONT. ON SHEET 1.1 SH. NO. 1.0

DCP-191 (REV 12-71)

File Copy 6 of 27

GENERAL ELECTRIC

PARTS, DRAWING OR MODEL LIST: IC3622GM1C3
CONT. ON SHEET 2.0 SH. NO. 1.1

REV NO	TITLE
CATALOG OR MODEL NUMBER: IC3622GM1C3	MULTIPLEX INPUT GM1C3
	FIRST MADE FOR PLASTIC MOLDING

PART NO.	IDENTIFICATION OF PART	REFERENCE	CLASS	SOURCE	QUANTITY
451	44A293756-001 TRANSISTOR PNP	Q13-Q24	UD		012
452	68A7715P1 TRANSISTOR	Q25-Q36	PM1		012
	NOTE: THIS BOARD TO BE CONFORMAL COATED ON BOTH SIDES.				

REVISIONS→

PRINTS TO

1338	1[illegible]1
53	2530
1RA1	30VN
29VN	

MADE BY C. Coffey, August 30, 1979
ISSUED SEP 04 1979

APPROVALS D Myers

DATA COMMUNICATION PRODUCTS DEPT.
WAYNESBORO, VIRGINIA

PARTS, DRAWING OR MODEL LIST: IC3622GM1C3
CONT. ON SHEET 2.0 SH. NO. 1.1

DCP-191 (REV 12-71)

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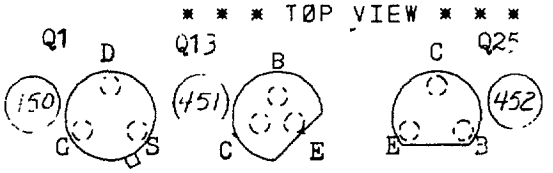
GENERAL ELECTRIC

IC3622GMIC3
CONT ON SHEET 3.0 SH NO 2.0

REV NO	TITLE
IC3622GMIC3 CONT ON SHEET 3.0 SH NO 2.0	MULTIPLEX INPUT FIRST MADE FOR PLASTIC MOLDING

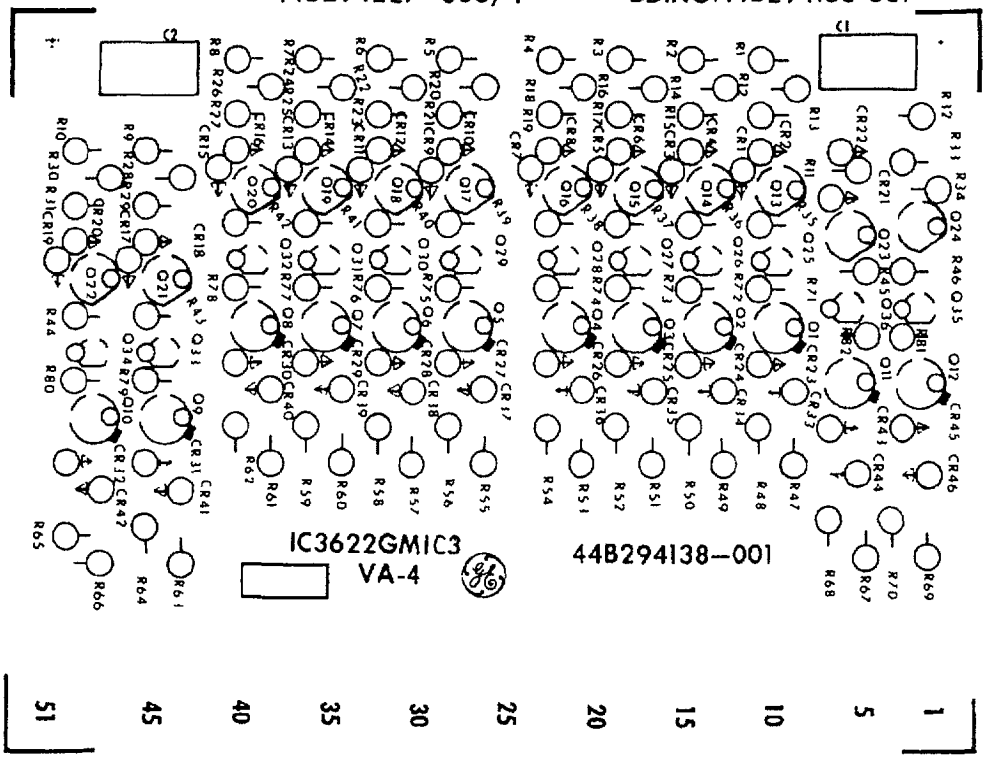
REVISIONS

* * * TØP VIEW * * *



ALL TRANSISTORS TO BE SUPPLIED AND MOUNTED AS SHOWN ABOVE.

44B294227-006/1 BD.NO.44B294138-001



NOTE: THIS BOARD TO BE CONFORMAL COATED ON BOTH SIDES.

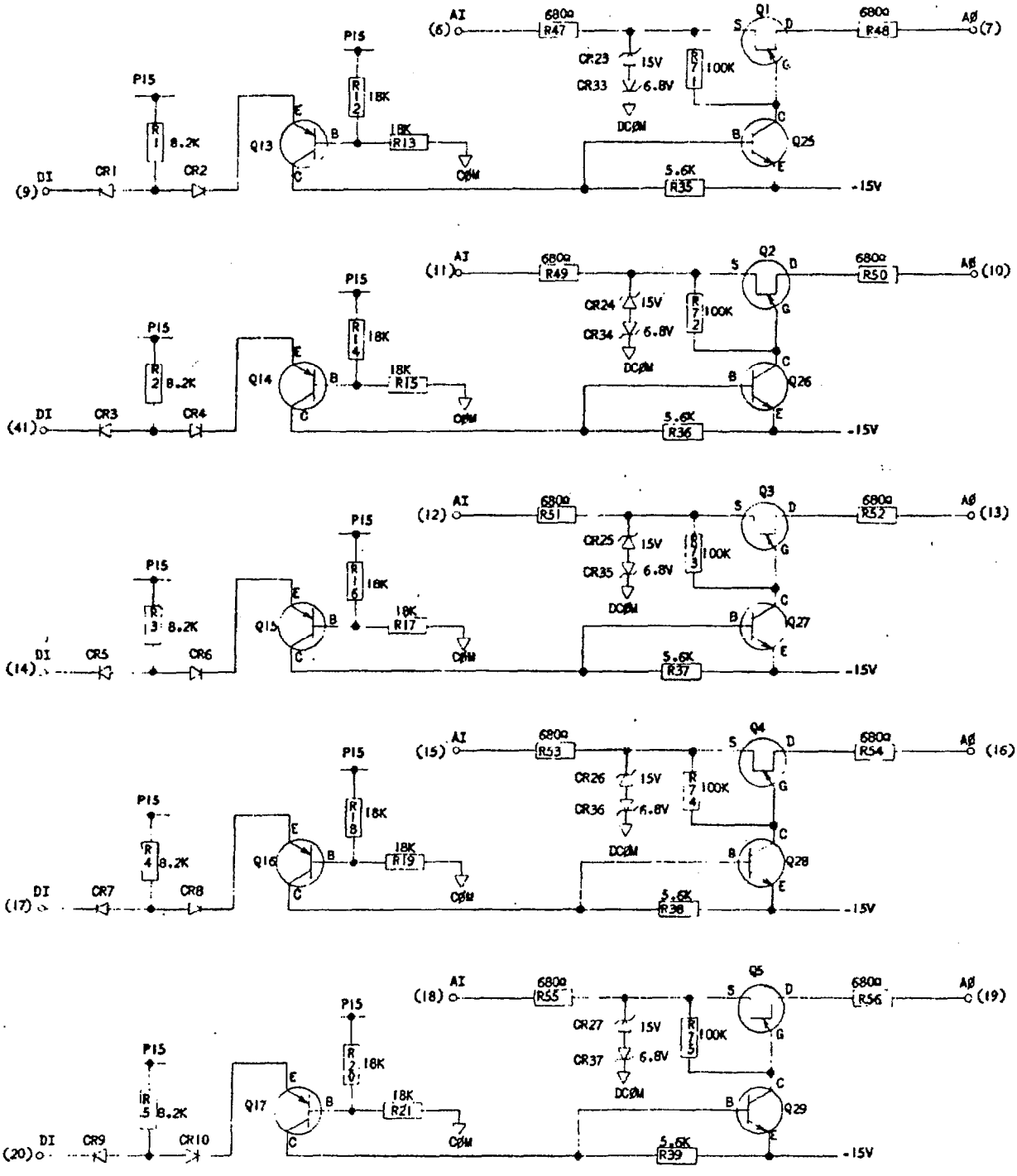
1330
1EL1
2530 2
CVN
2?VN
PRINTS TO

MADE BY	APPROVALS		
D. Moyer Nov. 16, 1978	J Myers	INDUSTRIAL CONTROL DEPT. (DIV OR DEPT)	IC3622GMIC3
ISSUED SEP 04 1979		CHARLOTTESVILLE, VA. (LOCATION)	CONT ON SHEET 3.0 SH NO 2.0

FF 803 WF 11-77
PRINTED IN U.S.A.

CODE IDENT NO

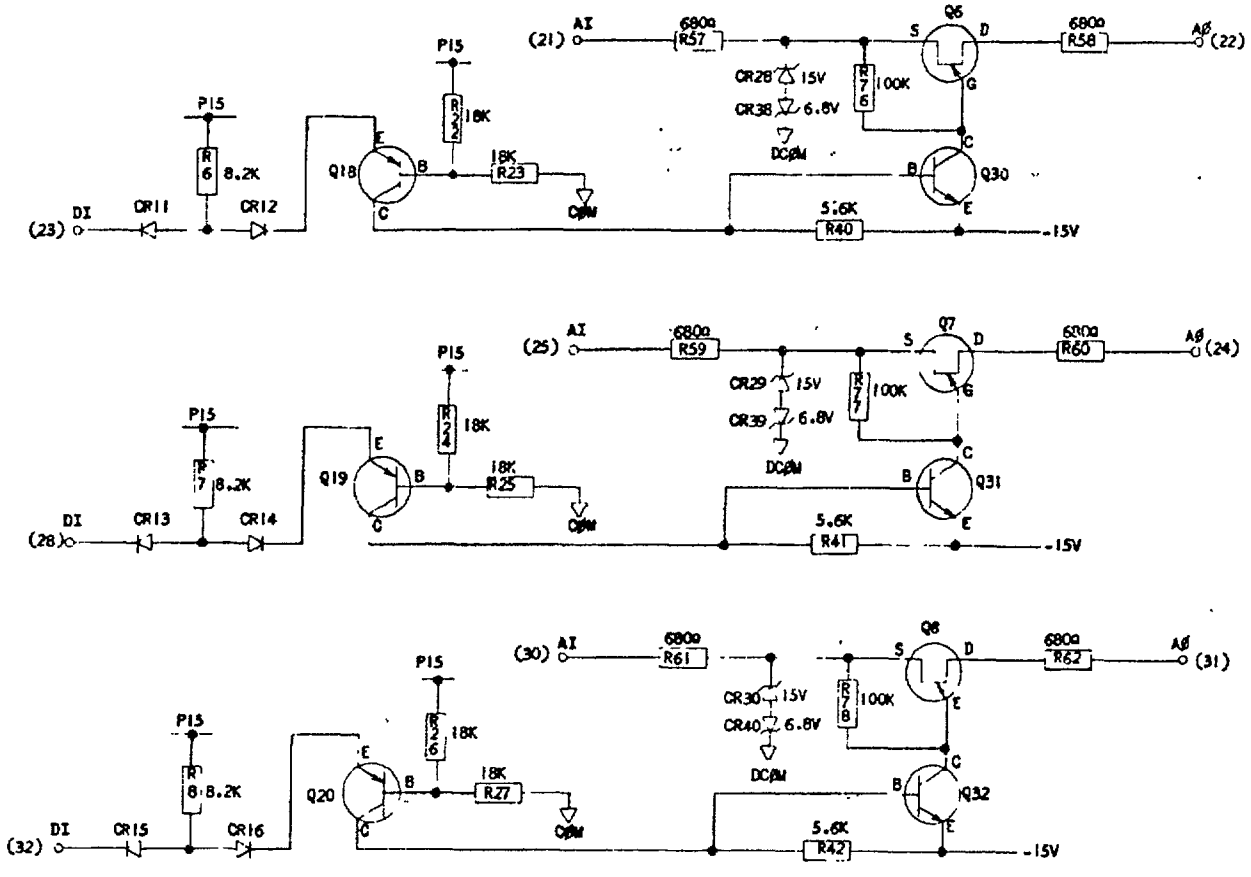
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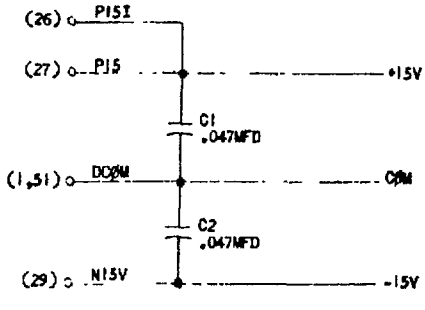


PROPRIETARY INFORMATION THE GENERAL ELECTRIC COMPANY

REV 1	REV 4	REV 5	PRINTS TC 1338, 53, 2530, 29VN, 30VN TEL	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV 2	ISSUED SEP 4 - 1979			FIRST MADE FOR PLASTIC MOLDING	INDUSTRY CONTROL DEPT. SALEM, VA., U.S.A.	MULTIPLEX INPUT
REV 3	MADE BY Aug 30, 1979			TC NO 7966		TC 2C411C3
						CONT. ON SH. 3.1 SH. NO. 3.0

File Copy 5 of 2T

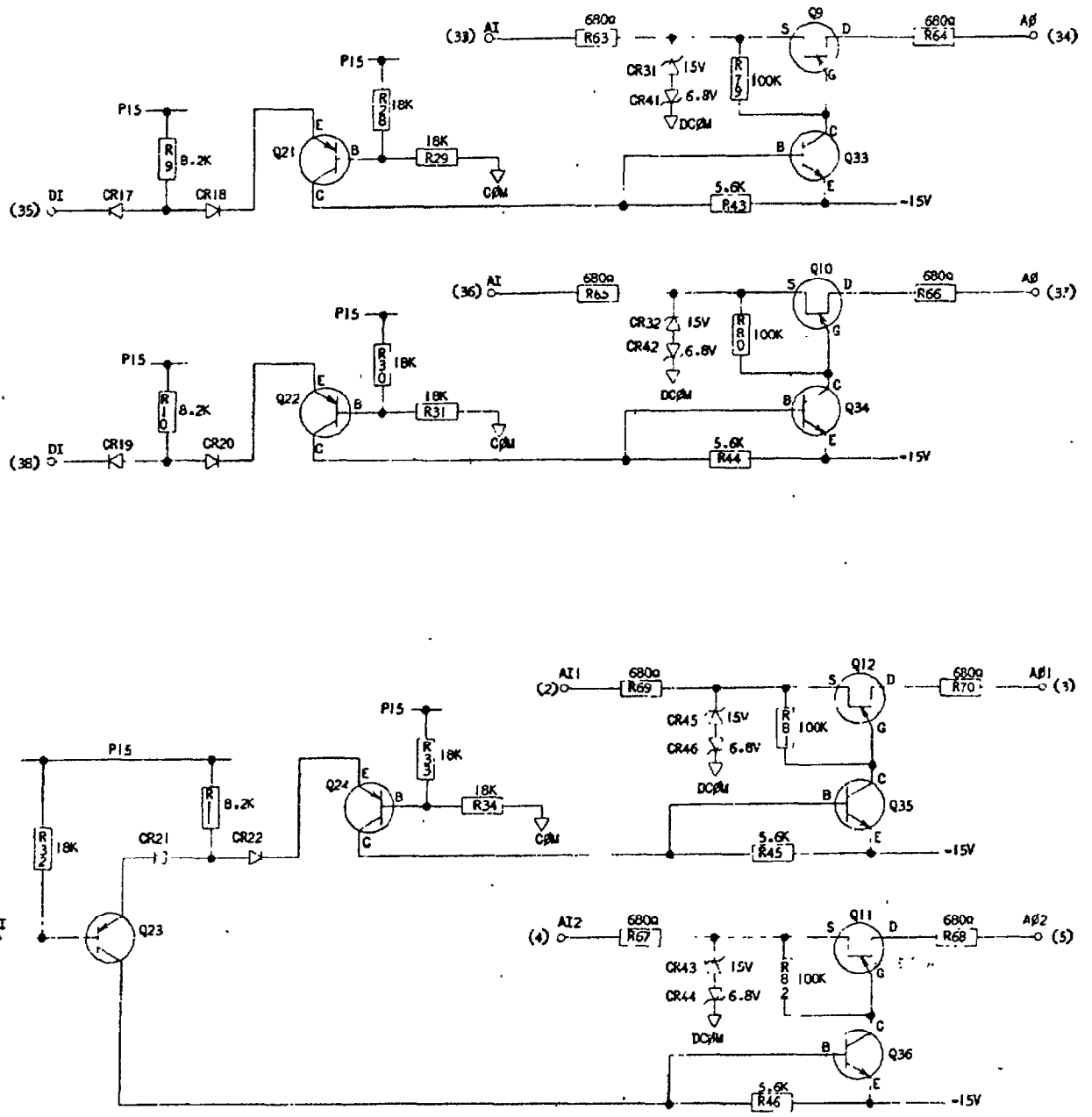




PROPRIETARY INFORMATION THE GENERAL ELECTRIC COMPANY

REV 1	REV 4	REV 5	PRINTS TO 1338,1EL1,53 2530,29VN,30VN	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM MULTIPLEX INPUT
REV 2	ISSUED SEP 4 - 1973			FIRST MADE FOR PLASTIC MOLDING	INDUSTRY CONTROL DEPT SALEM, VA, USA	
REV 3	MADE BY [illegible] May 31, 1979			TC NO 7966		CONT. ON SH. 3.2 SH. NO. 3.1

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PROPRIETARY INFORMATION THE GENERAL ELECTRIC COMPANY

REV1	REV4	REV 5	PRINTS TO 1330, FL1, 53, 2530, 29VN, 30V.	APPROVALS	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA., U.S.A.
REV2	ISSUED SEP 4 - 1979			FIRST MADE FOR PLASTIC MOLDING	
REV3	MADE BY [illegible] May 30, 1979			TC NO 7906	

ELEMENTARY DIAGRAM
MULTIPLEX INPUT

[illegible] SHEET NO.
CONT. ON SH. [illegible] SH. NO. 3.2

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1 NØ CØNTACT EACH	A CIRCUITS									
	1	2	3	4	5	6	7	8	9	10
AI	6	11	12	15	18	21	25	30	33	36
AØ	7	10	13	16	19	22	24	31	34	37
DI	9	41	14	17	20	23	26	32	35	38

1 NØ, 1 NC CØNTACT EACH	B CIRCUITS
	1 SEE FIG.2 FØR PIN #'S

SINCE THERE IS ØNLY ØNE TYPE "B" CIRCUIT, THE PIN NUMBERS ARE SHØWN BY FIGURE 2 AT THE RIGHT.

FIGURE 1: SYMBØL FØR THE TEN TYPE "A" CIRCUIT

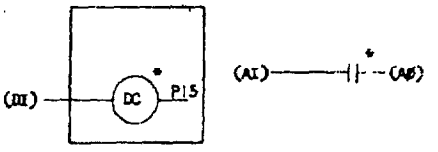
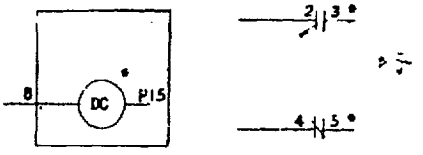


FIGURE 2: SYMBØL FØR THE ØNE TYPE B CIRCUIT



*CØNTACT FØR VARIABLE VØLTAGE

THEØRY ØF ØPERATIØN

THIS IS AN ANALØG MULTIPLEX CARD. IT CØNSITS ØF 10 SINGLE-PØLE JFET SWITCHES, 1 DØUBLE-PØLE JFET SWITCH, AND THEIR ASSØCIATED DRIVERS. ALL SWITCHES HAVE INDIVIDUAL ØUTPUTS WHICH MAY BE JUMPERED TØGETHER TØ FØRM A DESIRED GRØUPING. THE DØUBLE-PØLE SWITCH HAS ØNE NØRMALLY-ØPEN AND ØNE NØRMALLY-CLØSED CØNTACT WITH EACH PØLE HAVING A SEPARATE ØUTPUT.

THE DRIVERS ARE DESIGNED TØ ACCEPT A HIGH THRESHØLD LØGIC (HTL) DIGITAL INPUT. WITH LØGIC INPUT DI AT LØGIC "1", Q13 WILL CØNDUCT AND SATURATE Q25. Q25 SATURATING CØNNECTS -15V TØ GATE ØF Q1, THUS TURNING ØFF Q1; IN THE DØUBLE-PØLE CØNFIGURATIØN, Q23 AND Q36 WILL BE ØFF AND Q11 WILL BE ØN, Q12 WILL BE ØFF.

WITH DI AT LØGIC "0", Q13 AND Q25 WILL BE ØFF AND Q1 WILL BE ØN; IN THE DØUBLE PØLE CØNFIGURATIØN, Q23 AND Q36 WILL BE ØN AND Q11 WILL BE ØFF, Q12 WILL BR ØN.

SPECIFICATIØNS

DIGITAL INPUTS, DI - +4.6V MAX. "0" LEVEL, 10V MIN. "1" LEVEL. SINGLE-PØLE INPUTS ARE 2.5 HTL LØADS, DØUBLE-PØLE INPUTS ARE 1 HTL LØAD. ØNE HTL LØAD IS 18K ØHMS TØ +15V.

ANALØG INPUTS, AI - +15V MAX. PØSITIVE SIGNAL, -5V MIN. NEGATIVE SIGNAL.

ANALØG ØUTPUTS, AØ - WHEN THE SWITCH IS ØN THERE WILL BE A MAXIMUM ØF 1528 ØHMS BETWEEN AI AND AØ IF SWITCH CURRENT IS LIMITED TØ 4MA ØR LESS. MAXIMUM SWITCH IS 12MA.

PROPRIETARY INFORMATION THE GENERAL ELECTRIC COMPANY

REV 1	REV 4	REV 5	PRINTS TO 1335,2630,53 1FLT,30VN, 1RA1,29VN	APPROVALS	ELEMENTARY DIAGRAM
REV 2	ISSUED SEP 4 - 1979			FIRST MADE FOR PLASTIC MØLDING	DESIGN APPLICATION AND ØPERATIØN NOTES
REV 3	MADE BY [illegible] Aug 30, 1979			TC NO 7966	CONT. ON SH. 3.4 SH. NO. 3.3

GENERAL ELECTRIC
INDUSTRY CONTROL DEPT.
SALEM, VA., U.S.A.

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WHEN THE SWITCH IS ØFF, THERE WILL BE LESS THAN .2 MICRØ-AMPS BETWEEN AI AND AØ. WITH THE SWITCH ØFF, THE ØUTPUT IS SUBJECT TØ THE SAME VØLTAGE LIMITATIØNS AS THE INPUT, I. E. +15V TØ -5V.

POWER SUPPLY

+15V - 1.2MA NØMINAL PER EACH ØFF SWITCH (DI LØGIC "1"), 1.5MA MAX.
2.2MA NØMINAL PER EACH ØN SINGLE PØLE SWITCH, 2.7MA MAX.
3.0MA NØMINAL PER EACH ØN DØUBLE PØLE SWITCH, 3.6MA MAX.
-15V - .9MA NØMINAL PER EACH ØFF SWITCH (DI LØGIC "1"), 1.1MA MAX.
1.7MA NØMINAL PER EACH ØN DØUBLE PØLE SWITCH (DI LØGIC "0"), 2.1MA MAX.
ØMA PER EACH ØN SINGLE PØLE SWITCH (DI LØGIC "0")

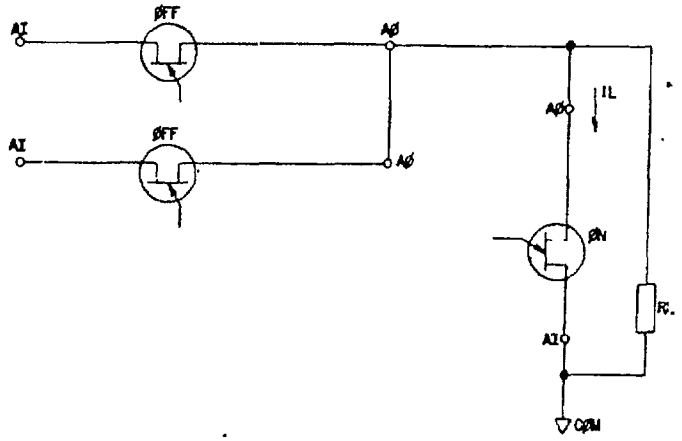
APPLICATIØN NØTES

AS PREVIØUSLY NØTED, MAXIMUM SWITCH CURRENT IS 12MA. THE SWITCH IS INTERNALLY PRØTECTED AGAINST AN ØUTPUT SHØRT-CIRCUIT.

NØTE ALSØ THAT WHEN A SWITCH IS ØFF, THIS CIRCUIT PRESENTS A 100K ØHM TØ -15V LØAD ØN THE INPUT SØURCE. IF THE SØURCE IS A PØTENTIØMETER, THIS LØADING WILL INTRØDUCE AN INPUT VØLTAGE ERRØR. THIS IMPLIES THAT EACH SWITCH HAVE ITS ØWN INPUT PØTENTIØMETER. IF IT IS DESIRED TØ DRIVE MØRE THAN ØNE SWITCH FRØM A CØMMØN SØURCE, USE ØF AN ØP-AMP RATHER THAN A PØT IS RECØMMENDED.

WHEN SEVERAL ØUTPUTS ARE TIED TØGETHER AND ALL SWITCHES ARE ØFF, THE LEAKAGE CURRENT ØF EACH SWITCH WILL FLØW THRØUGH THE LØAD RESISTØR. THIS WILL PRØDUCE AN ØFFSET ERRØR WHICH MAY NØT BE NEGLIGIBLE.

ØNE METHØD ØF MINIMIZING THIS ERRØR IS TØ SHUNT THE ØUTPUT ØF A SWITCH GRØUP WITH ANØTHER SWITCH AS SHØWN BELØW.



PROPRIETARY INFORMATION THE GENERAL ELECTRIC COMPANY

REV 1	REV 4	REV 5	PRINTS TO 1338,2530,53 ØAN, IFL, 30VN, IRA	APPROVALS J. Myers	GENERAL ELECTRIC INDUSTRY CONTROL DEPT. SALEM, VA, U.S.A.	ELEMENTARY DIAGRAM SECTION APPLICATIØN AND ØPERATIØN NØTES
REV 2	ISSUED SEP 4 - 1970			FIRST MADE FOR PLASTIC MØLDING		IC# 22 MIC 3
REV 3	MADE BY			IC NO 7466		CONT. ON SH. 5.0 SH. NO. 3.4

File Copy 4-27

APPLICATION SUMMARY

GENERAL DESCRIPTION

IC3622GMIC3 - THIS CARD DUPLICATES THE QUANTITY 8 SWITCHING FUNCTIONS PROVIDED BY THE GMIB2 CARD (PIN NUMBERS ARE THE SAME). IN ADDITION, THIS CARD PROVIDES QUANTITY 3 ADDITIONAL SWITCHING CIRCUITS, ONE OF WHICH IS A TWO POLE SWITCH WITH ONE NORMALLY OPEN CONTACT AND ONE NORMALLY CLOSED CONTACT. ALL OTHER CIRCUITS HAVE A SINGLE NORMALLY OPEN CONTACT.

THE ALLOWABLE INPUT VOLTAGE RATING OF THE GMIC3 CONTACT SWITCHES (+15V TO -5V) ENCOMPASSES THE RATING OF THE GMIB2 CONTACT SWITCHES (+15V TO 0V).

THIS CARD WAS INTRODUCED AS A FIELD REPLACEMENT FOR THE GMIB2 TO PROVIDE ADDITIONAL SWITCHING FUNCTIONS.

FIELD REPLACEMENT

IC3622GMIC3 - WILL FIELD REPLACE IC3622GMIA2 AND IC3622GMIB2 FOR ALL APPLICATIONS.

IC3622GMIB2 - CAN BE FIELD REPLACED WITH THE GMIC3 WHICH HAS MORE CIRCUITS.

IC3622GMIA2 - CAN BE FIELD REPLACED WITH THE GMIC3 WHICH HAS MORE CIRCUITS. IN ADDITION THE GMIC3 PROVIDES VOLTAGE SPIKE PROTECTION.

IC3622GMIA1 - CAN NOT BE FIELD REPLACED ACROSS THE BOARD WITH THE GMIC3 BECAUSE THE GMIC3 HAS THREE LESS CIRCUITS. IT CAN BE FIELD REPLACED BY THE GMIC3 IN THOSE APPLICATIONS WHERE THE THREE CIRCUITS WITH DIGITAL INPUTS ON PINS 42,43 AND 44 ARE NOT USED. THE GMIC3 PROVIDES NOISE SPIKE PROTECTION THAT THE GMIA1 DID NOT HAVE.

PROPRIETARY INFORMATION THE GENERAL ELECTRIC COMPANY

REV 1	REV 4	REV 5	PRINTS TO [33P, 530, 53 29VA, IFLT, 30VN, IRAI	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV 2	ISSUED SEP 4 - 1979			FIRST MADE FOR PLASTIC MOLDING	INDUSTRY CONTROL DEPT. SALEM, VA., U.S.A.	APPLICATION SUMMARY
REV 3	MADE BY [signature] Aug 30, 1979			IC NO 7966		IC3622GMIC3 CONT. ON SH. FNL SH. NO. 5.0

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